GSB518 Notes

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Preface

This is a selection of Handouts for Cal Poly GSB 518, Essential Statistics for Business Analytics, covering topics in Probability and Statistics.

1 Randomness and Probability

Probability comes up in a wide variety of situations. Consider just a few examples.

1. The probability that you roll doubles in a turn of a board game is 0.167.
2. The probability that a randomly selected Cal Poly student is a California resident is 0.84
3. The probability that the high temperature in San Luis Obispo, CA tomorrow is above 90 degrees F is 0.4
4. The probability that the Los Angeles Dodgers win the next World Series is 0.27.
5. The probability that extraterrestrial life currently exists somewhere in the universe.
6. The probability that you ate an apple on April 17, 2019.

Example 1.1. How are the situations above similar, and how are they different? What is one feature that all of the situations have in common? Is the interpretation of "probability" the same in all situations? The goal here is to just think about these questions, and not to compute any probabilities (or to even think about how you would).

- A phenomenon is **random** if there are multiple potential outcomes, and there is **uncertainty** about which outcome will occur.
- Uncertainty is understood in broad terms, and in particular does not only concern future occurrences.
- Many phenomena involve physical randomness, like flipping a coin or drawing powerballs at random from a bin, or in statistical applications of random sampling or random assignment.
- But in many other situations, randomness just vaguely reflects uncertainty.
- Random does *not* mean haphazard. In a random phenomenon, while individual outcomes are uncertain, there is a *regular distribution of outcomes over a large number of (hypothetical) repetitions.*
- Also, random does *not* necessarily mean equally likely. In a random phenomenon, certain outcomes or events might be more or less likely than others.

- The **probability** of an event associated with a random phenomenon is a number in the interval $[0, 1]$ measuring the event's likelihood or degree of uncertainty. A probability can take any value in the continuous scale from 0% to 100%.
- There are two main interpretations of probability.
  - **Long run relative frequency.** The probability of an event can be interpreted as the proportion of times that the event would occur in a very large number of hypothetical repetitions of the random phenomenon.
  - **Subjective probability.** There are many situations where the outcome is uncertain, but it does not make sense to consider the situation as repeatable. In such situations, a subjective (a.k.a., personal) probability describes the degree of likelihood a given individual ascribes to a certain event. Think of subjective probabilities as measuring relative degrees of likelihood rather than long run relative frequencies.
- Fortunately, the mathematics of probability work the same way regardless of the interpretation. In either case, the same basic logical consistency requirements must be satisfied.
- A **simulation** involves an artificial recreation of the random phenomenon, usually using a computer. The probability of an event can be approximated by simulating the random phenomenon a large number of times and determining the proportion of simulated repetitions on which the event occurred out of the total number of repetitions in the simulation.

Example 1.2. One of the oldest documented problems in probability is the following: If three fair six-sided dice are rolled, what is more likely: a sum of 9 or a sum of 10?

1. Conduct a *simulation* to investigate this question.

2. Use the simulation results to approximate the probability that the sum is 9; repeat for a sum of 10.

3. It can be shown that the theoretical probability that the sum is 9 is $25/216 = 0.116$. Write a clearly worded sentence interpreting this probability as a long run relative frequency.

4. It can be shown that the theoretical probability that the sum is 10 is $27/216 = 0.125$. How many times more likely is a sum of 10 than a sum of 9?

- A probability can be interpreted as a theoretical long run relative frequency.
- A probability can be approximated by a relative frequency from a large number of simulated repetitions, but there is some simulation margin of error, due to natural variability in the simulation.
- The margin of error when approximating a single probability based on a simulated relative frequency is roughly on the order $1/\sqrt{N}$, where N is the *number of independently simulated values used to calculate the relative frequency.* For example, if $N = 10000$ then the margin of error is roughly $1/\sqrt{10000} = 0.01$. (But be careful when approximating *conditional* probabilities.)

Example 1.3. What is your subjective probability that Professor Ross has ever attended a "famous" sporting event (like the Superbowl, a World Series game 7, a World Cup match, an NCAA tournament game with a famous shot, an Olympic final, etc)? Consider the following two bets, and suppose you must choose only one.

A. You win \$100 if Professor Ross has attended a famous sporting event, and you win nothing otherwise.
B. A box contains 40 green and 60 gold marbles that are otherwise identical. The marbles are thoroughly mixed and one marble is selected at random. You win \$100 if the selected marble is green, and you win nothing otherwise.

1. Which of the above bets would you prefer? Or are you completely indifferent? What does this say about your subjective probability that Professor Ross has attended a famous sporting event?

2. If you preferred bet B to bet A, consider bet C which has a similar setup to B but now there are 20 green and 80 gold marbles. Do you prefer bet A or bet C? What does this say about your subjective probability that Professor Ross has attended a famous sporting event?

3. If you preferred bet A to bet B, consider bet D which has a similar setup to B but now there are 60 green and 40 gold marbles. Do you prefer bet A or bet D? What does this say about your subjective probability that Professor Ross has attended a famous sporting event?

4. Continue to consider different numbers of green and gold marbles. Can you zero in on your subjective probability?

Example 1.4. Suppose your subjective probabilities for who will be the next World Series champion satisfy the following conditions.

- The Phillies and Mets are equally likely to win
- The Yankees are 1.5 times more likely than the Phillies to win
- The Dodgers are 2 times more likely than the Yankees to win
- The winner is as likely to be among these four teams—Phillies, Mets, Yankees, Dodgers—as not

1. Compute the subjective probability that each team wins.

2. Represent these probabilities in a spinner, like from a kids game.

3. Compute the probability that the winner is not the Dodgers.

4. How many times more likely are the Dodgers to not win than to win? (This is referred to as the "odds".)

5. Compute the probability that the winner is the Dodgers or the Yankees.

- We will use the two interpretations — long run relative frequencies and subjective probabilities — interchangeably.
- With subjective probabilities it is often helpful to consider what might happen in a simulation.
- It is also useful to consider long run relative frequencies in terms of relative degrees of likelihood.
- Fortunately, the mathematics of probability work the same way regardless of the interpretation.
- A probability takes a value in the sliding scale from 0 to 100%.
- Don't just focus on computation; always remember to properly interpret probabilities.

Example 1.5. In each of the following parts, which of the two probabilities, a or b, is larger, or are they equal? You should answer conceptually without attempting any calculations. Explain your reasoning.

1. Flip a coin *which is known to be fair* 10 times.

   a. The probability that the results are, in order, HHHHHHHHHH.
   b. The probability that the results are, in order, HHTHTTTHHT.

2. Flip a coin which is known to be fair 10 times.

   a. The probability that all 10 flips land on H.
   b. The probability that exactly 5 flips land on H.

- Warning! Your psychological judgment of probabilities is often inconsistent with the mathematical logic of probabilities.
- When interpreting probabilities, consider the conditions under which the probabilities were computed, in the proper direction

- When interpreting probabilities, be careful not to confuse "the particular" with "the general".
  - **"The particular:"** A very specific event, surprising or not, often has low probability.
  - **"The general:"** While a very specific event often has low probability, if there are many like events their combined probability can be high.
- Even if an event has extremely small probability, given enough repetitions of the random phenomenon, the probability that the event occurs on *at least one* of the repetitions is often high.

1.1 Exercises

Exercise 1.1. In each of the following parts, which of the two probabilities, a or b, is larger, or are they equal? You should answer conceptually without attempting any calculations. Explain your reasoning.

1. Consider a Cal Poly student who frequently has blurry, bloodshot eyes, generally exhibits slow reaction time, always seems to have the munchies, and disappears at 4:20 each day. Which of the following events, A or B, has a *higher* probability? (Assume the two probabilities are not equal.)
   a. The student has a GPA above 3.0.
   b. The student has a GPA above 3.0 and smokes marijuana regularly.
2. Randomly select a man.
   a. The probability that a randomly selected man is greater than six feet tall.
   b. The probability that a randomly selected man who plays in the NBA is greater than six feet tall.
3. In the Powerball lottery there are roughly 292 million possible winning number combinations, all equally likely.
   a. The probability you win the next Powerball lottery if you purchase a single ticket, 4-8-15-16-42, plus the Powerball number, 23
   b. The probability you win the next Powerball lottery if you purchase a single ticket, 1-2-3-4-5, plus the Powerball number, 6.
4. Continuing with the Powerball
   a. The probability that the numbers in the winning number are not in sequence (e.g., 4-8-15-16-42-23)
   b. The probability that the numbers in the winning number are in sequence (e.g., 1-2-3-4-5-6)

5. Continuing with the Powerball
    a. The probability that you win the next Powerball lottery if you purchase a single ticket.
    b. The probability that someone wins the next Powerball lottery. (FYI: especially when the jackpot is large, there are hundreds of millions of tickets sold.)

Exercise 1.2. Your favorite local weatherperson forecasts a 30% chance of rain tomorrow and a 60% chance of rain the next day in your city.

1. Explain how these probabilities are subjective.

2. You ask Donny Don't to interpret the 30% as a long run relative frequency. Donny says: "it will rain in 30% of the city tomorrow". You ask him to elaborate; he says: "Well, there are many different locations in the city. In some of the locations it will rain, in some it won't. It will rain in 30% of the locations, and not in the other 70%. That is, rain will cover 30% of the area of the city, and the other 70% won't have rain." Do you agree? If not, how would you interpret the 30% as a long run relative frequency?

3. You ask Donny Don't to interpret the values 30% and 60% as relative degrees of likelihood. Donny says: "Well, 30% is not that big, so it's not going to rain that hard tomorrow. Also, 60% is twice is big as 30%, so it's going to rain twice as hard two days from now as it will tomorrow". Do you agree? If not, how would you interpret the 30% and 60% as relative degrees of likelihood?

4. Donny says: "Can't we just look at the data from all the days with weather conditions similar to the ones forecast for tomorrow, and see how often it rained on those days to find the probability of rain tomorrow? No subjectivity about that!" How would you respond?

- A **probabilistic forecast** combines observed data and statistical or mathematical models to make predictions.
- Rather than providing a single prediction such as "it will rain tomorrow", probabilistic forecasts provide a range of scenarios and their relative likelihoods.
- Such forecasts are subjective in nature, relying upon the data used and assumptions of the model.
- Changing the data or assumptions can result in different forecasts and probabilities.

2 Probability Models

- Due to the wide variety of types of random phenomena, an **outcome** can be virtually anything. In particular, an outcome does *not* have to be a number.
- The **sample space** is the set of all possible outcomes of the random phenomenon..
- An event is something that could happen. That is, an *event* is a collection of outcomes that satisfy some criteria. If the sample space outcomes are represented by rows in a spreadsheet, then an event is a subset of rows that satisfies some criteria
- Events are typically denoted with capital letters near the start of the alphabet, with or without subscripts (e.g. A, B, C, A_1, A_2). Events can be composed from others using basic set operations like unions ($A \cup B$), intersections ($A \cap B$), and complements (A^c).
  - Read A^c as "not A".
  - Read $A \cap B$ as "A and B"
  - Read $A \cup B$ as "A or B". Note that unions ($\cup$, "or") are always inclusive. $A \cup B$ occurs if A occurs but B does not, B occurs but A does not, or both A and B occur.
- A collection of events $A_1, A_2, \ldots$ are **disjoint** (a.k.a. mutually exclusive) if $A_i \cap A_j = \emptyset$ for all $i \neq j$. That is, multiple events are disjoint if none of the events have any outcomes in common.
- A **probability measure**, typically denoted P, assigns probabilities to *events* to quantify their relative likelihoods according to the assumptions of the model of the random phenomenon.
- The probability of event A, computed according to probability measure $\mathrm{P}(A)$, is denoted $\mathrm{P}(A)$.
- A valid probability measure P must satisfy the following three logical consistency "axioms".
  - For any event A, $0 \leq \mathrm{P}(A) \leq 1$.
  - If Ω represents the sample space then $\mathrm{P}(\Omega) = 1$.
  - (*Countable additivity.*) If $A_1, A_2, A_3, \ldots$ are disjoint then

$$\mathrm{P}(A_1 \cup A_2 \cup A_2 \cup \cdots) = \mathrm{P}(A_1) + \mathrm{P}(A_2) + \mathrm{P}(A_3) + \cdots$$

- Additional properties of a probability measure follow from the axioms
  - *Complement rule.* For any event A, $\mathrm{P}(A^c) = 1 - \mathrm{P}(A)$.
  - *Subset rule.* If $A \subseteq B$ then $\mathrm{P}(A) \leq \mathrm{P}(B)$.
  - *Addition rule for two events.* If A and B are any two events

$$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$$

- *Law of total probability.* If $C_1, C_2, C_3 \ldots$ are disjoint events with $C_1 \cup C_2 \cup C_3 \cup \cdots = \Omega$, then
$$\mathrm{P}(A) = \mathrm{P}(A \cap C_1) + \mathrm{P}(A \cap C_2) + \mathrm{P}(A \cap C_3) + \cdots$$

- The axioms of a probability measure are minimal logical consistent requirements that ensure that probabilities of different events fit together in a valid, coherent way.
- A single probability measure corresponds to a particular set of assumptions about the random phenomenon.

Example 2.1. Roll a four-sided die twice, and record the result of each roll in sequence. For example, a 3 on the first roll and a 1 on the second is not the same outcome as a 1 on the first roll and a 3 on the second.

1. Identify the sample space.

2. We might be interested in the sum of the two rolls. Explain why it is still advantageous to define the sample space as in the previous part, rather than as just $2, \ldots, 8$.

Example 2.2. Regina and Cady are meeting for lunch. They will each definitely arrive between noon and 1, but their exact arrival times are uncertain. Rather than dealing with clock time, it is helpful to represent noon as time 0 and measure time as minutes after noon, including fractions of a minute, so that arrival times take values in the continuous interval [0, 60].

Draw a picture to represent the sample space.

Example 2.3. Roll a four-sided die twice, and record the result of each roll in sequence. Using the sample space from Example 2.1, identify the following events.

1. A, the event that the sum of the two dice is at most 4.

2. B, the event that the larger of the two rolls (or the common roll if a tie) is 3.

3. B^c (identify and interpret).

4. $A \cap B$ (identify and interpret).

5. C, the event that the first roll is a 3.

Example 2.4. Regina and Cady are meeting for lunch. They will each definitely arrive between noon and 1, but their exact arrival times are uncertain. Rather than dealing with clock time, it is helpful to represent noon as time 0 and measure time as minutes after noon, including fractions of a minute, so that arrival times take values in the continuous interval [0, 60].

Using the sample space from Example 2.2, draw a picture to represent each of the following events.

1. A, the event that Regina arrives after Cady.

2. B, the event that either Regina or Cady arrives before 12:30.

3. C, the event that they arrive within 15 minutes of each other.

4. D, the event that Regina arrives before 12:24.

- For a sample space Ω with finitely many possible outcomes, assuming **equally likely outcomes** corresponds to a probabiliy measure P which satisfies

$$\mathrm{P}(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } \Omega} \qquad \text{when outcomes are equally likely}$$

- Note that equally likely outcomes is an *assumption*, and by no means a requirement. In most situations equally likely outcomes is *not* a reasonable assumption.

Example 2.5. Roll a fair four-sided die twice, and record the result of each roll in sequence. Let P represent the probability measure which assumes equally likely outcomes.

1. How many possible outcomes are there? Why is it reasonable to assume they are equally likely?

2. Compute $\mathrm{P}(A)$, where A is the event that the sum of the two dice is at most 4.

3. Compute $\mathrm{P}(B)$, where B the event that the larger of the two rolls (or the common roll if a tie) is 3.

4. Compute and interpret $\mathrm{P}(A \cap B)$. (Is it equal to $\mathrm{P}(A)\mathrm{P}(B)$?)

5. Compute $\mathrm{P}(C)$, where C the event that the first roll is a 3.

- The continuous analog of equally likely outcomes is a **uniform probability measure**. When the sample space is uncountable, size is measured continuously (length, area, volume) rather that discretely (counting).

$$\mathrm{P}(A) = \frac{|A|}{|\Omega|} = \frac{\text{size of } A}{\text{size of } \Omega} \qquad \text{if P is a uniform probability measure}$$

- Note that a uniform probability measure is an *assumption*, and by no means a requirement. In most situations a uniform probability measure is *not* a reasonable assumption.

Example 2.6. Regina and Cady are meeting for lunch. Suppose they each arrive uniformly at random at a time between noon and 1:00, independently of each other. Record their arrival times as minutes after noon, so noon corresponds to 0 and 1:00 to 60.

Recall the sample space from Example 2.2 and assume a uniform probability measure P.

1. Find the probability that Regina arrives after Cady.

2. Find the probability that either Regina or Cady arrives before 12:30.

3. Find the probability that Cady arrives first and Regina arrives at most 15 minutes after Cady.

4. Find the probability that Regina arrives before 12:24.

2.1 Exercises

Exercise 2.1. The probability that a randomly selected U.S. household has a pet dog is 0.47. The probability that a randomly selected U.S. household has a pet cat is 0.25. (These values are based on the 2018 General Social Survey (GSS).)

1. Represent the information provided using proper symbols.

2. Donny Don't says: "the probability that a randomly selected U.S. household has a pet dog OR a pet cat is $0.47 + 0.25 = 0.72$." Do you agree? What must be true for Donny to be correct? Explain. (Hint: for the remaining parts it helps to consider two-way tables.)

3. What is the *smallest possible* value of the probability that a randomly selected U.S. household has a pet dog AND a pet cat? Describe the (unrealistic) situation in which this extreme case would occur.

4. What is the *largest possible* value of the probability that a randomly selected U.S. household has a pet dog AND a pet cat? Describe the (unrealistic) situation in which this extreme case would occur. What would be the probability that a randomly selected U.S. household has a pet dog OR a pet cat in this scenario?

5. Donny Don't says: "I remember hearing once that in probability OR means add and AND means multiply. So the probability that a randomly selected U.S. household has a pet dog AND a pet cat is $0.47 \times 0.25 = 0.1175$." Do you agree? Explain.

6. According to the GSS, the probability that a randomly selected U.S. household has a pet dog AND a pet cat is 0.15. Compute the probability that a randomly selected U.S. household has a pet dog OR a pet cat.

7. Compute and interpret $\mathrm{P}(C \cap D^c)$.

Exercise 2.2. Each question on a multiple choice test has four options. You know with certainty the correct answers to 70% of the questions. For 20% of the questions, you can eliminate two of the incorrect choices with certainty, but you guess at random among the remaining two options. For the remaining 10% of questions, you have no idea and guess one of the four options at random.

1. Randomly select a question from this test. What is the probability that you answer the question correctly? (Hint: make a two-way table.)

2. For any given question on the exam, your probability of answering it correctly is either 1, 0.5, or 0.25, depending on if you know it, can eliminate two choices, or are just guessing. How does your probability of correcting answering a randomly selected question relate to these three values? Which value — 1, 0.5, or 0.25 —is the overall probability closest to, and why?

3 Conditional Probability

Example 3.1. The probability[1] that a randomly selected American adult (18+) uses Snapchat is 0.24.

1. Suppose the randomly selected adult is under age 30. Do you think the probability that a randomly selected *adult who is under age 30* uses Snapchat is 0.24? What if the adult is over age 30?

2. The probability[2] that a randomly selected American adult is under age 30 is 0.20. Start to create a two-way table representing 10000 hypothetical U.S. adults. Is the probability that a randomly selected American adult both (1) is under age 30, *and* (2) uses Snapchat equal to 0.20×0.24? Explain.

3. Suppose that the probability that a randomly selected American adult both is use age 30 and uses Snapchat is 0.13. Construct an appropriate two-way table.

4. Find the probability that a randomly selected American *adult who is under age 30* uses Snapchat.

[1]The values in this problem are based on a April, 2021 report by the Pew Research Center.
[2]Based on data from the U.S. Census Bureau

5. How can the probability in the previous part be written in terms of the probabilities provided earlier?

6. Find the probability that a randomly selected American *adult who is over age 30* uses Snapchat.

7. Find the probability that a randomly selected American *adult who uses Snapchat* is under age 30.

- The **conditional probability of event** A given event B, denoted $\mathrm{P}(A|B)$, is defined as

$$\mathrm{P}(A|B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$$

- The conditional probability $\mathrm{P}(A|B)$ represents how the likelihood or degree of uncertainty of event A should be updated to reflect information that event B has occurred.
- Conditional probabilities are probabilities. There are analagous conditional versions of the properties of probability, e.g. $\mathrm{P}(A^c|B) = 1 - \mathrm{P}(A|B)$.
- In general, knowing whether or not event B occurs influences the probability of event A. That is,

$$\text{In general, } \mathrm{P}(A|B) \neq \mathrm{P}(A)$$

- Be careful: order is essential in conditioning. That is,

$$\text{In general, } \mathrm{P}(A|B) \neq \mathrm{P}(B|A)$$

- Within the context of two events, we have joint, conditional, and marginal probabilities.
  - Joint: unconditional probability involving both events, $\mathrm{P}(A \cap B)$.
  - Conditional: conditional probability of one event given the other, $\mathrm{P}(A|B)$, $\mathrm{P}(B|A)$.
  - Marginal: unconditional probability of a single event $\mathrm{P}(A)$, $\mathrm{P}(B)$.
- The relationship $\mathrm{P}(A|B) = \mathrm{P}(A \cap B)/\mathrm{P}(B)$ can be stated generically as

$$\text{conditional} = \frac{\text{joint}}{\text{marginal}}$$

- When dealing with probabilities (or proportions or percentages) be sure to ask "probability *of what*?" Thinking in fraction terms, be careful to identify the correct reference group which corresponds to the denominator.

Example 3.2. The following two phrases contain exactly the same words, just in different orders. Which is larger, the numerical value of 1 or 2?

1. The probability that a randomly selected man who is greater than six feet tall plays in the National Basketball Association (NBA).
2. The probability that a randomly selected man who plays in the National Basketball Association (NBA) is greater than six feet tall.

Example 3.3. Suppose

- 20% of adults are under age 30
- 63% of adults under age 30 use TikTok
- 31% of adults over age 30 TikTok

1. Use the information to set up a hypothetical two-way table.

2. Compute the probability that an adult uses TikTok.

3. Compute the probability that an adult who uses TikTok is under age 30.

- Rearranging the definition of conditional probability we get the **Multiplication rule:** the probability that two events both occur is

$$\begin{aligned} P(A \cap B) &= P(A|B)P(B) \\ &= P(B|A)P(A) \end{aligned}$$

- The multiplication rule says that you should think "multiply" when you see "and". However, be careful about *what* you are multiplying: to find a joint probability you need an unconditional and an appropriate conditional probability.
- You can condition either on A or on B, provided you have the appropriate marginal probability; often, conditioning one way is easier than the other.
- Be careful: the multiplication rule does *not* say that $P(A \cap B)$ is the same as $P(A)P(B)$.
- **Law of total probability** If C_1, C_2, C_3 ... are disjoint events with $C_1 \cup C_2 \cup C_3 \cup \cdots = \Omega$, then

$$\begin{aligned} P(A) &= P(A \cap C_1) + P(A \cap C_2) + P(A \cap C_2) + \cdots \\ &= P(A|C_1)P(C_1) + P(A|C_2)P(C_2) + P(A|C_3)P(C_3) + \cdots \end{aligned}$$

- The events $C_1, C_2, C_3, \ldots$, which represent the "cases", form a *partition* of the sample space; each outcome $\omega \in \Omega$ lies in exactly one of the C_i.
- The law of total probability says that we can interpret the unconditional probability $P(A)$ as a probability-weighted average of the case-by-case conditional probabilities $P(A|C_i)$ where the weights $P(C_i)$ represent the probability of encountering each case.

3.1 Exercises

Exercise 3.1. Continuing Exercise 2.1. The probability that a randomly selected U.S. household has a pet dog is 0.47. The probability that a randomly selected U.S. household has a pet cat is 0.25. the probability that a randomly selected U.S. household has a pet dog and a pet cat is 0.15.

In addition to computing, represent the probabilities using appropriate notation.

1. Compute the probability that a randomly selected U.S. household that has a pet dog also has a pet cat. Is it 0.25?

2. Compute the probability that a randomly selected U.S. household that does not have a pet dog has a pet cat. Is it the same as the previous part? Is it one minus the previous part?

3. Describe in words the probability that results from subtracting the answer to the first part from 1.

4. Compute the probability that a randomly selected U.S. household that has a pet cat also has a pet dog.

Exercise 3.2. Suppose that[3]

- 67% of Democrats believe in human-driven climate

[3]Probabilities are estimated based on this 2024 survey.

- 46% of Independents believe in human-driven climate
- 34% of Republicans believe in human-driven climate

Also suppose that[4]

- 28% of American adults are Democrats
- 42% of American adults are Independents
- 30% of American adults are Republicans

1. Define the event A to represent "believes in human-driven climate change" and D, I, R to correspond to affiliation in each of the parties. If the probability measure P corresponds to selecting an American adult uniformly at random, write all the percentages above as probabilities using proper notation.

2. Construct an appropriate two-way table of probabilities.

3. Now suppose that the randomly selected American believes in human-driven climate change. How does this information change the probability that the selected American belongs to each political party? Answer by computing appropriate probabilities (and using proper notation).

4. How many times more likely is it for an American *adult* to believe in human-driven climate change and be Independent than to:

    a. believe in human-driven climate change and be Democrat

[4]Estimate based on Gallup poll

b. believe in human-driven climate change and be Republican

5. How many times more likely is it for an American adult *who believes in human-driven climate change* to be Independent than to be:

   a. Democrat

   b. Republican

6. What do you notice about the answers to the two previous parts?

- The process of conditioning can be thought of as **"slicing and renormalizing".**
  - Extract the "slice" corresponding to the event being conditioned on (and discard the rest). For example, a slice might correspond to a particular row or column of a two-way table.
  - "Renormalize" the values in the slice so that corresponding probabilities add up to 1.
- Slicing determines the *shape*; renormalizing determines the *scale*.
- Slicing determines relative probabilities; renormalizing just makes sure they add up to 1.

Exercise 3.3. Continuing Exercise 3.2. Explain in detail how you could conduct a simulation, using only the information provided in the problem set up, and how you would use the results to approximate the conditional probability of each party given belief in human driven climate change.

- Conditional probabilities can be approximated by filtering (or subsetting) based on the condition
- To approximate $\mathrm{P}(A|B)$, simulate the random phenomenon for a set number of repetitions (say 10000), *discard those repetitions on which* B does not occur, and compute the relative frequency of A among the remaining repetitions (on which B does occur).

- Be careful! The margin of error is based on only the number of repetitions used to compute the relative frequency. So if you perform 10000 repetitions but B occurs only on 2000, then the margin of error for estimate $\mathrm{P}(A|B)$ is roughly on the order of $1/\sqrt{2000} = 0.022$ (rather than $1/\sqrt{10000} = 0.01$. Especially if $\mathrm{P}(B)$ is small, the margin of error could be large resulting in an imprecise estimate of $\mathrm{P}(A|B)$. (Advantage: not computationally intensive.)

4 Bayes Rule

- *Bayes' rule* describes how to update uncertainty in light of new information, evidence, or data.

Example 4.1. There exist multiple penguin species throughout Antarctica, including the *Adelie* (A), *Chinstrap* (C), and *Gentoo* (G). Suppose that among Antarctic penguins[1] of these three species, 44.2% of Adelie, 19.8% are Chinstrap, and 36.0% are Gentoo. Suppose that

- 83.4% of Adelie penguins are below average weight,
- 89.7% of Chinstrap penguins are below average weight, and
- 4.9% of Gentoo penguins are below average weight (that is, below 4200g).

1. Use the information to construct an appropriate two-way table of probabilities.

2. Compute the probability that a randomly selected penguin is below average weight. How is this related to the values 0.834, 0.897, 0.049?

3. If the selected penguin is below average weight, what is the probability that it is species A? Species C? Species G?

[1] Throughout "penguin" refers to an Antarctic penguin of one of these three species. This example is based on the famous Palmer penguins data set.

- **Bayes' rule for events** specifies how a prior probability $P(H)$ of event H is updated in response to the evidence E to obtain the posterior probability $P(H|E)$.

$$\mathrm{P}(H|E) = \frac{\mathrm{P}(E|H)\mathrm{P}(H)}{\mathrm{P}(E)}$$

- Event H represents a particular hypothesis (or model or case)
- Event E represents observed evidence (or data or information)
- $\mathrm{P}(H)$ is the unconditional or **prior probability** of H (prior to observing evidence E)
- $\mathrm{P}(H|E)$ is the conditional or **posterior probability** of H after observing evidence E.
- $\mathrm{P}(E|H)$ is the **likelihood** of evidence E given hypothesis (or model or case) H

Example 4.2. Continuing Example 4.1. Randomly select a penguin and suppose it is below average weight. We are interested in classifying the species of the penguin.

1. Frame this problem in the context of Bayes rule: Identify the hypotheses, prior probabilities, evidence, likelihood, and posterior probabilities.

2. Prior to observing the penguin's weight, how many times more likely is it to be Adelie than to be Gentoo?

3. How many times more likely is it for a penguin to be below average weight given that it is Adelie than given that it is Gentoo?

4. Compute the posterior probability of each species given that the penguin is below average weight. Given that the penguin is below average weight, how many times more likely is it to be Adelie than to be Gentoo?

5. How are the ratios in the three previous parts related?

6. Apply the same logic to compare species A and C.

- Bayes rule is often used when there are multiple hypotheses or cases. Suppose $H_1, \ldots, H_k$ is a series of distinct hypotheses which together account for all possibilities, and E is any event (evidence).
- Combining Bayes' rule with the law of total probability,

$$\begin{aligned}
P(H_j|E) &= \frac{P(E|H_j)P(H_j)}{P(E)} \\
&= \frac{P(E|H_j)P(H_j)}{\sum_{i=1}^k P(E|H_i)P(H_i)}
\end{aligned}$$

$$P(H_j|E) \propto P(E|H_j)P(H_j)$$

- The symbol $\propto$ is read "is proportional to". The relative *ratios* of the posterior probabilities of different hypotheses are determined by the product of the prior probabilities and the likelihoods, $P(E|H_j)P(H_j)$. The marginal probability of the evidence, $P(E)$, in the denominator simply normalizes the numerators to ensure that the updated probabilities sum to 1 over all the distinct hypotheses.
- **In short, Bayes' rule says**

$$\textbf{posterior} \propto \textbf{likelihood} \times \textbf{prior}$$

Example 4.3. Continuing Example 4.2. Randomly select a penguin. We are interested in classifying the species of the penguin depending on whether or not it is below average weight.

1. Before you know the weight of the penguin (or any other information) what specify would you predict it to be? Why? What is the probability that your classification is correct?

2. If the penguin is below average weight, what species would you classify it as? Why?

3. Now suppose that the penguin is not below average weight. Use a Bayes table to compute the posterior probability of each species given that the penguin is not below average weight. If the penguin is not below average weight, what species would you classify it as? Why?

4. Suppose that you classify any randomly selected penguin based on whether or not it is below average weight as in the two previous parts. What is the posterior probability that you classify the penguin correctly? How does this compare to your probability of being correct before you observed the penguin's weight?

- Bayesian analysis is often an iterative process.
- Posterior probabilities are updated after observing some information or data. These probabilities can then be used as prior probabilities before observing new data.
- Posterior probabilities can be sequentially updated as new data becomes available, with the posterior probabilities after the previous stage serving as the prior probabilities for the next stage.
- The final posterior probabilities only depend upon the cumulative data. It doesn't matter if we sequentially update the posterior after each new piece of data or only once after all the data is available; the final posterior probabilities will be the same either way. Also, the final posterior probabilities are not impacted by the order in which the data are observed.

4.1 Exercises

Exercise 4.1. A recent survey of American adults asked: "Based on what you have heard or read, which of the following two statements best describes the scientific method?"

- 70% selected "The scientific method produces findings meant to be continually tested and updated over time". (We'll call this the "iterative" opinion.)
- 14% selected "The scientific method identifies unchanging core principles and truths". (We'll call this the "unchanging" opinion).
- 16% were not sure which of the two statements was best.

How does the response to this question change based on education level? Suppose education level is classified as: high school or less (HS), some college but no Bachelor's degree (college), Bachelor's degree (Bachelor's), or postgraduate degree (postgraduate). The education breakdown is

- Among those who agree with "iterative": 31.3% HS, 27.6% college, 22.9% Bachelor's, and 18.2% postgraduate.
- Among those who agree with "unchanging": 38.6% HS, 31.4% college, 19.7% Bachelor's, and 10.3% postgraduate.
- Among those "not sure": 57.3% HS, 27.2% college, 9.7% Bachelor's, and 5.8% postgraduate

1. Consider the conditional probability that a randomly selected American adult agrees that the scientific method is "iterative" given that they have a postgraduate degree. Identify the prior probability, hypothesis, evidence, likelihood, and posterior probability, and use Bayes' rule to compute the posterior probability.

2. Find the conditional probability that a randomly selected American adult with a postgraduate degree agrees that the scientific method is "unchanging".

3. Find the conditional probability that a randomly selected American adult with a postgraduate degree is not sure about which statement is best.

4. How many times more likely is it for an *American adult* to agree with the “iterative” statement than to agree with the “unchanging” statement?

5. How many times more likely is it for an American adult to have a postgraduate degree when the adult agrees with the iterative statement than when the adult agree with the unchanging statement?

6. How many times more likely is it for an *American adult with a postgraduate degree* to agree with the “iterative” statement than to agree with the “unchanging” statement?

7. How are the values in the three previous parts related?

Exercise 4.2. Now suppose we want to compute the posterior probabilities for an American adult’s perception of the scientific method given that the randomly selected American adult has some college but no Bachelor’s degree (“college”).

1. Before computing, make an educated guess for the posterior probabilities. In particular, will the changes from prior to posterior be more or less extreme given the American has some college but no Bachelor’s degree than when given the American has a postgraduate degree? Why?

2. Construct a Bayes table and compute the posterior probabilities. Compare to the posterior probabilities given postgraduate degree from the previous examples.

Exercise 4.3. Within both the colleges of Agriculture and Architecture at Cal Poly, about 49% of admitted students are female, about 84% of admitted students went to high school in CA, and the median GPA of admitted students is about 4.1.

An orientation group of 100 newly admitted Cal Poly students includes 75 students in Agriculture and 25 students in Architecture. A student is randomly selected from this group. The selected student is Maddie, who is female, went to high school in CA, and had a high school GPA of 4.1.

Donny Don't says, "The information about Maddie applies equally well to Agriculture or Architecture and doesn't help us decide which college she's in, so it's just 50/50. Given the information about Maddie, the conditional probability that she is in Agriculture is 0.5." Do you agree? If not, what is the conditional probability that Maddie is in the college of Agriculture given the information about her?

5 Independence of Events

Example 5.1. Do Americans think it is important for federal judges to be impartial in how they decide court cases? Consider the following data from the Pew Research Center. (Indepedendents are classified as either lean Democrat or lean Republican.)

	Democrat/lean Dem (D)	Republican/lean Rep (D^c)	Total
Important (I)	1674	1531	3205
Not important (I^c)	146	133	279
Total	1820	1664	3484

Suppose a person is randomly selected from this sample. Consider the events

$$I = \{\text{person thinks it is important for judges to be impartial}\}$$
$$D = \{\text{person is a Democrat/leans Dem}\}$$

1. Compute and interpret $\mathrm{P}(I)$.

2. Compute and interpret $\mathrm{P}(I|D)$.

3. Compute and interpret $\mathrm{P}(I|D^c)$.

4. What do you notice about $\mathrm{P}(I)$, $\mathrm{P}(I|D)$, and $\mathrm{P}(I|D^c)$?

5. Compute and interpret $\mathrm{P}(D)$.

6. Compute and interpret $\mathrm{P}(D|I)$.

7. Compute and interpret $\mathrm{P}(D|I^c)$.

8. What do you notice about $\mathrm{P}(D)$, $\mathrm{P}(D|I)$, and $\mathrm{P}(D|I^c)$?

9. Compute and interpret $\mathrm{P}(D \cap I)$.

10. What is the relationship between $\mathrm{P}(D \cap I)$ and $\mathrm{P}(D)$ and $\mathrm{P}(I)$?

11. When randomly selecting a person from this particular group, would you say that events

D and I are independent? Why?

- Events A and B are **independent** if the knowing whether or not one occurs does not change the probability of the other.
- For events A and B (with $0 < \mathrm{P}(A) < 1$ and $0 < \mathrm{P}(B) < 1$) the following are equivalent. That is, if one is true then they all are true; if one is false, then they all are false.

$$
\begin{aligned}
A \text{ and } B &\text{ are independent}\\
\mathrm{P}(A \cap B) &= \mathrm{P}(A)\mathrm{P}(B)\\
\mathrm{P}(A^c \cap B) &= \mathrm{P}(A^c)\mathrm{P}(B)\\
\mathrm{P}(A \cap B^c) &= \mathrm{P}(A)\mathrm{P}(B^c)\\
\mathrm{P}(A^c \cap B^c) &= \mathrm{P}(A^c)\mathrm{P}(B^c)\\
\mathrm{P}(A|B) &= \mathrm{P}(A)\\
\mathrm{P}(A|B) &= \mathrm{P}(A|B^c)\\
\mathrm{P}(B|A) &= \mathrm{P}(B)\\
\mathrm{P}(B|A) &= \mathrm{P}(B|A^c)
\end{aligned}
$$

Example 5.2. This is a simplified example of *batch testing*[1] for a disease. In batch testing, specimens (like blood or saliva samples) from a group of people are pooled together into one batch, which then undergoes one test. If none of the people has the disease, then the batch test result will be negative, and no further tests are required. But if at least one person in the batch has the disease, then the batch test result will be positive, and then each person in the batch must be tested individually. (We're assuming no false positives and no false negatives for this example.)

Suppose that 8 people are to be tested, first in a single batch. Assume that each person has probability $p = 0.1$ of having the disease, independently from person to person. (The assumption of independence might be unreasonable if the people are in close contact with one another.)

1. Identify the possible values for the total number of tests conducted.

[1] Thanks to Allan Rossman for this example. His blog has more investigations into batch testing, including what happens as n and p change. Cal Poly implemented versions of batch testing for COVID through saliva samples and wastewater testing.

2. Compute the probability of each possible value.

- Events $A_1, A_2, A_3, \ldots$ are **independent** if:
  - any pair of events $A_i, A_j, (i \neq j)$ satisfies $\mathrm{P}(A_i \cap A_j) = \mathrm{P}(A_i)\mathrm{P}(A_j)$,
  - and any triple of events A_i, A_j, A_k (distinct i, j, k) satisfies $\mathrm{P}(A_i \cap A_j \cap A_k) = \mathrm{P}(A_i)\mathrm{P}(A_j)\mathrm{P}(A_k)$,
  - and any quadruple of events satisfies $\mathrm{P}(A_i \cap A_j \cap A_k \cap A_m) = \mathrm{P}(A_i)\mathrm{P}(A_j)\mathrm{P}(A_k)\mathrm{P}(A_m)$,
  - and so on.
- Intuitively, a collection of events is independent if knowing whether or not any combination of the events in the collection occur does not change the probability of any other event in the collection.
- When events are independent, the multiplication rule simplifies greatly.

$$\mathrm{P}(A_1 \cap A_2 \cap A_3 \cap \cdots \cap A_n) = \mathrm{P}(A_1)\mathrm{P}(A_2)\mathrm{P}(A_3) \cdots \mathrm{P}(A_n) \quad \text{if } A_1, A_2, A_3, \ldots, A_n \text{ are independent}$$

- When a problem involves independence, you will want to take advantage of it. Work with "and" events whenever possible in order to use the multiplication rule. For example, for problems involving "at least one" (an "or" event) take the complement to obtain "none" (an "and" event).

Example 5.3. Continuing Example 5.2. Now suppose that 8 people to be tested are split into two groups of 4 people each. Within each group of 4 people, specimens are combined into a batch to be tested. If anyone in the batch has the disease, then the batch test will be positive, and those 4 people will need to be tested individually. Assume that each person has probability $p = 0.1$ of having the disease, independently from person to person.

1. Identify the possible values for the total number of tests conducted.

2. Compute the probability of each possible value.

5.1 Exercises

Exercise 5.1. A certain system consists of four identical components. Suppose that the probability that any particular component fails is 0.1, and failures of the components occur independently of each other. Find the probability that the system fails if:

1. The components are connected in *parallel*: the system fails only if *all* of the components fail.

2. The components are connected in *series*: the system fails whenever *at least one* of the components fails.

3. Donny Don't says the answer to the previous part is $0.1 + 0.1 + 0.1 + 0.1 = 0.4$. Explain the error in Donny's reasoning.

Exercise 5.2. Roll two fair four-sided dice, one green and one gold. There are 16 total possible outcomes (roll on green, roll on gold), all equally likely. Consider the event $E = \{\text{the green die lands on 1}\}$. Answer the following questions by computing and comparing appropriate probabilities.

1. Consider $A = \{\text{the gold die lands on 4}\}$. Are A and E independent?

2. Consider $B = \{\text{the sum of the dice is } 3\}$. Are B and E independent?

3. Consider $C = \{\text{the sum of the dice is } 5\}$. Are C and E independent?

- Independence concerns whether or not the occurrence of one event affects the *probability* of the other.
- Given two events it is not always obvious whether or not they are independent.

- Independence depends on the underlying probability measure. Events that are independent under one probability measure might not be independent under another.
- Independence is often assumed. Whether or not independence is a valid assumption depends on the underlying random phenomenon.

6 Discrete Random Variables

- Roughly, a *random variable* assigns a number measuring some quantity of interest to each outcome of a random phenomenon.
- Mathematically, a **random variable (RV)** X is a *function* that takes an outcome in the sample space as input and returns a real number as output
- The random variable itself is typically denoted with a capital letter (X); possible values of that random variable are denoted with lower case letters (x).
  - Think of the capital letter X as a label standing in for a formula like "the number of heads in 4 flips of a coin" and
  - x as a dummy variable standing in for a particular value like 3.
- **Discrete random variables** take at most countably many possible values (e.g., $0, 1, 2, ...$). They are often counting variables (e.g., the number of Heads in 10 coin flips).
- **Continuous random variables** can take any real value in some interval (e.g., $[0, 1]$, $[0, \infty)$, $(-\infty, \infty)$.). That is, continuous random variables can take uncountably many different values. Continuous random variables are often measurement variables (e.g., height, weight, income).
- A function of a random variable is also a random variable: if X is a RV then so is $g(X)$
- Sums and products, etc., of random variables *defined on the same sample space* are random variables. IfX and Y are RVs defined on the same sample space then so are $X+Y$, $X-Y$, XY
- If the sample space outcomes are represented by rows in a spreadsheet, then random variables correspond to columns.
- Expressions like $X = x$ or $\{X = x\}$ represent *events*: for which outcomes is the value of the random variable X equal to the value x. (Remember, if the sample space outcomes are represented by rows in a spreadsheet, then an event corresponds to a subset of rows (outcomes) that satisfies some criteria.)
- The **(probability) distribution** of a collection of random variables identifies the possible values that the random variables can take and their relative likelihoods.
- We will see many ways of describing a distribution, depending on how many random variables are involved and their types (discrete or continuous).

Table 6.1: Two rolls of a fair four-sided die

First roll	Second roll	X	Y
1	1		
1	2		
1	3		
1	4		
2	1		
2	2		
2	3		
2	4		
3	1		
3	2		
3	3		
3	4		
4	1		
4	2		
4	3		
4	4		

Example 6.1. Roll a four-sided die twice, and record the result of each roll in sequence. Let X be the sum of the two dice, and let Y be the larger of the two rolls (or the common value if both rolls are the same). Table 6.1 represents the possible outcomes. Assume the die is fair and the rolls are independent.

1. Fill in the values of X and Y for each outcome in the table.

2. Identify the event $\{X = 4\}$ and compute its probability. Then interpret the probability both as a long relative frequency and as a relative likelihood.

3. Construct a table and plot of $\mathrm{P}(X = x)$ for each possible value x of X.

4. Construct a table and plot of $\mathrm{P}(Y = y)$ for each possible value y of Y.

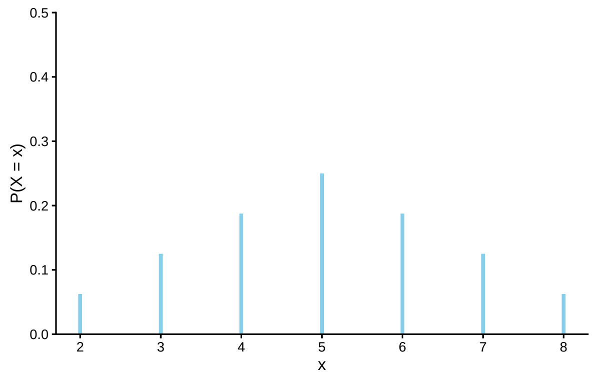


Figure 6.1: The marginal distribution of X, the sum of two rolls of a fair four-sided die.

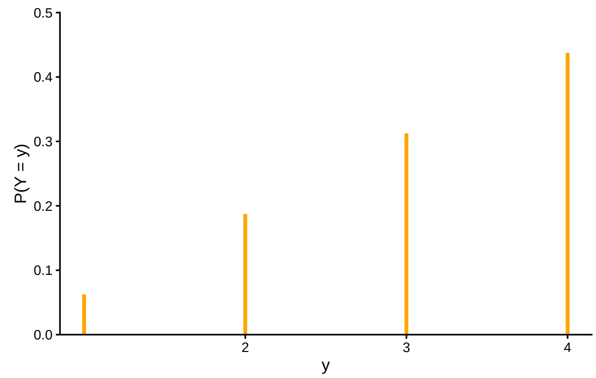


Figure 6.2: The marginal distribution of Y, the larger of two rolls of a fair four-sided die.

Example 6.2. Recall the batch testing scenarios from Example 5.2 and Example 5.3. Let X be the total number of tests required when we test all 8 people in a single batch first, and let Y be the total number of tests when we split into two batches of 4 people first.

1. Identify the distribution of X.

2. Identify the distribution of Y.

Example 6.3. The "matching problem" involves n distincts objects labeled $1, \ldots, n$ which are placed in n distinct boxes labeled $1, \ldots, n$, with exactly one object placed in each box. Suppose the objects are placed in the boxes uniformly at random, so that any possible arrangement is equally likely. Let X be the number of matches, that is, the number of objects for which their label matches the label of the box in which they are placed.

For example, suppose n people have a secret Santa gift exchange. Each person puts their name in a hat, the names are well shuffled, and everyone draws a name from the hat (to then purchase a gift for the selected person). Let X represent the number of people who draw their own name.

Consider the matching problem with $n = 3$. Specify the distribution of X, and interpret it.

6.1 Exercises

Exercise 6.1. Five individuals are interviewing for a single job opening, but only two are qualified for the position. Suppose the individuals are interviewed in a random order, and each individual is only interviewed once, until one of the qualified candidates is interviewed, and then no more candidates are interviewed. Let X count the number of interviews conducted.

Make a table to represent the distribution of X.

Exercise 6.2. Continuing Example 6.1.

1. Describe how you could simulate a single value of Y.

2. Construct a spinner (like from a kids game) to represent the distribution of Y.

3. Describe another way to simulate a single value of Y.

4. Describe how you could use simulation to approximate the distribution of Y.

- Don't confuse a random variable with its distribution!
  - A random variable measures a numerical quantity which depends on the outcome of a random phenomenon
  - The distribution of a random variable specifies the long run pattern of variation of values of the random variable over many repetitions of the underlying random phenomenon.
- Any marginal distribution can be represented by a single spinner (like from a kids game).
- In principle, there are always two ways of simulating a value x of a random variable X.
  1. *Simulate from the probability space.* Simulate an outcome ω from the underlying probability space and set $x = X(\omega)$.
  2. *Simulate from the distribution.* Construct a spinner corresponding to the distribution of X and spin it once to generate x.
- The second method requires that the distribution of X is known. However, as we will see in many examples, *it is common to specify the distribution of a random variable directly without defining the underlying probability space.*

7 Probability Mass Functions

- The **(probability) distribution** of a collection of random variables identifies the possible values that the random variables can take and their relative likelihoods.
- We will see many ways of describing a distribution, depending on how many random variables are involved and their types (discrete or continuous).
- The probability distribution of a single discrete random variable X is often displayed in a table containing the probability of the event $\{X = x\}$ for each possible value x.
- In some cases, a distribution has a "formulaic" shape. For a discrete random variable X, the **probability mass function (pmf)** p_X expresses $\mathrm{P}(X = x)$ as a function of x: $p_X(x) = \mathrm{P}(X = x)$.
- Be sure to specify the possible values of the random variable!

Example 7.1. Continuing Example 6.1. Roll a four-sided die twice, and record the result of each roll in sequence. Let X be the sum of the two dice, and let Y be the larger of the two rolls (or the common value if both rolls are the same). Assume the die is fair and the rolls are independent.

Verify that the pmf of X is

$$p_X(x) = \frac{4 - |x - 5|}{16}, \qquad x = 2, 3, \ldots, 8$$

Example 7.2. Suppose a very large number of applicants have applied for a job, and you'll interview one applicant at a time until you find one that is qualified, whom you'll hire, and then you stop interviewing people. Let X be the total number of applicants interviewed (including the hire)

Assume that 30% of applicants are qualified, and applicants are interviewed one at a time in random order, independently.

1. Compute $\mathrm{P}(X = 1)$.

2. Compute $\mathrm{P}(X = 2)$.

3. Compute $\mathrm{P}(X = 3)$.

4. Compute $\mathrm{P}(X = 4)$.

5. Specify the probability mass function of X.

- In a sequence of **Bernoulli(p) trials**
  - There are only two possible outcomes, "success" (1) or not (0, "failure"), on each trial.
  - The unconditional/marginal probability of success is the same on every trial, and equal to p.
  - The trials are independent.
- Perform Bernoulli(p) trials until a success occurs and then stop. Let X count the total number of trials, including the single success. The distribution of X is defined to be the **Geometric(p) distribution**.
- A discrete random variable X has a Geometric(p) distribution if and only if its probability

mass function is

$$p_X(x) = p(1-p)^{x-1}, \qquad x = 1, 2, 3, \ldots$$

- The Geometric(p) probability mass function follows from the fact that if X has a Geometric distribution, then $X = x$ if and only if
  - the first $x-1$ trials are failures, and
  - the xth (last) trial results in success.

Example 7.3. Randomly select a county in the U.S. Let X be the leading digit in the county's population. For example, if the county's population is 10,040,000 (Los Angeles County) then $X = 1$; if 3,170,000 (Orange County) then $X = 3$; if 283,000 (SLO County) then $X = 2$; if 30,600 (Lassen County) then $X = 3$. The possible values of X are $1, 2, \ldots, 9$. You might think that X is equally likely to be any of its possible values. However, a more appropriate model is to assume that X has pmf

$$p_X(x) = \begin{cases} \log_{10}(1 + \frac{1}{x}), & x = 1, 2, \ldots, 9, \\ 0, & \text{otherwise} \end{cases}$$

This distribution is known as Benford's law.

1. Construct a table specifying the distribution of X, and the corresponding spinner.

2. Find $\mathrm{P}(X \geq 3)$

- Certain common distributions have special names and properties.
- Learn to identify these special situations and use their "shortcut" formulas.

7.1 Exercises

Exercise 7.1. Continuing Example 7.1. Is the following the pmf of Y? Discuss.

$$p_Y(u) = \frac{2u-1}{16}$$

Exercise 7.2. Shishito peppers are typically relatively mild, but about 10% are spicy. Suppose you eat Shishito peppers one at a time until you eat a spicy one and then you stop. Let X be the total number of peppers you eat (including the spicy one).

1. Explain why it is reasonable to assume eating the peppers constitutes a sequence of Bernoulli trials.

2. Describe in full detail how you could use simulation to approximate the distribution of X.

3. Identify the distribution of X by name, including the values of any relevant parameters.

4. Specify the probability mass function of X.

5. Make a table representing the distribution of X. (You can cut it off after the probabilities get small.)

6. Compute $\mathrm{P}(X > 10)$ without summing terms.

Exercise 7.3. Recall the matching problem from Example 6.3 with a general n and let X be the number of matches. Finding the distribution of X by enumerating the $n!$ possible outcomes as we did for $n = 3$ is not feasible in general. But we can approximate the distribution of X with simulation, as we did in an earlier exercise using this applet. Simulation shows that the distribution of the number of matches X is approximately the same for all n (unless n is very small, i.e., $n \leq 5$). In particular, for any $n > 5$, the probability mass function of X is approximately

$$p_X(x) = \frac{e^{-1}}{x!}, \qquad x = 0, 1, 2, ...$$

We will study where this distribution—called the Poisson(1) distribution—comes from in more detail later. For now we'll just use the pmf formula.

1. Construct a table, plot, and spinner corresponding to the above pmf. Compare to the simulation results for general n.

2. Use the pmf to approximate the probability of at least one match, and compare to the simulation results for general n.

3. Interpret the probability both as a long relative frequency and as a relative likelihood.

8 Expected Value

- The distribution of a random variable specifies the possible values and the probability of any event that involves the random variable.
- The distribution of a random variable contains all the information about its long run behavior. It is also useful to summarize some key features of a distribution.
- One summary characteristic of a distribution is the **long run average value** of the random variable.
- We can approximate the long run average value by simulating many values of the random variable and computing the average (mean) in the usual way: sum the simulated values and divide by the number of simulated values.

Example 8.1. This is a very simplified example illustrating the basic idea of how insurance works. Every year an insurance company sells many thousands of car insurance policies to drivers within a particular risk class. Each policyholder pays a "premium" of \$1000 at the start of the year, and the insurance company agrees to pay for the cost of all damages that occur during the year. Suppose that each policy incurs damage of either \$0, \$5000, \$20000, or \$50000 with the following probabilities.

Amount of damage (\$)	Profit (\$)	Probability
0	1000	0.910
5000	-4000	0.070
20000	-19000	0.019
50000	-49000	0.001

The insurance company's profit on a policy at the end of the year is the difference between the premium of \$1000 and any damage paid out. For example, a policy that incurs no damage results in a profit of \$1000; a policy that incurs \$5000 in damage results in a profit of -\$4000 (that is, a loss of \$4000) for the insurance company.

1. Interpret the probabilities 0.91, 0.07, 0.019, and 0.001 as long run relative frequencies in this context.

2. Imagine 100,000 hypothetical policies. How many of these policies would you expect to result in a profit of \$1000? -\$4000? -\$19000? -\$49000?

3. What do you expect the total profit for these 100,000 policies to be?

4. What do you expect the average profit per policy for these 100,000 policies to be?

5. Compute the probability that a policy has a profit equal \$220.

6. Compute the probability that a policy has a profit greater than \$220.

7. Is \$220 the most likely value of profit for a single policy?

8. Is \$220 the profit you would expect for a single policy?

9. Explain in what sense \$220 is "expected".

- The long run average value of a random quantity is called its "expected value".
- Be careful: the term "expected value" is somewhat of a misnomer.
- The expected value is *not* necessarily the value we expect on a single repetition of the random phenomenon, nor the most likely value (or even a possible value).
- Rather, the expected is the value we expect to see on average in the long run over many repetitions.
- A probability can be interpreted as a long run relative frequency; an expected value can be interpreted as a long run average value.

Example 8.2. Continuing Example 8.1. We considered what we would expect for 100000 hypothetical policies, but what about an unspecified large number of policies?

1. Imagine that we have recorded the profit for each of a large number of policies (not necessarily 100000). Explain in words the process by which you would compute the average profit per policy. (In other, more general, words: how do you compute an average of a list of numbers?)

2. Given that the profit of any policy is either 1000, -4000, -19000, or -49000, how could we simplify the calculation of the sum in the previous part? Write a general expression for the average profit per policy in this scenario.

3. What do you think the expression in the previous part converges to in the long run?

4. Explain how the value in the previous part is a "probability-weighted average value".

- The **expected value** (a.k.a. *expectation* a.k.a. *mean*), of a random variable X is a number denoted $\mathrm{E}(X)$ representing the probability-weighted average value of X.
- The expected value of a *discrete* random variable with pmf p_X is defined as

$$\mathrm{E}(X) = \sum_x x p_X(x) = \sum \text{value} \times \text{probability}$$

- Note well that $\mathrm{E}(X)$ represents *a single number*.
- The expected value is the "balance point" (center of gravity) of a distribution.
- The "probability-weighted average value" is just a more compact way of computing an average in the usual way: add up all the values and divide by the number of values.
- The expected value of a random variable X is defined by the probability-weighted average according to the underlying probability measure. But the expected value can also be interpreted as the **long-run average value**, and so can be approximated via simulation.
- Read the symbol $\mathrm{E}(\cdot)$ as
  - Simulate lots of values of what's inside $(\cdot)$
  - Compute the average. This is a "usual" average; just sum all the simulated values and divide by the number of simulated values.

Example 8.3. Recall Example 7.2. Suppose a very large number of applicants have applied for a job, and you'll interview one applicant at a time until you find one that is qualified, whom you'll hire, and then you stop interviewing people. Let X be the total number of applicants interviewed (including the hire)

Assume that 30% of applicants are qualified, and applicants are interviewed one at a time in random order, independently.

1. Make an educated guess for the expected value of X. Suggest an intuitive formula for the expected value of a Geometric(p) distribution. (It might help to consider $p = 0.1$ first.)

2. Use the definition to compute $\mathrm{E}(X)$. Does it match your guess?

3. Interpret $\mathrm{E}(X)$ in this context.

- If X has a Geometric distribution with parameter p then $\mathrm{E}(X) = 1/p$.

8.1 Exercises

Exercise 8.1. Continuing Example 8.1 and Example 8.2. For a randomly selected policy let X be the amount of damage and let Y be the company's profit.

1. What is the relationship between X and Y?

2. Compute and interpret $\mathrm{E}(Y)$.

3. What is the relationship between $\mathrm{E}(X)$ and $\mathrm{E}(Y)$? Does this make sense? Why?

- Expected values and linear rescaling: If X is a random variable and a, b are non-random constants then
$$\mathrm{E}(aX + b) = a\mathrm{E}(X) + b$$

Exercise 8.2. Recall the matching problem from Example 6.3 and Exercise 7.3 with a general n and let X be the number of matches.

1. Compute and interpret $\mathrm{E}(X)$ when $n = 3$.

2. Compute and interpret $\mathrm{E}(X)$ for a general n.

9 Variance and Standard Deviation

- The expected value, a.k.a. long run average value, is just one feature of a distribution.
- Random variables vary, and the distribution describes the entire pattern of variability.
- Roughly, standard deviation measures overall degree of variability in a single number, as the average distance from the mean.
- Technically, the **variance** is the long run average squared distance from the mean, and the **standard deviation** is the square root of the variance.

Example 9.1. We'll motivate the calculation of variance by considering a small example. Say we have data on living area X, measure in thousands of square feet, for a small sample of 10 houses.

repetition	X	Deviation from mean	Squared deviation
1	1.12		
2	1.05		
3	1.67		
4	1.48		
5	0.43		
6	2.50		
7	1.71		
8	1.88		
9	0.85		
10	1.90		
Sum			
Average			

1. Compute the average of the X values.
2. Compute the deviation from the mean for each X value.
3. Compute the squared deviations.
4. Compute the average squared deviation. This is called the variance. What are the measurement units?
5. Compute the square root of the variance. This is called the standard deviation. What are the measurement units?

- The **variance** of a random variable X is

$$\begin{aligned}\mathrm{Var}(X) &= \mathrm{E}\left((X-\mathrm{E}(X))^2\right)\\ &= \mathrm{E}\left(X^2\right)-(\mathrm{E}(X))^2\end{aligned}$$

- The **standard deviation** of a random variable is

$$\mathrm{SD}(X)=\sqrt{\mathrm{Var}(X)}$$

- Variance is the long run average squared deviation from the mean.
- Standard deviation measures, roughly, the long run average distance from the mean. The measurement units of the standard deviation are the same as for the random variable itself.
- The definition $\mathrm{E}((X-\mathrm{E}(X))^2)$ represents the concept of variance. However, variance is usually computed using the following equivalent but slightly simpler formula.

$$\mathrm{Var}(X)=\mathrm{E}\left(X^2\right)-\left(\mathrm{E}\left(X\right)\right)^2$$

- That is, variance is the expected value of the square of X minus the square of the expected value of X.

- Variance has many nice theoretical properties. Whenever you need to compute a standard deviation, first find the variance and then take the square root at the end.

Example 9.2. An American roulette wheel has 18 black spaces, 18 red spaces, and 2 green spaces, all the same size and each with a different number on it. Xavier bets \$1 on black. If the wheel lands on black, Xavier wins his bet back plus an additional \$1; otherwise he loses the money he bet. Let X be Xavier's net winnings (net of the initial bet of \$1.)

1. Find the distribution of X.

2. Compute $\mathrm{E}(X)$.

3. Interpret $\mathrm{E}(X)$ in context.

4. An expected profit *for the casino* of 5 cents per \$1 bet seems small. Explain how casinos can turn such a small profit into billions of dollars.

5. Compute $\mathrm{Var}(X)$.

6. Compute and interpret $\mathrm{SD}(X)$.

Example 9.3. Continuing with roulette, Yasmin bets \$1 on number 7. If the wheel lands on 7, Yasmin wins her bet back plus an additional \$35; otherwise she loses the money she bet. Let Y be Yasmin's net winnings (net of the initial bet of \$1.)

1. Find the distribution of Y.

2. Compute $\mathrm{E}(Y)$.

3. How do the expected values of the two \$1 bets—bet on black versus bet on 7—compare? Does that mean that the two \$1 bets—bet on black versus bet on 7—are identical? If not, explain why not.

4. Before doing any calculations, determine if $\mathrm{SD}(Y)$ is greater than, less than, or equal to $\mathrm{SD}(X)$. Explain.

5. Compute $\mathrm{Var}(Y)$.

6. Compute and interpret $\mathrm{SD}(Y)$.

- Standardization measures values in terms of "standard deviations away from the mean"
- This idea is particularly useful when comparing random variables with different measurement units but whose distributions have similar shapes.

$$\text{Standardized value} = \frac{\text{Value - Mean}}{\text{Standard deviation}}$$

- If X is a random variable with expected value $\mathrm{E}(X)$ and standard deviation $\mathrm{SD}(X)$, then the **standardized random variable** is

$$Z = \frac{X - \mathrm{E}(X)}{\mathrm{SD}(X)}$$

Example 9.4. The SAT and ACT are standardized tests; scores range from 200 to 1600 for the SAT and from 1 to 36 for the ACT. SAT scores have, approximately, a symmetric bell-shaped distribution with a mean of 1050 and a standard deviation of 200. ACT scores have, approximately, a symmetric bell-shaped distribution with a mean of 21 and a standard deviation of 5.5. Troy's score on the SAT is 1500. Abed's score on the ACT is 31. Who scored relatively better on their test?

1. Compute and interpret the standardized value for Troy's SAT score.

2. Compute and interpret the standardized value for Abed's ACT score.

3. Who scored relatively better on their test?

9.1 Exercises

Exercise 9.1. The plots in Figure 9.1 summarize hypothetical distributions of quiz scores in six classes. All plots are on the same scale. Each quiz score is a whole number between 0 and 10 inclusive.

1. Donny Dont says that C represents the smallest SD, since there is no variability in the heights of the bars. Do you agree that C represents "no variability"? Explain.

2. What is the smallest *possible* value the SD of quiz scores could be? What would need to be true about the distribution for this to happen? (This scenario might not be represented by one of the plots.)

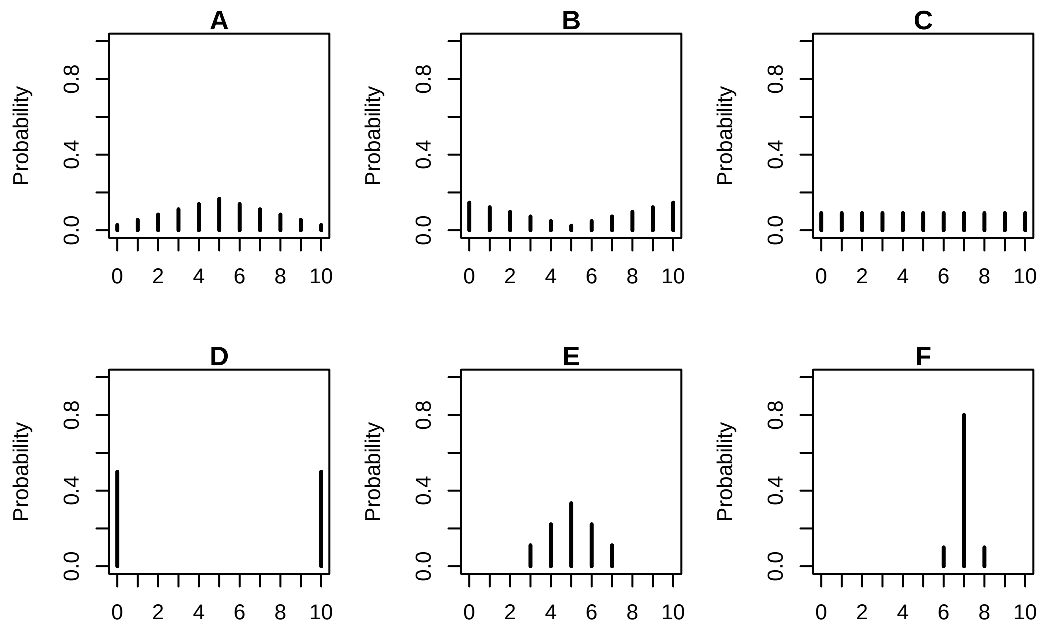


Figure 9.1: A few discrete distributions.

3. Without doing any calculations, arrange the classes in order based on their SDs from smallest to largest.

4. In one of the classes, the SD of quiz scores is 5. Which one? Why?

5. Is the SD in F greater than, less than, or equal to 1? Why?

6. Without doing any calculations, provide a ballpark estimate of SD in each case.

7. The standard deviation in plot E is about 1.15. Suppose that the scores are curved by adding 2 points to every score. What is the standard deviation and variance of the revised scores?

8. The standard deviation in plot E is about 1.15. Suppose that the scores are curved by multiplying every score by 2. What is the standard deviation and variance of the revised scores?

- Variance and linear rescaling: If X is a random variable and a, b are non-random constants then

$$\begin{aligned}
\mathrm{SD}(aX+b) &= |a|\mathrm{SD}(X) \\
\mathrm{Var}(aX+b) &= a^2\mathrm{Var}(X)
\end{aligned}$$

Exercise 9.2. Recall the matching problem from Example 6.3, Exercise 7.3, and Exercise 8.2 with a general n and let X be the number of matches.

1. Compute and interpret $\mathrm{Var}(X)$ and $\mathrm{SD}(X)$ when $n = 3$.

2. Compute and interpret $\mathrm{Var}(X)$ and $\mathrm{SD}(X)$ for a general n.

Exercise 9.3. Suppose that X has a Geometric(p) distribution.

1. Without doing any computations, determine $\mathrm{Var}(X)$ if $p = 1$.

2. As p decreases, does $\mathrm{Var}(X)$ increase or decrease? Why?

- If X has a Geometric distribution with parameter p then $\mathrm{Var}(X) = (1-p)/p^2$.

10 Binomial Distributions

- Some probabilistic situations are so common that they have special names.
- In a sequence of **Bernoulli(p) trials**
  - There are only two possible outcomes, "success" (1) or not (0, "failure"), on each trial.
  - The unconditional/marginal probability of success is the same on every trial, and equal to p.
  - The trials are independent.
- There are several commonly used distributions associated with Bernoulli trials, including Geometric and Binomial.

Example 10.1. Consider an extremely simplified model for the daily closing price of a certain stock. Every day the price either goes up or goes down, and the movements are independent from day-to-day. Assume that the probability that the stock price goes up on any single day is 0.55. Let X be the number of days on which the price goes up in a 5-day week.

1. Compute and interpret $\mathrm{P}(X = 0)$.

2. Compute the probability that the price goes up on the first day and then down on the following four days.

3. Why is $\mathrm{P}(X = 1)$ different from the probability in the previous part? Compute and interpret $\mathrm{P}(X = 1)$.

4. Compute and interpret $\mathrm{P}(X = 2)$.

5. Suggest a general formula for the probability mass function of X.

6. Construct a table and plot to represent the distribution of X, and interpret it.

- Consider a *fixed number*, n, of Bernoulli(p) trials and let X count the number of successes. The distribution of X is defined to be the **Binomial(n, p) distribution**.
- A Binomial distribution is specified by two parameters:
  - n (an integer): the fixed number of Bernoulli trials, or the sample size
  - p (in $[0, 1]$): the fixed marginal probability of success on each Bernoulli trial, or the population proportion of success
- A discrete random variable X has a **Binomial distribution** with parameters n, a nonnegative integer, and $p \in [0, 1]$ if and only if its probability mass function is

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, 2, \dots, n$$

  - In the Binomial(n, p) situation, there are $\binom{n}{x}$ S/F sequences that result in exactly x successes—and thus $n - x$ failures—in n trials,
  - and each of these sequences has probability $p^x(1-p)^{n-x}$
- The *binomial coefficient* (read "n choose x")

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

counts the number of success/failure sequences of length n in which there are exactly x successes. (Remember: by definition $0! = 1$.)

Example 10.2. Continuing Example 10.1.

1. Suggest a shortcut formula for $\mathrm{E}(X)$.

2. Compute $\mathrm{E}(X)$ using the distribution of X from Example 10.1. Did the shortcut formula work?

3. Interpret $\mathrm{E}(X)$.

4. Remember that we assumed $p = 0.55$. If instead p were 0.9, would $\mathrm{Var}(X)$ be greater or less? What if $p = 0.5$?

- If X has a Binomial(n, p) distribution then

$$\begin{aligned}\mathrm{E}(X) &= np\\ \mathrm{Var}(X) &= np(1-p)\end{aligned}$$

10.1 Exercises

Exercise 10.1. Continuing Example 10.1.

1. What does the random variable $5 - X$ represent? What is its distribution?

2. Suppose that the price is currently \$100 and each it either moves up \$2 or down \$2. Let S be the stock price after 5 days. How does S relate to X? Does S have a Binomial distribution?

3. Recall that X is the number of days on which the price goes up in the next five days. Suppose that Y is the number of days on which the price goes up in the ten days after that (days 6-15). What is the distribution of Y? What does $X + Y$ represent and what is its distribution? (Continue to assume independence between days, with probability 0.55 of an up movement on any day.)

- If X and Y are independent, X has a Binomial(n_X, p) distribution, and Y has a Binomial(n_Y, p) distribution then $X + Y$ has a Binomial($n_X + n_Y$, p)
- A Binomial(1, p) distribution is also known as a **Bernoulli(p)** distribution, taking a value of 1 with probability p and 0 with probability $1 - p$.
  - For a single trial, a Bernoulli random variable *indicates* if there is a success (1) or failure (0) on the trial
- If $X_1, X_2, \dots, X_n$ are independent each with a Bernoulli(p) distribution, then $X_1 + \cdots + X_n$ has a Binomial(n, p) distribution.
  - The random variable $X_1 + X_2 + ... + X_n$ incrementally counts the total number of successes in the n trials

Exercise 10.2. In each of the following situations determine whether or not X has a Binomial distribution. If so, specify n and p. If not, explain why not.

1. Roll a die 20 times; X is the number of times the die lands on an even number.

2. Roll a die 20 times; X is the number of times the die lands on 6.

3. Roll a die until it lands on 6; X is the total number of rolls.

4. Roll a die 20 times; X is the sum of the numbers rolled.

5. Shuffle a standard deck of 52 cards (13 hearts, 39 other cards) and deal 5 *without replacement*; X is the number of hearts dealt. (Hint: be careful about why.)

6. Roll a fair six-sided die 10 times and a fair four-sided die 10 times; X is the number of 3s rolled (out of 20).

7. Randomly select a sample of 35 Cal Poly students; X is the number of students in the sample who are CA residents.

- Consider selecting a random sample of size n from a population in which every individual is classified as "success" or "failure". Suppose that the proportion of individuals in the population classified as success is p.
- The individuals selected for the sample can be considered a sequence of Bernoulli(p) trials.
  - There are only two possible outcomes, success (1) and failure (0), on each trial.
  - The unconditional/marginal probability of success is the same on every trial, and equal to the population proportion of success p
  - The trials are independent
    * If sampling with replacement, or
    * If sampling without replacement but the population size is much larger than the sample size n (which is usually the case in practice)
- If X counts the number of successes in a random sample of size n from a large population with population proportion of success p then X has a Binomial(n, p) distribution.

Exercise 10.3. Donny Dont is thoroughly confused about the distinction between a random variable and its distribution. Help him understand by by providing a simple concrete example of *two different random variables* X and Y that have the *same distribution*. Can you think of X and Y that have the same distribution but for which $\mathrm{P}(X = Y) = 0$?

11 Poisson Distributions

- Binomial models have several restrictive assumptions that might not be satisfied in practice
- Poisson models are more flexible and are often used to model the distribution of random variables that count the number of "relatively rare" events that occur over a certain interval of time/in a certain location.

Example 11.1. Recall the matching problem from Example 6.3, Exercise 7.3, Exercise 8.2, and Exercise 9.2. There are n distinct objects that are shuffled and placed uniformly at random in n distinct spots with one object per spot, and X is the number of correct matches of object with spot. Note: n is a fixed and known number, but we are considering different values of it.

We have seen that $\mathrm{E}(X) = 1$ for any n and that unless n is small (e.g., $n \leq 6$), $\mathrm{Var}(X) = 1$ and the approximate distribution of X is given by Table 11.1.

Starting with $x = 0$, describe the distribution of X in terms of relative likelihoods:

1. 1 is [blank] times as likely as 0

2. 2 is [blank] times as likely as 1

3. 3 is [blank] times as likely as 2

4. 4 is [blank] times as likely as 3

Table 11.1: Poisson(1) distribution, the approximate distribution of X, the number of matches in the matching problem for general n and uniformly random placement of objects in spots.

x	P(X=x)
0	0.36788
1	0.36788
2	0.18394
3	0.06131
4	0.01533
5	0.00307
6	0.00051
7	0.00007

5. 5 is [blank] times as likely as 4

6. 6 is [blank] times as likely as 5

7. In general, x is [blank] times as likely as $x-1$, for $x = 1, 2, 3, ...$

- A random variable X has a **Poisson distribution** with parameter $\mu > 0$ if the possible values are $0, 1, 2, ...$ and the pmf follows the

$$\text{Poisson}(\mu) \text{ pattern:} \quad x \text{ is } \frac{\mu}{x} \text{ as likely as } x-1 \qquad (\text{for } x = 1, 2, 3, ...)$$

- As a function of $x = 0, 1, 2, ...$, a Poisson(μ) pmf increases for $x < \mu$ and then decreases for $x > \mu$
  - If μ is a whole number, then the Poisson pmf assigns the same probability to the values μ and $\mu - 1$.
  - If $\mu < 1$ then $x = 0$ has the highest probability and the pmf decreases from there.

- If X has a Poisson(μ) distribution then

$$\begin{aligned} \mathrm{E}(X) &= \mu \\ \mathrm{Var}(X) &= \mu \end{aligned}$$

Example 11.2. Let X be the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game. Technically, there is no fixed upper bound on what X can be, so mathematically it is convenient to consider $0, 1, 2, \ldots$ as the possible values of X. Assume that X has a Poisson distribution with parameter 2.4.

1. Interpret 2.4 in context.

2. Identify what the Poisson(2.4) patterns implies in context.

3. Sketch a plot of the distribution of X.

4. Assuming that the probability that X is greater than 9 is negligible, make a table representing the distribution of X.

5. The pmf of X is

$$\mathrm{P}(X = x) = e^{-2.4}\frac{2.4^x}{x!}, \quad x = 0, 1, 2, \ldots$$

Use the pmf to compute $\mathrm{P}(X = 0)$.

6. Use the pmf to compute $\mathrm{P}(X = 1)$.

7. Use the pmf to compute $\mathrm{P}(X = 2)$.

8. Use the pmf to construct a table and plot representing the distribution of X. Does it match your previous work?

9. Use the pmf to compute $\mathrm{E}(X)$ and verify that it is 2.4.

10. Compute and interpret $\mathrm{P}(X \leq 2)$.

- A discrete random variable X has a Poisson distribution with parameter $\mu > 0$ if and only if its probability mass function p_X satisfies

$$p_X(x) = \frac{e^{-\mu}\mu^x}{x!}, \quad x = 0, 1, 2, ...$$

- The Poisson pattern implies that the shape of a Poisson pmf as a function of x is given by $\mu^x/x!$.

$$p(x) = p(0)\frac{\mu^x}{x!}, \qquad x = 0, 1, 2, ...$$

- The constant $p(0) = e^{-\mu}$ simply renormalizes the pmf so that the probabilities sum[1] to 1.

Example 11.3. Let X be the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game. Assume that X has a Poisson(2.4) distribution.

1. In what ways is this like the Binomial situation?

2. In what ways is this NOT like the Binomial situation?

3. In what sense are home runs "relatively rare"?

- Binomial models have several restrictive assumptions that might not be satisfied in practice
- Poisson models are more flexible and are often used to model the distribution of random variables that count the number of "relatively rare" events that occur over a certain interval of time/in a certain location
- **Poisson paradigm.** Let $A_1, A_2, \dots, A_n$ be a collection of n events. Suppose event i occurs with marginal probability $p_i = \mathrm{P}(A_i)$. Let $N = \mathrm{I}_{A_i} + \mathrm{I}_{A_2} + \cdots + \mathrm{I}_{A_n}$ be the random variable which counts the number of the events in the collection which occur. Suppose
    - n is "large" (but not necessarily fixed),

[1] Recall the series expansion $e^{\mu} = \sum_{x=0}^{\infty} \frac{\mu^x}{x!}$

- $p_1, \dots, p_n$ are "comparably small" (but not necessarily the same), and
- the events $A_1, \dots, A_n$ are "not too dependent" (but not necessarily independent).

- Then N has an approximate Poisson distribution with parameter $\mathrm{E}(N) = \sum_{i=1}^{n} p_i$.

11.1 Exercises

Exercise 11.1. Continuing Example 11.1. Explain why X in the matching problem has a Poisson distribution when n is large.

Exercise 11.2. Suppose that the number of fatal crashes in SLO County in a week follows, approximately, a Poisson(0.53) distribution, independently from week to week. Let X be the number of fatal crashes in a period of 4 consecutive weeks (close enough to a "month").

1. Make an educated guess for $\mathrm{E}(X)$.

2. Compute $\mathrm{P}(X = 0)$.

3. Compute $\mathrm{P}(X = 1)$.

4. How could you use simulation to approximate the distribution of X?

5. Use simulation to approximate the distribution of X. Does the approximate distribution follow (approximately) the Poisson pattern?

6. Use a Poisson pmf to compute $\mathrm{P}(X = 5)$.

- **Poisson aggregation.** If X and Y are independent, X has a Poisson(μ_X) distribution, and Y has a Poisson(μ_Y) distribution, then $X + Y$ has a Poisson($\mu_X + \mu_Y$) distribution.
- If component counts are independent and each has a Poisson distribution, then the total count also has a Poisson distribution.

12 Continuous Random Variables

- A continuous random variable can take any value within some uncountable interval.
- Simulated values of a continuous random variable are usually plotted in a **histogram** which groups the observed values into "bins" and plots densities or frequencies for each bin.
- In a histogram *areas* of bars represent relative frequencies; the axis which represents the height of the bars is called "density".
- The continuous analog of a probability mass function (pmf) for a discrete random variable is a probability density function (pdf) However there are some fundamental differences between discrete and continuous random variables, and pmfs and pdfs.
- The distribution of a continuous random variable can be described by a **probability density function (pdf)**, for which *areas under the density curve determine probabilities.*

Example 12.1. Suppose that in your office everyone arrives to work between 8am and 9am. Randomly select an employee and let X be employee's arrival time measured in minutes after 8am, including fractions of a minute, so that X takes a value in the continuous interval [0, 60].

Data on many randomly selected employees suggests that the distribution of X is described by this probability density function (pdf), where c is a constant that we'll determine soon.

$$f_X(x) = \begin{cases} cx, & 0 < x < 60 \\ 0, & \text{otherwise.} \end{cases}$$

1. Sketch a plot of the pdf. Roughly, what does this say about arival times?

2. Find the value of c and specify the pdf of X.

3. Compute and interpret $P(X < 15)$.

4. Compute and interpret $P(X > 45)$.

- The continuous analog of a probability mass function (pmf) is a *probability density function (pdf)*.
- However, while pmfs and pdfs play analogous roles, they are different in one fundamental way; namely, a pmf outputs probabilities directly, while a pdf does not.
- A pdf of a continuous random variable must be *integrated* to find probabilities of related events.
- The **probability density function (pdf)** (a.k.a. density) of a *continuous* RV X is the function f_X which satisfies

$$P(a \le X \le b) = \int_a^b f_X(x)dx, \qquad \text{for all } -\infty \le a \le b \le \infty \tag{12.1}$$

- For a continuous random variable X with pdf f_X, the probability that X takes a value in the interval $[a, b]$ is the *area under the pdf over the region* $[a, b]$.
- The axioms of probability imply that a valid pdf must satisfy

$$f_X(x) \ge 0 \qquad \text{for all } x, \tag{12.2}$$

$$\int_{-\infty}^{\infty} f_X(x)dx = 1 \tag{12.3}$$

Example 12.2. Continuing Example 12.1, recall that X has pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 \le x \le 60, \\ 0, & \text{otherwise.} \end{cases}$$

1. Consider the probability that an employee arrives within 1 minute after 8:15. Compute $P(15 < X < 16)$, the probability that X, truncated to the nearest minute, is 15. How is the probability related to $f(15)$?

2. Consider the probability that an employee arrives within 1 minute after 8:45. Compute $\mathrm{P}(45 < X < 46)$, the probability that X, truncated to the nearest minute, is 45. How is the probability related to $f(45)$?

3. Compute and interpret the ratio of the two previous probabilities.

4. Consider the probability that an employee arrives within 1 *second* after 8:15. Compute $\mathrm{P}(15 < X < 15 + 1/60)$, the probability that X, truncated to the nearest *second*, is 15. (The measurement units of X are minutes, so $1/60$ minutes is 1 second.) How is the probability related to $f(15)$?

5. Consider the probability that an employee arrives within 1 *second* after 8:45. Compute $\mathrm{P}(45 < X < 45 + 1/60)$, the probability that X, truncated to the nearest *second*, is 45. How is the probability related to $f(45)$?

6. Compute and interpret the ratio of the two previous probabilities.

7. Consider the probability that an employee arrives within 1 *millisecond* after 8:15. Compute $\mathrm{P}(15 < X < 15 + 1/60000)$, the probability that X, truncated to the nearest *millisecond*, is 15. (The measurement units of X are minutes, so $1/60000$ minutes is 1 millisecond.) How is the probability related to $f(15)$?

8. Consider the probability that an employee arrives within 1 *millisecond* after 8:45. Compute $\mathrm{P}(45 < X < 45 + 1/60000)$, the probability that X, truncated to the nearest *millisecond*, is 45. How is the probability related to $f(45)$?

9. Compute and interpret the ratio of the two previous probabilities.

10. Compute and interpret $\mathrm{P}(X = 15)$.

11. Compute and interpret $\mathrm{P}(X = 45)$.

12. In practical terms, is any employee more likely to arrive "at 8:15" or "at 8:45"? Explain.

- **The probability that a continuous random variable X equals any particular value is 0**. That is, if X is continuous then $\mathrm{P}(X = x) = 0$ for all x.
- For a *continuous* random variable X, $\mathrm{P}(X \le x) = \mathrm{P}(X < x)$.
  - Be careful: for a *discrete* random variable X, $\mathrm{P}(X \le x)$ can be different than $\mathrm{P}(X < x)$.
- Even though any specific value of a continuous random variable has probability 0, *intervals* still can have positive probability.
- For continuous random variables, it doesn't really make sense to talk about the probability that the random value is *equal to* a particular value. In practical applications involving continuous random variables, "equal to" really means "close to", and "close to" probabilities correspond to intervals which can have positive probability.

- The density $f_X(x)$ at value x is *not* a probability.
- Rather, the density $f_X(x)$ at value x is related to the probability that the RV X takes a value "close to x" in the following sense

$$\mathrm{P}\left(x - \frac{\epsilon}{2} \le X \le x + \frac{\epsilon}{2}\right) \approx f_X(x)\epsilon, \qquad \text{for small } \epsilon$$

- The quantity ϵ is a small number that represents the desired degree of precision. For example, rounding to two decimal places corresponds to $\epsilon = 0.01$.
- Roughly,

$$\mathrm{P}(X \text{ is close to } x) \approx f_X(x) \times (\text{what counts as close to})$$

- What's important about a pdf is *relative* heights. For example, if $f_X(x_2) = 2f_X(x_1)$ then X is roughly "twice as likely to be close to x_2 than to be close to x_1" in the above sense.

$$\frac{f_X(x_2)}{f_X(x_1)} = \frac{f_X(x_2)\epsilon}{f_X(x_1)\epsilon} \approx \frac{\mathrm{P}\left(x_2 - \frac{\epsilon}{2} \le X \le x_2 + \frac{\epsilon}{2}\right)}{\mathrm{P}\left(x_1 - \frac{\epsilon}{2} \le X \le x_1 + \frac{\epsilon}{2}\right)} \approx \frac{\mathrm{P}(X \text{ is close to } x_2)}{\mathrm{P}(X \text{ is close to } x_1)}$$

Example 12.3. Suppose that X is a claim amount in excess of the decuctible for a car insurance policy that incurs a claim, measured in thousands of dollars. Assume that X has pdf

$$f_X(x) = (1/4.3)e^{-x/4.3}, \qquad x > 0$$

1. Sketch the pdf of X and interpret, roughly, its shape in context.

2. Without integrating, approximate the probability that X rounded to the nearest hundred dollars is \$500.

3. Compute and interpret the ratio $f(0.5)/f(5)$.

4. Compute and interpret $\mathrm{P}(X < 5)$.

5. Compute and interpret $\mathrm{P}(X > 10)$.

12.1 Exercises

Exercise 12.1. In a certain population, household income X (\$ thousands) follows the pdf

$$f_X(x) = cx^{-2.5}, \quad x \geq 30$$

for an appropriate constant c. Note that the measurement units are \$ thousands, so 1 represents 1 thousand dollars. (This is not a super realistic example, but just go with it.)

1. What are the possible values of X? Sketch the pdf of X.

2. Without integrating, find the probability that a household has an income, rounded to the nearest thousand dollars, of \$200K.

3. Without integrating, find the probability that a household has an income, rounded to the nearest *hundred* dollars, of \$200K.

4. Without integrating, determine how many times more likely it is for a household to have an income, rounded to the nearest *hundred* dollars, of \$100K than \$200K.

5. Compute the value of c. (For this and the following parts, set up any calculations you would need to perform, but you can use a calculator or software to do the computations.)

6. Compute the probability that a household has an income under \$100K.

7. Compute the probability that a household has an income over \$200K.

8. Find the probability that a household has an income exactly equal to \$200K.

Exercise 12.2. In Example 12.1, assume instead that employees are "equally likely" to arrive at any time between 8am and 9am. Let X be the arrival time of a randomly selected employee, measured in minutes after 8am. We'll assume that X has a "Uniform distribution" on [0, 60].

1. What shape would you want the pdf to have? Sketch that shape and then determine the pdf.

2. Compute and interpret $\mathrm{P}(X < 15)$.

3. Compute and interpret $\mathrm{P}(X > 45)$.

4. How do the two previous parts compare to each other? Why? How do they compare to the corresponding parts in Example 12.1? Why?

5. Compute $\mathrm{P}(X = 15)$.

6. Replace the previous part with a more sensible question, then solve it.

7. Make an educated guess for $\mathrm{E}(X)$.

8. Make an educated guess for $\mathrm{SD}(X)$. (We'll see how to compute expected value and variance for continuous random variables later, but for this example you should be able to make reasonable guesses without doing much computation.)

- A continuous random variable X has a **Uniform distribution** with parameters a and b, with $a < b$, if its probability density function f_X satisfies

$$f_X(x) \propto \text{constant}, \qquad a < x < b \tag{12.4}$$

$$= \frac{1}{b-a}, \qquad a < x < b. \tag{12.5}$$

- If X has a Uniform(a, b) distribution then

$$\mathrm{E}(X) = \frac{a+b}{2} \tag{12.6}$$

$$\mathrm{Var}(X) = \frac{|b-a|^2}{12} \tag{12.7}$$

$$\mathrm{SD}(X) = \frac{|b-a|}{\sqrt{12}} \approx 0.289|b-a| \tag{12.8}$$

13 Percentiles: Cumulative Distribution Functions and Quantile Functions

- The probability density function of a continuous random variable describes which values are more or less likely in the "close to" sense.
- However, the pdf does not provide probabilities directly; rather, it is areas under the pdf which provide probabilities.
- There are many ways of describing distributions. In particular, a distribution is basically just a collection of percentiles.
- Cumulative distribution functions and quantile functions describe distributions by specifying all the percentiles.
- Roughly,
  - The *cumulative distribution function (cdf)* of a random variable fills in the blank for any given x: x is the (blank) percentile.
  - The *quantile function* fills in the following blank for a given $p \in (0, 1)$: the $(100p)$th percentile is (blank).

Example 13.1. Explain what the percentiles mean in the following statements.

- 47cm is 12th percentile of heights for newborn girls.
- 620 is the 80th percentile of SAT Math scores.

- A distribution is characterized by its percentiles.
- Roughly, the value x is the p-th **percentile** (a.k.a. quantile) of a distribution of a random variable X if p percent of values of the variable are less than or equal to x: $\mathrm{P}(X \le x) = p$.
- A few commonly used percentiles
  - First (or lower) quartile is the 25th percentile ($p = 0.25$)
  - Median is the 50th percentile ($p = 0.50$)

- Third (or upper) quartile is the 75th percentile ($p = 0.75$)

Example 13.2. Recall Example 12.1 and Example 12.2 where office arrival times, measured in minutes after 8am, followed this pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

1. Compute $\mathrm{P}(X < 15)$ and phrase the result in terms of a percentile.

2. Compute $\mathrm{P}(X < 45)$ and phrase the result in terms of a percentile.

3. Find an expression for $\mathrm{P}(X \leq x)$ for general x. This function, denoted $F_X(x)$, is called the *cumulative distribution function (cdf)* of X.

4. How does the cdf relate to percentiles?

5. Use the cdf to determine what percentile 30 minutes after 8am is.

6. Compute and interpret $F(40) - F(30)$.

- The *cumulative distribution function (cdf)* of a random variable fills in the blank for any given x: x is the (blank) percentile. That is, for an input x, the cdf outputs $\mathrm{P}(X \le x)$.
- The **cumulative distribution function (cdf)** of a random variable X is the function F_X defined by $F_X(x) = \mathrm{P}(X \le x)$. A cdf is defined for all real numbers x regardless of whether x is a possible value of X.

Example 13.3. Continuing Example 13.2 where X (minutes after 8am) has pdf

$$f_X(x) = (2/3600)x, \qquad 0 < x < 60.$$

and cdf

$$F_X(x) = \begin{cases} 0, & x < 0, \\ (x/60)^2, & 0 < x < 60, \\ 1, & x > 60. \end{cases}$$

1. Find and interpret the 10th percentile.

2. Find and interpret the 90th percentile.

3. Find an equation for the $(100p)$th percentile (e.g., $p = 0.9$ for the 90th percentile).

- The *quantile function* (essentially the inverse cdf) fills in the following blank for a given $p \in (0, 1)$: the $100p$th percentile is (blank).
- For example, evaluating the quantile function at $p = 0.25$ outputs the 25th percentile.
- For a *continuous* random variable X with cdf F_X, the **quantile function** Q_X is the inverse of the cdf, $Q_X(p) = F_X^{-1}(p)$.
- A quantile function basically gives you the same information as a cdf, just in the other direction.

- Both a quantile function and a cdf define a distribution by specifying all the percentiles.

Example 13.4. Continuing Example 13.3 where X (minutes after 8am) has pdf

$$f_X(x) = (2/3600)x, \qquad 0 < x < 60.$$

and cdf

$$F_X(x) = \begin{cases} 0, & x < 0, \\ (x/60)^2, & 0 < x < 60, \\ 1, & x > 60. \end{cases}$$

and quantile function

$$Q(p) = 60\sqrt{p}, \qquad 0 < p < 1.$$

1. Construct a circular spinner for simulating values of X. Version 1: label the values 0, 10, 20, 30, 40, 50, 60. Careful: they will not be equally spaced.

2. Sketch a histogram with equal bin widths—0 to 10, 10 to 20, etc.—that you would expect to see after spinning the version 1 spinner many times. Does it have the proper shape?

3. Construct a circular spinner for simulating values of X. Version 2: label the circular axis so that your spinner has 12 sectors each with probability 1/12.

4. Sketch a histogram that you would expect to see after spinning the version 2 spinner many times. Start by sketching a histogram with *unequal* bin widths, then sketch the correspond equal bin width histogram. Does it have the proper shape?

- The cdf and the quantile function can be used to create a spinner for a distribution.
- A spinner basically describes a distribution by specifying all the percentiles. For example,
  - the 25th percentile goes 25% of the way around ("3 o clock")
  - the 50th percentile goes 50% of the way around ("6 o clock")
  - the 75th percentile goes 75% of the way around ("9 o clock")
- Intervals corresponding to regions of higher density ("more likely" values) are stretched out on the spinner boundary; intervals corresponding regions of lower density ("less likely" values) are shrunk.
- **Universality of the Uniform (or "one spinner to rule them all").** Let F be a cdf and Q its corresponding quantile function. Let U have a Uniform(0, 1) distribution and define the random variable $X = Q(U)$. Then the cdf of X is F.
- Universality of the uniform might look complicated but all it basically says is that you can construct a spinner by putting the 25th percentile 25% of the way around, the 75th percentile 75% of the way around, etc.
- Actually, universality of the uniform says we don't have to create a new spinner. We can just spin the Uniform(0, 1) spinner, with evenly spaced values from 0 to 1, and transform each resulting value by plugging it into the quantile function.

Example 13.5. Continuing Example 12.3. Suppose that X is a claim amount in excess of the decuctible for a car insurance policy that incurs a claim, measured in thousands of dollars. Assume that X has pdf

$$f_X(x) = (1/4.3)e^{-x/4.3}, \qquad x > 0$$

1. Compute $\mathrm{P}(X < 5)$ and phrase the result in terms of a percentile.

2. Find an expression for the CDF of X.

3. Compute the 25th percentile.

4. Find an expression for the quantile function of X.

5. Use the quantile function to find the 75th percentile.

13.1 Exercises

Exercise 13.1. Continuing Exercise 12.1. In a certain population, household income X ($ thousands) follows the pdf

$$f_X(x) = cx^{-2.5}, \quad x \geq 30$$

for an appropriate constant c. Note that the measurement units are $ thousands, so 1 represents 1 thousand dollars. (This is not a super realistic example, but just go with it.)

1. Compute $\mathrm{P}(X \leq 100)$ and phrase the result in terms of a percentile.

2. Find the cdf of X.

3. Find the 10th percentile.

4. Find the quantile function of X.

5. Sketch a circular spinner for simulating values of X.

Exercise 13.2. Let X be the number of heads in 3 flips of a fair coin.

1. Find the pmf of X and sketch a plot of it.

2. Find the cdf of X and sketch a plot of it.

3. Let Y be the number of tails in 3 flips of a fair coin. Find the cdf of Y.

- A cdf is a non-decreasing function: if $x_1 \leq x_2$ then $F_X(x_1) \leq F_X(x_2)$.
- A cdf approaches 0 as the input approaches $-\infty$: $\lim_{x\to-\infty} F_X(x) = 0$
- A cdf approaches 1 as the input approaches ∞: $\lim_{x\to\infty} F_X(x) = 1$
- The cdf of a *discrete* random variable is a step function.
  - The steps occur at the possible values of the random variable.
  - The height of a particular step corresponds to the probability of that value, given by the pmf.
- The cdf of a *continuous* random variable is a continuous function.
  - The cdf of a *continuous* random variable is obtained by integrating the pdf, so

- The pdf of a *continuous* random variable is obtained by differentiating the cdf

$$F_X' = f_X \qquad \text{if } X \text{ is continuous}$$

- For any random variable X with cdf F_X

$$F_X(b) - F_X(a) = \mathrm{P}(a < X \leq b)$$

  Whether the inequalities in the above event are strict ($<$) or not ($\leq$) matters for discrete random variables, but not for continuous.
- Random variables X and Y **have the same distribution** if their cdfs are the same, that is, if $F_X(u) = F_Y(u)$ for all u.
- That is, two random variables have the same distribution if all the percentiles are the same.

14 Expected Values of Continuous Random Variables

Example 14.1. Recall Example 12.1 and Example 12.2 where office arrival times, measured in minutes after 8am, followed this pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

1. Explain how you could use simulation to approximate $\mathrm{E}(X)$.

2. Explain how you could approximate $\mathrm{E}(X)$ as a sum (as if X were a discrete random variable). Hint: recall Example 12.2.

3. How could you obtain a better approximation than the sum in the previous part? What happens in the limit?

4. Compute $\mathrm{E}(X)$ as an integral. Compare to the approximations.

5. Interpret $\mathrm{E}(X) = 40$ in context.

6. Compute and interpret $\mathrm{P}(X \le \mathrm{E}(X))$.

- The **expected value** (a.k.a. *expectation* a.k.a. *mean*), of a random variable X is a number denoted $\mathrm{E}(X)$ representing the probability-weighted average value of X.
- The expected value of a *continuous* random variable with pdf f_X is defined as

$$\mathrm{E}(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

- Replace the generic bounds $(-\infty, \infty)$ with the possible values of the random variable
- Note well that $\mathrm{E}(X)$ represents *a single number*.
- The expected value is the "balance point" (center of gravity) of a distribution.
- The expected value of a random variable X is defined by the probability-weighted average according to the underlying probability measure. But the expected value can also be interpreted as the **long-run average value**, and so can be approximated via simulation.
- Read the symbol $\mathrm{E}(\cdot)$ as
  - Simulate lots of values of what's inside $(\cdot)$
  - Compute the average. This is a "usual" average, regardless of whether the variable is discrete or continuous; just sum all the simulated values and divide by the number of simulated values.

Example 14.2. Continuing Example 14.1 where X has pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

Now we want to compute variance and standard deviation.

1. What is the formula for computing variance?

2. Explain how you could use simulation to approximate $\mathrm{E}(X^2)$.

3. Explain how you could approximate $\mathrm{E}(X^2)$ as a sum (like a discrete random variable).

4. How could you obtain a better approximation than the sum in the previous part? What happens in the limit?

5. Compute $\mathrm{E}(X^2)$ as an integral. (You should get 1800.)

6. Compute $\mathrm{Var}(X)$.

7. Compute and interpret $\mathrm{SD}(X)$.

- The **"law of the unconscious statistician" (LOTUS)** says that the expected value of a transformed random variable can be found without finding the distribution of the transformed random variable, simply by applying the probability weights of the original random variable to the transformed values.

$$\text{Discrete } X \text{ with pmf } p_X: \qquad \mathrm{E}[g(X)] = \sum_x g(x)p_X(x)$$

$$\text{Continuous } X \text{ with pdf } f_X: \qquad \mathrm{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x)dx$$

- LOTUS says we don't have to first find the distribution of $Y = g(X)$ to find $\mathrm{E}[g(X)]$; rather, we just simply apply the transformation g to each possible value x of X and then apply the corresponding weight for x to $g(x)$.
- Whether in the short run or the long run, in general

$$\text{Average of } g(X) \neq g(\text{Average of } X)$$

- In terms of expected values, in general

$$\mathrm{E}(g(X)) \neq g(\mathrm{E}(X))$$

  The left side $\mathrm{E}(g(X))$ represents first transforming the X values and then averaging the transformed values. The right side $g(\mathrm{E}(X))$ represents first averaging the X values and then plugging the average (a single number) into the transformation formula.

Example 14.3. Continuing Example 12.3. Suppose that X is a claim amount in excess of the decuctible for a car insurance policy that incurs a claim, measured in thousands of dollars. Assume that X has pdf

$$f_X(x) = (1/4.3)e^{-x/4.3}, \qquad x > 0$$

1. Explain how you could use simulation to approximate $\mathrm{E}(X)$.

2. Donny Dont says $\mathrm{E}(X) = \int_0^\infty (1/4.3)e^{-x/4.3}dx = 1$. Do you agree?

3. Compute and interpret $\mathrm{E}(X)$.

4. Compute and interpret $\mathrm{P}(X \le \mathrm{E}(X))$.

5. Explain how you could use simulation to approximate $\mathrm{E}(X^2)$.

6. Donny Dont says: "I can just use LOTUS and replace x with x^2, so $\mathrm{E}(X^2)$ is $\int_{-\infty}^{\infty}(1/4.3)x^2 e^{-x^2/4.3}dx$". Do you agree?

7. Compute $\mathrm{E}(X^2)$.

8. Compute $\mathrm{Var}(X)$.

9. Compute and interpret $\mathrm{SD}(X)$.

14.1 Exercises

Exercise 14.1. Continuous Exercise 12.1. In a certain population, household income X (\$ thousands) follows the pdf

$$f_X(x) = cx^{-2.5}, \quad x \geq 30$$

for an appropriate constant c. Note that the measurement units are \$ thousands, so 1 represents 1 thousand dollars. (This is not a super realistic example, but just go with it.)

1. Compute $\mathrm{E}(X)$.

2. Interpret $\mathrm{E}(X)$ in context.

3. Compute and interpret $\mathrm{P}(X \leq \mathrm{E}(X))$.

4. Compute $\mathrm{Var}(X)$.

5. Compute and interpret $\mathrm{SD}(X)$.

15 Exponential Distributions

- Exponential distributions are often used to model the waiting times between events in a random process that occur continuously over time.

Example 15.1. A business is tracking when online orders are placed on their website. Suppose an order just occurred and let X be the time until the next order is placed, measured in hours, including fractions of an hour. Assume that X has an *Exponential distribution with rate parameter 2.5 per hour*, with pdf

$$f_X(x) = 2.5e^{-2.5x}, \qquad x > 0$$

1. Sketch the pdf of X. Roughly, what does this tell you about times between orders?

2. Compute and interpret $\mathrm{P}(X < 0.5)$.

3. Determine the cdf of X.

4. Determine what percentile 2 hours is.

5. Compute and interpret the median of X.

6. Make an educated guess for $\mathrm{E}(X)$. Then compute and interpret $\mathrm{E}(X)$; did your guess work?

7. Compute and interpret $\mathrm{P}(X < \mathrm{E}(X))$.

- A continuous random variable X has an **Exponential distribution** with *rate* parameter $\lambda > 0$ if its pdf is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0, \\ 0, & \text{otherwise} \end{cases}$$

- If X has an Exponential(λ) distribution then

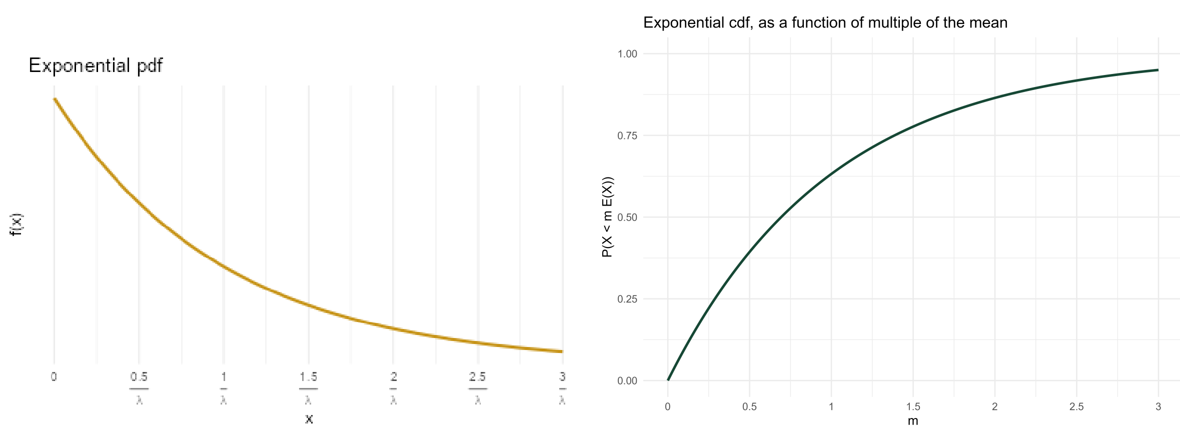
$$\begin{aligned} \text{cdf:} \quad F_X(x) &= 1 - e^{-\lambda x}, \quad x > 0 \\ \mathrm{P}(X > x) &= e^{-\lambda x}, \quad x > 0 \\ \text{quantile:} \quad Q(p) &= -(1/\lambda)\ln(1-p), \quad 0 < p < 1 \\ \mathrm{E}(X) &= \frac{1}{\lambda} \\ \mathrm{Var}(X) &= \frac{1}{\lambda^2} \\ \mathrm{SD}(X) &= \frac{1}{\lambda} \end{aligned}$$

- Percentiles for any Exponential distribution follow a particular pattern in terms of "multiples of the mean":
  - For any $m > 0$, $\mathrm{P}(X \le m\mathrm{E}(X)) = 1 - e^{-m}$.
    - For example, for any Exponential distribution the mean ($m = 1$) is the 63.2rd percentile ($1 - e^{-1} = 0.632$).
  - The pth percentile is $-\ln(1-p)$ times $\mathrm{E}(X)$.

  * For example, for any Exponential distribution the median ($p = 0.5$) is 0.693 times the mean ($-\ln(1 - 0.5) = 0.693$).

- See Table 15.1.

- Exponential distributions are often used to model the *waiting time* in a random process until some event occurs.
  - λ is the average *rate* at which events occur over time (e.g., 2 per hour)
  - $1/\lambda$ is the mean time between events (e.g., 1/2 hour)
- If X has an Exponential(λ) distribution and $a > 0$ is a constant, then aX has an Exponential(λ/a) distribution.
  - If X is measured in hours with rate $\lambda = 2$ per hour and mean 1/2 hour
  - Then $60X$ is measured in minutes with rate 2/60 per minute and mean $60(1/2) = 30$ minutes.
- If X has an Exponential(1) distribution and $\lambda > 0$ is a constant then X/λ has an Exponential(λ) distribution.
- If U has a Uniform(0, 1) distribution then $-\ln(1 - U)$ has Exponential(1) distribution, and $-(1/\lambda)\ln(1 - U)$ has an Exponential(λ) distribution.



(a) PDF (b) CDF

Figure 15.1: A generic Exponential distribution with rate parameter λ and mean $1/\lambda$. In the PDF plot, the x-axis is labeled in measurement units; in the CDF plot, the x-axis is labeled in terms of multiples of the mean.

Example 15.2. Continuing Example 15.1. Let $N(1)$ be the number of orders in a 1 hour period, a discrete random variable. Then the expected value of $N(1)$ is 2.5 orders. If $N(2)$ is the number of orders that happen in a 2 hour period then its expected value is $\mathrm{E}(N(2)) = 5$

Table 15.1: Percentiles for an Exponential distribution in terms of multiples of the mean

Percentile	Relative to mean
10%	0.11 times the mean
25%	0.29 times the mean
39.3%	0.50 times the mean
50%	0.69 times the mean
63.2%	1.00 times the mean
75%	1.39 times the mean
77.7%	1.50 times the mean
86.5%	2.00 times the mean
90%	2.30 times the mean
91.8%	2.50 times the mean
95%	3.00 times the mean
97%	3.50 times the mean
98.2%	4.00 times the mean
98.9%	4.50 times the mean
99.3%	5.00 times the mean

orders. In general, if $N(t)$ is the number of orders in a t hour period its expected value is $\mathrm{E}(N(t)) = 2.5t$ orders.

Assume that, for any t, the discrete random variable $N(t)$ has a Poisson distribution with parameter $2.5t$. Now we'll consider what this assumption implies about the time between orders. Suppose an order just occurred and let T be the time in hours until the next order; T is a continuous random variable.

1. Interpret the event $\{T > 0.5\}$. How can you express this as an equivalent event involving a number of order?

2. Compute and interpret $\mathrm{P}(T > 0.5)$.

3. Compute $\mathrm{P}(T > t)$ as a function of $t > 0$.

4. Identify by name the distribution of T; be sure to specify the values of any relevant parameters.

5. Compute and interpret $\mathrm{E}(T)$.

- Events occur continuously over time according to a **Poisson process** with *rate* $\lambda > 0$ (per unit time) if
  - The number of events that occur in any time interval of length t has a Poisson(λt) distribution.
    * In particular, the distribution does not depend on the starting time or ending time of the interval, just its length.
  - The numbers of events that occur in disjoint intervals of time are independent.
- A random process is a Poisson process with rate λ if and only if
  - it is a counting process, that is, it is a process which counts the number of occurrences of some event over time.
  - the *interarrival times* (times between events) $W_0, W_1, \ldots$ are independent Exponential(λ) random variables

15.1 Exercises

Exercise 15.1. Use the "multiples of the mean rule" to create a circular spinner for simulating values from a generic Exponential distribution.

1. Version 1: label the circular axis in increments of 0.5, 1.0, 1.5, 2.0, 2.5, etc., multiple of the mean. Careful: the values should not be evenly spaced.

2. Sketch a histogram with equal bin widths—0 to 0.5, 0.5 to 1.0, 1.0 to 1.5, etc.—that you would expect to see after spinning the version 1 spinner many times. Does it have an Exponential shape?

3. Version 2: label the circular axis so that your spinner has 10 sectors each with probability 0.1.

4. Sketch a histogram that you would expect to see after spinning the version 2 spinner many times. Start by sketching a histogram with *unequal* bin widths, then sketch the correspond equal bin width histogram. Does it have an Exponential shape?

5. Explain how could you use this spinner (either version) to simulate values from an Exponential distribution with rate parameter 2.5 (like in Example 15.1).

Exercise 15.2. Continuing Example 15.1. Let X be the waiting time (hours) until the next order and assume X has an Exponential(2.5) distribution.

1. Find the conditional probability that the waiting time from now until the next order is greater than 3 hours given that no orders occur in the next hour. Be sure to write a valid probability statement involving X before computing.

2. Compare to $\mathrm{P}(X > 2)$. What do you notice?

- **Memoryless property.** If W has an Exponential(λ) distribution then for any $w, h > 0$

$$\mathrm{P}(W > w + h \,|\, W > w) = \mathrm{P}(W > h)$$

- Given that we have already waited w units of time, the conditional probability that we wait at least an additional h units of time is the same as the unconditional probability that we wait at least h units of time.
- In this sense, the future waiting time does not "remember" how long we have already waited.
- A continuous random variable W has the memoryless property if and only if W has an Exponential distribution. That is, Exponential distributions are the *only* continuous[1] distributions with the memoryless property.
- If W has an Exponential(λ) distribution then the conditional distribution of the *excess* waiting time, $W - w$, given $\{W > w\}$ is Exponential(λ).

[1] Geometric distributions are the only discrete distributions with the discrete analog of the memoryless property.

16 Normal Distributions

- Normal distributions are probably the most important distributions in probability and statistics.
- A continuous random variable X has a **Normal** (a.k.a., Gaussian) distribution with mean $\mu \in (-\infty, \infty)$ and standard deviation $\sigma > 0$ if its pdf is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right), \quad -\infty < x < \infty$$

- If X has a Normal(μ, σ) distribution then

$$\begin{aligned} \mathrm{E}(X) &= \mu \\ \mathrm{SD}(X) &= \sigma \end{aligned}$$

- A Normal density is a particular "bell-shaped" curve which is symmetric about its mean μ. The mean μ is a *location* parameter: μ indicates where the center and peak of the distribution is.
- The standard deviation σ is a *scale* parameter: σ indicates the distance from the mean to where the concavity of the density changes. That is, there are inflection points at $\mu \pm \sigma$.

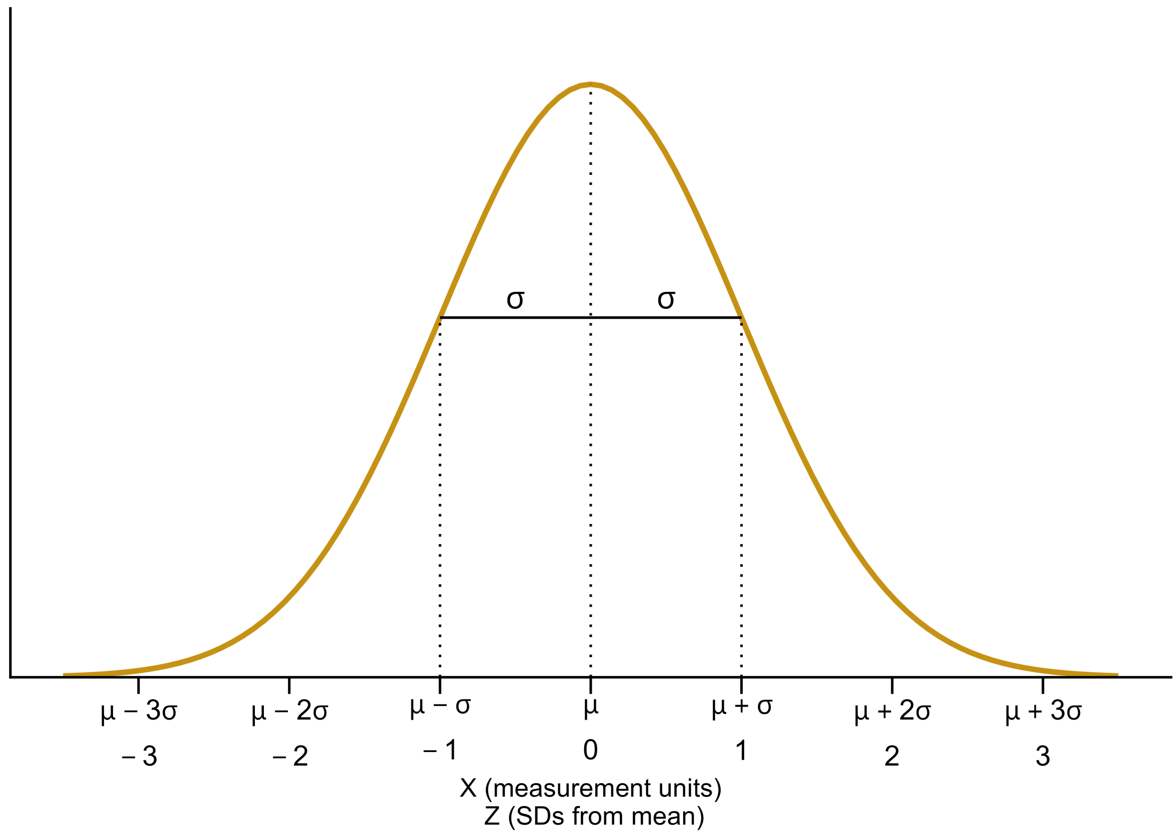


Figure 16.1: A generic Normal distribution

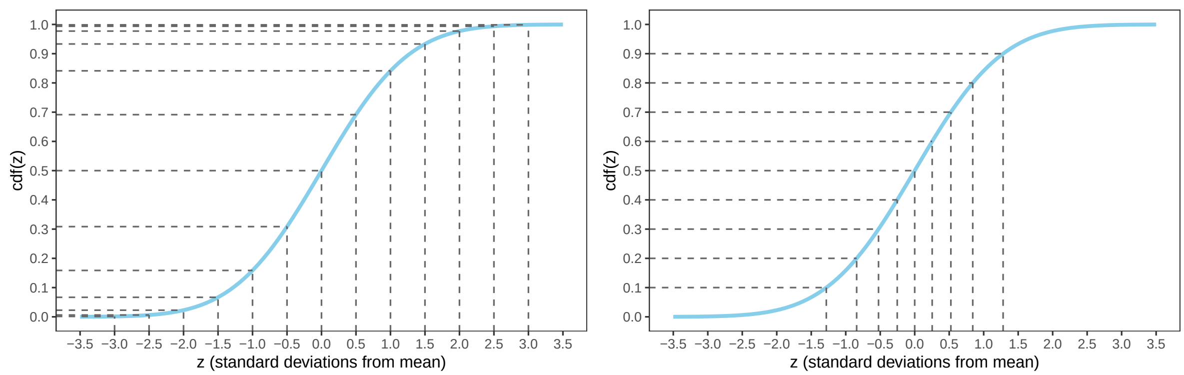
- The key to working with Normal distributions is to work in terms of standard deviations from the mean (that is, standardized values).
  - If X has a Normal(μ, σ) distribution then $Z = \frac{X-\mu}{\sigma}$ has a Normal(0, 1) distribution
  - If Z has a Normal(0, 1) distribution then $X = \mu + \sigma Z$ has a Normal(μ, σ) distribution
- Any Normal distribution follows an "empirical rule" which relates percentiles to standard deviation from the mean and defines the particular bell shape.
- A continuous random variable Z has a **Standard Normal** (a.k.a. Normal(0, 1)) distribution if its pdf is

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty,$$

- The cdf is a Normal(0, 1) distribution is

$$\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}, \quad -\infty < z < \infty,$$

but integrals of this type do not have a closed form and must be evaluated with software



(a) Percentiles highlighted in 0.5 increments of standard deviations from the mean

(b) Deciles highligted in terms of standard deviations from the mean

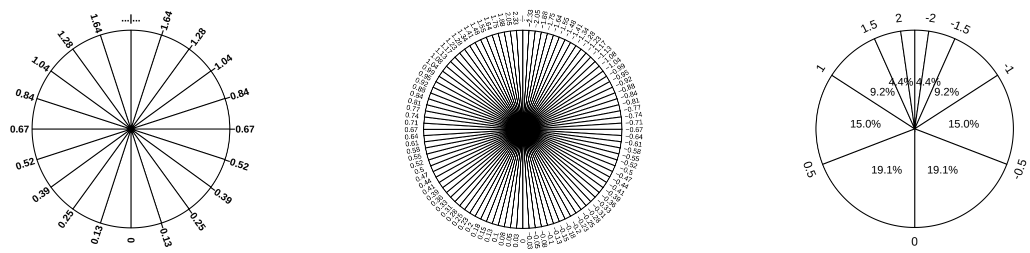
Figure 16.2: The Normal(0, 1) cdf. Provides the cdf for any Normal distribution if the variable is measured in *standard deviations from the mean*. It's the same cdf in both figures just with different values highlighted.

Table 16.1: Select percentiles for a Normal distribution in terms of standard deviations from the mean

(a) In 0.5 increments of standard deviations from the mean

(b) Deciles and quantiles as standard deviations from the mean

Percentile	Standard deviations from the mean	Percentile	Standard deviations from the mean
0.02%	-3.5	1%	-2.33
0.1%	-3.0	5%	-1.64
0.6%	-2.5	10%	-1.28
2.3%	-2.0	20%	-0.84
6.7%	-1.5	25%	-0.67
15.9%	-1.0	30%	-0.52
30.9%	-0.5	40%	-0.25
50%	0.0	50%	0.00
69.1%	0.5	60%	0.25
84.1%	1.0	70%	0.52
93.3%	1.5	75%	0.67
97.7%	2.0	80%	0.84
99.4%	2.5	90%	1.28
99.9%	3.0	95%	1.64
99.98%	3.5	99%	2.33



(a) Each sector represents an interval with 5% probability

(b) Each sector represents an interval with 1% probability

(c) Values labeled in increments of 0.5 standard deviations from the mean

Figure 16.3: A continuous *Normal(0, 1)* spinner. The same spinner is displayed in all figures, with different features highlighted. This spinner can be used to simulate values from any Normal distribution by simulating values in terms of standard deviations from the mean.

- Figure 16.2 defines a Normal distribution by specifying all of its percentiles.
- The empirical rule is often described in terms of "within [blank] standard deviations of the mean" as in the following:
  - 38% of values are within 0.5 standard deviations of the mean
  - 50% of values are withing 0.67 standard deviations of the mean
  - 68% of values are within 1 standard deviation of the mean
  - 87% of values are within 1.5 standard deviations of the mean
  - 95% of values are within 2 standard deviations of the mean
  - 99% of values are within 2.6 standard deviations of the mean
  - 99.7% of values are within 3 standard deviations of the mean
  - 99.99% of values are within 4 standard deviations of the mean

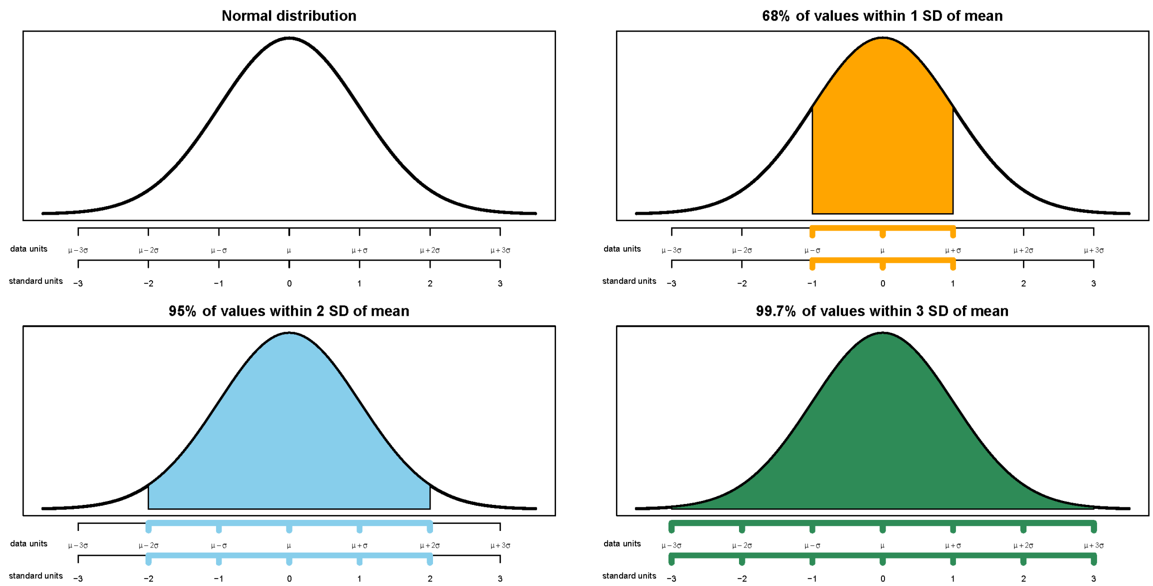


Figure 16.4: Illustrations of the empirical rule for Normal distributions

Example 16.1. The wrapper of a package of candy lists a weight of 47.9 grams. Naturally, the weights of individual packages vary somewhat. Suppose package weights have an approximate Normal distribution with a mean of 49.8 grams and a standard deviation of 1.3 grams.

1. Sketch the distribution of package weights. Carefully label the variable axis. It is helpful to draw two axes: one in the measurement units of the variable, and one in standardized units.

2. Why wouldn't the company print the mean weight of 49.8 grams as the weight on the package?

3. Estimate $\mathrm{P}(X < 47.9)$ and interpret the value in context.

4. Estimate $\mathrm{P}(47.9 < X < 53.0)$ and interpret the value in context.

5. Suppose that the company only wants 1% of packages to be underweight. Find the weight that must be printed on the packages.

6. Find the 99th percentile of package weights.

16.1 Exercises

Exercise 16.1. A bank uses an applicant's score on some criteria to decide whether or not to approve their loan application. Based on past history, the bank has determined that

- Scores for those who repay the loan follow a Normal distribution with mean 60 and SD 8
- Scores for those who do NOT repay the loan follow a Normal distribution with mean 40 and SD 12

When someone applies for a loan the bank does not know whether the applicant will eventually repay the loan. How can the bank use the applicant's score to decide whether or not to approve the loan?

1. Draw sketches of these two normal curves on the same axis.

2. Suggest a general form for the decision rule, based on an applicant's score, for deciding whether or not to approve the applicant's loan. ("Approve the loan if the score…") Just give the general idea; you're not computing anything yet.

3. Describe in words the two kinds of errors the bank could make in this situation.

4. Determine the probability that the bank incorrectly rejects the application of someone who would repay. Shade in the corresponding region under the Normal curve, and interpret this probability.

5. Determine the probability that the bank incorrectly approves the application of someone who would NOT repay. Shade in the corresponding region under the Normal curve, and interpret this probability.

6. In which direction—smaller or larger—would you need to change the decision rule's cutoff value in order to decrease the probability that an applicant who would repay the loan is incorrectly rejected? Would the probability of the other kind of error—incorrectly approving a loan for an applicant who would not repay it—increase or decrease with this new cutoff value?

7. Determine the cutoff value needed to decrease the probability that an applicant who would repay the loan is incorrectly rejected to 0.05.

8. Determine the other error probability with this new cut-off rule.

9. Now repeat the two previous parts with the goal of decreasing to 0.05 the probability of incorrectly approving an applicant who would not repay the loan.

10. If you consider the two kinds of errors to be equally serious, how might you decide which of the three decision rules considered so far is the best?

11. Which error do you think the bank would consider more serious? In which direction would that move the threshold?

- Making decisions/predictions based on data often involves trade-offs.
- Decreasing the probability of one kind of error often comes at the expense of increasing the probability of another kind of error.

17 Joint and Conditional Distributions

- The *joint distribution* of random variables X and Y is a probability distribution on (x, y) *pairs*, and describes how the values of X and Y vary together or jointly.
- Marginal distributions can be obtained from a joint distribution by "stacking"/"collapsing"/"aggregating" out the other variable.
- **In general, marginal distributions alone are not enough to determine a joint distribution.** (The exception is when random variables are *independent*.)

Example 17.1. Roll a fair four-sided die twice. Let X be the sum of the two dice, and let Y be the larger of the two rolls (or the common value if both rolls are the same). Recall Table 6.1.

1. Compute and interpret $p_{X,Y}(5, 3) = \mathrm{P}(X = 5, Y = 3)$.

2. Construct a "flat" table displaying the distribution of (X, Y) pairs, with one pair in each row.

3. Construct a two-way displaying the joint distribution on X and Y.

Table 17.1: Two-way table representing the joint distribution of the sum (X) and larger (Y) of two rolls of a four-sided die.

$x \setminus y$	1	2	3	4
2	1/16	0	0	0

3	0	2/16	0	0
4	0	1/16	2/16	0
5	0	0	2/16	2/16
6	0	0	1/16	2/16
7	0	0	0	2/16
8	0	0	0	1/16

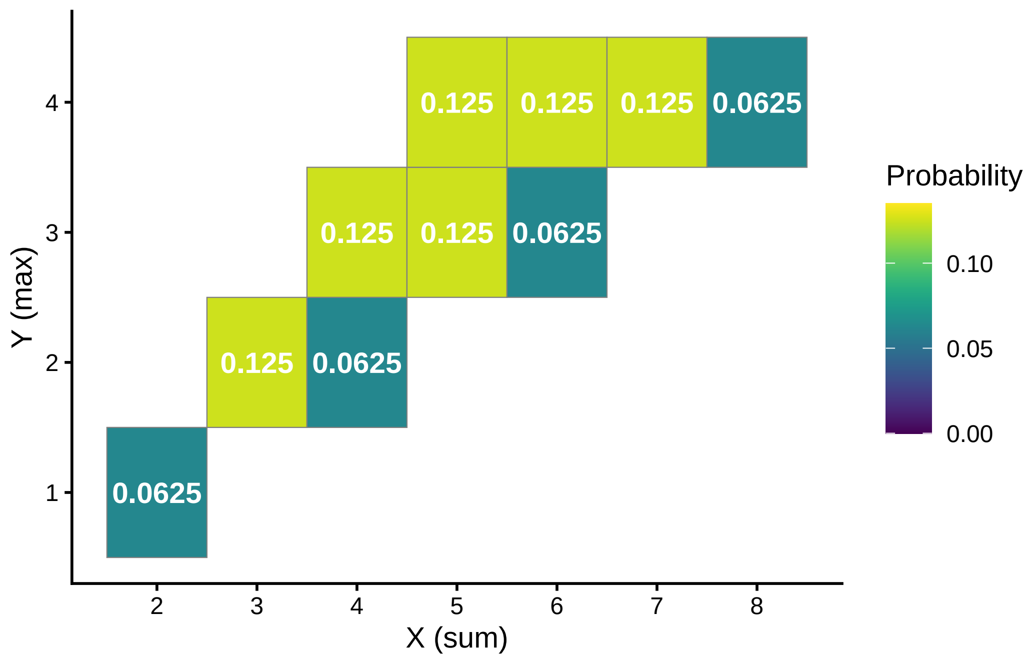


Figure 17.1: Tile plot representing the joint distribution of the sum (X) and larger (Y) of two rolls of a four-sided die.

- The **conditional distribution of** Y given $X = x$ is the distribution of Y values over only those outcomes for which $X = x$. It is a distribution on values of Y only; treat x as a fixed constant when conditioning on the event $\{X = x\}$.
- Conditional distributions can be obtained from a joint distribution by *slicing and renormalizing*. The conditional distribution of Y given $X = x$, where x represents a particular number, can be thought of as:
  - the slice of the joint distribution corresponding to $X = x$, a distribution on values of Y alone with $X = x$ fixed
  - renormalized so that the slice accounts for 100% of the probability over the values of Y

- The *shape* of the conditional distribution of Y given $X =$ is determined by the shape of the slice of the joint distribution over values of Y for the fixed x.
- For each fixed x, the conditional distribution of Y given $X = x$ is a different distribution on values of the random variable Y. There is not one "conditional distribution of Y given X", but rather a *family of conditional distributions of* Y given different values of X.
- Each conditional distribution is a distribution, so we can summarize its characteristics like mean and standard deviation. The conditional mean and standard deviation of Y given $X = x$ represent, respectively, the long run average and variability of values of Y over only (X, Y) pairs with $X = x$.
- Since each value of x typically corresponds to a different conditional distribution of Y given $X = x$, the conditional mean and standard deviation will typically be functions of x.

Example 17.2. Continuing Example 17.1.

1. Make a table representing the conditional distribution of X given $Y = 4$.

2. Make a table representing the conditional distribution of X given $Y = 3$.

3. Make a table representing the conditional distribution of X given $Y = 2$.

4. Make a table representing the conditional distribution of X given $Y = 1$.

- The **conditional expected value** (a.k.a. *conditional expectation* a.k.a. *conditional mean*), of a random variable Y given the event $\{X = x\}$, defined on a probability space with measure P, is a *number* denoted $\mathrm{E}(Y|X = x)$ representing the probability-weighted

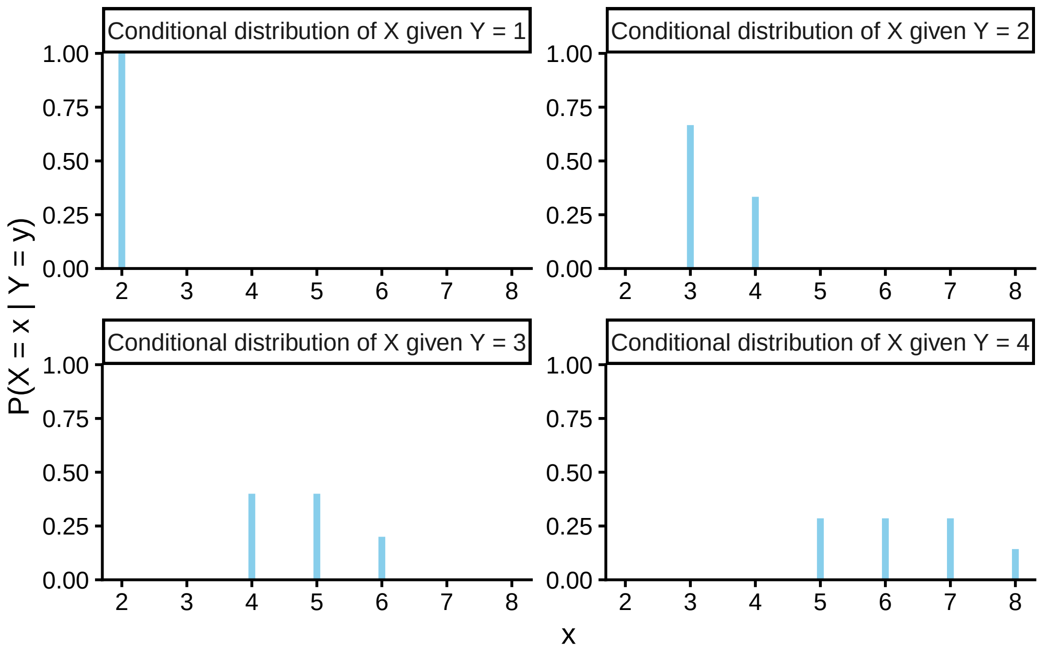


Figure 17.2: Impulse plots representing the family of conditional distributions of X given Y for the dice rolling example. Each plot represents a conditional distribution of X given $Y = y$ for a particular value of $y = 1, 2, 3, 4$.

average value of Y, where the weights are determined by the conditional distribution of Y given $X = x$.

$$\text{Discrete } X, Y \text{ with conditional pmf } p_{Y|X}: \qquad \mathrm{E}(Y|X = x) = \sum_y y p_{Y|X}(y|x)$$

- Remember, when conditioning on $X = x$, x is treated as a fixed constant. The conditional expected value $\mathrm{E}(Y|X = x)$ is a *number* representing the mean of the conditional distribution of Y given $X = x$.
- The conditional expected value $\mathrm{E}(Y|X = x)$ is the long run average value of Y over only those outcomes for which $X = x$.
- To approximate $\mathrm{E}(Y|X = x)$, simulate many (X, Y) pairs, discard the pairs for which $X \neq x$, and average the Y values for the pairs that remain.

Example 17.3. Continuing Example 17.2.

1. Compute and interpret $\mathrm{E}(X|Y = 4)$.

2. Without computing, will $\mathrm{E}(X|Y = 3)$ be greater than, less than, or equal to $\mathrm{E}(X|Y = 4)$?

- The **joint probability mass function (pmf)** of two *discrete* random variables (X, Y) defined on a probability space is the function defined by

$$p_{X,Y}(x, y) = \mathrm{P}(X = x, Y = y) \qquad \text{for all } x, y$$

- Remember to specify the possible (x, y) pairs when defining a joint pmf.

Example 17.4. Let X be the number of home runs hit by the home team, and Y the number of home runs hit by the away team in a randomly selected Major League Baseball game. Suppose that X and Y have joint pmf

$$p_{X,Y}(x, y) = \begin{cases} e^{-2.4}\frac{1.21^x 1.19^y}{x!y!}, & x = 0, 1, 2, \ldots; y = 0, 1, 2, \ldots, \\ 0, & \text{otherwise.} \end{cases}$$

1. Compute and interpret $\mathrm{P}(X = 2, Y = 1)$.

2. Construct a two-way table representation of the joint pmf (you can use software or a spreadsheet).

3. Compute and interpret $\mathrm{P}(X \leq 3, Y \leq 3)$.

4. Compute and interpret $\mathrm{P}(X + Y = 3)$.

5. Compute and interpret $\mathrm{P}(X = Y)$.

6. Compute and interpret $\mathrm{P}(X = 2)$.

7. Compute and interpret $\mathrm{P}(Y = 1)$.

- Two random variables X and Y defined on a probability space with probability measure P are **independent** if $\mathrm{P}(X \le x, Y \le y) = \mathrm{P}(X \le x)\mathrm{P}(Y \le y)$ for all x, y. That is, two random variables are independent if their joint cdf is the product of their marginal cdfs.
- Random variables X and Y are independent if and only if the joint distribution factors into the product of the marginal distributions. The definition is in terms of cdfs, but analogous statements are true for pmfs and pdfs.

$$\begin{aligned} &\text{Discrete RVs } X \text{ and } Y \text{ are independent} \\ &\iff p_{X,Y}(x,y) = p_X(x)p_Y(y) \qquad \text{for all } x, y \end{aligned}$$

$$\begin{aligned} &\text{Continuous RVs } X \text{ and } Y \text{ are independent} \\ &\iff f_{X,Y}(x,y) = f_X(x)f_Y(y) \qquad \text{for all } x, y \end{aligned}$$

17.1 Exercises

Exercise 17.1. Continuing Example 17.1.

1. Starting with the two-way table representing the joint distribution of X and Y, how could you obtain $\mathrm{P}(X = 5)$?

2. Starting with the two-way table, how could you obtain the marginal distribution of X? of Y?

3. Starting with the marginal distribution of X and the marginal distribution of Y, could you necessarily construct the two-way table of the joint distribution? Explain.

4. Remember that you constructed a spinner corresponding to the marginal distribution of Y in Example 6.1. Suppose you also construct a separate spinner corresponding to the marginal distribution of X. Donny Don't says: "to simulate an (X, Y) pair I can just

spin the X spinner once to get X and spin the Y spinner once to get Y." Do you agree? Explain.

- Recall that we can obtain marginal distributions from a joint distribution.
- *Marginal pmfs* are determined by the joint pmf via the law of total probability.
- If we imagine a plot with blocks whose heights represent the joint probabilities, the marginal probability of a particular value of one variable can be obtained by "stacking" all the blocks corresponding to that value.
- For discrete random variables X and Y:

$$
\begin{aligned}
p_X(x) &= \sum_y p_{X,Y}(x,y) && \text{a function of } x \text{ only}\\
p_Y(y) &= \sum_x p_{X,Y}(x,y) && \text{a function of } y \text{ only}
\end{aligned}
$$

Exercise 17.2. Continuing Exercise 17.1.

1. Construct a spinner that could be used to simulate (X, Y) pairs with the proper joint distribution.

2. How could you use Example 17.2 to construct a series of spinners to simulate (X, Y) pairs with the proper joint distribution? (Hint: consider simulating Y first; then how would you simulate X?)

3. Explain how you could use simulation to approximate the conditional distribution of Y given $X = 6$.

4. Explain how you could use simulation to approximate $\mathrm{E}(Y|X=6)$.

- Rather than directly simulating from a joint distribution, we can simulate an (X, Y) pair in two stages:
  - Simulate a value of X from its marginal distribution. Call the simulated value x.
  - Given x, simulate a value of Y from the conditional distribution of Y given $X = x$. There will be a different distribution (spinner) for each possible value of x.
- This "marginal then conditional" process is essentially implementing the multiplication rule

$$\text{joint} = \text{conditional} \times \text{marginal}$$

- In many problems a joint distribution is naturally described by specifying the marginal distribution of X and the family of conditional distributions of Y given values of X. (This is sometimes called a "hierarchical model".)

18 Covariance and Correlation

Quantities like expected value and variance summarize characteristics of the *marginal* distribution of a single random variable. When there are multiple random variables their *joint* distribution is of interest. *Covariance* and *correlation* summarize a characteristic of the joint distribution of two random variables, namely, the degree to which they "co-deviate from the their respective means".

Example 18.1. We'll motivate the calculation of covariance by considering a small example. Say we have data on the sale price X (measured in millions of dollars) and the living area Y (measured in thousands of square feet) for a small sample of 10 houses.

repetition	X	Y	X deviation	Y deviation	Product
1	0.28	1.12			
2	0.63	1.05			
3	0.76	1.67			
4	0.56	1.48			
5	0.36	0.43			
6	0.89	2.50			
7	0.47	1.71			
8	0.82	1.88			
9	0.71	0.85			
10	0.75	1.90			
Sum	6.23	14.59			
Average	0.62	1.46			

1. Compute the average of the X values and the average of the Y values.
2. Compute the deviation from its mean for each X value, and the deviation from its mean for each Y value.
3. Compute the product of the paired deviations.
4. Compute the average of the products of the deviations. This is called the covariance. What are the measurement units?

- The **covariance** between two random variables X and Y is

$$\begin{aligned}\mathrm{Cov}(X,Y) &= \mathrm{E}\left[(X-\mathrm{E}[X])\,(Y-\mathrm{E}[Y])\right]\\ &= \mathrm{E}(XY)-\mathrm{E}(X)\mathrm{E}(Y)\end{aligned}$$

- Covariance is the long run average of the products of the paired deviations from the mean
- Covariance is the expected value of the product minus the product of expected values.

Example 18.2. Roll a fair four-sided die twice and let X be the sum of the two rolls, and Y the larger of the two rolls. Recall Example 17.1 and Table 17.1.

1. Will $\mathrm{Cov}(X,Y)$ be positive, negative, or close to 0? Explain without computing.

2. Compute $\mathrm{Cov}(X,Y)$ using the definition.

3. Compute $\mathrm{Cov}(X,Y)$ using the "shortcut" formula.

- $\mathrm{Cov}(X,Y) > 0$ (positive association): above average values of X tend to be associated with above average values of Y
- $\mathrm{Cov}(X,Y) < 0$ (negative association): above average values of X tend to be associated with below average values of Y
- $\mathrm{Cov}(X,Y) = 0$ indicates that the random variables are **uncorrelated**: there is no overall positive or negative association. But be careful: if X and Y are uncorrelated there can still be a relationship between X and Y; there is just no overall positive or negative association.. We will see examples that demonstrate that being uncorrelated does not necessarily imply that random variables are independent.

Example 18.3. We'll motivate the calculation of correlation by considering a small example. Continuing Example 18.1, say we have data on the sale price X (measured in millions of dollars) and the living area Y (measured in thousands of square feet) for a small sample of 10 houses.

repetition	X	Y	Standardized X	Standardized Y	Product
1	0.28	1.12			
2	0.63	1.05			
3	0.76	1.67			
4	0.56	1.48			
5	0.36	0.43			
6	0.89	2.50			
7	0.47	1.71			
8	0.82	1.88			
9	0.71	0.85			
10	0.75	1.90			
Sum	6.23	14.59	0	0	
Average	0.62	1.46	0	0	
SD	0.19	0.57	1	1	NA

1. Compute the average and SD of the X values and the average and SD of the Y values.
2. Compute the standardized value of each X value, and the standardized value of each Y value.
3. Compute the product of the paired standardized values.
4. Compute the average of the products of the standardized values. This is called the correlation.
5. Divide the covariance from Example 18.1 by the product of the two standard deviations. What do you notice?

- The **correlation (coefficient)** between random variables X and Y is

$$\begin{aligned}\text{Corr}(X,Y) &= \text{Cov}\left(\frac{X-\text{E}(X)}{\text{SD}(X)}, \frac{Y-\text{E}(Y)}{\text{SD}(Y)}\right)\\ &= \frac{\text{Cov}(X,Y)}{\text{SD}(X)\text{SD}(Y)}\end{aligned}$$

- The correlation for two random variables is the covariance between the corresponding standardized random variables. Therefore, correlation is a *standardized* measure of the association between two random variables.

- A correlation coefficient has no units and is measured on a universal scale. Regardless of the original measurement units of the random variables X and Y

$$-1 \le \text{Corr}(X,Y) \le 1$$

- $\text{Corr}(X,Y) = 1$ if and only if $Y = aX + b$ for some $a > 0$
- $\text{Corr}(X,Y) = -1$ if and only if $Y = aX + b$ for some $a < 0$
- Therefore, correlation is a standardized measure of the strength of the *linear* association between two random variables.
- Covariance is the correlation times the product of the standard deviations.

$$\text{Cov}(X,Y) = \text{Corr}(X,Y)\text{SD}(X)\text{SD}(Y)$$

18.1 Exercises

Exercise 18.1. Continuing Example 18.2. Let X be the sum of the two rolls, Y the larger of the two rolls, W the number of rolls equal to 4, and Z the number of rolls equal to 1. You should explain your answers without doing any computation.

1. Is $\text{Cov}(X,W)$ positive, negative, or zero? Why?

2. Is $\text{Cov}(X,Z)$ positive, negative, or zero? Why?

3. Let $V = W + Z$; what does this represent? Is $\text{Cov}(X,V)$ positive, negative, or zero? Why?

4. Is $\text{Cov}(W,Z)$ positive, negative, or zero? Why?

Exercise 18.2. What is another name for $\mathrm{Cov}(X, X)$?

Exercise 18.3. Sketch a few scatterplots where X and Y have a correlation of 0, but they are not independent.

Exercise 18.4. Try the Guessing Correlations applet! Since we're interested in the long run, set number of points to 1000. There is nothing to submit for this exercise but play around with the applet to get a feel for the difference between a correlation of 0.8-ish versus 0.5-ish versus 0.2-ish.

19 Linear Combinations of Random Variables

- In general, the distribution of a sum (or linear combination) of random variables can be difficult to obtain and depends on the joint distribution of the random variables.
- But there are some special cases where the distribution of a sum (or linear combination) works out nicely.
- And there are shortcut formulas that allow us to find the expected value and variance of a sum (or linear combination) without first finding its distribution.

Example 19.1. At a small hospital, let X be the number of babies born in a 5 day work week and let Y be the number of babies born on a weekend. Then $W = X + Y$ is the total number of babies born in a week. Assume that

- X and Y are independent
- X has a Poisson(2.1) distribution
- Y has a Poisson(0.4) distribution

Table 19.1 displays the marginal distributions of X and Y. (The probabilities have been rounded a little.)

Table 19.1: Distributions in Example 19.1

x, y	0	1	2	P(X = x)
0				0.122
1				0.257
2				0.270
3				0.189
4				0.099
5				0.042
6				0.021
P(Y = y)	0.670	0.268	0.062	

1. Create a table for the joint distribution of X and Y.

2. Compute and interpret $\mathrm{P}(W = 0)$.

3. Compute and interpret $\mathrm{P}(W = 1)$.

4. Compute and interpret $\mathrm{P}(W = 2)$.

5. Find the distribution of W.

6. Compute and interpret $\mathrm{E}(W)$. How does it relate to $\mathrm{E}(X)$ and $\mathrm{E}(Y)$?

7. Compute $\mathrm{Var}(W)$. How does it relate to $\mathrm{Var}(X)$ and $\mathrm{Var}(Y)$?

- **Poisson aggregation.** If X and Y are independent, X has a Poisson(μ_X) distribution, and Y has a Poisson(μ_Y) distribution, then $X + Y$ has a Poisson($\mu_X + \mu_Y$) distribution.
- If component counts are independent and each has a Poisson distribution, then the total count also has a Poisson distribution.

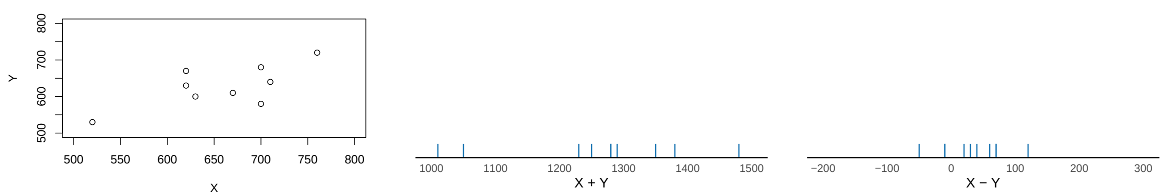
Example 19.2. We'll illustrate some ideas with a small sample of simulated values. Say we have SAT Math scores (X) and Reading scores (Y) for 10 students. We're interested in the sum ($T = X + Y$) and difference (Math minus Reading, $D = X - Y$) of the scores.

The 10 X values are the same in each scenario, and the 10 Y values are the same in each scenario, but the (X, Y) values are paired in different ways:

- Scenario 1: correlation is 0.78
- Scenario 2: correlation is about 0 (actually -0.04)
- Scenario 3: correlation is -0.94

Table 19.2: Example 19.2 scenario 1, correlation is 0.78

Student	X	Y	T	D
1	520	530	1050	-10
2	540	470	1010	70
3	620	670	1290	-50
4	620	630	1250	-10
5	630	600	1230	30
6	670	610	1280	60
7	700	680	1380	20
8	700	580	1280	120
9	710	640	1350	70
10	760	720	1480	40
Mean	647	613	1260	34
SD	72	70	134	47
Corr(X, Y)	0.78			

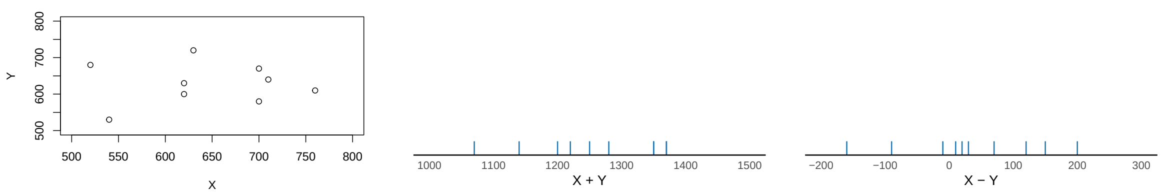


(a) Scatterplot of (X, Y) (b) Rug plot of $X + Y$ (c) Rug plot of $X - Y$

Figure 19.1: Example 19.2 scenario 1, correlation is 0.78

Table 19.3: Example 19.2 scenario 2, correlation is about 0

Student	X	Y	T	D
1	520	680	1200	-160
2	540	530	1070	10
3	620	630	1250	-10
4	620	600	1220	20
5	630	720	1350	-90
6	670	470	1140	200
7	700	670	1370	30
8	700	580	1280	120
9	710	640	1350	70
10	760	610	1370	150
Mean	647	613	1260	34
SD	72	70	98	103
Corr(X, Y)	-0.04			



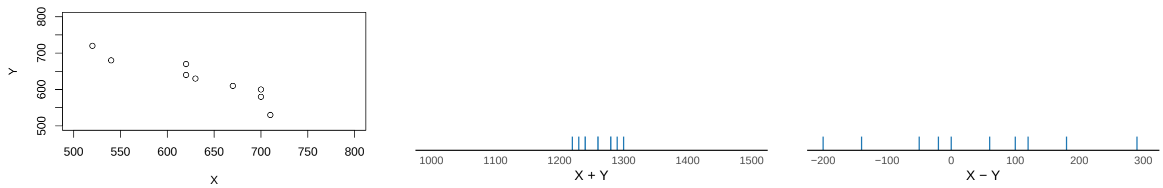
(a) Scatterplot of (X, Y) (b) Rug plot of $X + Y$ (c) Rug plot of $X - Y$

Figure 19.2: Example 19.2 scenario 2, correlation is about 0

Table 19.4: Example 19.2 scenario 3, correlation is -0.94

Student	X	Y	T	D
1	520	720	1240	-200
2	540	680	1220	-140
3	620	670	1290	-50
4	620	640	1260	-20
5	630	630	1260	0
6	670	610	1280	60
7	700	600	1300	100
8	700	580	1280	120
9	710	530	1240	180

Student	X	Y	T	D
10	760	470	1230	290
Mean	647	613	1260	34
SD	72	70	26	140
Corr(X, Y)	-0.94			



(a) Scatterplot of (X, Y) (b) Rug plot of $X + Y$ (c) Rug plot of $X - Y$

Figure 19.3: Example 19.2 scenario 3, correlation is -0.94

In each scenario:

1. Compute the mean of total scores.

2. How does the mean of $X + Y$ relate to the means of X and Y?

3. Compute the mean of the difference in scores (Math – Reading)

4. How does the mean of $X - Y$ relate to the means of X and Y?

- **Linearity of expected value.** For *any* two random variables X and Y,

$$\text{E}(X+Y) = \text{E}(X) + \text{E}(Y)$$

- That is, the expected value of the sum is the sum of expected values, regardless of how the random variables are related.
- Therefore, you only need to know the marginal distributions of X and Y to find the expected value of their sum. (But keep in mind that the *distribution* of $X+Y$ will depend on the joint distribution of X and Y.)
- Whether in the short run or the long run,

$$\text{Average of } X+Y \; = \text{Average of } X + \text{Average of } Y$$

  regardless of the joint distribution of X and Y.
- A **linear combination** of two random variables X and Y is of the form $aX+bY$ where a and b are non-random constants. Combining properties of linear rescaling with linearity of expected value yields the expected value of a linear combination.

$$\text{E}(aX+bY) = a\text{E}(X) + b\text{E}(Y)$$

- For example ($a=1, b=-1$): $\text{E}(X-Y) = \text{E}(X) - \text{E}(Y)$.
- Linearity of expected value extends naturally to more than two random variables.

$$\text{E}(a_1X_1 + \cdots + a_nX_n) = a_1\text{E}(X_1) + \cdots + a_n\text{E}(X_n)$$

Example 19.3. Continuing with the three scenarios in Example 19.2.

In each scenario:

1. Compute the variance of total scores.

2. How does correlation affect $\text{Var}(X+Y)$? In which of the three scenarios is $\text{Var}(X+Y)$ the largest? The smallest? Can you explain why? In which scenario is $\text{Var}(X+Y)$ roughly equal to the sum of $\text{Var}(X)$ and $\text{Var}(Y)$?

3. Compute the variance of the difference in scores (Math - Reading)

4. How does correlation affect $\text{Var}(X-Y)$? In which of the three scenarios is $\text{Var}(X-Y)$ the largest? The smallest? Can you explain why? In which scenario is $\text{Var}(X-Y)$ roughly equal to the *sum* of $\text{Var}(X)$ and $\text{Var}(Y)$?

- Variance of sums and differences of random variables

$$
\begin{aligned}
\text{Var}(X+Y) &= \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X,Y) \\
\text{Var}(X-Y) &= \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X,Y)
\end{aligned}
$$

- If X and Y are uncorrelated:

$$
\begin{aligned}
\text{Var}(X+Y) &= \text{Var}(X) + \text{Var}(Y) \qquad \text{if } X, Y \text{ are uncorrelated} \\
\text{Var}(X-Y) &= \text{Var}(X) + \text{Var}(Y) \qquad \text{if } X, Y \text{ are uncorrelated}
\end{aligned}
$$

- Variance of linear combinations: If a, b, c are non-random constants and X and Y are random variables then

$$
\text{Var}(aX + bY + c) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X,Y)
$$

Example 19.4. Let X be the sales of beer and Y the sales of salty snacks in a week at a certain supermarket, measured in thousands of dollars.
Suppose

- $\text{E}(X) = 22$, $\text{SD}(X) = 6$
- $\text{E}(Y) = 14$, $\text{SD}(Y) = 4$
- $\text{Corr}(X,Y) = 0.55$

1. Compute and intepret $\text{E}(X+Y)$.

2. Compute and intepret $\mathrm{SD}(X+Y)$.

Example 19.5. Let X be the sales of beer and Y the sales of wine in a week at a certain supermarket, measured in thousands of dollars.
Suppose

- $\mathrm{E}(X) = 22$, $\mathrm{SD}(X) = 6$
- $\mathrm{E}(Y) = 14$, $\mathrm{SD}(Y) = 4$
- $\mathrm{Corr}(X, Y) = -0.55$

1. Compute and intepret $\mathrm{E}(X+Y)$.

2. Compute and intepret $\mathrm{SD}(X+Y)$.

19.1 Exercises

Exercise 19.1. Consider a random variable X with $\mathrm{Var}(X) = 1$. What is $\mathrm{Var}(2X)$?

- Walt says: $\mathrm{SD}(2X) = 2\mathrm{SD}(X)$ so $\mathrm{Var}(2X) = 2^2\mathrm{Var}(X) = 4(1) = 4$.
- Jesse says: Variance of a sum is a sum of variances, so $\mathrm{Var}(2X) = \mathrm{Var}(X+X)$ which is equal to $\mathrm{Var}(X) + \mathrm{Var}(X) = 1 + 1 = 2$.

Who is correct? Why is the other wrong?

Exercise 19.2. Percent returns for assets X, Y, and Z follow a joint distribution with

- Mean 10 and standard deviation 15 for asset X,
- Mean 5 and standard deviation 3 for asset Y,
- Correlation of -0.6 between asset X and asset Y
- Asset Z yields a constant return of 1 percent.

An investment portfolio has 60% of its funds in asset X, 30% in asset Y, and 10% in asset Z.

1. Let R be the portfolio return. Express R in terms of X, Y, Z.

2. Compute $\mathrm{E}(R)$.

3. Compute $\mathrm{Cov}(X, Y)$.

4. Compute $\mathrm{Cov}(X, Z)$.

5. Compute $\mathrm{SD}(R)$.

20 Joint Normal Distributions

- Jointly continuous random variables X and Y have a **Bivariate Normal** distribution with parameters μ_X, μ_Y, $\sigma_X > 0$, $\sigma_Y > 0$, and $-1 < \rho < 1$ if the joint pdf is {

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}\exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right]\right)$$

}

- If the pair (X,Y) has a BivariateNormal(μ_X, μ_Y, σ_X, σ_Y, ρ) distribution

$$\begin{aligned}
\mathrm{E}(X) &= \mu_X\\
\mathrm{E}(Y) &= \mu_Y\\
\mathrm{SD}(X) &= \sigma_X\\
\mathrm{SD}(Y) &= \sigma_Y\\
\mathrm{Corr}(X,Y) &= \rho
\end{aligned}$$

- A Bivariate Normal Density has elliptical contours. For each height $c > 0$ the set $\{(x,y) : f_{X,Y}(x,y) = c\}$ is an ellipse. The density decreases as (x,y) moves away from (μ_X, μ_Y), most steeply along the minor axis of the ellipse, and least steeply along the major of the ellipse.
- A scatterplot of (x,y) pairs generated from a Bivariate Normal distribution will have a rough linear association and the cloud of points will resemble an ellipse.
- If X and Y have a Bivariate Normal distribution, then the marginal distributions are also Normal: X has a Normal(μ_X, σ_X) distribution and Y has a Normal(μ_Y, σ_Y).
- If X and Y have a Bivariate Normal distribution and $\mathrm{Corr}(X,Y) = 0$ then X and Y are independent. (Remember, in general it is possible to have situations where the correlation is 0 but the random variables are not independent.)
- X and Y have a Bivariate Normal distribution if and only if every linear combination of X and Y has a Normal distribution. That is, X and Y have a Bivariate Normal distribution if and only if $aX + bY + c$ has a Normal distribution for all a, b, c.

Example 20.1. Suppose that SAT Math (M) and Reading (R) scores have (roughly) a Bivariate Normal distribution. Math scores have mean 500 and SD 140, Reading scores have mean 520 and SD 110, and the correlation between scores is 0.6.

1. Compute and interpret $\mathrm{P}(M > 700)$.

2. Compute and interpret $\mathrm{P}(M + R > 1500)$.

3. Compute and interpret $\mathrm{P}(M > R)$.

- If X and Y have a Bivariate Normal distribution then any conditional distribution of one variable given the value of the other is Normal. The conditional distribution of Y given $X = x$ is
$$N\left(\mu_Y + \frac{\rho\sigma_Y}{\sigma_X}(x - \mu_X)\,,\ \sigma_Y\sqrt{1-\rho^2}\right)$$
- The conditional expected value of Y given $X = x$ is a *linear* function of x, called the *regression line of Y on X*:
$$\mathrm{E}(Y|X = x) = \mu_Y + \rho\sigma_Y\left(\frac{x - \mu_X}{\sigma_X}\right)$$
  - The regression line passes through the point of means (μ_X, μ_Y) and has slope
$$\frac{\rho\sigma_Y}{\sigma_X}$$
  - The regression line estimates that if the given x value is z SDs above the mean of X, then the corresponding Y values will be, on average, ρz SDs away from the mean of Y
$$\frac{\mathrm{E}(Y|X = x) - \mu_Y}{\sigma_Y} = \rho\left(\frac{x - \mu_X}{\sigma_X}\right)$$
  - Since $|\rho| \le 1$, for a given x value the corresponding Y values will be, on average, relatively closer to the mean of Y than the given x value is to the mean of X. This is known as *regression to the mean*.
- For Bivariate Normal distributions, the conditional variance of Y given $X = x$ does not depend on x:
$$\mathrm{SD}(Y|X = x) = \sigma_Y\sqrt{1-\rho^2}$$

Example 20.2. Continuing Example 20.1. Suppose that SAT Math (M) and Reading (R) scores have (roughly) a Bivariate Normal distribution. Math scores have mean 500 and SD 140, Reading scores have mean 520 and SD 110, and the correlation between scores is 0.6. Now we'll consider the conditional distribution of Math scores given a Reading score of 700.

1. Compute and interpret $\mathrm{E}(M|R = 700)$.

2. Compute and interpret $\mathrm{SD}(M|R = 700)$.

3. Compute and interpret $\mathrm{P}(M > 700|R = 700)$.

20.1 Exercises

Exercise 20.1. The moral of this exercise is more important than the details, so just explain intuitively without doing a lot of computations.

Suppose that X has a Normal(0, 1) distribution. Flip a coin and if it lands on heads let $Y = X$, and if tails let $Y = -X$. For example, say you spin a Normal(0, 1) spinner; the result would be X. If a spin lands on 1.234, then Y is either 1.234 (if the coin lands on heads) or -1.234 (if tails).

1. Sketch a scatter plot of many (X, Y) pairs. (Remember: either $Y = X$ or $Y = -X$.)

2. Make an educated guess for the distribution of Y. (Hint:you know the distribution of X; what is the distribution of $-X$?)

3. Is the conditional distribution of Y given $X = 1.234$ Normal?

4. Make an educated guess for $\mathrm{Corr}(X, Y)$.

5. Are X and Y independent?

6. Does $X + Y$ have a Normal distribution? (Hint: what is $\mathrm{P}(X + Y = 0)$ in this exercise and what would it need to be if $X + Y$ had a Normal distribution?)

7. Moral: In this exercise X has a Normal distribution and Y has a Normal distribution, but does the pair (X, Y) have a Normal distribution?

- If the pair (X, Y) has a joint Normal distribution then each of X and Y has a Normal distribution.
- But the converse is not true. That is, if each of X and Y has a Normal distribution, it is not necessarily true that the pair (X, Y) has a joint Normal distribution
- However, if X and Y are *independent* and each of X and Y has a Normal distribution, then the pair (X, Y) has a joint Normal distribution. (But joint Normal is much more general that two independent Normal random variables.)

21 Statistical Estimation: From Probability to Statistics

Example 21.1. Consider the following two scenarios. Which is a *probability* scenario and which is a *statistics* scenario? Discuss some features of each scenario

- Assume that the number of cars sold daily at a certain used car dealership follows a Poisson distribution with parameter 2.3, independently from day to day. How likely is it for the dealership to sell more than 100 cars in a month?
- The car dealership observes how many cars they sell each day for the next month. Given their data, is it reasonable to assume daily car sales follow a Poisson distribution? If so, how could they use the observed data to estimate the Poisson parameter?

- **Probability:** Assume a model for a random (uncertain) process and evaluate probabilities of potential outcomes or events — what might the data look like?
  - In probability, a model is assumed and parameters are assumed to be known, but the data is unknown.
  - In probability, we typically call the model a "distribution" and the data "random variables".
- **Statistical inference:** Observe sample data and make conclusions about the process that generated the data
  - In statistics, the data is observed but parameters are unknown.
  - In statistics, we typically call the model the "population" and the data the "sample".
- *Statistical inference* involves using sample data to make conclusions about the population.
  - **Point estimation:** Estimate an unknown population parameter with a single number based on sample data

  - **Interval estimation:** Estimate an unknown population parameter with a plausible range of values based on sample data
  - **Hypothesis testing:** Assess the evidence that the sample data provides regarding a particular claim about unknown population parameters.

- The *population distribution* is the distribution of individual values of the variable of interest. A *parameter*, generically denoted θ, is a number that summarizes the population distribution.
- The **population distribution**, denoted p_θ or $p_\theta(x)$, is a probability model for the distribution of values of a variable[1] X over all *individuals* in the population.
  - If the variable X is discrete, then p_θ represents a probability mass function (pmf): $p_\theta(x) = \mathrm{P}(X = x)$
  - If the variable X is continuous, then p_θ represents a probability density function (pdf): $p_\theta(x) \approx \frac{1}{\epsilon}\mathrm{P}(x - \epsilon/2 < X < x + \epsilon/2)$
- A **parameter**, generically denoted θ, is a characteristic of the population distribution, often computed as an expected value of some function of X.
  - In some cases θ represents a vector of multiple parameters.
  - For example, if p_θ represents a Normal distribution, then θ might represent the pair $\theta \equiv (\mu, \sigma)$ consisting of the population mean and the population standard deviation.
- A parameter is a *fixed number* but its value is *unknown*[2].
- Many statistical studies involve *random sampling* from a population.
- A **(simple) random sample** of size n is a collection of random variables $X_1, \dots, X_n$ that are *independent* and *identically distributed* (i.i.d.)
  - Identically distributed — each individual X_i has the same marginal distribution — the population distribution — since they are sampled from the same population:

$$X_i \sim p_\theta \quad \text{for all } i = 1, \dots, n$$

[1]Source: http://www.nejm.org/doi/full/10.1056/NEJMoa013259

[2]There are several ways to do this. The simplest is a *Bonferroni* adjustment, which splits the error rate evenly across all intervals. For example, for simultaneous 95% confidence in 10 intervals (i.e. a total error rate of 0.05), use the t^* factor for 99.5% confidence ($0.995 = 1 - 0.05/10$), $t^* = 2.8$, instead of 2 when computing the intervals. But the Tukey method is more widely used when comparing means.

- Independent — the values are sampled independently, so the joint distribution[3] is the product of marginal distributions:

$$p_\theta(x_1, \dots, x_n) = p_\theta(x_1)p_\theta(x_2)\cdots p_\theta(x_n) \quad \text{for all } x_1, \dots, x_n$$

- A **statistic** is a characteristic of the *sample* which can be computed from the data. More precisely, a statistic is a function of $X_1, \dots, X_n$, but not of θ.
  - That is, a statistic is itself a *random variable*, and therefore has its own probability distribution, which describes how values of the statistic would vary from sample-to-sample over many (hypothetical) samples.
  - The probability distribution of a statistic is called a **sampling distribution**.
  - The *distribution* of a statistic will depend on θ, but the *value* of a statistic will not.
- A statistic $\hat{\theta} \equiv T(X_1, \dots, X_n)$ used to estimate a parameter θ is called a **(point) estimator**.
  - Naturally, we hope that there is some correspondence between our estimator $\hat{\theta}$ and the parameter it is estimating θ — for example, estimating the population mean with the sample mean — but the definition does not require this.
  - When the numerical value of a statistic is calculated from observed sample data, its value is called a *(point) estimate* of the parameter.
  - The estimator $\hat{\theta}$ is a RV whose value changes from sample to sample; an estimate is the observed value computed based on the results of a single sample.

Example 21.2. Jerry has just opened a car dealership in a new location. He assumes that the number of cars he'll sell in a day has a Poisson(μ) distribution. But he doesn't know the value of μ, so he plans to use data from his first few days of business to estimate μ. Let $X_1, \dots, X_n$ represent the number of cars sold for a sample of n days.

1. What plays the role of the population?

[3]This number assumes independence of confidence intervals — $0.95^{10} \approx 0.60$ — but similar ideas apply more generally.

2. What is the variable of interest?

3. What is the population distribution?

4. What is the main parameter of interest? How do you interpret it?

5. Suggest one estimator of μ.

6. Suppose Jerry sells 3 cars on the first day, 0 cars on the second day, and 2 on the third day (that is, $x_1 = 3, x_2 = 0, x_3 = 2$). What would your estimate of μ be based on the (3, 0, 2) sample?

7. Suggest another estimator of μ. What would your estimate of μ be based on the (3, 0, 2) sample?

Example 21.3. Continuing Example 21.2, we'll consider a few different estimators of μ, which yield a few different estimates of μ based on the (3, 0, 2) sample.

1. Ahsoka suggests that since μ is the expected value or population mean of a Poisson distribution, our estimator should be the sample mean

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Compute Ahsoka's estimate of μ based on the (3, 0, 2) sample.

2. Boba suggests that since μ is the population variane of a Poisson distribution, our estimator should be the sample variance

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

Compute Boba's estimate of μ based on the (3, 0, 2) sample.

3. Cassian likes Boba's idea, but he heard that you should divide by $n-1$ instead of n when computing the sample variance, so Cassian says our estimator should be this sample variance

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

Compute Cassian's estimate of μ based on the (3, 0, 2) sample.

4. Darth has data from 100 days of sales at another car dealership where the sample mean was 2.3, and he thinks that information should be able to help us. He decides to take a weighted average of the sample mean we observe at the new dealership ($\bar{X}$) and 2.3, the average from the other dealership. Darth's estimator is

$$\frac{n}{n+100}\bar{X} + \frac{100}{n+100}(2.3),$$

reasoning that the more data we collect from the new dealership (represented by n), the less weight the 2.3 from the other dealership should get. Compute Darth's estimate of μ

based on the (3, 0, 2) sample.

5. Ezra remembers the Poisson pattern. In particular, he remembers that the probability of 0 cars sold in a day is $p_0 = e^{-\mu}$, and rearranging shows that $\mu = -\log p_0$. Ezra suggests that we estimate p_0 with the sample proportion of days with 0 cars sold, $\hat{p}_0$, so that

$$-\log \hat{p}_0$$

is Ezra's estimator of μ. Compute Ezra's estimate of μ based on the (3, 0, 2) sample.

6. We have seen a few different estimators of μ, each with a seemingly reasonable rationale behind it. Each estimator produces a different estimate of μ based on the same (3, 0, 2) sample data. Which of these 5 numbers is the best *estimate* of μ? That is, which of 5 values is closest to the true value of μ?

7. How might you investigate which of the *estimators* corresponds to the best estimation *procedure* of estimaing μ for a Poisson distribution?

- Because the true value of the parameter θ is unknown, we never know if the sample data provides a reasonable estimate of θ
- Rather, we evaluate the estimation *procedure* for potential values of θ: How does the estimation procedure perform over many possible samples given potential values of θ?
- Some questions of interest when estimating a parameter θ
  - How do we find an estimator of θ?
  - What are properties of "good" estimators?
  - How can we tell if one estimator is "better" than another?

- Can we find a "best" estimator of θ? (Short answer: no)
- How do we determine the margin of error for an estimator?

21.1 Exercises

Exercise 21.1. In each of the following scenarios, one of (i) and (ii) is a *probability* and one is a *statistics* question. Classify each question as "probability" or "statistics". Then discuss some general features of the probability questions, and some general features of the statistics questions

1. U.S. Satisfaction
   - Assume that 30% of Americans are satisfied with the way things are going in the U.S. If a random sample of 1000 Americans is selected, what are the chances that more than 330 Americans in the sample are satisfied with the way things are going in the U.S?
   - In a random sample of 1000 Americans, 330 Americans are satisfied with the way things are going in the U.S. What percent of all Americans are satisfied with the way things are going in the U.S?
2. Great white shark length
   - Assume that lengths of female great white sharks follow a Normal distribution with mean 16 ft and standard deviation 3 ft. If a random sample of 40 female great white sharks is selected, what are the chances that the average length in the sample is between 16.5 and 17.5 ft?
   - In a sample of 40 female great white sharks, the sample mean length is 17.0 ft. What is the average length of female great white sharks?
3. In a clinical trial, 200 subjects are randomly assigned to receive either an experimental vaccine or no treatment (e.g., a placebo), and then each subject is tested to see if they developed immunity.
   - If the vaccine is in reality not effective, what are the chances that the proportion of subjects who develop immunity is twice as large for those who received the vaccine than for those who did not?
   - Among the 100 subjects who received the vaccine, 20 developed immunity; among the 100 subjects who did not receive the vaccine, 10 developed immunity. Is there evidence that the vaccine is effective?
4. Snap streaks

- Assume that lengths of streaks on Snapchat follow an Exponential distribution, with mean 75 for teen users and mean 40 for adult users. A random sample of 200 Snapchat users is selected, 100 teens and 100 adults. What are the chances that the sample mean streak length for teens is with within 20 of the sample mean streak length for adults?
- In a random sample of 100 teen Snapchat users, the sample mean streak length is 80. In a random sample of 100 adult Snapchat users, the sample mean streak length is 35. In general, how much longer do streak lengths for teen Snapchat users tend to be than for adult Snapchat users?

Exercise 21.2. Continuing Exercise 21.2. In each scenario, identify

1. What plays the role of the population(s)?

2. What is the variable(s) of interest?

3. What is the population distribution(s)?

4. What is the main parameter(s) of interest?

22 Maximum Likelihood Estimation

Example 22.1. Suppose we are willing to assume a Poisson(μ) model for some variable X. But we don't know the value of μ, so we'll collect data and use it to estimate μ. Suppose first that we have just a single observation of X. (We'll look at a sample of size n soon.)

For context, think of μ as the long run average number of cars sold per day over many days at a new car dealership, and we're trying to estimate μ based on just the number of cars sold in a single day. But what we really want is an estimation procedure that we can use any time we have data from a Poisson distribution, regardless of the context.

1. Does X take values on a discrete or continuous scale? What about μ?

2. Write the formula for the Poisson(μ) pmf.

3. Now remember that μ is unknown. Suppose that a single observation yields $x = 3$ (e.g. 3 cars are sold in a single day.) Now write the pmf plugging in everything we know; what is this a function of?

4. Compute and intepret $\mathrm{P}(X = 3)$ if $\mu = 2.3$.

5. Compute and intepret $\mathrm{P}(X = 3)$ if $\mu = 1$.

6. Compute and intepret $\mathrm{P}(X = 3)$ if $\mu = 4.8$.

7. Suppose that $x = 3$. Based just on this observation, which value — 1, 2.3, or 4.8 — would you choose as your estimate for μ? Why?

8. We obviously have more choices for our estimate of μ than just 1, 2.3, or 4.8. Describe in principle the process you would follow to find the estimate of μ based on a single observation $x = 3$.

- Consider a single observation X from a population p_θ with parameter θ: Let $X \sim p_\theta$. The **likelihood function** given $X = x$ is defined as

$$L(\theta) \equiv L_x(\theta) = p_\theta(x)$$

- That is, the likelihood function is the probability (or density for continuous data X) of observing the given sample data x viewed as a *function of the unknown parameter(s) θ*. The observed value of x (data) is treated as a fixed constant.
- The *maximum likelihood estimate* of θ is the value of θ for which the observed data has the highest likelihood of occurring.
- The **maximum likelihood estimator (MLE)** of θ is the RV $\hat{\theta}$ which maximizes the likelihood, i.e. the value $\hat{\theta}$ which satisfies

$$L(\hat{\theta}) \geq L(\theta) \text{ for all } \theta.$$

- Remember that an estimator $\hat{\theta}$ is a function of X, the observed data.

- It is often convenient to consider the **log-likelihood[1] function** of θ

$$\ell(\theta) \equiv \ell_x(\theta) = \log L_x(\theta)$$

- Because log is an increasing function it follows that $\hat{\theta}$ is the MLE of θ if and only if

$$\ell(\hat{\theta}) \geq \ell(\theta) \text{ for all } \theta.$$

Example 22.2. Suppose we are willing to assume a Poisson(μ) model for some variable X. But we don't know the value of μ, so we'll collect data and use it to estimate μ. Let $X_1, \ldots, X_n$ be the data in a random sample of size n. For example, suppose $n = 4$ and we observe $x_1 = 3$, $x_2 = 0$, $x_3 = 2$, $x_4 = 0$.

For context, think of μ as the long run average number of cars sold per day over many days at a new car dealership, and we're trying to estimate μ based on just the number of cars sold in n days (e.g., 3 cars sold on the first day, 0 on the second, 2 on the third, and 0 on the fourth). But what we really want is an estimation procedure that we can use any time we have data from a Poisson distribution, regardless of the context.

1. Write an expression for the likelihood function. What is this a function of?

2. Compute and intepret the likelihood if $\mu = 2.3$.

3. Compute and intepret the likelihood if $\mu = 1$.

4. Compute and intepret the likelihood if $\mu = 4.8$.

[1] "log" always means natural log.

5. Suppose we observe the sample (3, 0, 2, 0). Based just on this sample, which value — 1, 2.3, or 4.8 — would you choose as your estimate for μ? Why?

6. We obviously have more choices for our estimate of μ than just 1, 2.3, or 4.8. Describe in principle the process you would follow to find the estimate of μ based on the (3, 0, 2, 0) sample.

- Consider a random sample of n values from a population p_θ with parameter θ: Let $X_1, \dots, X_n$ be i.i.d. $\sim p_\theta$.
- The **likelihood function** of θ, given $X_1 = x_1, \dots, X_n = x_n$, is

$$L(\theta) \equiv L_{x_1,\dots,x_n}(\theta) = \prod_{i=1}^{n} p_\theta(x_i) = p_\theta(x_1) \times \cdots \times p_\theta(x_n)$$

- That is, the likelihood function is the joint density[2] of the sample $X_1, \dots, X_n$, evaluated at the observed data values $x_1, \dots, x_n$, viewed as *a function of the (unknown) parameter(s)* θ.
- Note that $x_1, \dots, x_n$ represent distinct values (like $x_1 = 3, x_2 = 0, x_3 = 2$ in the example).
- The MLE of θ is the value of θ for which the observed sample data has the highest likelihood of occurring.
- The **maximum likelihood estimator (MLE)** of θ is the RV $\hat{\theta}$ which maximizes the likelihood, i.e. the value $\hat{\theta}$ which satisfies

$$L(\hat{\theta}) \geq L(\theta) \text{ for all } \theta.$$

- Remember that an estimator $\hat{\theta}$ is a function of $X_1, \dots, X_n$.
- The **log-likelihood function** of θ is

$$\begin{aligned} \ell(\theta) \equiv \ell_{x_1,\dots,x_n}(\theta) &= \log L(\theta) \\ &= \sum_{i=1}^{n} \log p_\theta(x_i) = \log p_\theta(x_1) + \cdots + \log p_\theta(x_n) \end{aligned}$$

- Because log is an increasing function it follows that $\hat{\theta}$ is the MLE of θ if and only if

$$\ell(\hat{\theta}) \geq \ell(\theta) \text{ for all } \theta.$$

[2]Source: http://www.nejm.org/doi/full/10.1056/NEJMoa013259

Example 22.3. We have seen that $\bar{X}$ is the MLE of μ in a Poisson(μ) model. (Example 22.4 proves this fact using calculus.) Now we'll consider estimating some other parameters in the Poisson situation. Again, our goal is to determine procedures for Poisson models regardless of the context, but you can think in terms of the car dealership context.

1. Consider $\sqrt{\mu}$. Is $\sqrt{\mu}$ a parameter or a statistic? How would you interpret $\sqrt{\mu}$ in this context?

2. What do you think the MLE of $\sqrt{\mu}$ is, in general and for the observed (3, 0, 2, 0) sample?

3. Consider $e^{-\mu}$. Is $e^{-\mu}$ a parameter or a statistic? How would you interpret $e^{-\mu}$ in this context?

4. What do you think the MLE of $e^{-\mu}$ is, in general and for the observed (3, 0, 2, 0) sample

5. Consider $\mu e^{-\mu}$. Is $\mu e^{-\mu}$ a parameter or a statistic? How would you interpret $\mu e^{-\mu}$ in this context?

6. What do you think the MLE of $\mu e^{-\mu}$ is, in general and for the observed (3, 0, 2, 0) sample?

7. We started by assuming a Poisson model, but how do we know that is a reasonable assumption? Suppose we observe data for a random sample of size n. Suggest you might use the sample data to investigate whether a Poisson model is appropriate.

- A parameter is any characteristic of a population distribution
- For many populations, there are a few "main" parameters (like μ for Poisson(μ), or (μ, σ) for Normal(μ, σ), but any characteristic of a population is parameter and can be estimated
- **Invariance property of MLEs.** If $\hat{\theta}$ is the MLE of θ, then $g(\hat{\theta})$ is the MLE of $g(\theta)$.
- That is, the "MLE of a function is the function of the MLE"
- The invariance principle says that if you find the MLE of the "main parameters" then you automatically get the MLE of all the parameters that are based on it

Example 22.4. We wish to estimate the parameter μ for a Poisson(μ) distribution based on a random sample of n values $X_1 \ldots, X_n$.

1. Suppose that $n = 4$ and the sample is (3, 0, 2, 0). Carefully write the likelihood function.

2. Suppose that $n = 4$ and the sample is (3, 0, 2, 0). Carefully write the log-likelihood function.

3. Use calculus to find the MLE of μ given $n = 4$ and the sample (3, 0, 2, 0). Compare to what we found previously using graphical methods.

4. Now consider a general sample of size n. Carefully write the likelihood function.

5. Now consider a general sample of size n. Carefully write the log-likelihood function.

6. Use calculus to find the MLE of μ given a sample of size n. Compare to what we found previously using graphical methods.

- In many situations, the full sample data is not necessary to find the MLE; rather, a few descriptive statistics are often sufficient to evaluate the likelihood and determine the MLE
- Poisson(μ): the MLE of μ is the sample mean $\bar{X}$
- Binomial(n, p), with n known: the MLE of p is the sample proportion $\hat{p} = X/n$, where X is the number of "successes" in the sample
- Exponential(λ): the MLE of λ is the sample rate $1/\bar{X}$; the usual situation is where the X's measure times between "events" and so $\bar{X}$ is the sample mean time between "events".
- Normal(μ, σ), with both μ and σ unknown: the MLEs of μ, σ^2, and σ, are, respectively

$$\begin{aligned}
\hat{\mu} &= \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \\
\hat{\sigma}^2 &= \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 \\
\hat{\sigma} &= \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2}
\end{aligned}$$

- Maximum likelihood estimation procedures are widely applicable and maximum likelihood estimators have many nice theoretical properties.
- However, there are some drawbacks:

- May require numerical maximization, which can be computationally intensive (especially if many parameters)
- May not be reliable estimators in small samples.

22.1 Exercises

Exercise 22.1. Suppose that times (minutes) between earthquakes (of any magnitude in a certain region) follow an Exponential distribution with rate parameter λ. Let $X_1, \dots, X_n$ be a random sample of times (minutes) between earthquakes for n earthquakes. Suppose the data for a sample of size $n = 5$ is: 20, 37, 13, 10, 25 minutes.

1. Carefully write the likelihood function for the observed sample. (Hint: For a random sample from a continuous distribution, to find the likelihood function you plug each observed value into the *pdf* and then form the product.)

2. Use any software you want to carefully plot the likelihood function from the previous part. Then use your plot—you can just zoom in as needed—to find the MLE of λ based on this sample. (Hint: you should find that the MLE is 0.0476, which is equal to $1/\bar{x} = 5/105 = 0.0476$.)

3. Write a clearly worded sentence reporting in context your estimate of λ from the previous part.

Exercise 22.2. Suppose that within a certain population, credit scores follow a Normal(μ, σ) distribution. In a random sample of 5 individuals from this population, the credit scores are 600, 630, 680, 700, 770.

1. Compute the MLE of μ.

2. Compute the MLE of σ.

3. Explain in words what it means for these parameters to be the MLE.

4. Compute the MLE of the 10th percentile of credit scores for this population

5. Compute the MLE of the population proportion of individuals with a credit score above 750.

23 Bias of Estimators

- MLE is one procedure for finding an estimator of a parameter θ. But when several estimators of θ are available, how do we decide which is "better"?
- The value of a parameter θ is unknown, so it is impossible to determine if any single estimate of θ is good or bad.
- An estimator is a random variable that returns different estimates of θ for different samples. So even if an estimator produces "good" estimates of θ for some samples, it might produce "bad" estimates of θ for others.
- Therefore, we can never determine for any particular sample if an estimator produces a good estimate of the true unknown value of θ.
- Rather, we evaluate the estimation *procedure*: Over many samples, does an estimator tend to produce reasonable estimates of θ for a variety of potential values of θ?
- Remember: a statistic/estimator is a random variable whose sampling distribution describes how values of the estimator vary from sample-to-sample over many (hypothetical) samples.
  - We can estimate the degree of this variability by *simulating many hypothetical samples* from an assumed population distribution and computing the value of the statistic for each sample.
  - However, *in practice usually only a single sample is selected* and a single value of the statistic is observed, and we don't know what the population distribution is.

Example 23.1. Recall that in the car dealership problem we were trying to estimate μ based on a random sample $X_1, \ldots, X_n$ from a Poisson(μ) distribution. We considered a few different estimators of μ.

$$\begin{aligned}
\bar{X} &= \frac{1}{n}\sum_{i=1}^{n} X_i \\
S^2 &= \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 \\
\hat{\sigma}^2 &= \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 \\
\hat{\mu} &= \frac{n}{n+100}\bar{X} + \frac{100}{n+100}(2.3)
\end{aligned}$$

Let's see how these estimators perform when $\mu = 2$ for samples of size $n = 3$.

1. Describe in detail how you could use simulation to approximate the distribution of $\bar{X}$ for samples of size $n = 3$ when $\mu = 2$.

2. Conduct the simulation and approximate $\mathrm{E}(\bar{X})$. Interpret this value.

3. Repeat parts 1 and 2 for S^2.

4. Repeat parts 1 and 2 for $\hat{\sigma}^2$.

5. Repeat parts 1 and 2 for $\hat{\mu}$.

6. Identify if each estimator tends to overestimate μ, underestimate μ, or neither, when $\mu = 2$. But then explain why this information by itself itself helpful, and what we should do next.

- $\hat{\theta}$ is an **unbiased** estimator of θ if

$$\mathrm{E}\left(\hat{\theta}\right) = \theta \quad \text{for } \textit{all} \text{ potential values of } \theta$$

- Roughly speaking, an unbiased estimator does not tend to systematically overestimate or underestimate the parameter of interest, regardless of the true value of the parameter.
- The **bias (function)** of an estimator[1] is the difference between its expected value (over many hypothetical samples) and the true (but unknown) value of the parameter

$$\text{bias}(\hat{\theta}) = \mathrm{E}(\hat{\theta} - \theta) = \mathrm{E}(\hat{\theta}) - \theta, \qquad \text{a function of } \theta$$

- If an estimator is unbiased its bias function is 0
- In general, the sample mean $\bar{X}$ is an *unbiased* estimator of the population mean μ.

$$\mathrm{E}(\bar{X}) = \mu$$

- In general, the "$1/(n-1)$" sample variance S^2 is an *unbiased* estimator of the population variance σ^2

$$\mathrm{E}(S^2) = \sigma^2$$

Example 23.2. Consider estimating μ for a Poisson(μ) distribution. Determine which of the following estimators are unbiased estimators of μ based on a random sample from a Poisson(μ) distribution. If the estimator is not unbiased, identify and plot its bias function.

1. $\bar{X}$.

2. S^2

3. $\hat{\sigma}^2$. (Hint: $\hat{\sigma}^2 = \frac{n-1}{n}S^2$.)

[1] We are only considering bias in *estimation*: is there bias in how we use the data to estimate the parameter? There could also be *bias in data collection*, in how the sample is selected and how the variables are measured. In this course we always assume that we have a "perfect" random sample from the population; in practice, that will never be true, but it might be a reasonable assumption based on how the data are collected. In any case, the bias function only quantifies bias in estimation, not bias in data collection.

4. $\hat{\mu} = \frac{n}{n+100}\bar{X} + \frac{100}{n+100}(2.3)$.

Example 23.3. Continuing the Poisson(μ) problem. Let $\theta = e^{-\mu}$. We know $\bar{X}$ is an unbiased estimator of μ. We also know that $\hat{\theta} = e^{-\bar{X}}$ is the MLE of $\theta = e^{-\mu}$. Is $\hat{\theta}$ an unbiased estimator of θ?

1. Describe in detail how you would use simulation to approximate the bias of $\hat{\theta}$ as an estimator of θ when $\mu = 2$.

2. Describe in detail how you would use simulation to approximate the bias *function* of $\hat{\theta}$.

3. Explain the reason why $e^{-\bar{X}}$ is a biased estimator of $e^{-\mu}$ even though $\bar{X}$ is an unbiased estimator of μ.

- In general, if $\hat{\theta}$ is an unbiased estimator of θ, $g(\hat{\theta})$ is usually NOT an unbiased estimator of $g(\theta)$ (unless g is linear).
  - Because in general $\mathrm{E}(g(\hat{\theta})) \neq g(\mathrm{E}(\hat{\theta}))$.
- For example, even though the "$1/(n-1)$" sample variance S^2 is an unbiased estimator of the population variance σ^2, the corresponding sample standard deviation S is a biased estimator of the population standard deviation σ

Example 23.4. Continuing the Poisson(μ) problem. Consider the estimator $\hat{\sigma}^2$. Recall that the bias function is $\text{bias}(\hat{\sigma}^2) = -\frac{\mu}{n}$. What happens to the bias function as n increases. What does this mean?

- An estimator $\hat{\theta}_n$ is an **asymptotically unbiased** estimator of θ if $\mathrm{E}(\hat{\theta}_n) \to \theta$ as $n \to \infty$ for all θ, that is, if $\mathrm{bias}(\hat{\theta}_n) \to 0$ as $n \to \infty$ for all θ.
- For an asymptotically unbiased estimator, if the sample size is large $\mathrm{E}(\hat{\theta}) \approx \theta$, regardless of the true value of θ. That is, if the sample size is large the estimator is "nearly unbiased".
- MLEs are often biased, but in general, MLEs are asymptotically unbiased

23.1 Exercises

Exercise 23.1. Consider estimating the rate parameter λ of an Exponential population based on data $X_1, \dots, X_n$ from a random sample of size n. (For example, you want to estimate the rate at which earthquakes occur based on a sample of times between earthquakes.)

We have seen that the MLE of λ is $1/\bar{X}$.

1. Determine $\mathrm{E}(\bar{X})$ in this case. Your answer should be an expression involving λ.

2. Is $1/\bar{X}$ an unbiased estimator of λ? Justify your reasoning by considering expected values.

3. Explain in full detail how you could conduct a simulation to approximate the bias of $1/\bar{X}$ when $\lambda = 2$ for a given n.

4. Explain in full detail how you could conduct a simulation to approximate the bias *function* of $1/\bar{X}$ for a given n.

Exercise 23.2. Consider estimating the population mean μ and population variance σ^2 for a Normal population based on data $X_1, \dots, X_n$ from a random sample of size n. (For example, assume credit scores follow a Normal distribution and you want to use sample data to estimate the mean and variance of credit scores.)

We have seen the MLEs of μ and σ^2 in previous handouts.

1. Is the MLE of μ an unbiased estimator of μ? If so, explain why. If not, find the bias function.

2. Is the MLE of σ^2 an unbiased estimator of σ^2? If so, explain why. If not, find the bias function.

24 Variance and Mean Square Error of an Estimator

- The bias function of an estimator measures the estimator's tendency to overestimate or underestimate the parameter, as a function of potential values of the parameter.
- But bias is only one consideration. Bias only considers the average value of the estimator over many samples, which is just one feature of the estimator's sampling distribution.
- Statistics exhibit *sample-to-sample variability*: the value of a statistic varies from sample to sample.
- When choosing between two *unbiased* estimators, the one with smaller variance is generally preferred.
- Remember that the variance of an estimator will be a function of the unknown parameter θ, so we need to consider the variance *function* for various potential values of θ.

Example 24.1. Consider again estimating μ for a Poisson(μ) distribution based on a random sample $X_1, \dots, X_n$ of size n. We have seen that both $\bar{X}$ and S^2 are unbiased estimators of μ. If we want to choose between these two estimators, how do we decide?

1. Assume $n = 3$ and $\mu = 2.3$. Describe in full detail how you could conduct a simulation to approximate the sample-to-sample distribution of $\bar{X}$ and its and its expected value and standard deviation. Then conduct the simulation and record the results. What does the standard deviation measure?

2. Repeat part 1 for S^2.

3. Compare the simulation results for $\bar{X}$ and S^2 when $n = 3$ and $\mu = 2.3$. Based on the simulation results, which estimator of μ is preferred when $\mu = 2.3$ — $\bar{X}$ or S^2? Why? But then explain why this information isn't very helpful.

- For an i.i.d. random sample $X_1, \ldots, X_n$, the **sample mean** is the *statistic*

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{X_1 + \cdots + X_n}{n}$$

- The sample mean $\bar{X}$ is an *unbiased* estimator of the population mean μ:

$$\mathrm{E}(\bar{X}_n) = \mu$$

- Sample-to-sample variability of sample means decreases as the sample size increases

$$\mathrm{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$$

$$\text{sample-to-sample SD of sample means} = \frac{\text{unit-to-unit SD of values of the variable}}{\sqrt{\text{sample size}}}$$

  - The more variable the individual values of the variable are (i.e. the larger the population SD of the variable, σ), the more variable sample means will be (i.e. the larger than the SE).
  - Sample means are less variable than individual values of the variable
  - Over many random samples, sample means from larger random samples vary less, from sample to sample, than sample means from smaller random samples.
  - The SD of the sample means decreases as the sample size increases, but this reduction in SD follows a "square root rule". For example, if the sample size is increased by a factor of 4, then the SD is reduced by a factor of 2 (not 4).

Example 24.2. Continuing Example 24.1 where the population is Poisson(μ).

1. Assuming $n = 3$ and $\mu = 2.3$, compute $\mathrm{E}(\bar{X})$ and $\mathrm{SD}(\bar{X})$, and compare to the simulation results.

2. For a general n and $\mu > 0$, find expressions for $\mathrm{E}(\bar{X})$ and $\mathrm{SD}(\bar{X})$.

3. It can be shown that

$$\text{Var}(S^2) = \frac{2\mu^2}{n-1} + \frac{\mu}{n}$$

Sketch a plot of the variance functions of both $\bar{X}$ and S^2 as a function of μ. Regardless of the value of μ, which estimator has smaller variance? Which of these two unbiased estimators of Poisson μ, $\bar{X}$ or S^2, is preferred?

4. Suppose $n = 3$ and the sample is $(3, 0, 2)$. For this sample $\bar{x} = 1.67$ and $s^2 = 2.33$. Which number, 1.67 or 2.33, is a better estimate of μ? Explain.

5. Suppose $n = 3$ and the sample is $(3, 0, 2)$. For this sample $\bar{x} = 1.67$ and $s^2 = 2.33$. Which number, 1.67 or 2.33, would you choose as the estimate of μ based on this sample? Why?

- MLE is one procedure for finding an estimator of a parameter θ. But when several estimators of θ are available, how do we decide which is "better"?
- The value of a parameter θ is unknown, so it is impossible to determine if any single estimate of θ is good or bad.
- An estimator is a random variable that returns different estimates of θ for different samples. So even if an estimator produces "good" estimates of θ for some samples, it might produce "bad" estimates of θ for others.
- Therefore, we can never determine for any particular sample if an estimator produces a good estimate of the true unknown value of θ.
- Rather, we evaluate the estimation *procedure*: Over many samples, does an estimator tend to produce reasonable estimates of θ for a variety of potential values of θ?
- Remember: a statistic/estimator is a random variable whose sampling distribution distribution describes how values of the estimator vary from sample-to-sample over many (hypothetical) samples.
  - We can estimate the degree of this variability by *simulating many hypothetical samples* from an assumed population distribution and computing the value of the statistic for each sample.

- However, *in practice usually only a single sample is selected* and a single value of the statistic is observed, and we don't know what the population distribution is.

Example 24.3. Continuing the situation of estimating μ for a Poisson(μ) distribution based on a random sample $X_1, \dots, X_n$ of size n. Consider the estimators

$$
\begin{aligned}
S^2 &= \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 \\
\hat{\sigma}^2 &= \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 = \left(\frac{n-1}{n}\right) S^2
\end{aligned}
$$

We have seen that S^2 is an unbiased estimator of μ but $\hat{\sigma}^2$ is not. Does that mean we should prefer S^2 over $\hat{\sigma}^2$?

1. Assume $n = 3$ and $\mu = 2.3$. Use simulation to approximate the distribution of $\hat{\sigma}^2$ and its expected value and standard deviation.

2. Compare the simulation results to those for S^2 (from earlier). For which estimator is the bias smaller? For which estimator is the variance smaller? Is there a clear preference between the two estimators when $\mu = 2.3$?

- Estimators can be evaluated based on both bias and variability.
- Mean square error is a combined measure of both an estimator's bias and its variability.
- The **mean square error (MSE)** of an estimator $\hat{\theta}$ of a parameter θ is

$$
\text{MSE}_\theta(\hat{\theta}) = \text{E}\left(\left(\hat{\theta} - \theta\right)^2\right), \qquad \text{a function of } \theta
$$

- Mean square error measures, on average, how far the estimator deviates from the parameter it is estimating.
- The MSE of an estimator is a function of the parameter θ. It can be shown that[1]

$$
\begin{aligned}
\text{MSE}_\theta(\hat{\theta}) &= \text{Var}(\hat{\theta}) + \left(\text{E}(\hat{\theta}) - \theta\right)^2 \\
&= \text{Var}(\hat{\theta}) + \left(\text{bias}_\theta(\hat{\theta})\right)^2
\end{aligned}
$$

[1]Source: http://www.nejm.org/doi/full/10.1056/NEJMoa013259

- There can be many "reasonable" estimators of a parameter. Given two estimators, it's often the situation that neither one has smaller MSE for all potential values of θ. Choosing an estimator often involves a tradeoff between bias and variability.

Example 24.4. Continuing Example 24.3.

1. Use the simulation results to compute the MSE of S^2 and $\hat{\sigma}^2$ when $\mu = 2.3$. Determine which estimator has smaller MSE when $\mu = 2.3$, but then explain why this information is not very useful.

2. Plot the MSE functions of S^2 and $\hat{\sigma}^2$. Which estimator is preferred? Discuss.

Example 24.5. Continuing estimating μ for a Poisson(μ) distribution. Consider $\bar{X}$ and the constant estimator 4.2.

1. Esplain what it means to use 4.2 as an estimation procedure.

2. Find the MSE of the constant estimator 4.2.

3. Find the MSE of $\bar{X}$.

4. Suppose $n = 3$. Plot the MSE functions of the two estimators. Does either estimator have a better MSE? Explain.

5. Donny Don't says: "neither estimator has smaller MSE for all values of μ. So we'll use the constant estimator 4.2 when μ is near 4.2 (between 3.17 and 5.56 if $n = 3$) and we'll use $\bar{X}$ for other values of μ". Do you agree with Donny's strategy? Do you see any problems with it?

6. What happens to the MSEs as n increases? In particular, what happens to the range of values of μ for which the constant estimator 4.2 has smaller MSE than $\bar{X}$?

- Given a parameter θ, it is never possible to find a single estimator that has the smallest MSE for all potential values of θ.
- Choosing an estimator often involves a tradeoff between bias and variability.
- MLEs generally have good properties. For many situations, roughly,
  - The MLE is asymptotically unbiased; that is, the bias is small when the sample size is large
  - The variance of an MLE is about as small as it can be for an unbiased estimator
- But there are many reasonable estimation procedures besides MLE (e.g., Bayes estimators).
- Bias is estimation is not necessarily bad. Being willing to accept a little bit of bias can often lead to a beneficial reduction in variability.
  - On the other hand, try to minimize bias in *data collection*: how the sample is selected, how the variables are measured, etc
  - A lot of the theory assumes we have a perfect random sample, which is never true in practice

24.1 Exercises

Exercise 24.1. Consider estimating the rate parameter λ of an Exponential population based on data $X_1, \dots, X_n$ from a random sample of size n. (For example, you want to estimate the rate at which earthquakes occur based on a sample of times between earthquakes.)

We have seen that the MLE of λ is $1/\bar{X}$.

1. Explain in full detail how you could conduct a simulation to approximate the MSE of $1/\bar{X}$ when $\lambda = 2$ for a given n.

2. Explain in full detail how you could conduct a simulation to approximate the MSE *function* of $1/\bar{X}$ for a given n.

Exercise 24.2. Recall Darth's estimator of μ in the Poisson(μ) car dealership problem of Example 21.3

$$\hat{\mu} = \frac{n}{n+100}\bar{X} + \frac{100}{n+100}(2.3),$$

1. Compute the bias, variance, and MSE of $\hat{\mu}$ when $\mu = 2$ for a sample of size $n = 3$.

2. Compute the bias, variance, and MSE *functions* of $\hat{\mu}$ for a sample of size $n = 3$.

25 Central Limit Theorem

- A **statistic** is a characteristic of the *sample* which can be computed from the data.
    - That is, a statistic is itself a *random variable*, and therefore has its own probability distribution, which describes how values of the statistic would vary from sample-to-sample over many (hypothetical) samples.
    - The probability distribution of a statistic is called a **sampling distribution**.
- Statistics exhibit **sample-to-sample variability**: the value of a statistic varies from sample to sample.
    - We can estimate the degree of this variability by *simulating many hypothetical samples* and computing the value of the statistic for each sample.
    - The resulting standard deviation, called the **standard error**, measures the sample-to-sample variability of the statistic over many (hypothetical) samples of the same size.
    - However, *in practice usually only a single sample is selected* and a single value of the statistic is observed.
- For many statistics (e.g., proportions, means, differences in means) — but not all —
    - Statistics from larger random samples vary less, from sample to sample, than statistics from smaller random samples.
    - The pattern of sample-to-sample variability of the statistic follows, approximately, a Normal distribution

Example 25.1. A random sample of n customers at a snack bar is selected, independently. Let $\bar{X}_n$ represent the sample mean dollar amount spent by the n customers in the sample.

Assume that

- 40% of customers spend 5 dollars
- 40% of customers spend 6 dollars
- 20% of customers spend 7 dollars

Let X denote the amount spent by a single randomly selected customer.

1. Compute $\mathrm{E}(X)$. We'll call this value the population mean μ; explain what this means.

2. Compute $\mathrm{SD}(X)$. We'll call this value the population standard deviation σ; explain what this means.

- The **population distribution** describes the distribution of values of a variable over all *individuals* in the population.
  - The **population mean**, denoted μ, is the average of the values of the variable over all individuals in the population.
  - The **population standard deviation**, denoted σ, is the standard deviation of all the individual values in the population.
- A **(simple) random sample** of size n is a collection of random variables $X_1, \ldots, X_n$ that are *independent* and *identically distributed* (*i.i.d.*)

Example 25.2. Continuing Example 25.1. Now we'll consider samples of size $n = 2$. Randomly select two customers, independently, and let X_1 and X_2 denote the amounts spent by the two people selected. Let $\bar{X}_2$ represent the sample mean.

1. Make a table of all possible (X_1, X_2) pairs and their probabilities, and use the table to find the distribution of $\bar{X}_2$. Interpret in words in context what this distribution represent.

2. Compute and interpret $\mathrm{P}(\bar{X}_2 > 6)$.

3. Compute $\mathrm{E}(\bar{X}_2)$. How does it relate to μ?

4. There are 3 "means" in the previous part. What do they all mean?

5. Compute $\mathrm{Var}(\bar{X}_2)$ and $\mathrm{SD}(\bar{X}_2)$. How do these values relate to the population variance σ^2 and population standard deviation σ?

6. Describe in words in context what $\mathrm{SD}(\bar{X}_2)$ measures variability of.

- The **sample mean** is
$$\bar{X}_n = \frac{X_1 + \cdots + X_n}{n},$$
- Because the sample is randomly selected, the sample mean $\bar{X}_n$ is a random variable that has a distribution. This distribution describes how sample means vary from sample-to-sample over many random samples of size n.
- Over many random samples, sample means do not systematically overestimate or underestimate the population mean.
$$\mathrm{E}(\bar{X}_n) = \mu$$
- Variability of sample means depends on the variability of individual values of the variable; the more the values of the variable vary from individual-to-individual in the population, the more the sample means will vary from sample-to-sample. However, sample means are less variable than individual values of the variable. Furthermore, the sample-to-sample

variability of sample means decreases as the sample size increases

$$\mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$
$$\mathrm{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$$

Example 25.3. Continuing Example 25.2. Now suppose we take a random sample of 20 customers; consider $n = 20$ and $\bar{X}_{20}$.

1. Find $\mathrm{E}(\bar{X}_{20})$, $\mathrm{Var}(\bar{X}_{20})$, and $\mathrm{SD}(\bar{X}_{20})$.

2. Describe in words in context what $\mathrm{SD}(\bar{X}_{20})$ measures variability of.

3. Use simulation to determine the approximate shape of the distribution of $\bar{X}_{20}$.

4. Simulation shows that $\bar{X}_{20}$ has an approximate Normal distribution. Use this Normal distribution to approximate $\mathrm{P}(\bar{X}_{20} > 6)$, and interpret the probability.

- The **(Standard) Central Limit Theorem**. Let $X_1, X_2, ...$ be independent and identically distributed (i.i.d.) random variables with finite mean μ and finite standard deviation σ. Then

$$\bar{X}_n \text{ has an approximate } N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \text{ distribution, if } n \text{ is large enough.}$$

- The CLT says that if the sample size is large enough, the sample-to-sample distribution of sample means is approximately Normal, *regardless of the shape of the population distribution.*

Example 25.4. Continuing Example 25.2. Now suppose that each customer spends

- 5 dollars with probability 0.4
- 6 with probability 0.4
- 7 with probability 0.19
- 30 with probability 0.01 (maybe they treat a few friends).

For this distribution, the population mean is $\mu = 6.03$ and the population standard deviation is $\sigma = 2.52$.

1. Use simulation to approximate the distribution of $\bar{X}_{30}$; is it approximately Normal?

2. Use simulation to approximate the distribution of $\bar{X}_{100}$; is it approximately Normal?

3. Use simulation to approximate the distribution of $\bar{X}_{300}$; is it approximately Normal?

- The CLT says that if the sample size is large enough, the sample-to-sample distribution of sample means is approximately Normal, *regardless of the shape of the population distribution.*
- However, the shape of the population distribution — which describes the distribution of individual values of the variable — does matter in determining *how large* is large enough.
  - If the population distribution is Normal, then the sample-to-sample distribution of sample means is Normal for any sample size.
  - If the population distribution is "close to Normal" — e.g., symmetric, light tails, single peak — then smaller samples sizes are sufficient for the sample-to-sample distribution of sample means to be approximately Normal.
  - If the population distribution is "far from Normal" — e.g., severe skewness, heavy tails, extreme outliers, multiple peaks — then larger sample sizes are required for the sample-to-sample distribution of sample means to be approximately Normal.

- You should certainly be aware that there is no magic number and that some population distributions require a large sample size for the CLT to kick in. However, in many situations the sample-to-sample distributions of sample means (or other statistics) is approximately Normal even for small or moderate sample sizes.

Example 25.5. Suppose a business is tracking when online orders are placed on their website. We times between orders, in hours, follow an Exponential distribution with rate parameter 3.2 per hour. Approximate the probability that it takes more than a week for the 500th order to be placed.

Hints:

- What plays the role of the population distribution?
- What is the population mean?
- What is the population SD?
- How can you express the probability in question in terms of a sample mean?
- What does the CLT say about sample means from this population?

Example 25.6. Suppose that average annual income for U.S. households is about \$120,000, and that the standard deviation of income is about \$160,000.

1. What can you say about the probability that a single randomly selected household has an income above \$200,000?

2. Donny Don't says: "Since a sample size of one million is extremely large, the CLT says that the incomes in a sample of one million households should follow a Normal distribution". Do you agree? If not, explain to Donny what the CLT does say.

3. Use the CLT to approximate the probability that in a sample of 1000 households the sample mean income is above \$130,000.

4. Donny says: "Wait, the distribution of household income is highly skewed to the right. Wouldn't it be more appropriate to use the sample median than the sample mean"? Do you agree? Explain.

- Be careful to distinguish between what the CLT does say, and what it doesn't say.
- The CLT does NOT say anything about the shape of the population distribution of individual values of the variable; it can be anything.
  - Randomly selected samples tend to be representative of the population they're sampled from, so we expect the shape of the distribution of individual values in the sample to resemble the shape of the distribution of individual values in the population, regardless of the sample size.
  - Over many samples, the distribution of individual values within each sample will resemble the population distribution. For example, if the population distribution has an Exponential shape, then the distribution of values in any single randomly selected sample will tend to have a roughly Exponential shape.
- The CLT applies to how sample *means* behave over many samples, not what happens within any single sample.
  - If n is large enough, then over many *samples* of size n the *sample-to-sample distribution of sample means* will be approximately Normal, regardless of the shape of the population distribution of individual values of the variable.

25.1 Exercises

Exercise 25.1. The number of free throws that a particular basketball player attempts in a game has a Poisson distribution with parameter 2.7, independently from game to game. Approximate the probability that the player attempts more than 60 free throws in the next 20 games.

Exercise 25.2. A fair six-sided die is rolled until the total sum of all rolls exceeds 300. Approximate the probability that at least 80 rolls are necessary.

Exercise 25.3. This applet can used to investigate the CLT (and other ideas). Enter 100 for the population mean and 25 for the population standard deviation. (These are just made up numbers, but for some context you can imagine that the variable is income in thousands of dollars.)

1. Choose a super-small sample size, like $n = 2$. Choose a population shape, and then run the simulation to generate many values of the sample mean. Then repeat for different population shapes. When the sample size is small, does the *shape* of the sample-to-sample distribution of sample means (plot on right) depend on the shape of the population (plot on the left)?

2. Now increase the sample size, and repeat for the several populations. What happens to the *shape* of the sample-to-sample distribution of sample means as the sample size increases?

3. Now choose a not-super-small sample size, like $n = 500$. Choose a population, and simulate *a single sample*. Look at the simulated sample (the middle plot), then repeat to simulate a few samples. Then repeat for the different populations. Does the *distribution of individual values within the sample* depend on the shape of the population?

4. When the sample size is large, does the *sample-to-sample distribution of sample means* depend on the shape of the population?

26 Confidence Intervals

- The estimators we have considered so far are *point* estimators; we estimate the population parameter θ, an unknown number, by the observed sample statistic $\hat{\theta}$, a number which can be computed based on sample data $X_1, \dots, X_n$
- For a particular sample a particular numerical value of $\hat{\theta}$ is observed, but there are many possible randomly selected samples and many possible values of the random variable $\hat{\theta}$.
- While for any particular sample we hope the observed value of $\hat{\theta}$ is close to the true value of θ, there will be some error in estimation due to natural sample-to-sample variability.
- Because of this natural sample-to-sample variability, it is usually better to provide a range of values when estimating a parameter
- In determining a plausible range of values, there is a trade-off between
  - Accuracy: the wider the interval the more confident we are that it actually contains the true parameter value
  - Precision: but the wider the interval the less precise of an estimate it provides
- The usual approach is to specify a desired degree of accuracy. Specifically, interval estimate procedures are constructed so that they will be accurate "most of the time".
- A *confidence interval* estimates a population parameter based on sample data with
  - a plausible range of values and
  - a *confidence level* that the estimated range contains the true parameter value.
    - By far, the most commonly used confidence level is 95%.
    - But see Exercise 26.2 and related material about multiple confidence intervals

Example 26.1. In a recent survey of a representative sample of U.S. adults aged 18-29, 67% saw a movie in the theater in the last 12 months. The survey reports a margin of error for 95% confidence of 4.2 percentage points.

1. Use the information provided to compute and interpret a 95% confidence interval.

2. The study also surveyed 3257 U.S. adults aged 30-49, and 60% saw a movie in the theater in the last year. Without computing, will the margin of error for 95% confidence for the 30-49 age group be greater than, less than, or equal to 4.2 percentage points?

3. For the 30-49 age group the margin of error for 95% confidence is 2.2 percentage points. Compute and interpret a 95% confidence interval.

- A $1-\alpha$ **confidence interval** for θ is an *interval estimator* of θ based on the sample $X_1, \dots, X_n$ of the form $[L(X_1, \dots, X_n), U(X_1, \dots, X_n)]$ which satisfies

$$\mathrm{P}\left(L(X_1, \dots, X_n) \le \theta \le U(X_1, \dots, X_n)\right) = 1-\alpha$$

- $L(X_1, \dots, X_n)$ and $U(X_1, \dots, X_n)$ represent, respectively, the lower and upper endpoints of the interval. L and U are *random variables* that vary from sample-to-sample.
- Typical values of α include 0.05 and 0.01. We express $1-\alpha$ as a percentage. For example, $\alpha = 0.05$ corresponds to a 95% confidence interval.
- Often $L = \hat{\theta} - M$ and $U = \hat{\theta} + M$, where the **margin of error** M — where M can be a RV — accounts for both the sample-to-sample variability of $\hat{\theta}$ and the confidence level.
  - The margin of error is often of the form: multiple of standard error
  - The multiple is determined by the confidence level and the distribution of the point estimator $\hat{\theta}$ (e.g., 2 for 95% confidence when the estimator follows a Normal distribution.)
    - The larger the confidence level, the larger the margin of error — and the wider the range of plausible values — required to achieve that level of confidence.
  - The **standard error** is the estimated sample-to-sample SD of the estimator, $\mathrm{SE}(\hat{\theta}) = \widehat{\mathrm{SD}}(\hat{\theta})$.
    - The less variable the statistic (from sample to sample) the smaller the margin of error and the narrower the interval.
    - In general, the larger the sample size, the smaller the SE of the statistic, the smaller the margin of error, and the narrower the interval.

Example 26.2. Consider the general problem of estimating a population mean μ based on a random sample of size n. A natural point estimator of μ is the sample mean $\bar{X}$. The central limit theorem says that, if n is large enough, $\bar{X}$ has an approximate Normal(μ, $\sigma/\sqrt{n}$) distribution, where σ is the population standard deviation. We'll use this theory to derive a confidence interval for μ.

1. Fill in the blank: for 95% of samples, the sample mean $\bar{X}$ is within [blank] of the population mean μ.

2. Fill in the blank: for 95% of samples, the interval with endpoints $\bar{X}\pm$ [blank] will contain the population mean μ.

3. The previous part suggests a formula for a CI for μ. However, this is not a practically useful formula; esplain why. Then suggest how we can fix it.

4. Everything else being equal, how does the sample size affect the confidence interval and its margin of error: the larger the sample size...? Does this make sense?

5. Everything else being equal, how does the confidence level (e.g. 95%) affect the confidence interval and its margin of error: the larger the confidence level...? Does this make sense?

- A **one-sample t confidence interval for a population mean μ** has endpoints

$$\bar{X} \pm t^* \frac{S}{\sqrt{n}}$$

  where S is the sample standard deviation $S = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2}$ and t^*, the multiple corresponding to the confidence level, is from the t *distribution with* $n-1$ *degrees of freedom*.
- For all practical purposes you can treat t distributions like Normal distributions and use the usual empirical rule
  - Technically, t^* is a little bigger than z^* to account for the additional sample-to-sample variability in S.
  - But in most situations encountered in practice, t^* and z^* are essentially the same.
- Assumptions:
  - Simple random sample from a large population, that is, independent observations
  - No bias in data collection
  - Population mean is the parameter you want to estimate

Confidence level	90%	95%	99%	99.7%	99.99%
Multiple ($z^* \approx t^*$)	1.7	2.0	2.6	3.0	4.0

Example 26.3. A frequently used example data set is the Ames housing data set which consists of residential properties sold in Ames, Iowa from 2006 to 2010. For more information about the variables in this data set, refer to the data documentation.

Suppose we wish to use this data set to estimate μ, the *population mean sale price of single family homes.*

In the sample, there are 2425 single family homes with a sample mean sale price ($K) of 185 and sample standard deviation 83.

1. Compute a 95% confidence interval.

2. Write a clearly worded sentence containing the conclusion of the confidence interval in context.

3. To what extent can you generalize your results? That is, what is the relevant "population"? What are some issues to consider?

4. What if we wanted 95% confidence, but we wanted the margin of error to be no bigger than 1K. How could we achieve this?

5. Assuming that this is a random sample from the appropriate population, does the 95% confidence interval you computed in the previous part provide an accurate estimate of the corresponding parameter? That is, is μ contained within the CI? Explain.

- Because the value of the parameter is *unknown*, we never know for sure whether any one confidence interval contains the parameter or not.
- But the procedure guarantees that in the long run — that is, over many unbiased random samples — "most" confidence intervals will contain the parameter, regardless of what the true value of the parameter is. "Most" is determined by the level of confidence.
- Any value within the confidence interval is a plausible value of the parameter
- Values outside of the confidence interval are not plausible values of the parameter
- The CI could be wrong; we are confident that it is correct, but we'll never know for sure
- There are two sources of differences between a statistic and a corresponding parameter
  - systematic errors due to **bias**
  - "chance errors" due to **sampling variability**
- While there is no bias inherent in simple random sampling from a population, bias can be present in other ways, including undercoverage, nonresponse, wording of questions, etc.
- Bias concerns the *center* of the sampling distribution of a statistic. A biased sampling procedure produces values which consistently overestimate or consistently underestimate the respective parameter.

- The inference procedures — confidence intervals and hypothesis tests — we will encounter assume that the data are collected in an unbiased way from a random sample.
- If there is bias present in data collection then the observed statistic (i.e. the center of the confidence interval) is likely to be a poor estimate of the parameter.
- The margin of error in a confidence interval only accounts for sampling variability, NOT bias.
- To see if bias is present — in which case the confidence interval cannot be trusted — ask: how was the data collected?

Example 26.4. Continuing Example 26.3.

1. Consider the distribution of sale price. What shape do you expect this distribution to have? Why?

2. A student says, "The distribution of sale price is clearly skewed to the right. Since it's not Normal the confidence interval we computed, and the resulting conclusion about the population mean, is not valid." Do you agree with this critique? Explain.

3. Considering that the distribution of sale price is skewed to the right, suggest a valid criticism of the confidence interval we computed. Hint: what parameter is it estimating?

- The t interval for the population *mean* is valid as long as the sample size is large enough, regardless of the shape of the population distribution.
- However, if the population distribution is highly skewed, then the population *median* might be a more appropriate parameter to consider as a measure of center.
- Which is more appropriate, the population mean or the population median, depends on the purpose for the inference.
- ALWAYS PLOT YOUR DATA, and consider the distribution of the variable. If the values in a representative sample are skewed or include large outliers, the same is also plausibly true for the population it's supposed to represent.

26.1 Exercises

Exercise 26.1. For each of the following statements, identify if it is a valid interpretation of the confidence interval from Example 26.3. If not, explain why.

1. We estimate that 95% of single family homes in the population have sale price ($K) between 181 and 188.

2. We estimate with 95% confidence that the sample mean sale price ($K) of single family homes is between 181 and 188.

3. If many samples of this size were selected about 95% would produce a CI of (181, 188).

4. If many hypothetical unbiased random samples of this size were selected, and a 95% confidence interval computed based on each sample, about 95% of these confidence intervals would contain the population mean sale price.

- A CI is a conclusion about a population parameter — a number that *summarizes* the population (e.g. the population mean) — and not about individual values of a variable.
- The conclusions from inference regard the *population* from which the random sample was selected; there is nothing to conclude for the sample.
- No one particular sample (e.g. the one observed) or CI is necessarily good or representative of the many other possible samples or CIs.
- The randomness is in the *sampling procedure*, which determines L and U, and not the parameter. The parameter θ is a fixed number. After selecting the sample, there is no

randomness left. Either the CI contains the parameter or not; you just don't know which it is.

- We are confident in our observed estimate because we are confident in our estimation *procedure*
  - By considering what would happen over many hypothetical random samples we are able to determine the necessary margin of error so that our CIs would contain the parameter in 95% of samples.
  - Because the value of the parameter is unknown, we never know for sure whether any one confidence interval contains the parameter or not.
  - But the procedure guarantees that in the long run — that is, over many unbiased random samples — "most" confidence intervals will contain the parameter, regardless of what the true value of the parameter is. "Most" is determined by the level of confidence.

Exercise 26.2. Suppose we wish to estimate population mean GPA of students at each of the 23 CSU campuses. For each campus, you select a random sample of students, collect the GPAs of the students, and then use the sample data to compute a 95% confidence interval. That is you compute 23 separate 95% confidence intervals, based on 23 independent random samples, for each of 23 population means.

1. First just consider Cal Poly. What is the probability that your confidence interval based on the Cal Poly sample contains the true population mean GPA at Cal Poly?

2. What is the probability that all 23 of your 95% confidence intervals contain their respective population mean?

3. How confident are you that all 23 of your 95% confidence intervals contain their respective population mean?

- When computing multiple confidence intervals, the multiple (z^* or t^*) of each confidence interval should be increased to account for the **issue of multiple comparisons**.
  - For example, with a multiple of 2 in a *single* confidence interval, we are 95% confident that the interval contains the corresponding parameter.
  - However, if we use a multiple of 2 for each of 10 confidence intervals, we are only 60% confident[1] that *all* of the 10 95% confidence intervals contain their respective parameters.
- A **Bonferroni** adjustment, splits the error rate evenly across all intervals. For example, for simultaneous 95% confidence in 10 intervals (i.e. a total error rate of 0.05), use the z^*/t^* multiple for 99.5% confidence when constructing each interval.
  - A Bonferroni adjustment guarantees **simultaneous confidence**
  - But it is conservative, and tends to produce wider intervals than necessary
  - There are other methods; for example, the *Tukey-Kramer method* is often used for *all pairwise comparisons of means* (e.g., in ANOVA)

4. Suppose we want simultaneous 95% confidence that all 23 CIs contain their respective mean. Find the z^* multiple corresponding to a Bonferroni adjustment. How many times wider will each CI be than the unadjusted intervals?

5. How confident are you that all 23 of your adjusted confidence intervals contain their respective population mean?

Exercise 26.3. Continuing with the Ames housing data set. Now suppose we want to compare sale price of single family homes with other homes (including condos, townhouses, etc). In particular, suppose we wish to estimate $\mu_S - \mu_N$, the difference in population mean sale price between single family homes and non-single family homes. The table below summarizes the sample data.

[1] This number assumes independence of confidence intervals — $0.95^{10} \approx 0.60$ — but similar ideas apply more generally.

	count	mean	std	min	25%	50%	75%	max
SingleFamily								
False	505.0	161.511398	60.394023	55.000	120.0	147.4	190.0	392.5
True	2425.0	184.812041	82.821802	12.789	130.0	165.0	220.0	755.0

1. Suggest a point estimator, compute its variance, and suggest a formula for a CI for $\mu_S - \mu_N$.

2. Compute a 95% confidence interval.

3. Write a clearly worded sentence interpreting the confidence interval from the previous part in context.

4. Are we reasonably confindent that single family homes tend to have higher sale prices than other homes? Explain.

- For many statistics (e.g. means, differences in means, proportions, differences in proportions) the sample-to-sample pattern of variability of values of the statistic over many unbiased random samples has an approximate Normal shape centered at the true value of the population parameter
  - In such situations confidence intervals take the form

$$\text{statistic} \pm \text{multiple of its SE}$$

- If two samples are selected *independently* from each of two populations with parameters θ_1 and θ_2, then an approximate $1-\alpha$ CI for $\theta_1 - \theta_2$ has endpoints

$$\hat{\theta}_1 - \hat{\theta}_2 \pm t^* \,\mathrm{SE}(\hat{\theta}_1 - \hat{\theta}_2)$$

  where

$$\mathrm{SE}(\hat{\theta}_1 - \hat{\theta}_2) = \sqrt{\mathrm{SE}(\hat{\theta}_1)^2 + \mathrm{SE}(\hat{\theta}_2)^2}$$

- The above assumes that $\hat{\theta}_1$ and $\hat{\theta}_2$ each have an approximate Normal distribution, and so, since they are independent, $\hat{\theta}_1 - \hat{\theta}_2$ also has an approximate Normal distribution.
- A difference provides an absolute comparison between groups.
  - Ratios can be used to provide relative comparisons; however, the CI formulas are more complicated, but we'll see another approach soon

Exercise 26.4. You'll use a simulation applet that randomly generates confidence interals for a population mean to help you understand some ideas. The applet calculates a confidence interval for each set of randomly generated data.

- In the box for "Statistic" select means "Means".
- In the box next to "Method" select "t".
- Start with "Number of intervals" equal to 1 at first to see what is happening, but then change it to lots.

Experiment with different simulations; be sure to change

- Distribution
- Population Mean
- Population SD
- Sample size
- Confidence level

Then write a paragraph or two or some bullet points of your main observations. Based on this applet, what can you say about how confidence intervals work? How does changing each of the inputs affect the confidence intervals and the Results? Include a few screenshots from the applet to support your conclusions.

27 Bootstrap Confidence Intervals

The value of any statistic varies from sample to sample. Statistical inference procedures such as confidence intervals are based on sampling distributions which describe how the values of a statistic vary from sample to sample over many possible samples. In particular, the standard error of a statistic measures its degree of sample-to-sample variability.

If a population distribution is assumed, then we can approximate the sampling distribution of a statistic and characteristics like SE via simulation. But what if the population distribution is completely unknown? For some statistics — most notably means, proportions and certain transformations of them — the Central Limit Theorems and its relatives (e.g., Delta Method) guarantee that the sampling distribution will be approximately Normal for large samples, regardless of the population distribution. However, not every statistic has an approximate Normal distribution, so we need other methods for approximating sampling distributions. But how do we estimate the sampling distribution of a statistic based only on sample data? Bootstrapping provides a simple but extremely versatile and powerful method for doing so.

Example 27.1. For purposes of illustration, we'll start with a small data set. The table below contains data on all the quidditch matches (for which the final score was discernible) in the *Harry Potter* books[1]. The sample mean is $\bar{x} = 340$ and the sample SD is $\hat{\sigma} = 134.8$ points[2]. Say we want a confidence interval for μ, the population mean quidditch score, based on this sample.

200	230	250	260	380	470	590

1. Warmup exercise: If we assume that quidditch scores follow a $N(345, 140)$ distribution, explain how you would use simulation to approximate the sampling distribution of the sample mean. Note: we really want to approximate the sampling distribution without assuming anything about the population distribution; this is just to remind you of how we have done things so far. Also note: the values 345 and 140 are just made up; we're assuming three things: (1) scores follow a Normal distribution, (2) population mean is 345, (3) population SD is 140.

[1] Source: http://www.nejm.org/doi/full/10.1056/NEJMoa013259

[2] There are several ways to do this. The simplest is a *Bonferroni* adjustment, which splits the error rate evenly across all intervals. For example, for simultaneous 95% confidence in 10 intervals (i.e. a total error rate of 0.05), use the t^* factor for 99.5% confidence ($0.995 = 1 - 0.05/10$), $t^* = 2.8$, instead of 2 when computing the intervals. But the Tukey method is more widely used when comparing means.

2. Warmup exercise: How could you estimate the SE of the sample mean based on the above simulation? Compare to the theoretical formula.

Example 27.2. Continuing Example 27.1.

1. Now suppose we make no assumptions about the population distribution. How could you estimate the population distribution based only on the observed sample data? Construct a spinner corresponding to this distribution.

2. Continuing the previous part, explain how you would use simulation to approximate the sampling distribution of the sample mean *without knowing the population distribution.*

3. Suppose that instead of using a spinner, you wrote each observed value on a piece of paper. How would you use the pieces of paper to simulate a sample?

4. Conduct by hand a few repetitions of the simulation from the previous part.

5. Use software to perform the simulation. How would you use the simulation results to approximate the SE?

6. How could you find a confidence interval for the population mean?

- The sampling distribution of a statistic can be approximated by simulating many samples *from the population* and computing the value of the statistic for each simulated sample.
- The idea behind **bootstrapping** is that if the sample data is representative of the population, then we can approximate the process of selecting a sample from the population by *treating the observed sample as if it were the population and selecting a sample — called a resample or bootstrap sample — from the observed sample.*
- A **bootstrap sample** *resamples* the data from the *observed sample*, drawing the same number of observations, but *with replacement*.
- The **bootstrap distribution** of a statistic is obtained by simulating many bootstrap samples and computing the value of the statistic for each sample
  - Simulate a (re)sample of size n with replacement from the observed sample
  - Compute the value of the statistic for the simulated bootstrap sample
  - Repeat many times, recording the value of the statistic for each repetition
- The **bootstrap standard error** of a statistic is the standard deviation of the bootstrap distribution of the statistic.
- The bootstrap distribution of a statistic approximates some, but not all, features of the sampling distribution of the statistic.
  - The shape of the bootstrap distribution usually approximates the shape of the sampling distribution
  - The variability of the bootstrap distribution usually approximates the variability of the sampling distribution. That is, the bootstrap SE usually approximates the true SE.
  - The center of the bootstrap distribution does NOT accurately approximate the center of the sampling distribution
  - The bootstrap estimate of bias usually approximates the bias of the statistic.
- There are many methods for constructing bootstrap confidence intervals for parameters; we'll only study a few.

- **Bootstrap Normal interval**: If the bootstrap distribution for the corresponding statistic is approximately Normal, then an approximate 95% confidence interval has endpoints

$$\text{observed statistic} \pm 2 \times \text{bootstrap SE}$$

- **Bootstrap percentile interval**: The 95% bootstrap percentile interval is the interval with endpoints equal to the 2.5th and 97.5th percentiles of the bootstrap distribution.

- One of the main advantages of bootstrapping is that it can be used to approximate the SE of *any statistic*.

Example 27.3. Continuing Example 27.2, say we want to estimate the population *median* quidditch score.

1. Describe in detail how would use simulation to approximate the bootstrap distribution. Then conduct the simulation. What are the possible values in the bootstrap distribution? Why?

2. Which bootstrap confidence interval would be more appropriate, Normal or percentile? Find and interpret the appropriate confidence interval.

- The accuracy of bootstrap procedures is primarily influenced by the size of the original sample.
- Bootstrapping methods do not overcome the weaknesses of small samples.

27.1 Exercises

Exercise 27.1. Using the Ames housing data set, suppose we wish to estimate the population *median* sale price of single family homes.

1. Describe in detail how you would use bootstrapping to find a confidence interval for the population median.

2. Conduct the simulation and find and interpret the confidence interval.

Exercise 27.2. Continuing with the Ames housing data set. Now suppose we want to compare sale price of single family homes with other homes.

1. Describe in detail how you would use bootstrapping to find a confidence interval for the *difference in population means.*

2. Conduct the simulation and find and interpret the confidence interval.

3. Describe in detail how you would use bootstrapping to find a confidence interval for the *ratio of population medians.*

4. Conduct the simulation and find and interpret the confidence interval.

Exercise 27.3. Continuing with the Ames housing data. Suppose we want to estimate the *correlation* between sale price and square footage ("Gr Liv Area").

1. Describe in detail how you would use bootstrapping to find a confidence interval for the *population correlation coefficient* between sale price and square footage for single family homes.

2. Conduct the simulation and find and interpret the confidence interval.

28 Null Hypothesis Testing

- Statisticians view conjectures about a population as competing *hypotheses*. These hypotheses usually constitute claims about the values of certain parameters.
- A statistical **(null) hypothesis test** consists of using data to make a decision regarding two explanations (hypotheses).
- The **null hypothesis**, denoted H_0, represents the "status quo/no difference/no effect/no association" hypothesis.
  - If the null hypothesis is true then any observed difference/effect/association is just due to "random chance" resulting from natural sampling variability
- The **alternative hypothesis**, denoted H_a (or H_1), typically corresponds to what the researchers are hoping to gather evidence to support, and thus is a statement in line with the research conjecture
- A statistical hypothesis test assesses if the data provide evidence to rule out the "random chance" explanation (which corresponds to the null hypothesis) in favor of the explanation provided by the research conjecture (alternative hypothesis).
- The null hypothesis is often of the form

$$H_0 : \text{parameter} = \text{hypothesized value}$$

  where the hypothesized value is a specific number determined by the research question (often 0 for "no difference").
- The alternative hypotheses is often of one of the following forms

$$\begin{aligned}
\text{Case 1a:} \quad & H_a : \text{parameter} > \text{hypothesized value} \\
\text{Case 1b:} \quad & H_a : \text{parameter} < \text{hypothesized value} \\
\text{Case 2:} \quad & H_a : \text{parameter} \neq \text{hypothesized value (“two-sided”)}
\end{aligned}$$

  Which of the three cases above is appropriate is determined by the research conjecture. When in doubt, use a two-sided ($\neq$) alternative.

Example 28.1. Did the New England Patriots deliberately deflate footballs in an NFL playoff game against the Indianapolis Colts on January 18, 2015? The following table[1] summarizes the decrease in air pressure (psi), from pre-game to halftime, for the footballs from each team that were checked at halftime by officials.

[1]Source: http://www.nejm.org/doi/full/10.1056/NEJMoa013259

Team	Deflation (psi)											Size	Mean	SD
Patriots	1.00	1.65	1.35	1.80	1.40	0.90	0.65	1.40	1.55	2.00	1.60	11	1.391	0.402
Colts	0.30	0.25	0.50	0.45								4	0.375	0.119

1. State the null and the alternative hypotheses in words.

2. Compute the observed difference in sample means.

3. If there were no difference between the Patriots and the Colts footballs, what would you expect the difference in sample means to be? Is the fact that we observed another value necessarily evidence that the Patriots footballs tended to deflate more than the Colts?

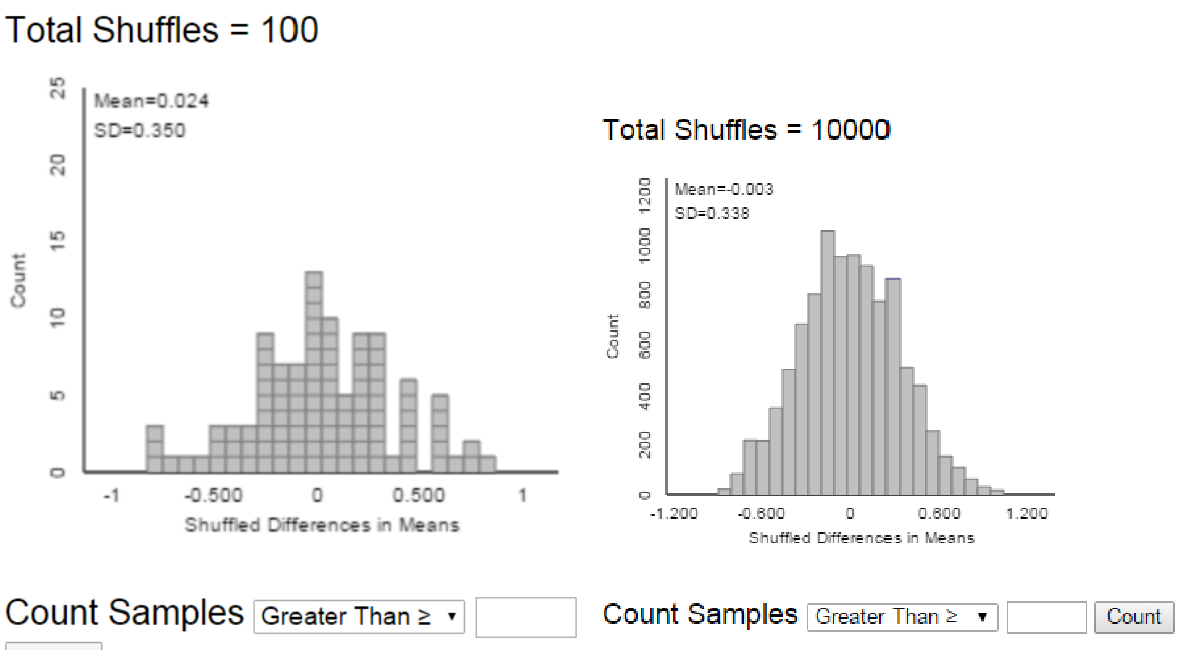
4. Suppose we want to assess if it is plausible that the Patriots didn't interfere with the footballs, and the observed difference in sample means is just due to chance variability. Specify how the "null distribution" of the statistic could be simulated with cards. Note: this is NOT bootstrapping; how is it different?

5. Use cards to perform one repetition of the above simulation, record the results here, and compute the appropriate difference in means.

Team	Deflation (psi)								Size	Mean	SD
Patri-ots									11		
Colts		X	X	X	X	X	X	X	4		

6. How could we use the results of many repetitions to assess if the data provides evidence that the Patriots footballs really tended to deflate more than the Colts?

7. The following plot displays results from an applet used to run 10000 repetitions of the simulation (the plot on the left displays the results from just the first 100 repetitions). Estimate the probability of observing a difference in sample means as extreme as the observed difference just by chance if there were no real difference between the Patriots and Colts footballs.



(a) First 100 repetitions

(b) 10000 repetitions

Figure 28.1: Permutation simulation of null distribution in Example 28.1.

8. The simulation outputs the p-value, which is close to 0. Interpret this p-value.

9. One explanation is "there is no real difference and the observed difference of 1.016 is just random chance". Is that a plausible explanation? If not, what is a more plausible explanation?

- To assess the plausibility of the "random chance" explanation for the observed result — corresponding to the null hypothesis being true — we need to know the pattern of sampling variability of the statistic that would be expected just due to random chance if the null hypothesis were true. This is called the **null distribution** of the statistic.
- If the actual observed value of the statistic is not consistent with this pattern, then we have convincing evidence that the results did not occur by random chance alone, and we have evidence to reject the null hypothesis
- The (simulated) null distribution provides a picture of what typical values of the statistic would be expected just by random chance alone if the null hypothesis were true.
- The question is then: how surprising is the result we actually observed from the perspective of the null distribution? If the result would be very surprising, then "random chance" is not a plausible explanation for the observed result.
- This gives us evidence to rule out a "random chance" explanation like in favor of a "research conjecture" explanation.
- In other words, the more surprising the observed result would be just due to random chance, the stronger the evidence to **reject the null hypothesis** in favor of the alternative hypothesis.
- We quantify "how surprising" with a p-value.
  - The **p-value** is the probability of observing a value at least as extreme as the value actually observed, when the null hypothesis is true.
  - The p-value can be estimated by the proportion (out of the total number of repetitions) of values in the simulated null distribution that are *at least as extreme* as the actual observed value of the statistic.
    - Which tail(s) constitutes "at least as extreme" is determined by the form of the *alternative hypothesis*

- The smaller the p-value — that is, the closer the p-value is to 0 — the less plausible it is that the observed result occurred just by random chance alone.
- Thus, **the smaller the p-value the stronger the evidence to reject the null hypothesis in favor of the alternative hypothesis.**
- If the observed results provide strong evidence that the data did not arise by random chance alone — that is, when the p-value is small — then there is evidence of a **statistically discernible difference**.
- How small is a small p-value? You should think of a sliding scale: the closer the p-value gets to 0 the stronger the evidence to reject the null hypothesis in favor of the alternative. Some guidelines are below, but these should *not* be followed too strictly (especially when performing many hypothesis tests simultaneously):
- Do NOT rely strictly on arbitrary thresholds like p-value < 0.05.

p-value	Evidence to reject the null hypothesis?
Above 0.10	Little to no evidence
Between 0.01 and 0.10	Weak-ish evidence
Between 0.001 and 0.01	Moderate-ish evidence
Between 0.0001 and 0.001	Strong-ish evidence
Less than 0.0001	Very strong-ish evidence

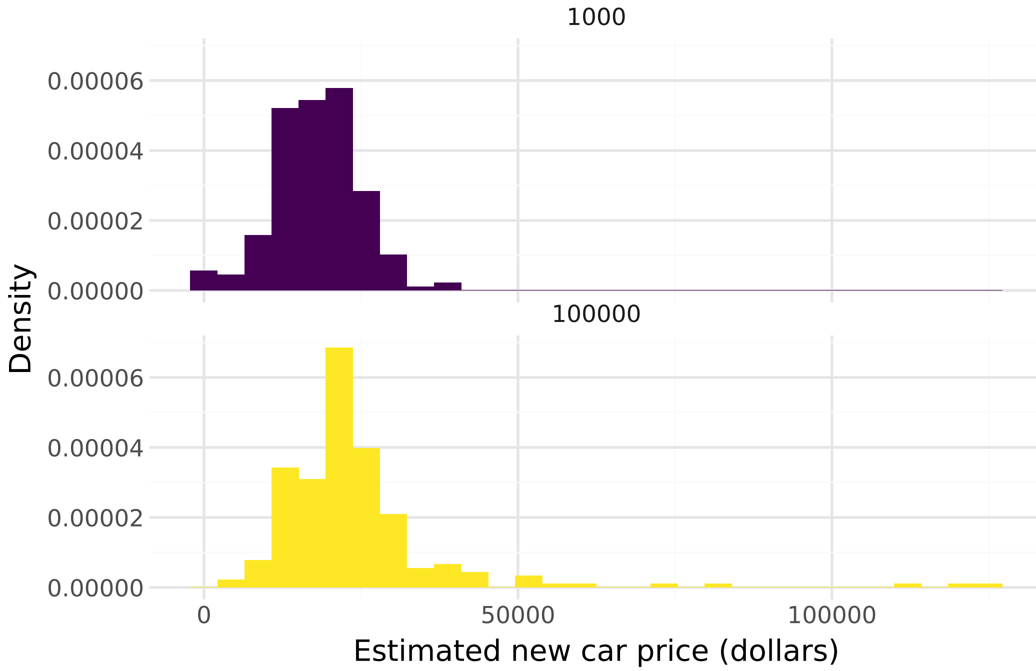
Example 28.2. In a study published in 2016, participants were asked "How much does an average new small car cost?" But participants were randomly assigned to one of two versions of this question:

- Does an average new car cost less or more than **100 hundred thousand** dollars? How much does an average new small car cost?
- Does an average new car cost less or more than **1 thousand** dollars? How much does an average new small car cost?

Did people tend to responded with different amounts for the new car price based on the wording of the question?

The following summarizes the sample data.

	count	mean	std	min	25%	50%	75%	max
anchor								
1000	205.0	18010.960976	6399.902625	13.0	15000.0	18000.0	20000.0	40000.0
100000	210.0	24288.971429	15345.328365	4000.0	18000.0	20000.0	25000.0	125000.0



1. Identify the null hypothesis.

2. Describe in detail how to use simulation to compute the p-value.

3. Describe in detail how to use theory to compute the p-value.

4. The p-value is <0.0001. State a conclusion in context.

5. Does the p-value by itself provide evidence that the people who saw the 100 thousand dollar “anchor” tended to give *much* higher estimates than those who saw the 1 thousand dollar “anchor”? Explain.

6. How could we address the question in the previous part? Perform a relevant analysis and interpret the results in context.

Two-sample t test for a difference in population (or treatment) means.

- Situation: binary explanatory variable (population/treatment 1, 2) and numerical response variable
- Parameter: $\mu_1 - \mu_2$ is the difference in population means (treatment means)
- Goal: assess the strength of the evidence against the “no difference” or “no treatment effect” hypothesis
- The null hypothesis is $H_0 : \mu_1 - \mu_2 = 0$
- The alternative hypothesis is one of $H_a : \mu_1 - \mu_2 > 0$, $H_a : \mu_1 - \mu_2 < 0$, or $H_a : \mu_1 - \mu_2 \neq 0$
  - When in doubt, use a two-sided alternative and a two-tailed p-value.
- Statistic: $\bar{x}_1 - \bar{x}_2$ is the difference in sample means for samples (groups) of size n_1, n_2
- The SE of the difference in sample means is

$$\mathrm{SE}(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\mathrm{SE}^2(\bar{x}_1) + \mathrm{SE}^2(\bar{x}_2)}$$

- Test statistic:

$$t = \frac{\text{observed} - \text{hypothesized}}{\text{SE}} = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

- The p-value is the probability of observing a value at least as extreme as t, in the tail direction indicated by the alternative hypothesis, for the t-distribution with (complicated) degrees of freedom.
  - In almost all situations encountered in practice you can use the usual empirical rule for Normal distributions

- Assumptions:
  - Either
    - The data are obtained through two unbiased and independent random samples from large populations
      - Or from one unbiased random sample which is classified according to two variables, and the two groups can be considered independent of each other.
    - The data are obtained from an experiment in which subjects were randomly assigned to two treatments.
  - No bias in data collection
  - Difference in population means is the parameter you want to estimate
- The t procedures for population means are valid as long as the sample size is large enough, regardless of the shape of the population distribution.
- **However, if the population distributions are highly skewed, then the population *medians* might be more appropriate parameters to consider as measures of center.**
- Which is more appropriate, the population means or the population medians, depends on the purpose for the inference.
- Also remember that sample means and SDs, and hence t procedures, are sensitive to outliers and extreme values.

Example 28.3. Continuing Example 28.2. In this data set there are some outliers that influence the sample means. What if instead we wanted to compare sample medians? There is no theory-based test for comparing medians like the two-sample t test for comparing means.

1. Explain in full detail how you could perform a simulation to approximate the p-value based on the difference in sample medians.

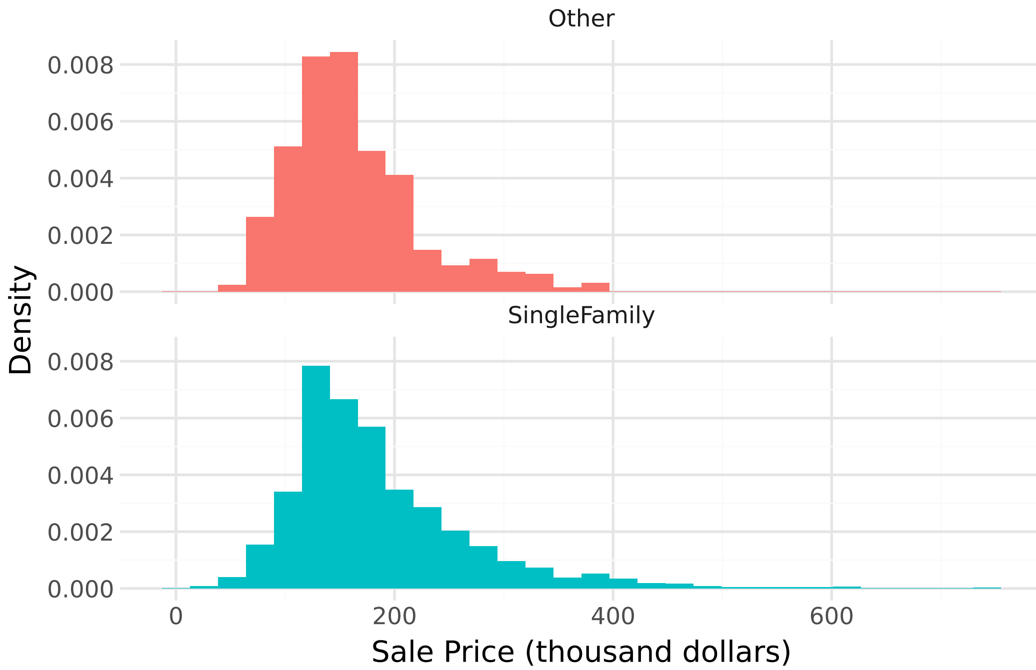
2. Conduct the simulation and use it to compute the p-value. Then interpret the result in context.

28.1 Exercises

Exercise 28.1. Continuing with the Ames housing data set. Now suppose we want to compare sale prices of single family homes with other homes (including condos, townhouses, etc). In particular, do single family homes tend to have higher sale prices on average than other homes?

The following summarizes the sample data.

SingleFamily	count	mean	std	min	25%	50%	75%	max
Other	505.0	161.511398	60.394023	55.000	120.0	147.4	190.0	392.5
SingleFamily	2425.0	184.812041	82.821802	12.789	130.0	165.0	220.0	755.0



1. Explain in detail how you could use simulation to approximate the p-value.

2. Use the sample data to conduct by hand an appropriate hypothesis test.

3. Write a clearly worded sentence interpreting the p-value in context.

4. Does the p-value by itself provide evidence that single family homes tend to have higher sale prices on average than other homes? Explain.

5. Does the p-value by itself provide evidence that single family homes tend to have *much* higher sale prices on average than other homes? Explain.

6. How could we address the question in the previous part? Perform a relevant analysis and interpret the results in context.

- Confidence intervals and hypothesis tests are commonly used in conjunction, but they address different questions
  - Hypothesis test: How strong is the evidence that there is a difference (or association or effect)?
  - Confidence interval: How large is the difference (or association or effect)?
- A confidence interval
  - provides a range of plausible values
  - but the cut-off between plausible and implausible is arbitrary (e.g. 95% confidence corresponds to $\alpha = 0.05$)
    * But be especially careful about interpreting confidence when conducting multiple confidence intervals.

- A null hypothesis test
  - Considers the plausibility of only a single hypothesized value
  - But provides a precise measure—the p-value—of the plausibility of the particular hypothesized value in light of the observed data
    * But be especially careful about interpreting p-values when conducting multiple hypothesis tests.

Exercise 28.2. A 2007 paper[2] suggests that in coin-tossing there is a particular "dynamical bias" that causes a coin to be slightly more likely to land the same way up as it started. That is, the paper claims that a coin is more likely to land facing Heads up if it starts facing Heads up than if it starts facing Tails up; likewise if it starts facing Tails up. Is this really true? Two students at Berkeley investigated this phenomenon by tossing a coin many, many times.

1. Suppose that out of 10000 flips, 5083 landed the same way they started, for a sample proportion of 0.5083. Based on this data, can you conclude that a coin is more likely to land the same way it started using a strict "significance level" of $\alpha = 0.05$? The p-value is 0.0495. State your conclusion at strict level $\alpha = 0.05$ in context. Note: One goal of the example is to show you one reason why using a strict α is a bad idea.

2. Repeat the previous part assuming 5082 out of 10000 flips landed the same way they started; the sample proportion is 0.5082 and the p-value is now 0.0515.

3. Compare the two previous parts. Give one reason why you should be wary of using a "significance level" (like 0.05) too strictly.

4. The students actually tossed the coin 40,000 times![3]. Of the 40,000 flips, 20,245 landed the same way they started, for a sample proportion of 0.5062. The p-value is 0.0072 and state a conclusion in context. (Do NOT use a strict α level.) Which result—20245 out of

[2] http://statweb.stanford.edu/~cgates/PERSI/papers/dyn_coin_07.pdf

[3] https://www.stat.berkeley.edu/~aldous/Real-World/coin_tosses.html

40000 or 5083 out of 10000—gives stronger evidence to reject the null hypothesis?

5. Would you say the results of the real study, in the previous part, are newsworthy? Or important? Or "significant"? Using this problem as an example, explain why "statistically significant" is a poor choice of words. In other words, explain why "statistical significance" is not the same as practical importance.

- Remember: the smaller the p-value, the stronger the evidence to reject the null hypothesis.
  - But how small is small enough?
- A "significance level", usually denoted α, is a pre-specified cut-off for how small the p-value must be in order to reject the null hypothesis. The most common value is $\alpha = 0.05$.
  - HOWEVER, in general, do NOT rely too strictly on a significance level α.
  - In fact, for years the the American Statistical Association has said to stop using values like 0.05 to make binary reject/not reject decisions.
- If the p-value is small the result is sometimes referred to as "statistically significant".
  - HOWEVER, this is a poor choice of words.
  - In fact, for years the American Statistical Association has said to stop using the term "statistically significant". Quote: " 'statistically significant'—don't say it and don't use it."
- Reporting the actual p-value and interpreting it as a sliding scale of weak/moderate/strong evidence to reject the null hypothesis is is much more informative than just reporting "reject H_0 at level $\alpha = 0.05$" or "p-value < 0.05".
- Remember that hypothesis testing and p-values are just small parts of a much bigger picture. A hypothesis test provides some information, but there are many important questions that hypothesis testing and p-values do not have the ability to address.
- Beware the meaning of "significant".
- A "statistically significant" result is not necessarily practically important or meaningful. Rather, it is just is more extreme than what would be expected to occur just due to chance if the null hypothesis were true, but not necessarily large in practical terms.

- Especially when the sample size is large, even a small observed difference can result in a small p-value.
- On the other hand, an observed result might have practical significance even when the p-value is not small.
- A confidence interval estimates the size of the difference, which helps in judging practical significance. It is also important to consider *effect size*.
- In short: Do NOT use $\alpha = 0.05$ and do NOT use the terminology "statistically significant" do NOT pay too much attention to hypothesis tests.

Exercise 28.3. (For this exercise, you'll need to believe it's possible for a person to have ESP.) "Ganzfeld studies" test psychic ability (like ESP). Ganzfeld studies involve two people, a "sender" and a "receiver", who are placed in separate acoustically-shielded rooms. The sender looks at a "target" image on a television screen (which may be a static photograph or a short movie segment playing repeatedly) and attempts to *transmit with their mind* information about the target to the receiver. The receiver is then shown four possible choices of targets, one of which is the correct target and the other three are "decoys." The receiver must choose the one transmitted by the sender[4].

1. We will now conduct some Ganzfeld tests to see if anyone in the class has ESP.
   - Find a partner. Identify one person as the sender and one as the receiver.
   - The sender rolls a four-sided die, looks at it (without letting the receiver see), and then tries to transmit the number rolled to the mind of the receiver.
   - After receiving the mental image, the receiver identifies the number. Record whether the answer is correct or not.
   - Repeat the above process for a total of 9 trials with the same receiver. The receiver should record the number of trials (out of 9) they got correct.
   - Then switch sender/receiver roles and do it again. Each receiver should record the number of trials (out of 9) that they got correct.

2. Use your results to conduct a hypothesis test to see if you have evidence that you did better than just guessing (and so you have some kind of ESP!)

[4]Source: Singh R, Meier T, Kuplicki R, Savitz J, et al., "Relationship of Collegiate Football Experience and Concussion With Hippocampal Volume and Cognitive Outcome," JAMA, 311(18), 2014.

3. Suppose we had set a cut-off of 0.05. Based on this strict cutoff, do you have enough evidence to reject the null hypothesis? That is, based on this cutoff, is your p-value small enough to conclude that you have ESP? (Hint: if you got at least 5 out of 9 correct your answer should be "yes"; the p-value for 5 correct out of 9 guesses is 0.0489, and it's even smaller if you get more than 5 correct out of 9.)

4. Consider a student who rejects the null hypothesis. Could their conclusion be wrong? How?

5. Consider a student who fails to reject the null hypothesis. Could their conclusion be wrong? How? (Again, you'll need to believe it's possible for a person to have ESP.)

6. Using the cutoff of 0.05, did any receiver in our class (of about 30 students) discover evidence that they have ESP? Should we be surprised that this happened?

- The null hypothesis H_0 is "innocent until 'proven' guilty". That is, the null hypothesis is not rejected unless there is convincing evidence against it.
- When making a conclusion — reject H_0, or fail to reject H_0 — there are two errors that could be made:
    - **Type I error:** Reject H_0 when H_0 is actually true (unbeknownst to the researcher)
        * "False alarm" or "false discovery"
        * Analogous to sending an innocent person to jail.
    - **Type II error:** Fail to reject H_0 when H_0 is actually false (unbeknownst to the researcher)
        * "Missed opportunity"

        - Analogous to letting a guilty person go free.

- The probability of making a Type I error can be managed by setting a "significance" level α.
- If decisions are made according to the rule "reject the null hypothesis only when p-value $< \alpha$" then the probability of making a Type I error is (at most) α.
- But there is a tradeoff: decreasing the probability of making a Type I error will increase probability of making a Type II error.
- The **power** of a test is the probability of correctly rejecting the null hypothesis when the the null hypothesis is false.

    - The power is the probability of making a "true discovery".

		What is true (unbeknownst to the jury)	
		Null hypothesis is true (defendant is innocent)	**Null hypothesis is false (defendant is guilty)**
What jury decides (based on evidence)	**Reject null hypothesis (Jury finds defendant guilty)**	Type I error (false alarm)	Correct Decision
	Do not reject null hypothesis (Jury finds defendant not guilty)	Correct Decision	Type II error (missed opportunity)

- Hypothesis tests are appropriate when there is a clearly defined research question and conjecture (i.e. alternative hypothesis), and sample data is collected for this purpose.
- If many hypothesis tests are performed, eventually you will reject a null hypothesis — just by chance — even if all the null hypotheses are really true.

    - For example, in a set of 35 tests all conducted using strict level $\alpha = 0.05$, there is about an 84 percent chance of rejecting at least one of the null hypotheses when all 35 null hypotheses are true.

- Beware of **p-hacking**, which involves repeating hypothesis tests until a small p-value is found
- The issue of multiple testing/p-hacking is another reason why you should not rely too strictly on a "significance" level α.
- Do NOT use a strict level α to make a strict reject H_0/fail to reject H_0 conclusion based on observed sample data about a particular hypothesized value or research question

    - Rather, report the p-value and assess strength of evidence in favor of rejecting the null hypothesis on a sliding scale.
    - But remember the p-value doesn't tell you anything about how big or meaningful the difference is—but rather just if it's not due to random chance—especially if the sample size is large

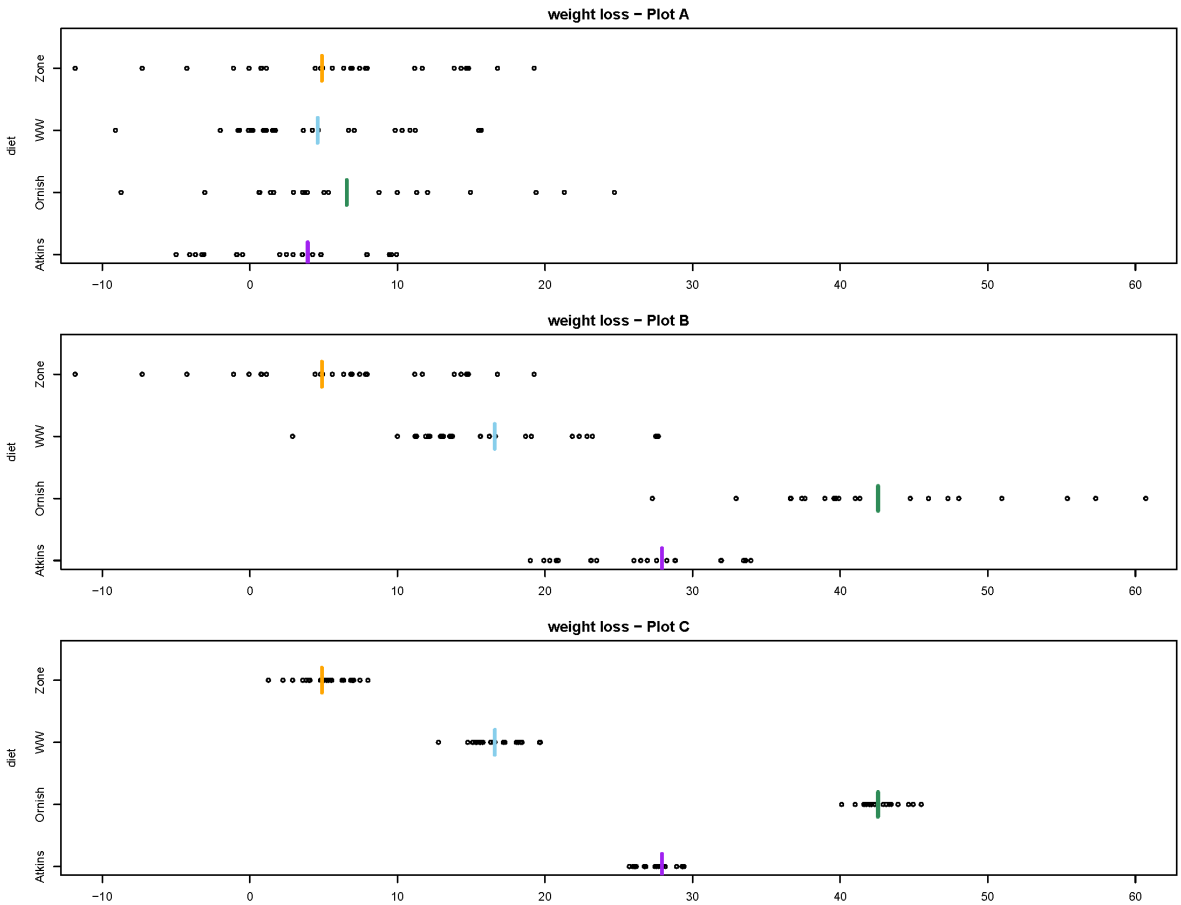
    - Be especially careful about using α or interpreting p-values when conducting multiple hypothesis tests on the same data. Beware p-hacking.
- Do USE "significance" levels
    - For research planning purposes — to manage probability of Type I error, to investigate power, and to determine an appropriate sample size — *before* collecting data
    - When determining a *range* of plausible values for the parameter, e.g., to determine the cutoff between plausible/not plausible in a confidence interval.
        * A 95% confidence is essentially the range of hypothesized values that would not be rejected at level 0.05 based on the observed data.
- Remember that hypothesis testing and p-values are just small parts of a much bigger picture. A hypothesis test provides some information, but there are many important questions that hypothesis testing and p-values do not have the ability to address.

29 Comparing multiple means: ANOVA F procedures

When dealing with a numerical variable, we often summarize the variable with its mean. We have seen how to perform inference for a single mean and for comparing two means. Now we will see how to compare multiple means. The process is called "analysis of variance" or ANOVA, but that name is a little misleading. Our conclusions will still be about means.

Example 29.1. Consider the following plots, which are all on the same scale. (This example just illustrates some general ideas so the context isn't important, but the data relate to a study on different types of diet plans (Atkins, etc) and weight loss. Plot A displays the study data. Plots B and C show hypothetical data sets with the same group sizes as in Plot A.) Group means are indicated by the colored vertical lines.

- The SD for each group in Plot B equals the SD for the corresponding group in Plot A.
- The mean for each group in Plot C equals the mean for the corresponding group in Plot B.



1. Rank the three scenarios (A, B, C) in order from weakest to strongest evidence to reject the null hypothesis of no difference in means between groups. That is, rank the plots in order from largest to smallest p-value.

2. Compare plots A and B.

   - The variability *between* groups in Plot B is **(choose one: $>$, $<$, $=$)** the variability *between* groups in Plot A.
   - The variability *within* groups in Plot B is **(choose one: $>$, $<$, $=$)** the variability *within* groups in Plot A.

3. Compare plots B and C.

   - The variability *between* groups in Plot C is **(choose one: $>$, $<$, $=$)** the variability *between* groups in Plot B.
   - The variability *within* groups in Plot C is **(choose one: $>$, $<$, $=$)** the variability *within* groups in Plot B.

4. Rank the plots in order from smallest to largest F statistic.

- Example 29.1 illustrates the idea behind the F statistic: we reject the null hypothesis of no difference in means between groups/populations/treatments if the observed variability *between* groups is large relative to the observed variability *within* groups.
- Roughly, we reject the null hypothesis if the ratio

$$\frac{\text{variability between groups}}{\text{variability within groups}}$$

  is large enough. This idea is known as **analysis of variance (ANOVA)**.
- The name ANOVA refers to *what* we are doing, and not *why* we are doing it.
- Remember, we are analyzing the variance to make a conclusion about the *means*
- The F statistic quantifies the above ratio. The larger the value of the F statistic the stronger the evidence to reject the null hypothesis (of no difference) in favor of the alternative hypothesis (of some difference). That is, the largest the F statistic the smaller the p-value.

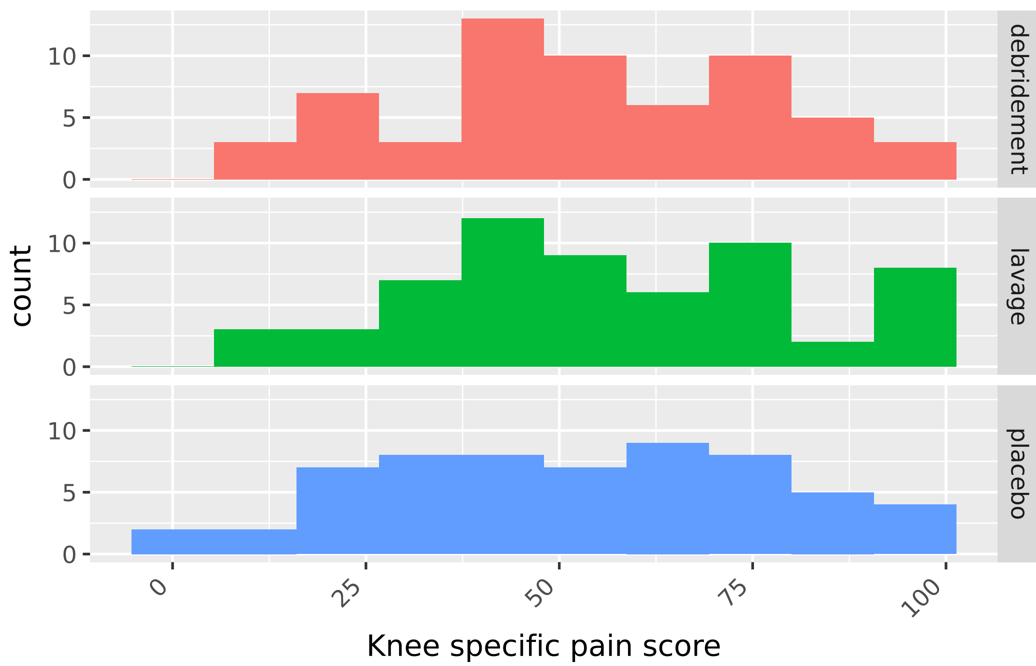
Example 29.2. When medical therapy fails to relieve the pain of osteoarthritis of the knee, arthroscopic lavage or debridement is often recommended. More than 650,000 such procedures are performed each year at a cost of roughly $5,000 each. A recent double blind experiment investigated the effectiveness of these procedures[1]. 180 subjects with osteoarthritis of the knee were randomly assigned to receive one of three treatments: arthroscopic lavage (which involved flushing the knee joint with fluid), arthroscopic debridement (which involved both lavage and shaving of cartilage and removal of debris), and placebo surgery (which involved incisions in the knee, but no insertion of instruments). Various outcomes were measured; the summaries below summarize scores two years after the procedure on a knee-specific pain scale (0-100, with with higher scores indicating more severe pain). Does any procedure tend to result in better pain scores on average?

The following summarizes the sample data. Note that the sample size is the same within each group, which somewhat simplifies the computation of the ANOVA F statistic.

	count	mean	std	min	25%	50%	75%	max
treatment								
debridement	60.0	53.983333	23.304536	10.0	40.75	52.0	76.25	97.0

[1]Source: http://www.nejm.org/doi/full/10.1056/NEJMoa013259

treatment	count	mean	std	min	25%	50%	75%	max
lavage	60.0	56.733333	24.150564	8.0	41.00	53.5	73.50	99.0
placebo	60.0	52.450000	25.079146	3.0	31.00	54.0	70.25	97.0



1. What statistic can be used to measure the variability *within* the Debridement group? Lavage? Placebo? How could you combine these numbers to to measure, in a single number, *variability within groups*?

2. If the null hypothesis is true, what would you expect about the group means?

3. What statistic can be used to measure the observed variability *between* groups?

4. Regardless of whether or not the null hypothesis is true, which of the values from parts 1 nand 3 would you expect to be larger? Why?

5. How could use measure the expected variability between groups if the null hypothesis is true?

6. Compute the F statistic as the value from part 3 divided by the value from part 5.

7. What value would you expect F to be if the null hypothesis is true?

8. Specify in detail how you could use simulation to approximate the null distribution of the F statistic.

9. Figure 29.1 displays results from an applet used to run 10000 repetitions of the simulation. How could you use the simulation results to approximate the p-value? Could you use the empirical rule for Normal distributions?

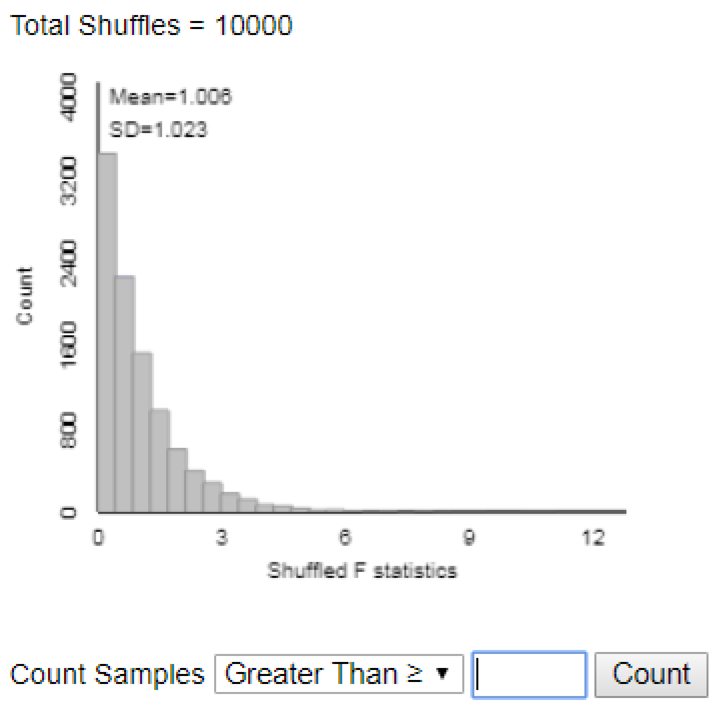


Figure 29.1: Permutation null distribution of F statistic for Example 29.2.

10. The p-value is about 0.6. State a conclusion in context. Are the results "significant"?

Hypothesis test for comparing multiple means: "One-way ANOVA F test"

- Situation: N observations of numerical response variable; categorical explanatory variable with g groups
- Parameters: μ_j's are the population/process/treatment response means; σ is the common group SD of responses
- Goal: test for any differences in means
- The null hypothesis is $H_0 : \mu_1 = \cdots = \mu_g$ ("no difference" or "no treatment effect")
- The alternative hypothesis is not H_0; that is, there is some difference in means between groups, $H_1 : \mu_j \neq \mu_k$ for some $j \neq k$
- Test statistic: one-way ANOVA F statistic

$$F = \frac{\sum_{j=1}^{g} n_j \left(\bar{x}_j - \bar{x}\right)^2 / (g-1)}{\sum_{j=1}^{g} (n_j - 1) s_j^2 / (N - g)}$$

- The p-value is the probability of observing a value at least as large as the observed F based on an F distribution with $g-1$ degrees of freedom in the numerator and $N-g$ degrees of freedom in the denominator
- Assumptions:
  - Either:
    - The data are obtained through multiple unbiased and independent random samples from large populations
      - Or from one unbiased random sample which can be classified into multiple groups, and the groups can be considered independent of each other.
    - The data are obtained from an experiment in which subjects were to two treatments.
  - No bias in data collection
  - Comparing *means* is an appropriate analysis
  - Equal variance: True SD of responses is same, σ, for each population/process/treatment.
    - Typically satisfied through random assignment.
    - Might not be satisfied if the data are sampled from different populations/processes.
    - One rule of thumb is that the largest observed sample SD should be no more than twice the smallest.
- Special case: when the group size n is the same for all g groups is

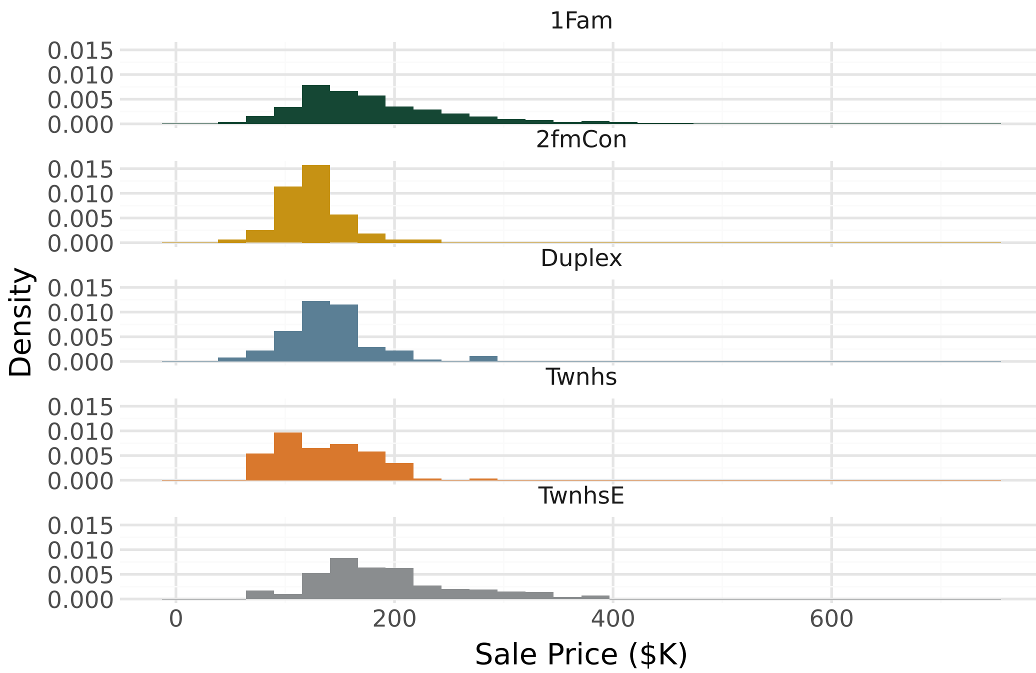
$$\begin{aligned} F &= \frac{\text{group size} \times \text{variance of group means}}{\text{mean of group variances}} \\ &= \frac{n \sum_{j=1}^{g} \left(\bar{x}_j - \bar{x}\right)^2 / (g-1)}{\sum_{j=1}^{g} s_j^2 / g} \end{aligned}$$

- The calculations are traditionally summarized in an **ANOVA table**

Source	df	SS (Sum of squares)	MS (Mean Square)	F
Between groups (a.k.a. "treament")	$g-1$	SS(between) $= \sum_{j=1}^{g} n_j \left(\bar{x}_j - \bar{x}\right)^2$	$\frac{\text{SS(between)}}{g-1}$	$\frac{\text{MS(between)}}{\text{MS(within)}}$
Within groups (a.k.a. "error")	$N-g$	SS(within) $= \sum_{j=1}^{g} (n_j - 1) s_j^2$	$\frac{\text{SS(within)}}{N-g}$	
Total	$N-1$	(SS(between) + SS(within)) $= \sum_{i=1}^{N} \left(x_i - \bar{x}\right)^2$		

Example 29.3. Continuing with the Ames housing data set. Now suppose we want to compare sale price of homes by building type: single-family (1Fam), two-family (2fmCon), duplex, townhouse end unit (TwnhsE), townhouse inside unit (Twnhs). The following summarizes the sample data.

BldgType	count	mean	std	min	25%	50%	75%	max
1Fam	2425.0	184.812041	82.821802	12.789	130.0000	165.000	220.000	755.00
2fmCon	62.0	125.581710	31.089240	55.000	106.5625	122.250	140.000	228.95
Duplex	109.0	139.808936	39.498974	61.500	118.8580	136.905	153.337	269.50
Twnhs	101.0	135.934059	41.938931	73.000	100.5000	130.000	170.000	280.75
TwnhsE	233.0	192.311914	66.191738	71.000	145.0000	180.000	222.000	392.50



1. Table 29.1 is the ANOVA table. The F statistic is 26.1 and the p-value is basically 0. State the conclusion of the ANOVA F test in context.

2. What are some natural follow-up questions? How might you use the data to answer them?

Table 29.1: ANOVA table for Example 29.4.

	sum_sq	df	F	PR(>F)
C(BldgType)	6.454111e+05	4.0	26.151358	2.475923e-21
Residual	1.804713e+07	2925.0	NaN	NaN

- Rejecting the null hypothesis based on an F statistic only tells you that "the mean for at least one of the treatments (populations) is not the same as the mean for at least one of the other treatments (populations)".
- The natural follow-up questions are: which treatment (population) means differ? and by how much?
- One approach when you have rejected the null hypothesis is to calculate — using software — all pairwise confidence intervals. In particular, which intervals do not contain zero?
- When computing multiple confidence intervals, the multiple (z^* or t^*) should be increased[2] to account for the **issue of multiple comparisons** (recall Exercise 26.2 and related notes).
- **Tukey's method** is a widely used method for constructing simultaneous confidence intervals for *all pairwise comparisons of means.* The confidence intervals computed using Tukey's method already contain an adjustment to account for multiple comparisons.

Example 29.4. Continuing Example 29.3. Table 29.2 contains the results of Tukey pairwise 95% confidence intervals.

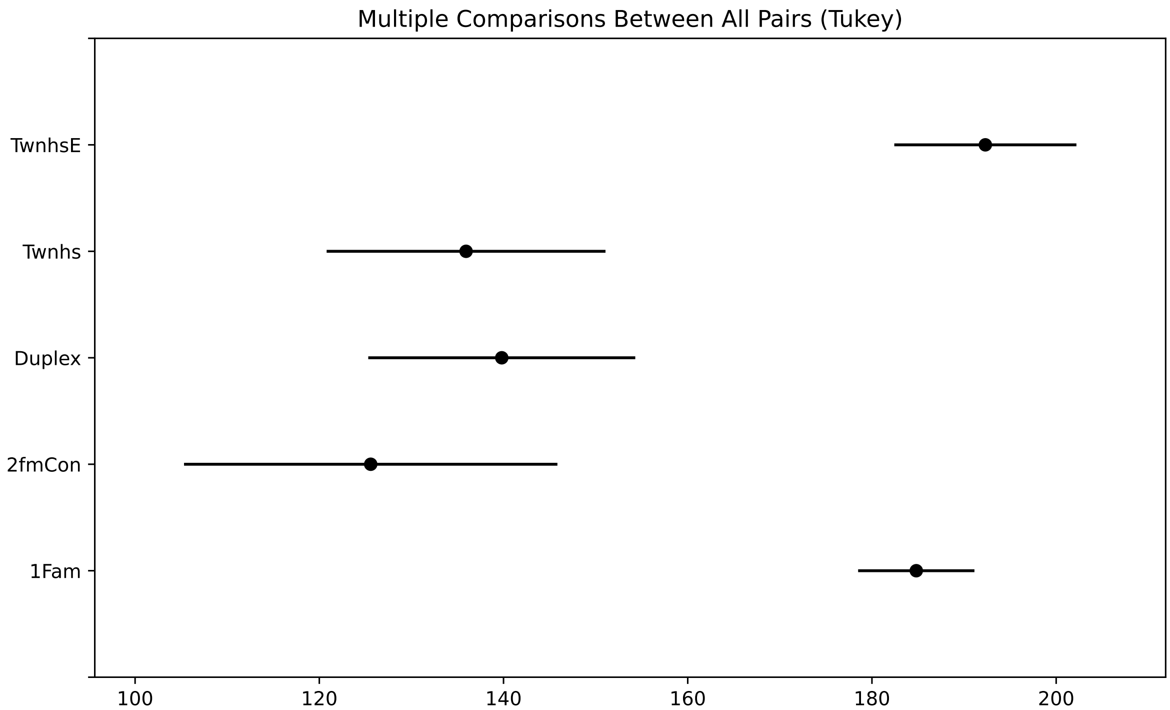
1. For which pairs of building types is there reasonable evidence of a difference in mean sale price? For which pairs is there NOT reasonable evidence of a difference?

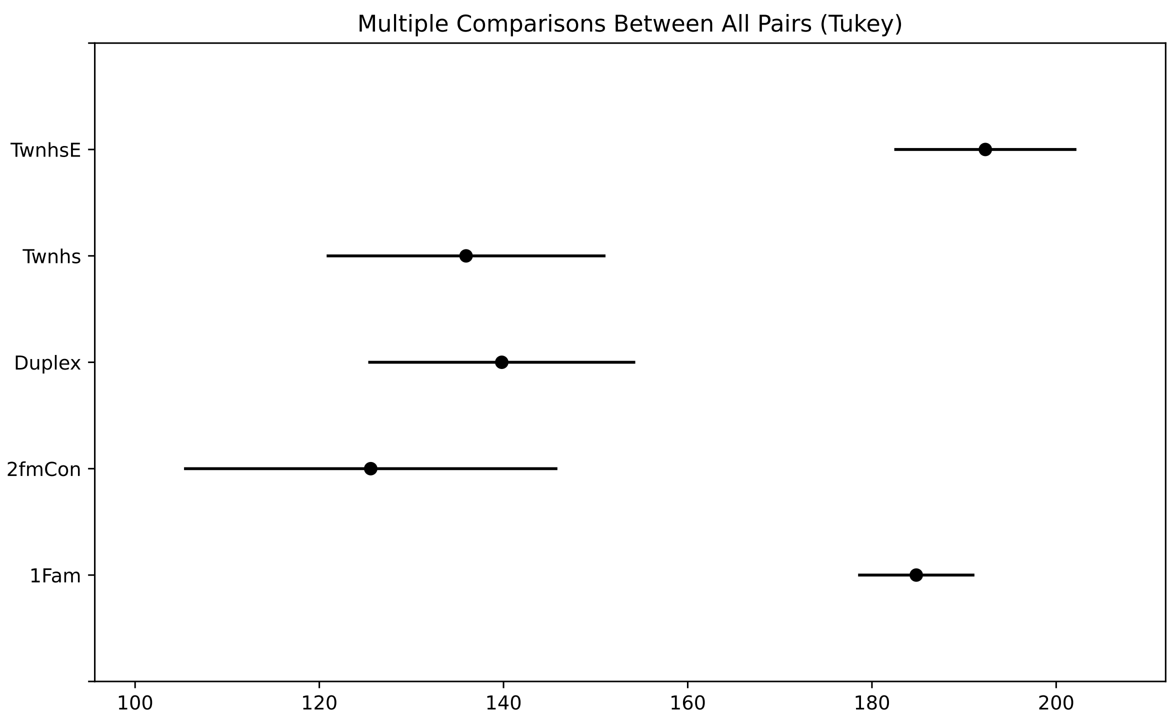
[2]There are several ways to do this. The simplest is a *Bonferroni* adjustment, which splits the error rate evenly across all intervals. For example, for simultaneous 95% confidence in 10 intervals (i.e. a total error rate of 0.05), use the t^* factor for 99.5% confidence ($0.995 = 1 - 0.05/10$), $t^* = 2.8$, instead of 2 when computing the intervals. But the Tukey method is more widely used when comparing means.

Table 29.2: Tukey pairwise 95% confidence intervals for Example 29.4.

```
 Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====================================================
group1 group2 meandiff p-adj   lower    upper   reject
-----------------------------------------------------
  1Fam 2fmCon -59.2303    0.0 -86.8048 -31.6559   True
  1Fam Duplex -45.0031    0.0 -65.9952 -24.0111   True
  1Fam  Twnhs  -48.878    0.0 -70.6511 -27.1049   True
  1Fam TwnhsE   7.4999 0.6328  -7.2051  22.2048  False
2fmCon Duplex  14.2272  0.786 -19.8771  48.3316  False
2fmCon  Twnhs  10.3523 0.9255 -24.2382  44.9429  False
2fmCon TwnhsE  66.7302    0.0  36.0924   97.368   True
Duplex  Twnhs  -3.8749 0.9965 -33.4861  25.7363  False
Duplex TwnhsE   52.503    0.0  27.6234  77.3825   True
 Twnhs TwnhsE  56.3779    0.0  30.8358  81.9199   True
-----------------------------------------------------
```

2. Interpret in context each of the confidence intervals that does NOT contain 0.


Multiple Comparisons Between All Pairs (Tukey)
TwnhsE
Twnhs
Duplex
2fmCon
1Fam
100
120
140
160
180
200


Multiple Comparisons Between All Pairs (Tukey)
TwnhsE
Twnhs
Duplex
2fmCon
1Fam
100
120
140
160
180
200

29.1 Exercises

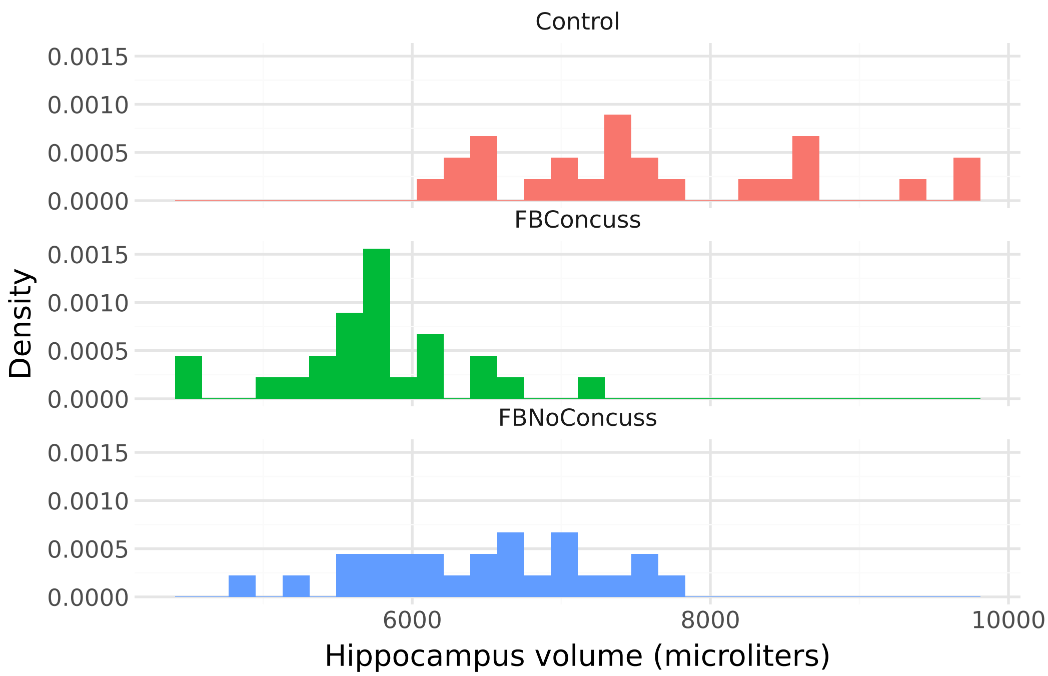
Exercise 29.1. Continuing Example 29.2. If we were to compute a series of pairwise confidence intervals for the difference in group means — Debridement − Lavage, Debridement − Placebo, Lavage − Placebo — what value would each of the confidence intervals contain? Why?

Exercise 29.2. A study[3] on brain trauma in football players included 3 groups, with 25 cases in each group. The control group consisted of healthy individuals with no history of brain trauma who were comparable to the other groups in age, sex, and education. The second group consisted of NCAA Division 1 college football players with no history of concussion, while the third group consisted of NCAA Division 1 college football players with a history of concussion. High resolution MRI was used to collect brain hippocampus volume (microliters).

Plots and summary statistics of hippocampus volume (microliters) for all three groups

	count	mean	std	min	25%	50%	75%	max
Group								
Control	25.0	7602.6	1074.022928	6175.0	6780.0	7385.0	8510.0	9710.0
FBConcuss	25.0	5734.6	593.397562	4490.0	5505.0	5710.0	6035.0	7160.0
FBNoConcuss	25.0	6459.2	779.739112	4810.0	5965.0	6515.0	7020.0	7790.0

[3] Source: Singh R, Meier T, Kuplicki R, Savitz J, et al., "Relationship of Collegiate Football Experience and Concussion With Hippocampal Volume and Cognitive Outcome," JAMA, 311(18), 2014.



1. Specify in words the null hypothesis.
2. The SD of the numbers 7602.6, 6459.2, and 5734.6 is 941.8. Compute by hand the ANOVA F statistic.
3. The ANOVA F statistic is 31.5. Describe in full detail how (in principle) you could use index cards to simulate the null distribution of the ANOVA F statistic and how you would use the simulation results to find the p-value. (You don't have to compute the p-value; just explain how you could find it after you performed the simulation.)
4. Use this applet to conduct the simulation from the previous part.
    - You will first need to copy and paste the data from this study into the applet.
    - In the Select Statistic box in the lower left, select F-statistic. Check that the Observed F-statistic is 31.473, and that your summary statistics match those from the summaries.
    - Click the "Show Shuffle Options" box and then run the simulation.
    - Enter the appropriate value in the "Count Samples" box and click "Count" to compute the p-value.
5. The p-value is basically 0 (see Table 29.3). Write a clearly worded sentence containing the conclusion of the hypothesis test in the context of the problem.
6. Table 29.4 contains the Tukey confidence intervals. Which of the confidence intervals contain 0, and which do not? Explain what this means.
7. Write a clearly worded sentence interpreting each of the three confidence intervals in context.

Table 29.3: ANOVA table for Exercise 29.2.

```
              sum_sq    df          F        PR(>F)
C(Group)  44348606.0   2.0  31.473165  1.507201e-10
Residual  50727336.0  72.0        NaN           NaN
```

Table 29.4: Tukey pairwise 95% confidence intervals for Exercise 29.2.

```
       Multiple Comparison of Means - Tukey HSD, FWER=0.05
==================================================================
  group1     group2   meandiff p-adj    lower      upper    reject
------------------------------------------------------------------
  Control   FBConcuss  -1868.0    0.0 -2436.1525 -1299.8475   True
  Control FBNoConcuss  -1143.4    0.0 -1711.5525  -575.2475   True
FBConcuss FBNoConcuss    724.6 0.0088   156.4475  1292.7525   True
------------------------------------------------------------------
```

Homework 1

1. In each of the following, write a clearly worded sentence interpreting the numerical value of the probability as a long run relative frequency in context. (Just take the numerical values as given for now. We'll see how to compute probabilities like these later.)

   a. The probability of rolling doubles when you roll two fair six-sided dice is 1/6.

   b. The probability of rolling doubles on three consecutive rolls of two fair six-sided dice is 0.00463.

   c. The probability that the sum of 100 rolls of a fair six-sided die is less than 370 is 12%.

   d. Roll a fair six-sided die until you roll a 6 three times and then stop. The probability that you roll the die at least 10 times is 0.822.

2. Suppose that at some point in 2023 your subjective probabilities for who would win the 2024 U.S. Presidential Election satisfied the following. (Imagine a time before Trump had been declared the Republican nominee, and while Biden was the presumptive Democratic nominee there was still speculation that he should step aside.)

   - The Democratic nominee and the Republican nominee are equally likely to be the winner of the Presidential Election
   - Joe Biden is 5 times more likely to win than Kamala Harris, and no other Democratic candidate has a chance of winning
   - Donald Trump is twice as likely to win as any other Republican candidate.

   Create a table of your subjective probabilities.

3. Suppose that you have applied to two graduate schools, A and B. Your subjective probability of being accepted is 0.6 for school A and 0.7 for school B. Hint: the fact that these are subjective probabilities does not change the way you solve the problem. Remember, the mathematics of probability is the same regardless of whether the probabilities represent long run relative frequencies or subjective degrees of relative likelihood. Just take the 0.6 and 0.7 values as given and use them to solve the following parts.

   Let A represent the event that you are accepted at school A, B the event that you are accepted at school B, and P the probability measure that represents your subjective probabilities. In addition to computing, represent each of the probabilities below with appropriate notation.

a. Interpret the subjective probabilities: How many times more likely than not are you to be accepted at school A? How many times more likely are you to be accepted at school B than at school A?
b. What is the largest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.
c. What is the smallest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.
d. Explain why your subjective probability of being accepted by both schools is not necessarily 0.42.
e. **For the remaining parts, suppose your subjective probability of being accepted at both schools is 0.55.** If you are accepted at school A, what is your probability of also being accepted at school B? For this and the remaining parts in addition to computing the probability represent it with proper notation.
f. If you are accepted at school A, what is your probability of not being accepted at school B?
g. If you are not accepted at school A, what is your probability of being accepted at school B?
h. If you are accepted at school B, what is your probability of also being accepted at school A?
i. If you are not accepted at school B, what is your probability of being accepted at school A?

4. For each of the following, create your own "which of (a) or (b) is larger?" example. You can use the ones from the handout as examples, but you should choose your own contexts. Identify the correct answer and write a sentence or two explaining the answer to a student who is confused. You can use illustrations in your explanations, but don't do any calculations.

   a. Reverse the direction of conditioning (like the 6 foot tall and NBA example)
   b. Compare conditional and unconditional probabilities (like man, versus man who is greater than 6 feet tall)
   c. Two "particulars" that people might think are not equally likely but are (like HHHHH versus HTHTT).
   d. Comparing "the particular" and "the general" (HTHTT versus 2 H in 5 flips).
   e. Comparing the probability of happening once versus at least once (like you winning the lottery versus someone winning the lottery)

5. The "matching problem" involves n distincts objects labeled $1, \ldots, n$ which are placed in n distinct boxes labeled $1, \ldots, n$, with exactly one object placed in each box. Suppose the objects are placed in the boxes uniformly at random, so that any possible arrangement is equally likely. We're interested in the number of matches (the number of objects for

which their label matches the label of the box in which they are placed.) Let C be the event that at least one object is placed in the correct spot (at least one match).

For example, suppose n people have a secret Santa gift exchange. Each person puts their name in a hat, the names are well shuffled, and everyone draws a name from the hat (to then purchase a gift for the selected person). Then C represents the event that at least one person draw their own name.

a. Describe in full detail how, in principle, you would physical objects (like cards) to simulate a single repetition of the placement of objects in boxes, and the number of matches.
b. Describe in full detail how, in principle, you could conduct a simulation and use the results to approximate $\mathrm{P}(C)$.
c. This applet conducts a simulation of the matching problem (in the context of babies in a hospital being mixed up and returned to parents uniforomly at random.) In the applet, "Number of babies" represents n and "Number of trials" represents the number of simulated repetitions. Starting with $n = 4$, try different values for the number of repetitions and hit the "Randomize" button to run the simulation. It helps to start with just a few repetitions so that you can see how the simulation is working, before jumping to thousands of repetitions. Use the simulation results to approximate $\mathrm{P}(C)$ when $n = 4$.
d. How do you expect $\mathrm{P}(C)$ to depend on n? As n increases, do you expect $\mathrm{P}(C)$ to increase, decrease, or stay about the same? Just think about it before proceeding; what does your intuition say?
e. Now try a few different values of n (say $n = 5$, $n = 10$, and $n = 20$, but you're free to choose other values). For each value of n run the simulation and use the results to approximate $\mathrm{P}(C)$. Record your results.
f. What do the simulation results suggest about the relationship between $\mathrm{P}(C)$ and n? (Remember that there is simulation margin of error, based on the number of repetitions. Two numerical approximations that are within the margion of error of each other are essentially the same approximation.)

6. Katniss throws a dart at a circular dartboard with radius 1 foot. Suppose that Katniss's dart lands at a uniformly random location on the dartboard (and she never misses the dartboard).

   a. Compute the probability that Katniss's dart lands within 1 inch of the center of the dartboard.
   b. Compute the probability that Katniss's dart lands more than 1 inch but less than 2 inches away from the center of the dartboard.

7. Imagine a light that flashes every few seconds[4]. The light randomly flashes green with probability 0.75 and red with probability 0.25, independently from flash to flash.

[4] Thanks to Allan Rossman for this example.

a. Write down a sequence of G's (for green) and R's (for red) to predict the colors for the next 40 flashes of this light. Before you read on, please take a minute to think about how you would generate such a sequence yourself.
b. Most people produce a sequence that has 30 G's and 10 R's, or close to those proportions, because they are trying to generate a sequence for which each outcome has a 75% chance for G and a 25% chance for R. That is, they use a strategy in which they predict G with probability 0.75, and R with probability 0.25. How well does this strategy do? Compute the probability of correctly predicting any single item in the sequence using this strategy.
c. Describe a better strategy. (Hint: can you find a strategy for which the probability of correctly predicting any single flash is 0.75?)

Homework 2

Problem 1

A standard deck of playing cards has 52 cards, 13 cards (2 through 10, jack, king, queen, ace) in each of 4 suits (hearts, diamonds, clubs, spades). Shuffle a deck and deals cards one at a time without replacement.

1. What is the probability that the first card dealt is a heart?
2. What is the probability that the second card dealt is a heart? Just make a guess now before proceeding, but you'll revisit this a few parts down.
3. What is the probability that the second card dealt is a heart if the first card dealt is a heart?
4. What is the probability that the second card dealt is a heart if the first card dealt is not a heart?
5. Revisit part 2. What is the probability that the second card dealt is a heart? Hint: your answer should be a single number, but show all your work.
6. Find the probability that the first two cards dealt are hearts.
7. Find the probability that the first two cards dealt are hearts and the third card dealt is a diamond.

Problem 2

The ELISA test for HIV was widely used in the mid-1990s for screening blood donations. As with most medical diagnostic tests, the ELISA test is not perfect. If a person actually carries the HIV virus, experts estimate that this test gives a positive result 97.7% of the time. (This number is called the *sensitivity* of the test.) If a person does not carry the HIV virus, ELISA gives a negative (correct) result 92.6% of the time (the *specificity* of the test). Estimates at the time were that 0.5% of the American public carried the HIV virus.

Suppose that a randomly selected American tests positive; we are interested in the conditional probability that the person actually carries the virus.

1. Before proceeding, make a guess for the probability in question. Which range do you think it's in: 0-20%, 20-40%, 40-60%, 60-80%, or 80-100%?
2. Denote the probabilities provided in the setup using proper notation

3. Construct an appropriate two-way table and use it to compute the probability of interest.
4. Construct a Bayes table and use it to compute the probability of interest.
5. Explain why this probability is small, compared to the sensitivity and specificity.
6. By what factor has the probability of carrying HIV increased, given a positive test result, as compared to before the test?
7. Now suppose that 5% of individuals in a high-risk group carry the HIV virus. Consider a randomly selectd person from this group who takes the test. Given that the test is positive, how many times more likely is it for the person not to have HIV than to have it? Answer without first computing a two-way or Bayes table.
8. Using the result from the previous part, compute the conditional probability that a person in this risk group who tests positive has HIV.
9. Is the posterior probability influenced by the prior probability? Discuss.

Problem 3

Consider three tennis players Arya ("A"), Brienne ("B"), and Cersei ("C"). One of these players is better than the other two, who are equally good/bad. When the best player plays either of the others, she has a 2/3 probability of winning the match. When the other two players play each other, each has a 1/2 probability of winning the match. But you do not know which player is the best. Based on watching the players warm up, you start with subjective probabilities of 0.5 that A is the best, 0.35 that B is the best, and 0.15 that C is the best. A and B will play the first match.

1. Suppose that A beats B in the first match. Compute your posterior probability that each of A, B, C is best given that A beats B in the first match.
2. Compare the posterior probabilities from the previous part to the prior probabilities. Explain how your probabilities changed, and why that makes sense.
3. **Coding required.** Code and run a simulation to simulate (1) who is the best player according to your prior distribution, (2) who wins the first match given who is the best player; then use the results to approximate the probabilities from the first part.
4. Suppose instead that B beats A in the first match. Compute your posterior probability that each of A, B, C is best given that B beats A in the first match.
5. Compare the posterior probabilities from the previous part to the prior probabilities. Explain how your probabilities changed, and why that makes sense.
6. Now suppose again that A beats B in the first match, and also that A beats C in the second match. Compute your posterior probability that each of A, B, C is best given the results of the first two matches. (Hint: use as the prior your posterior probabilities from the previous part.) Explain how your probabilities changed, and why that makes sense.

Problem 4

In the Powerball lottery, a player picks five different whole numbers between 1 and 69, and another whole number between 1 and 26 that is called the Powerball. In the drawing, the 5 numbers are drawn without replacement from a "hopper" with balls labeled 1 through 69, but the Powerball is drawn from a separate hopper with balls labeled 1 through 26. The player wins the jackpot if both the first 5 numbers match those drawn, in any order, and the Powerball is a match. Under this set up, there are 292,201,338 possible winning numbers.

1. What is the probability the next winning number is 6-7-16-23-26, plus the Powerball number, 4.
2. What is the probability the next winning number is 1-2-3-4-5, plus the Powerball number, 6.
3. The Powerball drawing happens twice a week. Suppose you play the same Powerball number, twice a week, every week for over 50 years. Let's say you purchase a ticket for 6000 drawings in total. What is the probability that you win at least once?
4. Instead of playing for 50 years, you decide only to play one lottery, but you buy 6000 tickets, each with a different Powerball number. What is the probability that at least one of your tickets wins? How does this compare to the previous part? Why?
5. Each ticket costs 2 dollars, but the jackpot changes from drawing to drawing. Suppose you buy 6000 tickets for a single drawing. How large does the jackpot need to be for your "expected" profit to be positive? To be $100,000? (We're ignoring inflation, taxes, transaction costs, and any changes in the rules.)

Problem 5

Consider a "best-of-5" series of games between two teams: games are played until one of the teams has won 3 games (requiring at most 5 games total). Suppose one team, team A, is better than the other, having a 0.55 probability of winning any particular game. Assume the results of the games are independent (and ignore advantage, etc). Let X represent the number of games played in the series. (Hint: It's helpful to first construct a two-way table of probabilities with the number of games played and which team wins, and then use it to answer the following questions. It will also help to list some outcomes, like AABA (team A wins game 1, 2, and 4, and B wins game 3).)

1. Compute the probability that team A wins the series in 3 games.
2. Compute the probability that the series ends in 3 games.
3. Compute the probability that team A wins the series.
4. Are the events "team A wins the series" and "the series ends in 3 games" independent? Explain by comparing relevant probabilities.
5. Let X represent the number of games played in the series. Find the distribution of X.

Homework 3

Problem 1

Maya is a basketball player who makes 40% of her three point field goal attempts. Suppose that at the end of every practice session, she attempts three pointers until she makes one and then stops. Let X be the total number of shots she attempts in a practice session. Assume shot attempts are independent, each with probability of 0.4 of being successful.

1. Identify by name the distribution of X. Be sure to specify the values of any relevant parameters.
2. Specify the probability mass function of X. Be sure to specify the possible values.
3. Construct a table, plot, and spinner corresponding to the distribution of X.
4. Compute $\mathrm{P}(X > 5)$ without summing.
5. Compute and interpret $\mathrm{E}(X)$.
6. Compute and interpret $\mathrm{SD}(X)$.

Problem 2

Ron and Leslie agree to the following bet. They'll ask Professor Ross if he saw the Eras Tour live. If he did, Leslie will pay Ron \$200; if not, Ron will pay Leslie \$100. (Neither has any direct information about whether or not Prof Ross saw the Eras Tour live.)

1. Given this setup, which of the following is being judged as more likely: that Prof Ross saw the Eras Tour, or that he did not? Why?
2. Ron and Leslie agree that this is a fair bet, and neither would accept worse odds. What is their subjective probability that Professor Ross saw the Eras Tour?
3. Suppose they were to hypothetically repeat this bet many times, say 3000 times. Given the probability from the previous part, how many times would you expect Leslie to win? To lose? What would you expect Leslie's net dollar winnings to be? In what sense is this bet "fair"?
4. Let X be Leslie's net winnings. Identify the distribution of X. (Remember: Leslie's winnings are Ron's losses and vice versa.)
5. Compute and interpret $\mathrm{E}(X)$.

Problem 3

Suppose that a total of 350 students at a college are taking a particular statistics course. The college offers five sections of the course, each taught by a different instructor. The class sizes are shown in the following table.

Section	A	B	C	D	E
Number of students	35	35	35	35	210

We are interested in: What is the average class size?

1. Suppose we randomly select one of the 5 *instructors*. Let X be the class size for the selected instructor. Specify the distribution of X. (A table is fine.)
2. Compute and interpret $\mathrm{E}(X)$.
3. Compute and interpret $\mathrm{P}(X = \mathrm{E}(X))$.
4. Suppose we randomly select one of the 350 *students*. Let Y be the class size for the selected student. Specify the distribution of Y. (A table is fine.)
5. Compute and interpret $\mathrm{E}(Y)$.
6. Compute and interpret $\mathrm{P}(Y = \mathrm{E}(Y))$.
7. Comment on how these two expected values compare, and explain why they differ as they do. Which average would you say is more relevant?

Problem 4

(This is an overly simplified example from actuarial science.) A life insurance company sells a term insurance policy to a 21 year old man (the "insured") that pays \$100,000 to a designated beneficiary if the insured dies within the next 5 years, and pays nothing if the insured does not die before age 26. The probability that a randomly chosen 21-year-old man will die each year can be found in mortality tables. The company collects a premium of \$250 at the beginning of each year of the 5 years of the term, as long as the insured is alive, as payment for the insurance. The amount Y (in dollars) that the company earns on this policy is \$250 per year, less the \$100,000 that it pays out in the event the insured dies. The company doesn't pay out anything if the insured does not die in the 5 year term.

Age at death	Earnings Y	Probability
21	$-99{,}750$	0.00183
22		0.00186
23		0.00189
24		0.00191
25		0.00193

Age at death	Earnings Y	Probability
26 or older		

1. Find the distribution of Y by completing the above table. For example, the value -99,750 provides an example of how Y is calculated when the insured dies in the first year of the policy.
2. Compute E(Y).
3. Write a sentence explaining in this context in what sense the number from the previous part is "expected".
4. Would you be willing to sell such an insurance policy to one of your 21 year old friends? Explain why this is good business for the insurance company, but not for you.
5. Compute Var(Y)
6. Compute SD(Y).
7. Compute the probability that Y is more than 1 standard deviation above its mean.

Problem 5

In a sample of 50 Olympic women's gymnasts the heights have a symmetric bell-shaped distribution with mean 60 inches and standard deviation 3 inches. In a sample of 50 Olympic men's basketball players the heights have symmetric bell-shaped distribution with mean 76 inches and the standard deviation 5 inches. Suppose the two samples are combined to obtain a sample with 100 heights. Choose one of the options below to complete the following sentence; explain your reasoning without doing any calculations (hint: drawing a picture might help; also, what is the mean of the combined sample?). The standard deviation of the combined sample is:

a. Less than 3 inches
b. Equal to 3 inches
c. Between 3 and 5 inches
d. Equal to 5 inches
e. Greater than 5 inches

Homework 4

Problem 1

Xavier bets on red on each of 3 spins of a roulette wheel. After the third spin, Xavier leaves and Zander arrives and bets on black on the next two spins. Let X be the number of bets that Xavier wins and let Z be the number that Zander wins.

1. Identify the distribution of X and the distribution of Z.
2. Are X and Z independent? Explain.
3. Compute and interpret $\mathrm{P}(X = 2, Z = 1)$.
4. What does $X + Z$ represent in this context? Identify the distribution of $X + Z$ without doing any calculations. Be sure to specify the values of any parameters. Explain your reasoning.
5. Compute $\mathrm{P}(X + Z = 3)$ without summing many terms.
6. Suppose instead that Yolanda bets on black on all 5 spins of the wheel. Does $X + Y$ have a Binomial distribution? Answer yes or no and explain.
7. Suppose instead that Zander bet on the number 7 in his two bets. Does $X + Z$ have a Binomial distribution in this case? Answer yes or no and explain.

Problem 2

Suppose the number of earthquakes per hour, for a certain range of magnitudes in a certain region, follows a Poisson distribution with parameter 0.7, independently from hour to hour.

1. Compute and interpret the probability that there is at least one earthquake of this size in the region in any given hour.
2. Compute and interpret the probability that there are exactly 3 earthquakes of this size in the region in any given hour.
3. Interpret the value 0.7 in context.
4. Construct a table, plot, and spinner corresponding to a Poisson(0.7) distribution.
5. Let X be the number of earthquakes in a 2 hour period. Identify by name the distribution of X. Be sure to specify the values of any relevant parameters.
6. Compute and interpret $\mathrm{P}(X = 3)$.

Problem 3

Recall Example 2.6. Regina and Cady are meeting for lunch. Suppose they each arrive uniformly at random at a time between noon and 1:00, independently of each other. Record their arrival times as minutes after noon, so noon corresponds to 0 and 1:00 to 60. Recall the sample space from Example 2.2 and assume a uniform probability measure P.

Let R be Regina's arrival time and Y Cady's. Then $W = |R - Y|$ is the amount of time (minutes) the first person to arrive waits for the second person to arrive.

1. Compute and interpret $\mathrm{P}(W < 15)$. (Hint: we computed this already in Example 2.6.)
2. Compute and interpret $\mathrm{P}(W > 45)$. (Hint: use a method similar to the previous part.)
3. It can be shown that W has pdf

$$f_W(w) = (2/3600)(60 - w), \qquad 0 < w < 60.$$

   Sketch a plot of the pdf. Describe roughly what this means in context.

4. Use the pdf to compute $\mathrm{P}(W < 15)$. Set up an integral, but sketch a picture and use geometry to compute.
5. Use the pdf to compute $\mathrm{P}(W > 45)$. Set up an integral, but sketch a picture and use geometry to compute.

Problem 4

Continuing with the setup of Problem 3. Consider $T = \min(R, Y)$ is the time (minutes after noon) at which the first person arrives.

1. Is T the same random variable as W from Problem 3? Explain.
2. Compute and interpret $\mathrm{P}(T < 15)$. (Hint: we computed something similar in Example 2.6.)
3. Compute and interpret $\mathrm{P}(T > 45)$. (Hint: use a method similar to the previous part.)
4. It can be shown that T has pdf

$$f_T(t) = (2/3600)(60 - t), \qquad 0 < t < 60.$$

   Does T have the same distribution as W from Problem 3? Explain what this means.

Homework 5

Problem 1

In a certain region, times (minutes) between occurrences of earthquakes (of any magnitude) have a distribution with percentiles displayed in the table below.

1. Construct a spinner corresponding to this distribution.
2. What percent of times are between 26.8 and 110.0 minutes?
3. Let F be the cdf. Evaluate and interpret $F(42.8)$.
4. Let Q be the quantile function. Evaluate and interpret $Q(0.9)$.
5. Sketch (by hand) a histogram of this distribution.

Problem 2

Recall Example 2.6. Regina and Cady are meeting for lunch. Suppose they each arrive uniformly at random at a time between noon and 1:00, independently of each other. Record their arrival times as minutes after noon, so noon corresponds to 0 and 1:00 to 60. Recall the sample space from Example 2.2 and assume a uniform probability measure P.

Let R be Regina's arrival time and Y Cady's. Then $W = |R - Y|$ is the amount of time (minutes) the first person to arrive waits for the second person to arrive. It can be shown that W has pdf

$$f_W(w) = (2/3600)(60 - w), \qquad 0 < w < 60.$$

Percentile	Value (minutes)
10th	12.6
20th	26.8
30th	42.8
40th	61.3
50th	83.2
60th	110.0
70th	144.5
80th	193.1
90th	276.3

1. Let F be the cdf. Evaluate and interpret $F(15)$.
2. Find an expression for the cdf of W. Set up an integral, but sketch a picture and use geometry to compute.
3. Find and interpret the 25th percentile of W. (You can do the next part first if you want, but it might help to start with a specific number like in this part.)
4. Find the quantile function of W.
5. Sketch a spinner corresponding to the distribution of W. Label the 25th, 50th, and 75th percentiles.

Problem 3

Continuing Problem 2. For this problem you should set up integrals when appropriate but you can use software to compute them.

1. Explain in full detail how you could use simulation to approximate
   a. $\mathrm{E}(W)$
   b. $\mathrm{E}(W^2)$
   c. $\mathrm{Var}(W)$ (explain how you can do it directly without using the previous part.)
2. Compute and interpret $\mathrm{E}(W)$.
3. Compute $\mathrm{E}(W^2)$.
4. Compute $\mathrm{Var}(W)$
5. Compute and interpret $\mathrm{SD}(W)$.
6. Compute the probability that W is more than 1 SD away from its mean.

Problem 4

Lifetimes of a certain type of component are Exponentially distributed with mean 20 thousand hours.

1. Approximate the probability that the lifetime, rounded to the nearest thousand hours, of a randomly selected component is 30 thousand hours.
2. Compute the probability that a randomly selected component has a lifetime greater than 30 thousand hours.
3. Compute the standard deviation of component lifetimes.
4. Suppose a device contains 2 such components connected in series, so that the system functions only if both of the components are functioning. Suppose the component lifetimes are independent of each other. Compute the probability that the *device* has a lifetime greater than 30 thousand hours.
5. Let T be the lifetime of the *device*. Compute $\mathrm{P}(T > t)$.
6. Identify the distribution of T by name, including the values of any relevant parameters.

Homework 6

Problem 1

For many animal species, the lengths of particular bones follow a Normal distribution. For adult male humans, the length of the femur (thigh bone, the longest bone in humans) has an approximate Normal distribution with mean 48.2 cm and standard deviation of 4.5 cm. Complete each of the following parts by standardizing, drawing appropriate plots, and then providing an educated ballpark estimate based on the empirical rule. Then you can use software to get more precise values. But software should be your last step, not the first.

1. What percent of adult male humans have a femur length above 50cm?
2. What percent of adult male humans have a femur length between 40cm and 50cm?
3. What is the 99th percentile of femur lengths for adult male humans?

Problem 2

Daily high temperatures (degrees Fahrenheit) in San Luis Obispo in August follow (approximately) a Normal distribution with a mean of 76.9 degrees F. The temperature exceeds 100 degrees Fahrenheit on about 1.5% of August days.

1. Compute the standard deviation.
2. Suppose the mean increases by 2 degrees Fahrenheit. On what percentage of August days will the daily high temperature exceed 100 degrees Fahrenheit? (Assume the standard deviation does not change.)
3. A mean of 78.9 is 1.02 times greater than a mean of 76.9. By what (multiplicative) factor has the percentage of 100-degree days increased? What do you notice?

Problem 3

The latest series of collectible Lego Minifigures contains 3 different Minifigure prizes (labeled 1, 2, 3). Each package contains a single unknown prize. Suppose we only buy 3 packages and we consider as our sample space outcome the results of just these 3 packages (prize in package 1, prize in package 2, prize in package 3). For example, 323 (or (3, 2, 3)) represents prize 3 in the first package, prize 2 in the second package, prize 3 in the third package. Let X be the number of distinct prizes obtained in these 3 packages. Let Y be the number of these 3 packages that contain prize 1. Suppose that each package is equally likely to contain any of the 3 prizes, regardless of the contents of other packages; let P denote the corresponding probability measure. There are 27 possible, equally likely outcomes

box1	box2	box3	X	Y
1	1	1		
2	1	1		
3	1	1		
1	2	1		
2	2	1		
3	2	1		
1	3	1		
2	3	1		
3	3	1		
1	1	2		
2	1	2		
3	1	2		
1	2	2		
2	2	2		
3	2	2		
1	3	2		
2	3	2		
3	3	2		
1	1	3		
2	1	3		
3	1	3		
1	2	3		
2	2	3		
3	2	3		
1	3	3		
2	3	3		
3	3	3		

1. Evaluate X and Y for each of the outcomes.
2. Construct a two-way table representing the joint distribution of X and Y.
3. Specify the marginal distribution of X and compute $\mathrm{E}(X)$.
4. Specify the marginal distribution of Y and compute $\mathrm{E}(Y)$.
5. Compute $\mathrm{Cov}(X, Y)$
6. Find the conditional distribution of Y given $X = x$ for each possible value of x of X.
7. Compute and interpret $\mathrm{E}(Y|X = x)$ for each possible value of x of X.

8. Find the conditional distribution of X given $Y = y$ for each possible value of y of Y.
9. Compute and interpret $\mathrm{E}(X|Y = y)$ for each possible value of y of Y.
10. Are X and Y independent? Explain; support your answer with appropriate numbers.

Problem 4

Xavier and Yolanda are playing roulette. They both place bets on red on the same 3 spins of the roulette wheel before Xavier has to leave. (Remember, the probability that any bet on red on a single spin wins is 18/38.) After Xavier leaves, Yolanda places bets on red on 2 more spins of the wheel. Let X be the number of bets that Xavier wins (out of 3) and let Y be the number that Yolanda wins (out of 5).

1. Identify by name the marginal distribution of X. Be sure to specify the values of any relevant parameters. Compute $\mathrm{E}(X)$.
2. Identify by name the marginal distribution of Y. Be sure to specify the values of any relevant parameters. Compute $\mathrm{E}(Y)$.
3. The joint distribution of X and Y is represented in the table below. Explain why $p_{X,Y}(1, 4) = 0$ and $p_{X,Y}(2, 1) = 0$.
4. Compute $p_{X,Y}(2, 3)$. (Yes, the table tells you it's 0.1766, but you have to show how this number can be computed based on the assumptions of the problem.)
5. Compute $\mathrm{Cov}(X, Y)$.
6. Are X and Y independent? Explain; support your answer with appropriate numbers.

x, y	0	1	2	3	4	5
0	0.0404	0.0727	0.0327	0	0	0
1	0	0.1090	0.1963	0.0883	0	0
2	0	0	0.0981	0.1766	0.0795	0
3	0	0	0	0.0294	0.0530	0.0238

Homework 7

Problem 1

Xavier and Yolanda are playing roulette. They both place bets on red on the same 3 spins of the roulette wheel before Xavier has to leave. (Remember, the probability that any bet on red on a single spin wins is 18/38.) After Xavier leaves, Yolanda places bets on red on 2 more spins of the wheel. Let X be the number of bets that Xavier wins and let Y be the number that Yolanda wins.

1. Compute and interpret $\mathrm{P}(X + Y = 4)$.
2. Make a table representing the marginal distribution of $X + Y$.
3. Use the table from the previous part to compute $\mathrm{E}(X + Y)$, and verify that it is equal to $\mathrm{E}(X) + \mathrm{E}(Y)$. (You computed $\mathrm{E}(X)$ and $\mathrm{E}(Y)$ in a previous homework.)
4. Without doing any computation, determine if $\mathrm{Var}(X + Y)$ will be greater than, less than, or equal to $\mathrm{Var}(X) + \mathrm{Var}(Y)$. Explain your reasoning.
5. Use the marginal distribution of $X + Y$ (from earlier in this problem) to compute $\mathrm{Var}(X + Y)$.
6. In a previous HW you identified the marginal distribution of X by name. Use facts about that distribution to compute $\mathrm{Var}(X)$.
7. In a previous HW you identified the marginal distribution of Y by name. Use facts about that distribution to compute $\mathrm{Var}(Y)$.
8. Verify that $\mathrm{Var}(X + Y)$ is equal to $\mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y)$. (You computed the covariance in a previous homework.)

x, y	0	1	2	3	4	5
0	0.0404	0.0727	0.0327	0	0	0
1	0	0.1090	0.1963	0.0883	0	0
2	0	0	0.0981	0.1766	0.0795	0
3	0	0	0	0.0294	0.0530	0.0238

Problem 2

A vendor at a weekend farmers market sells three products: jars of honey, loaves of sourdough bread, and bags of ice. Let X be the number of jars of honey sold, Y the number of loaves sold, and Z the number of bags of ice sold. Based on several seasons of sales data, the number of units sold on a given Saturday follows a joint distribution with

- Mean 40 and standard deviation 8 for honey jars sold,
- Mean 25 and standard deviation 5 for bread loaves sold,
- Mean 15 and standard deviation 6 for ice bags sold,
- Correlation of 0.5 between honey jars sold and bread loaves sold,
- Correlation of -0.4 between honey jars sold and ice bags sold. (Cooler days tend to bring out more shoppers browsing specialty foods like honey, but fewer people buying ice.)

Honey jars sell for \$12 each, bread loaves sell for \$7 each, and ice bags sell for \$4 each.

1. Let $R_1 = 12X + 7Y$ be the vendor's revenue from honey and bread combined. Compute $E(R_1)$ and $SD(R_1)$.
2. Let $R_2 = 12X + 4Z$ be the vendor's revenue from honey and ice combined. Compute $E(R_2)$ and $SD(R_2)$.
3. Without doing any additional calculation, explain whether you'd expect each of $SD(R_1)$ and $SD(R_2)$ to be larger or smaller if the variables were uncorrelated.

Problem 3

Devi and Paxton are meeting. Arrival times are measured in minutes after noon, with negative times representing arrivals before noon. Devi's arrival time follows a Normal distribution with mean 20 and SD 15 minutes, and Paxton's arrival time follows a Normal distribution with mean 25 and SD 10 minutes.

All of the probability calculations will involve a Normal distribution, but make sure you clearly identify the correct mean and SD. You should also compute standardized values and use a table to make a ballpark estimate of probabilities before jumping straight to software/calculator.

You will compute all of the following probabilities in each of the 3 scenarios below.

a. The probability that Devi arrives before noon.
b. The probability that Devi arrives before Paxton.
c. The probability that Devi arrives before Paxton given that Paxton arrives at 12:10.
d. The probability that the first person to arrive has to wait more than 15 minutes for the second person to arrive.

1. Scenario 1: Assume the arrival times are independent
2. Scenario 2: Assume the pairs of arrival times follow a Bivariate Normal distribution with correlation 0.8
3. Scenario 3: Assume the pairs of arrival times follow a Bivariate Normal distribution with correlation -0.7

Problem 4

Consider the general problem of estimating a population proportion p for some binary (success/failure) categorical variable of interest. Select a sample of size n and let X be the number of successes in the sample.

1. We are considering a general problem here, but describe a specific context where you might want to do this: what is the population and its rough size, what is the variable of interest and what is "success", what is the sample, how do you measure X for the sample? (For example, the handouts covered the general problem of estimating μ for a Poisson population, but we thought of it in terms of the car dealership problem.)
2. Suggest an estimator of p; it should be an expression involving X and n. Describe in words what your estimator is. (Hint: what is the most obvious estimator you can think of?)
3. Suppose you observe $x = 6$ successes in a sample of size $n = 10$. Use your estimator to compute a point estimate of p based on this sample.
4. Consider the estimator (called the "plus four" estimator)

$$\frac{X+2}{n+4} = \left(\frac{4}{n+4}\right)(0.5) + \left(\frac{n}{n+4}\right)\left(\frac{X}{n}\right)$$

   Explain in words what this estimator does. (The two expressions in the equation above are just different ways of expressing the estimator.)
5. Suppose you observe $x = 6$ successes in a sample of size $n = 10$. Use the plus four estimator to compute a point estimate of p based on this sample.
6. If you observe $x = 6$ successes in a sample of size $n = 10$, which of your two estimates is a better estimate of p? Explain.

Homework 8

Problem 1

Consider the general problem of estimating a population proportion p for some binary (success/failure) categorical variable of interest in a large population. Select a sample of size n and let X be the number of successes in the sample. (We are considering a general problem here, but recall your particular context from a previous homework.)

We'll consider two estimators of p:

$$
\begin{aligned}
\text{Sample proportion:} \qquad & \hat{p} = \frac{X}{n} \\
\text{Plus four:} \qquad & \tilde{p} = \frac{X+2}{n+4}
\end{aligned}
$$

Suppose we observe $x = 6$ successes in a sample of size $n = 10$. For this sample $\hat{p} = 0.6$ and $\tilde{p} = 0.571$.

1. Compute and interpret the probability of observing $x = 6$ successes in a sample of size $n = 10$ if $p = 0.6$.
2. Compute and interpret the probability of observing $x = 6$ successes in a sample of size $n = 10$ if $p = 0.571$.
3. Which of the estimates, 0.6 or 0.571, is a *better estimate* of p based on a sample of size $n = 10$ with $x = 6$ successes? That is, which estimate, 0.6 or 0.571, is closer to the true p? Explain.
4. Which of the estimates, 0.6 or 0.571, would you choose as your point estimate of p based on a sample of size $n = 10$ with $x = 6$ successes? Explain. Hint: consider parts 1 and 2.
5. Suppose you observe $x = 6$ successes in a sample of size $n = 10$. Carefully write the likelihood function. Hint: what is the distribution of X?
6. Use any software of your choice to plot the likelihood function. Based on your plot, what value appears to be the maximum likelihood estimate of p based on this sample? (You can do this just by zooming in on the plot.)
7. Carefully write the log-likelihood function.
8. Use calculus to find the maximum likelihood estimate of p based on $x = 6$ successes in a sample of size $n = 10$.

Problem 2

Continuing Problem 1, now consider a general sample of size n with x successes.

1. Carefully write the likelihood function. (The expression should contain x and n.)
2. Continuing the previous part, carefully write the log-likelihood function. (The expression should contain x and n.)
3. Use calculus to find a general expression for the maximum likelihood estimate of p. Your answer should be an expression involving x and n.

Problem 3

Assume a Poisson(μ) model for the number of home runs hit (in total by both teams) in a MLB game. Let $X_1, \dots, X_n$ be a random sample of home run counts for n games. (The context doesn't matter, but it helps to have something in mind rather than just the general problem.)

Suppose we want to estimate $\theta = \mu e^{-\mu}$, the probability that any single game has exactly 1 HR (for Poisson(μ), $P(X = 1) = e^{-\mu} u^1/1! = \mu e^{-\mu}$). Consider two estimators of θ:

- $\hat{\theta} = \bar{X} e^{-\bar{X}}$
- $\hat{p}$ = sample proportion of 1s = $\frac{\text{number of games in the sample with 1 HR}}{\text{sample size}}$

1. Compute the value of $\hat{\theta}$ based on the sample (3, 0, 1, 4, 0). Write a clearly worded sentence reporting in context this estimate of θ.
2. Compute the value of $\hat{p}$ based on the sample (3, 0, 1, 4, 0). Write a clearly worded sentence reporting in context your estimate of θ.
3. Which of these two estimators is the MLE of θ in this situation? Explain, without doing any calculations.
4. It can be shown that $\hat{p}$ is an unbiased estimator of θ. Explain in words what this means.
5. Is $\hat{\theta}$ an unbiased estimator of θ? Explain. (You don't have to derive anything; just apply a general principle.)
6. Suppose $\mu = 3.4$ and $n = 5$. Explain in full detail how you would use simulation to approximate the bias of $\hat{\theta}$ in this case. In particular, specify the spinner you would use to simulate values.
7. Explain in full detail how you would use simulation to approximate the bias *function* of $\hat{\theta}$ when $n = 5$.

Problem 4

Continuing Problem 3.

1. It can be shown that $\text{Var}(\hat{p}) = \frac{\theta(1-\theta)}{n}$. Compute $\text{Var}(\hat{p})$ when $\mu = 3.4$ and $n = 5$. Then write a clearly worded sentence interpreting this value.
2. Suppose $\mu = 3.4$ and $n = 5$. Explain in full detail how you would use simulation to approximate the variance of $\hat{\theta}$.
3. Explain in full detail how you would use simulation to approximate the variance function of $\hat{\theta}$ (if $n = 5$).

Problem 5

Continuing Problems 3 and 4

1. Compute $\text{MSE}(\hat{p})$ when $\mu = 3.4$ and $n = 5$. (You can do the next part first if you want, but it helps to work with specific numbers first.)
2. Derive the MSE function of $\hat{p}$. (Hint: use facts from previous parts.)
3. Suppose $\mu = 3.4$ (and $n = 5$). Explain in full detail how you would use simulation to approximate the MSE of $\hat{\theta}$.
4. Suppose $\hat{\theta}$ has smaller MSE than $\hat{p}$ when $\mu = 3.4$ (and $n = 5$). Which estimator would you prefer? Answer, but then explain why this information alone is not really useful.
5. Explain in full detail how you would use simulation to approximate the MSE *function* of $\hat{\theta}$ (if $n = 5$).
6. Suppose you determine that $\hat{\theta}$ has smaller MSE for almost all values of μ regardless of n. Recall parts 1 and 2 of Problem 3. Which value of these two estimates (the values from parts 1 and 2 of Problem 3) is a better estimate of θ? That is, which of the two estimates is closer to the true θ? Explain.
7. Suppose you determine that $\hat{\theta}$ has smaller MSE for almost all values of μ regardless of n. Recall parts 1 and 2 of Problem 3. Which value of these two estimates (the values from parts 1 and 2 of Problem 3) would you choose as your point estimate of θ? Explain.

Homework 9

Problem 1

In roulette, a bet on a single number has a 1/38 probability of success and pays 35-to-1. That is, if you bet 1 dollar, your net winnings are -1 with probability 37/38 and +35 with probability 1/38. Consider betting on a single number on each of n spins of a roulette wheel. Let $\bar{X}_n$ be your average net winnings per bet.

For each of the values $n = 10$, $n = 100$, $n = 1000$:

1. Compute $\mathrm{E}(\bar{X}_n)$
2. Compute $\mathrm{SD}(\bar{X}_n)$
3. Run a simulation to determine if the distribution of $\bar{X}_n$ is approximately Normal. (It's fine if you use an applet, but you can also try coding)
4. Use simulation to approximate $\mathrm{P}(\bar{X}_n > 0)$, the probability that you come out ahead after n bets.
5. If $n = 1000$ use the Central Limit Theorem to approximate $\mathrm{P}(\bar{X}_{1000} > 0)$, the probability that you come out ahead after 1000 bets.
6. The casino wants to determine how many bets on a single number are needed before they have (at least) a 99% probability of making a profit. (Remember, the casino profits if you lose; that is, if $\bar{X}_n < 0$.) Use the Central Limit Theorem to determine the minimum number of bets (keeping in mind that n must be an integer). You can assume that whatever n is, it's large for the CLT to kick in.

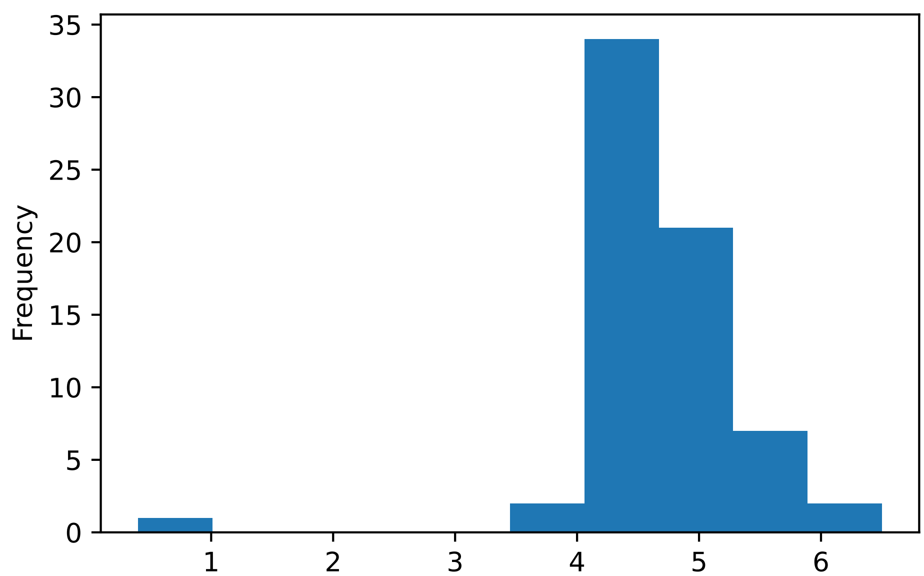
Problem 2

The standard measurement of the alcohol content of drinks is alcohol by volume (ABV), which is given as the volume of ethanol as a percent of the total volume of the drink. In a sample of 67 brands of beer, the mean ABV is 4.61 percent and the standard deviation of ABV is 0.754 percent.

We should never analuze data without first looking at some plots, but we're going to that now.

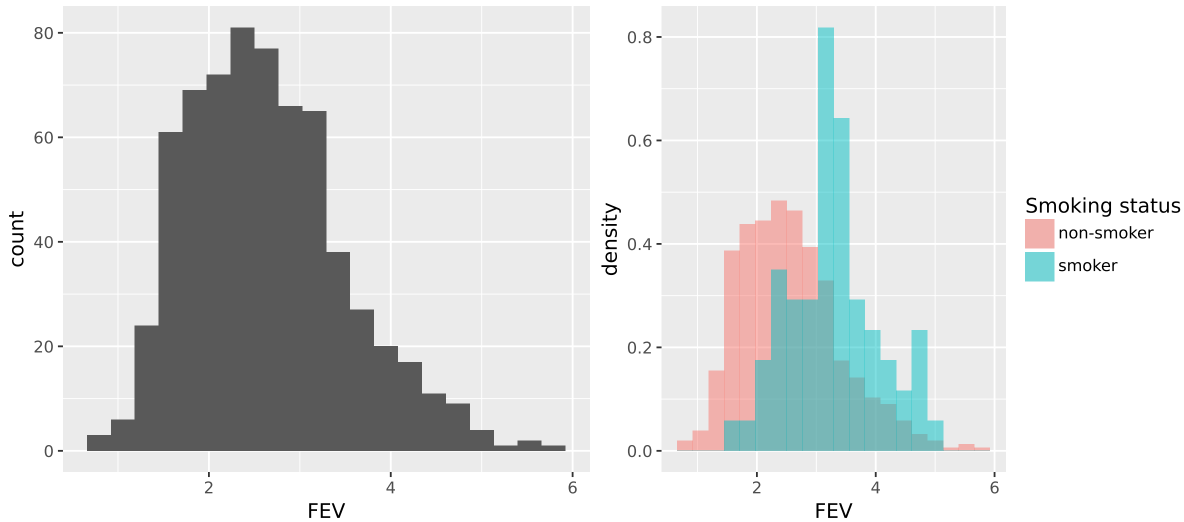
1. Compute a 95% confidence interval for the appropriate population mean.

2. Write a clearly worded sentence reporting your confidence interval in context.
3. Is 4.5% a plausible value of the parameter? Explain briefly.
4. When we plot the data (which we should have done first!) we notice an outlier (see plot below). One of the brands of beer in the sample is O'Doul's, a non-alcoholic beer. The ABV for O'Doul's is 0.4% (it has a bit of alcohol.) Suppose O'Doul's is removed from the data set. Compute the sample mean ABV of the remaining 66 brands.
5. The sample SD of ABV of the remaining 66 brands is 0.55 percent. Explain intuitively why this value is smaller than the sample SD of all 67 brands.
6. Compute the 95% confidence interval based on the sample with O'Doul's removed. Compare to the original interval, both in terms of center of the CI and its width. Explain briefly.
7. Based on the interval based on the sample with O'Doul's removed, is 4.5% a plausible value of the parameter? Explain briefly.
8. Which of the analyses is more appropriate: with or without O'Doul's? Explain your reasoning.



Problem 3

Do non-smokers have better lungs than smokers? One measure of lung function is forced expiratory volume (FEV), the amount of air (in liters) an individual can exhale in the first second of forceful breath. Larger values of FEV are indicative of better functioning lungs. In a study, FEV was measured for a group of 654 subjects, along with whether they smoked and other variables like age (years). Note: I'm purposely not yet giving you much information about how the sample was collected.



(a) (b)

1. Suppose you want to fill in the blanks in the following sentence: 95% of FEV values in the population are between [blank] and [blank]. Assuming this sample is representative, provide reasonable values that fill in the blanks, and explain your reasoning. Hint: it is better to give a correct rough estimate than a precise but incorrect answer, so think before trying to compute anything. (Since you don't have much information about the sample, you don't have to clarify yet what "population" is appropriate.)
2. Using appropriate summary statistics, compute by hand a 95% confidence for the appropriate difference in population means.
3. Write a clearly worded sentence interpreting the confidence interval in context.
4. Assuming this is a representative sample, are we reasonably confident that one group (smokers or non-smokers) tends to have better FEV? Why? Which group?
5. You should have observed something surprising in the previous parts. You decide to look closer at the data (you should have done this in the first place) and discover that the 654 subjects in this study were all children! Their ages varied from 3 to 19 years old. How does this change your conclusions? Can you suggest an explanation for the surprising result?

Problem 4

The following table displays the time (measured continuously in minutes) until the first goal was scored in each of 20 professional hockey games. The sample mean is 12.1 minutes, the sample median is 8.9 minutes, and the sample SD is 13.1 minutes.

4.4	14.9	2.8	1.4	1.1	8.1	11.7	3.9	2.4	15.7
8.8	0.6	5.1	10.4	9.1	13.7	27.8	9.0	46.0	44.7

1. You wish to estimate the population *median* time until the first goal is scored. Describe in detail in words how you could use the sample data and simulation to find an appropriate 95% bootstrap percentile confidence interval.
2. The following summarizes the bootstrap distribution of the sample median based on the sample data and an appropriate simulation.

Min	2.5th percentile	Median	97.5th percentile	Max	Mean	SD
1.25	4.15	8.90	12.70	19.75	8.52	2.10

   Using the above information, compute the endpoints of each of the following bootstrap 95% confidence intervals for the population median.

   a. Normal interval
   b. Percentile interval

3. Write a clearly worded sentence reporting the bootstrap percentile confidence interval from the previous part in context.

Problem 5

Two different machines (A and B) that fill packages of a certain candy are calibrated to a weight of 50 grams. Naturally, the weights of individual packages vary somewhat in the production process, but too much variation is undesirable. A small sample of packages is taken from each machine; the weights (grams) are in the following table.

Machine						Mean	SD
A	47.1	48.7	50.1	50.2	50.5	49.3	1.4
B	48.6	50.5	50.6	51.4	52.0	50.6	1.3

1. Describe in detail in words how you could use the sample data and simulation to find a 95% bootstrap percentile confidence interval for the *ratio of the variances* of packages weights for the two machines.
2. The confidence interval from the previous part is (0.01, 14.4). Write a clearly worded sentence reporting the confidence interval from the previous part in context.
3. The confidence interval was based on 10000 bootstrap samples. Explain why the interval is so imprecise.

Homework 1 Solutions

Problem 1

In each of the following, write a clearly worded sentence interpreting the numerical value of the probability as a long run relative frequency in context. (Just take the numerical values as given for now. We'll see how to compute probabilities like these later.)

a. The probability of rolling doubles when you roll two fair six-sided dice is 1/6.

b. The probability of rolling doubles on three consecutive rolls of two fair six-sided dice is 0.00463.

c. The probability that the sum of 100 rolls of a fair six-sided die is less than 370 is 12%.

d. Roll a fair six-sided die until you roll a 6 three times and then stop. The probability that you roll the die at least 10 times is 0.822.

Solution

a. In about 16.7% sets of two rolls both rolls in the set are 6.
b. In about 0.463% of sets of three rolls all three rolls are 6. (If the sets consisted of more rolls, say 10 rolls, the probability of at least 3 consecutive 6s somewhere in the set would be greater.)
c. In about 12% of sets of 100 rolls the sum of the 100 rolls is less than 370.
d. In about 82.2% of sets of 9 rolls there are at most 2 6s.

Problem 2

1. Suppose that at some point in 2023 your subjective probabilities for who would win the 2024 U.S. Presidential Election satisfied the following. (Imagine a time before Trump had been declared the Republican nominee, and while Biden was the presumptive Democratic nominee there was still speculation that he should step aside.)

- The Democratic nominee and the Republican nominee are equally likely to be the winner of the Presidential Election
- Joe Biden is 5 times more likely to win than Kamala Harris, and no other Democratic candidate has a chance of winning
- Donald Trump is twice as likely to win as any other Republican candidate.

Create a table of your subjective probabilities.

Solution

It helps to designate one outcome as the "baseline". It doesn't matter which one; we'll choose Kamala Harris.

- Suppose Harris accounts for 1 "unit".
- Biden accounts for 5 units
- Other Democrat candidates account for 0 units
- Democrats in total account for 6 units
- Republicans in total account for 6 units
- Trump accounts for 4 units
- Other Republican accounts for 2 units

Candidate	"Units"	Probability	As Percent
Biden	5	$5/12 = 0.417$	41.7%
Harris	1	$1/12 = 0.083$	8.3%
Other Democrat	0	0	0%
Trump	4	$4/12 = 0.333$	33.3%
Other Republican	2	$2/12 = 0.167$	16.7%
Total	12	1	100%

Problem 3

Suppose that you have applied to two graduate schools, A and B. Your subjective probability of being accepted is 0.6 for school A and 0.7 for school B. Hint: the fact that these are subjective probabilities does not change the way you solve the problem. Remember, the mathematics of probability is the same regardless of whether the probabilities represent long run relative frequencies or subjective degrees of relative likelihood. Just take the 0.6 and 0.7 values as given and use them to solve the following parts.

a. Interpret the subjective probabilities: How many times more likely than not are you to be accepted at school A? How many times more likely are you to be accepted at school B than at school A?

b. What is the largest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.
c. What is the smallest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.
d. Explain why your subjective probability of being accepted by both schools is not necessarily 0.42.
e. **For the remaining parts, suppose your subjective probability of being accepted at both schools is 0.55.** If you are accepted at school A, what is your probability of also being accepted at school B? For this and the remaining parts in addition to computing the probability represent it with proper notation.
f. If you are accepted at school A, what is your probability of not being accepted at school B?
g. If you are not accepted at school A, what is your probability of being accepted at school B?
h. If you are accepted at school B, what is your probability of also being accepted at school A?
i. If you are not accepted at school B, what is your probability of being accepted at school A?

Solution

a. The probability of being accepted at both schools can be no more than the probability of being accepted at either school, 0.6 and 0.7. So we try 0.6 and see if we get valid probabilities.

	Accepted by A	Not accepted by A	Total
Accepted by B	0.6	0.1	0.7
Not accepted by B	0	0.3	0.3
Total	0.6	0.4	1.00

All the probabilities above are valid, so the largest the probability of being accepted at both schools can be is 0.6. This occurs if getting into school A guarantees that you will also get into school B.

b. Trying a subjective probability of being accepted at both schools leads to a negative probability of not being accepted at either school. So try a probability of 0 for the probability of not being accepted at either school.

	Accepted by A	Not accepted by A	Total
Accepted by B	0.3	0.4	0.7
Not accepted by B	0.3	0	0.3

	Accepted by A	Not accepted by A	Total
Total	0.6	0.4	1.00

So the smallest the probability of being accepted at both schools can be is 0.3. This only occurs if you're guaranteed of getting into at least one of the two schools.

c. Your chances of getting into one school might change depending if you've gotten into the other. For example, your chances of getting to Cal Poly are probably higher if you know you've already gotten into Stanford.

d. For the remaining parts, suppose your subjective probability of being accepted at both schools is 0.55. If you are accepted at school A, what is your probability of also being accepted at school B?

	Accepted by A	Not accepted by A	Total
Accepted by B	0.55	0.15	0.70
Not accepted by B	0.05	0.25	0.30
Total	0.60	0.40	1.00

If you are accepted at school A, then your probability of also being accepted at school B is $\mathrm{P}(B|A) = 0.55/0.60 = 0.917$. (91.7% of students like you who get into school A also get into school B.)

e. If you are accepted at school A, what is your probability of not being accepted at school B?

If you are accepted at school A, then your probability of not being accepted at school B is $\mathrm{P}(B^c|A) = 0.05/0.60 = 0.083$. (8.3% of students like you who get into school A do not get into school B.)

f. If you are not accepted at school A, what is your probability of being accepted at school B?

If you are not accepted at school A, then your probability of not being accepted at school B is $\mathrm{P}(B|A^c) = 0.15/0.40 = 0.375$. (37.5% of students like you who do not get into school A do get into school B.)

g. If you are accepted at school B, what is your probability of also being accepted at school A?

If you are accepted at school B, then your probability of being accepted at school A is $\mathrm{P}(A|B) = 0.55/0.7 = 0.786$. (78.6% of students like you who get into school B also get into school A.)

h. If you are not accepted at school B, what is your probability of being accepted at school A?

   If you are not accepted at school B, then your probability of being accepted at school A is $\mathrm{P}(A|B^c) = 0.05/30 = 0.167$. (16.7% of students like you who do not get into school B do get into school A.)

Problem 4

For each of the following, create your own "which of (a) or (b) is larger?" example. You can use the ones from the handout as examples, but you should choose your own contexts. Identify the correct answer and write a sentence or two explaining the answer to a student who is confused. You can use illustrations in your explanations, but don't do any calculations.

a. Reverse the direction of conditioning (like the 6 foot tall and NBA example)
b. Compare conditional and unconditional probabilities (like man versus man who is greater than 6 feet tall)
c. Two "particulars" that people might think are not equally likely but are (like HHHHH versus HTHTT).
d. Comparing "the particular" and "the general" (HTHTT versus 2 H in 5 flips).
e. Comparing the probability of happening once versus at least once (like you winning the lottery versus someone winning the lottery)

Solution

Results vary. See Section 1.6 of my probability textbook for discussion.

Problem 5

The "matching problem" involves n distincts objects labeled $1, \ldots, n$ which are placed in n distinct boxes labeled $1, \ldots, n$, with exactly one object placed in each box. Suppose the objects are placed in the boxes uniformly at random, so that any possible arrangement is equally likely. We're interested in the number of matches (the number of objects for which their label matches the label of the box in which they are placed.) Let C be the event that at least one object is placed in the correct spot (at least one match).

a. Describe in full detail how, in principle, you would physical objects (like cards) to simulate a single repetition of the placement of objects in boxes, and the number of matches.
b. Describe in full detail how, in principle, you could conduct a simulation and use the results to approximate $\mathrm{P}(C)$.

c. This applet conducts a simulation of the matching problem (in the context of babies in a hospital being mixed up and returned to parents uniforomly at random.) In the applet, "Number of babies" represents n and "Number of trials" represents the number of simulated repetitions. Starting with $n = 4$, try different values for the number of repetitions and hit the "Randomize" button to run the simulation. It helps to start with just a few repetitions so that you can see how the simulation is working, before jumping to thousands of repetitions. Use the simulation results to approximate $\mathrm{P}(C)$ when $n = 4$.
d. How do you expect $\mathrm{P}(C)$ to depend on n? As n increases, do you expect $\mathrm{P}(C)$ to increase, decrease, or stay about the same? Just think about it before proceeding; what does your intuition say?
e. Now try a few different values of n (say $n = 5$, $n = 10$, and $n = 20$, but you're free to choose other values). For each value of n run the simulation and use the results to approximate $\mathrm{P}(C)$. Record your results.
f. What do the simulation results suggest about the relationship between $\mathrm{P}(C)$ and n? (Remember that there is simulation margin of error, based on the number of repetitions. Two numerical approximations that are within the margion of error of each other are essentially the same approximation.)

Solution

a. Use a box with 4 tickets, labeled 1, 2, 3, 4. Shuffle the tickets and draw all 4 *without* replacement and record the tickets drawn *in order*. Let X be the number of tickets that match their spot in order. (For example, if the tickets are drawn in the order 2431 then the realized value of X is 1 since only ticket 3 matches its spot in the order.) Since $C = \{X \geq 1\}$, event C occurs if $X \geq 1$ and does not occur if $X = 0$. We could record the realization of event C as "True" or "False".
b. Repeat the previous part many times. Count the number of repetitions on which C occurs and divide by the total number of repetitions to approximate $\mathrm{P}(C)$.
c. See the applet.
d. You might think the probability might decrease with n, since the probability that any particular object goes in its correct spot decreases with n. On the other hand, as n increases, the more chances there are to get at least one object in the correct spot. These considerations move in opposite directions; what happens as n increases?
e. When $n = 4$, $\mathrm{P}(X = 0)$ is approximately 0.375; when $n = 5$, $\mathrm{P}(X = 0)$ is approximately 0.367; but for $n \geq 6$, $\mathrm{P}(X = 0)$ is approximately 0.368 for any value of n.
f. The probability of no matches essentially does not depend on n! We'll investigate a little more in class.

Problem 6

Katniss throws a dart at a circular dartboard with radius 1 foot. Suppose that Katniss's dart lands at a uniformly random location on the dartboard (and she never misses the dartboard).

a. Compute the probability that Katniss's dart lands within 1 inch of the center of the dartboard.
b. Compute the probability that Katniss's dart lands more than 1 inch but less than 2 inches away from the center of the dartboard.

Solution

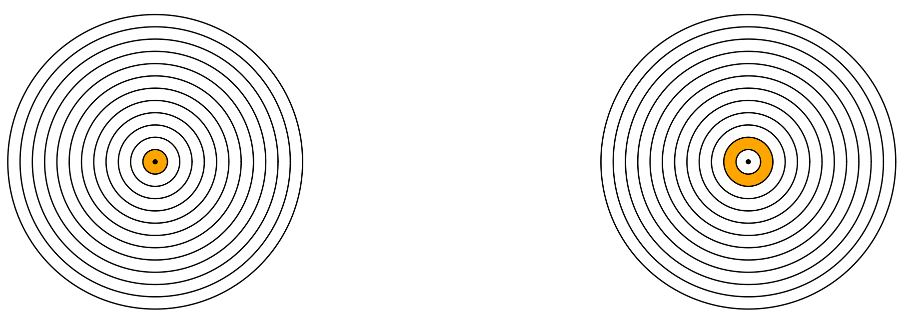
See Figure 29.3 for pictures.

1. Figure 29.3a: A, Katniss's dart lands within 1 inch of the center of the dartboard.
2. Figure 29.3b: B, Katniss's dart lands more than 1 inch but less than 2 inches away from the center of the dartboard.

Since the dart lands uniformly at random anywhere on the dartboard, probabilities are computed as the ratio between the area corresponding to the event of interest divided by the area of the total dartboard ($12^2\pi$ square-inches.)

Find the area of the shaded pieces by subtracting areas of circles.

1. $\mathrm{P}(A) = \frac{1^2\pi}{12^2\pi} = 1/144 = 0.00694$
2. $\mathrm{P}(B) = \frac{2^2\pi - 1^2\pi}{12^2\pi} = 3/144 = 0.021$



(a) Within 1 inch of center

(b) More than 1 but less than 2 inches from center

Figure 29.3: Events in dartboard problem

Problem 7

1. Imagine a light that flashes every few seconds[5]. The light randomly flashes green with probability 0.75 and red with probability 0.25, independently from flash to flash.

   a. Write down a sequence of G's (for green) and R's (for red) to predict the colors for the next 40 flashes of this light. Before you read on, please take a minute to think about how you would generate such a sequence yourself.
   b. Most people produce a sequence that has 30 G's and 10 R's, or close to those proportions, because they are trying to generate a sequence for which each outcome has a 75% chance for G and a 25% chance for R. That is, they use a strategy in which they predict G with probability 0.75, and R with probability 0.25. How well does this strategy do? Compute the probability of correctly predicting any single item in the sequence using this strategy.
   c. Describe a better strategy. (Hint: can you find a strategy for which the probability of correctly predicting any single flash is 0.75?)

Solution

a. Most people write a sequence with about 30 G and 10 R.
b. Use the law of total probability. There are two cases for your guesses: G or R.

   - If you guess G then you are correct if it really flashes G, and it flashes G with probability 0.75.
   - If you guess R then you are correct if it really flashes R, and it flashes R with probability 0.25.

   If you use a strategy where you guess G with probability 0.75, your probability of being correct is

$$\begin{aligned}\mathrm{P(correct)} &= \mathrm{P(correct|guess\ G)P(guess\ G) + P(correct|guess\ R)P(guess\ R)} \\ &= (0.75)(0.75) + (0.25)(0.25) = 0.625.\end{aligned}$$

c. If you get predict G every time, you are correct any time it is G, so the probability that you are correct is 0.75.

[5] Thanks to Allan Rossman for this example.

Homework 2 Solutions

In the solutions some of the parts have the wrong numbers, but all solutions should be there.

Problem 1

A standard deck of playing cards has 52 cards, 13 cards (2 through 10, jack, king, queen, ace) in each of 4 suits (hearts, diamonds, clubs, spades). Shuffle a deck and deals cards one at a time without replacement.

1. What is the probability that the first card dealt is a heart?
2. What is the probability that the second card dealt is a heart? Just make a guess now before proceeding, but you'll revisit this a few parts down.
3. What is the probability that the second card dealt is a heart if the first card dealt is a heart?
4. What is the probability that the second card dealt is a heart if the first card dealt is not a heart?
5. Revisit part 2. What is the probability that the second card dealt is a heart? Hint: your answer should be a single number, but show all your work.
6. Find the probability that the first two cards dealt are hearts.
7. Find the probability that the first two cards dealt are hearts and the third card dealt is a diamond.

Solution

1. If the cards are well shuffled, then any of the cards in the deck is equally likely to be the first card dealt. There are 13 hearts out of 52 cards in the deck, so the probability that the first card is a heart is $13/52 = 1/4 = 0.25$.
2. It's still $13/52$. If the cards are well shuffled, then any of the cards in the deck is equally likely to be the second card in the deck. There are 13 hearts out of 52 cards in the deck, so the probability that the second card is a heart is $13/52 = 1/4 = 0.25$. See the following parts for more discussion.

3. If the first card dealt is a heart, there are 51 cards left in the deck, 12 of which are hearts (and all the remaining cards in the deck are equally likely to be the next one drawn). So the conditional probability that the second card is a heart given that the first card is a heart is $12/51 = 0.2353$.
4. If the first card dealt is not a heart, there are 51 cards left in the deck, 13 of which are hearts (and all the remaining cards in the deck are equally likely to be the next one drawn). So the conditional probability that the second card is a heart given that the first card is NOT a heart is $13/51 = 0.2549$.
5. The probability that the second card is a heart is $13/52 = 0.25$. Don't let the fact that it is "second" fool you; if you do not know what the first card is, then you have no information that would revise your probability of what happens on the second card. If you do not know what the first card is, you need to consider both cases: either the first card is a heart or it's not. Use the law of total probability:

$$\begin{aligned}
\mathrm{P}(H_2) &= \mathrm{P}(H_2|H_1)\mathrm{P}(H_1) + \mathrm{P}(H_2|H_1^c)\mathrm{P}(H_1^c) \\
&= (12/51)(13/52) + (13/51)(39/52) = 13/52 = 0.25
\end{aligned}$$

Imagine 2652 repetitions (52×51, a convenient choice given these fractions) of drawing two cards, summarized in a two-way table.

	First is H	First is NOT H	Total
Second is H	156	507	663
Second is NOT H	507	1482	1989
Total	663	1989	2652

1. Use the multiplication rule: the probability the both cards are hearts is the product of the probability that the first card is a heart and the conditional probability that the second card is a heart given that the first card is a heart, $(13/52)(12/51) = 0.0588$. If we imagine 2652 repetitions, then we would expect the first card to be a heart in 663=2652(13/52) repetitions, and among these 663 repetitions we would expect the second card to be a heart in 156=663(12/51) repetitions, so the proportion of repetitions in which both cards are hearts is $156/2652 = 0.0588$.
2. The third card adds a third "stage" but the multiplication rule extends naturally. The probability the first two cards dealt are hearts and the third card dealt is a diamond is $(13/52)(12/51)(13/50)= 0.0153$, the product of:
   - $13/52$, the probability that the first card is a heart,
   - $12/51$, the conditional probability that the second card is a heart given that the first card is a heart, and
   - $13/50$, the conditional probability that the third card is a diamond given that the first two cards are hearts. (If the first two cards are hearts, then there are 50 cards remaining in the deck, of which 13 are diamonds.)

Problem 2

The ELISA test for HIV was widely used in the mid-1990s for screening blood donations. As with most medical diagnostic tests, the ELISA test is not perfect. If a person actually carries the HIV virus, experts estimate that this test gives a positive result 97.7% of the time. (This number is called the *sensitivity* of the test.) If a person does not carry the HIV virus, ELISA gives a negative (correct) result 92.6% of the time (the *specificity* of the test). Estimates at the time were that 0.5% of the American public carried the HIV virus.

Suppose that a randomly selected American tests positive; we are interested in the conditional probability that the person actually carries the virus.

1. Before proceeding, make a guess for the probability in question. Which range do you think it's in: 0-20%, 20-40%, 40-60%, 60-80%, or 80-100%?
2. Denote the probabilities provided in the setup using proper notation
3. Construct an appropriate two-way table and use it to compute the probability of interest.
4. Construct a Bayes table and use it to compute the probability of interest.
5. Explain why this probability is small, compared to the sensitivity and specificity.
6. By what factor has the probability of carrying HIV increased, given a positive test result, as compared to before the test?
7. Now suppose that 5% of individuals in a high-risk group carry the HIV virus. Consider a randomly selectd person from this group who takes the test. Given that the test is positive, how many times more likely is it for the person not to have HIV than to have it? Answer without first computing a two-way or Bayes table.
8. Using the result from the previous part, compute the conditional probability that a person in this risk group who tests positive has HIV.
9. Is the posterior probability influenced by the prior probability? Discuss.

Solution

1. We don't know what you guessed, but from experience many people guess 80-100%. Afterall, the test is correct for most of people who carry HIV, and also correct for most people who don't carry HIV, so it seems like the test is correct most of the time. But this argument ignores one important piece of information that has a huge impact on the results: most people do not carry HIV.
2. Let H denote the event that the person carries HIV (hypothesis), and let E denote the event that the test is positive (evidence). Therefore, H^c is the event that the person does not carry HIV, another hypothesis. We are given

- prior probability: $P(H) = 0.005$
- likelihood of testing positive, if the person carries HIV: $P(E|H) = 0.977$
- $P(E^c|H^c) = 0.926$

- likelihood of testing positive, if the person does not carry HIV: $P(E|H^c) = 1 - P(E^c|H^c) = 1 - 0.926 = 0.074$
- We want to find the posterior probability $P(H|E)$.

1. Assuming 1000000 Americans

	Tests positive	Does not test positive	Total
Carries HIV	4885	115	5000
Does not carry HIV	73630	921370	995000
Total	78515	921485	1000000

Among the 78515 who test positive, 4885 carry HIV, so the probability that an American who tests positive actually carries HIV is $4885/78515 = 0.062$.

2. See Table 29.5.

3. The result says that only 6.2% *of Americans who test positive* actually carry HIV. It is true that the test is correct for most Americans with HIV (4885 out of 5000) and incorrect only for a small proportion of Americans who do not carry HIV (73630 out of 995000). But since so few Americans carry HIV, the sheer *number* of false positives (73630) swamps the *number* of true positives (4885).

4. Prior to observing the test result, the prior probability that an American carries HIV is $P(H) = 0.005$. The posterior probability that an American carries HIV given a positive test result is $P(H|E) = 0.062$.

$$\frac{P(H|E)}{P(H)} = \frac{0.062}{0.005} = 12.44$$

An American who tests positive is about 12.4 times more likely to carry HIV than an American whom the test result is not known.
So while 0.067 is still small in absolute terms, the posterior probability is much larger relative to the prior probability.

5. In this risk group

- a person is 19 times more likely to not have HIV than to have it $(0.95/0.05 = 19)$.
- A positive test is 13.2 times *less* likely when the person does not have HIV than when they have it $(0.074/0.977 = 1/13.2)$.
- The product of these ratios is $19(1/13.2) = 1.44$.

Since posterior is proportional to the product of prior and likelihood, a person in this risk group who tests positive is 1.44 times more likely to not have HIV than to have HIV.

1. The posterior probabilities of not having HIV and having HIV are in a 1.44 to 1 ratio, the so the posterior probability of not having HIV is $1.44/(1+1.44) = 0.59$ and the posterior probability of having HIV is 0.41.
2. Yes, the posterior probability is influenced by the prior probability. Even though the prior probability of 5% is still relatively small in absolute terms, the posterior probability given a positive test is not close to 50/50.

Table 29.5: Bayes table for ELISA problem

hypothesis	prior	likelihood	product	posterior
Carries HIV	0.005	0.977	0.0049	0.0622
Does not carry HIV	0.995	0.074	0.0736	0.9378
sum	1.000	NA	0.0785	1.0000

Problem 3

Consider three tennis players Arya ("A"), Brienne ("B"), and Cersei ("C"). One of these players is better than the other two, who are equally good/bad. When the best player plays either of the others, she has a 2/3 probability of winning the match. When the other two players play each other, each has a 1/2 probability of winning the match. But you do not know which player is the best. Based on watching the players warm up, you start with subjective probabilities of 0.5 that A is the best, 0.35 that B is the best, and 0.15 that C is the best. A and B will play the first match.

1. Suppose that A beats B in the first match. Compute your posterior probability that each of A, B, C is best given that A beats B in the first match.
2. Compare the posterior probabilities from the previous part to the prior probabilities. Explain how your probabilities changed, and why that makes sense.
3. **Coding required.** Code and run a simulation to simulate (1) who is the best player according to your prior distribution, (2) who wins the first match given who is the best player; then use the results to approximate the probabilities from the first part.
4. Suppose instead that B beats A in the first match. Compute your posterior probability that each of A, B, C is best given that B beats A in the first match.
5. Compare the posterior probabilities from the previous part to the prior probabilities. Explain how your probabilities changed, and why that makes sense.
6. Now suppose again that A beats B in the first match, and also that A beats C in the second match. Compute your posterior probability that each of A, B, C is best given the results of the first two matches. (Hint: use as the prior your posterior probabilities from the previous part.) Explain how your probabilities changed, and why that makes sense.

best_player	prior	likelihood_A_beats_B	product	posterior
A	0.50	0.6667	0.3333	0.6349
B	0.35	0.3333	0.1167	0.2222
C	0.15	0.5000	0.0750	0.1429
Total	1.00	1.5000	0.5250	1.0000

Solution

1. Hypotheses are which player is best (A, B, C). Evidence is that A beats B. The likelihood is the probability that A beats B given each of the best players.

- If A is best, probability A beats B is 2/3.
- If B is best, probability A beats B is 1/3.
- If C is best, probability A beats B is 1/2.

1. A's probability of being the best increased, which makes sense because A won the match. B's probability of being the best decreased considerably, which makes sense because B lost the match. C's probability of being the best decreased slightly, despite C not being involved in the match.
2. Description of simulation

    - Spinner that returns "A is best", "B is best", "C is best" with probability 0.5, 0.35, 0.15.
    - Three additional spinners:
        - "Given A is best": returns "A beats B" with probability 2/3 and "B beats A" with probability 1/3
        - "Given B is best": returns "A beats B" with probability 1/3 and "B beats A" with probability 2/3
        - "Given C is best": returns "A beats B" with probability 1/2 and "B beats A" with probability 1/2
    - Spin the first spinner once to determine who is best. Given the result of the first spin, spin the appropriate spinner to determine whether or not A beats B. The pair of spins is one repetition.
    - Repeat many times to simulate many pairs.
    - To find the conditional probabilities for which player is best given A beats B, only consider the repetitions in which A beats B, and among those repetitions find the proportion of repetitions where A is best, etc.

3. Hypotheses are which player is best (A, B, C). Evidence is that B beats A. The likelihood is the probability that B beats A given each of the best players.

    - If A is best, probability B beats A is 1/3.

best_player	prior	likelihood_B_beats_A	product	posterior
A	0.50	0.3333	0.1667	0.3509
B	0.35	0.6667	0.2333	0.4912
C	0.15	0.5000	0.0750	0.1579
Total	1.00	1.5000	0.4750	1.0000

best_player	prior	likelihood_A_beats_C	product	posterior
A	0.6349	0.6667	0.4233	0.7273
B	0.2222	0.5000	0.1111	0.1909
C	0.1429	0.3333	0.0476	0.0818
Total	1.0000	1.5000	0.5820	1.0000

- If B is best, probability B beats A is 2/3.
- If C is best, probability B beats A is 1/2.

1. A's probability of being the best decreased, which makes sense because A lost the match. B's probability of being the best increased, which makes sense because B won the match. C's probability of being the best changed slightly, despite C not being involved in the match.

2. The prior is the posterior from the first part. Evidence is that A beats C. The likelihood is the probability that A beats C given each of the best players.

- If A is best, probability A beats C is 2/3.
- If B is best, probability A beats C is 1/2.
- If C is best, probability A beats C is 1/3.

By winning both matches, A's probability of being the best has increased considerably. By losing their only matches, B's and C's probabilities of being the best have decreased considerably.

Problem 4

In the Powerball lottery, a player picks five different whole numbers between 1 and 69, and another whole number between 1 and 26 that is called the Powerball. In the drawing, the 5 numbers are drawn without replacement from a "hopper" with balls labeled 1 through 69, but the Powerball is drawn from a separate hopper with balls labeled 1 through 26. The player wins the jackpot if both the first 5 numbers match those drawn, in any order, and the Powerball is a match. Under this set up, there are 292,201,338 possible winning numbers.

1. What is the probability the next winning number is 6-7-16-23-26, plus the Powerball number, 4.

2. What is the probability the next winning number is 1-2-3-4-5, plus the Powerball number, 6.
3. The Powerball drawing happens twice a week. Suppose you play the same Powerball number, twice a week, every week for over 50 years. Let's say you purchase a ticket for 6000 drawings in total. What is the probability that you win at least once?
4. Instead of playing for 50 years, you decide only to play one lottery, but you buy 6000 tickets, each with a different Powerball number. What is the probability that at least one of your tickets wins? How does this compare to the previous part? Why?
5. Each ticket costs 2 dollars, but the jackpot changes from drawing to drawing. Suppose you buy 6000 tickets for a single drawing. How large does the jackpot need to be for your "expected" profit to be positive? To be \$100,000? (We're ignoring inflation, taxes, transaction costs, and any changes in the rules.)

Solution

1. Each of the possible winning numbers is equally likely, so the probability is $1/292,201,338 \approx 3 \times 10^{-9}$.
2. Each of the possible winning numbers is equally likely. Remember, don't confuse a general event with a specific outcome.
3. The probability that you lose in any single drawing is $(1 - 1/292201338)$. The drawings are independent so the probability that you lose all 6000 is $(1 - 1/292201338)^{6000}$. The probability that you win at least once is $1 - (1 - 1/292201338)^{6000} \approx 0.00002$. If many people each play 6000 drawings, about 2 in every 100,000 people win will at least once.
4. If you play 6000 different numbers, the events that each different number wins are disjoint. So the probability you win at least once is $6000/292201338 \approx 0.00002$. This is about the same as the probability in the previous part. When you play 6000 different independent drawings, there is a possibility that you win multiple times, so the events of winning in each different drawing are not disjoint. But the probability of winning *multiple* lotteries is so small that it's negligible. The probability of winning any single drawing is about 1 in 300 million. The probability of winning twice in any two drawings is about 1 in 85 quadrillion.
5. You pay \$12,000 in total. Let w be the value of the jackpot. You win either 0 or w so your "expected" profit is $w(6000/292201338) - 12000$. But this not what you expect in a single repetition. Rather, it is the profit you would expect to see on average in the long run. You probably won't be buying 6000 tickets for a large number of drawings, so your long run average isn't really relevant. In any case, we must have $w > 584,402,676$ for the expected profit to be positive. Sometimes, but not often, the jackpot does get this high; even so, this just guarantees that your expected profit is positive. In order for your expected long run average profit to be greater than just \$100,000, the jackpot must be over 5 billion dollars, and the largest jackpot ever was 1.6 billion. The moral: there are better things to do with \$12,000 dollars.

Problem 5

Consider a "best-of-5" series of games between two teams: games are played until one of the teams has won 3 games (requiring at most 5 games total). Suppose one team, team A, is better than the other, having a 0.55 probability of winning any particular game. Assume the results of the games are independent (and ignore advantage, etc). Let X represent the number of games played in the series. (Hint: It's helpful to first construct a two-way table of probabilities with the number of games played and which team wins, and then use it to answer the following questions. It will also help to list some outcomes, like AABA (team A wins game 1, 2, and 4, and B wins game 3).)

1. Compute the probability that team A wins the series in 3 games.
2. Compute the probability that the series ends in 3 games.
3. Compute the probability that team A wins the series.
4. Are the events "team A wins the series" and "the series ends in 3 games" independent? Explain by comparing relevant probabilities.
5. Let X represent the number of games played in the series. Find the distribution of X.

Solution

1. It is helpful to construct a two-way table to answer the following questions.

	x			
	3	4	5	Total
A wins	0.553 = 0.166	3(0.553)(0.45) = 0.225	6(0.553)(0.45)2 = 0.202	0.593
A does not win	0.453 = 0.091	3(0.453)(0.55) = 0.150	6(0.453)(0.55)2 = 0.165	0.407
Total	0.258	0.375	0.368	1

   There is only one outcome for which A wins in 3 games: AAA. There are three outcomes in which A wins in 4 games: AABA, ABAA, BAAA. (Not AAAB because then the series would be over in 3 games.) Since the games are independent, an outcome like AABA has probability $(0.55^3)(0.45)$, so the probability that A wins in 4 games is $3(0.55^3)(0.45)$. There are six outcomes in which A wins in 5 games: AABBA, ABABA, BAABA, ABBAA, BABAA, BBAAA. (Not outcomes like AAABB or AABAB because then the series would be over in 3 or 4 games.) Since the games are independent, an outcome like AABBA has probability $(0.55^3)(0.45)^2$, so the probability that A wins in 5 games is $6(0.55^3)(0.45)$. You can fill in the rest of the table similarly.

   The probability that team A wins the series in 3 games is $\mathrm{P}(X = 3, A) = 0.55^3 = 0.166$.

2. Either the stronger team wins 3 in a row or loses 3 in a row. The probability is $0.55^3 + (1 - 0.55)^3 = 0.2575$.

3. The probability that team A wins the series is the sum of first row: $\mathrm{P}(A) = 0.593$.

4. No. $\mathrm{P}(X = 3, A) = 0.55^3 = 0.166 \neq 0.152 = (0.2575)(0.593) = \mathrm{P}(X = 3)\mathrm{P}(A)$. Alternatively, $\mathrm{P}(X = 3|A) = 0.166/0.593 = 0.2799 \neq 0.2575 = \mathrm{P}(X = 3)$. Given that A wins the series the series it more likely to end in 3 games than when B wins the series.

5. Total row above. X can take values 3, 4, or 5. Consider first the ways in which the stronger team wins in 4 games: AABA, ABAA, BAAA. (For example AABA, means the stronger team wins game 1, 2, and 4, and the weaker team wins game 3). Each of these outcomes has probability $0.55^3(0.45)$ so the probability that team A wins in 4 games in $3(0.55)^3(0.45)$. Similarly, the probability that team B wins in 4 games is $3(0.45)^3(0.55)$. So
$$\mathrm{P}(X = 4) = 3(0.55)^3(0.45) + 3(0.45)^3(0.55) = 0.3750$$
You could find $\mathrm{P}(X = 5)$ in a similar way, or just use the fact that the probabilities have to add up to 1. The distribution of X is given by the following table.

x	$\mathrm{P}(X = x)$
3	0.2575
4	0.3750
5	0.3675

Homework 3 Solutions

Problem 1

Maya is a basketball player who makes 40% of her three point field goal attempts. Suppose that at the end of every practice session, she attempts three pointers until she makes one and then stops. Let X be the total number of shots she attempts in a practice session. Assume shot attempts are independent, each with probability of 0.4 of being successful.

1. Identify by name the distribution of X. Be sure to specify the values of any relevant parameters.
2. Specify the probability mass function of X. Be sure to specify the possible values.
3. Construct a table, plot, and spinner corresponding to the distribution of X.
4. Compute $\mathrm{P}(X > 5)$ without summing.
5. Compute and interpret $\mathrm{E}(X)$.
6. Compute and interpret $\mathrm{SD}(X)$.

Solution

1. X counts the number of successes until the first success in a sequence of Bernoulli(0.4) trials, so X has a Geometric(0.4) distribution.

2. X can take values 1, 2, 3, ... Even though it is unlikely that X is very large, there is no fixed upper bound. Even though X can take infinitely many values, X is a discrete random variables because it takes *countably* many possible values. In order for X to take value x, the first success must occur on attempt x, so the first $x-1$ attempts must be failures.

$$p_X(x) = (1-0.4)^{x-1}(0.4), \qquad x = 1, 2, 3, ...$$

3. See below. $\mathrm{P}(X = 1) = 0.4$. If Maya does this every practice, then in about 40% of practices she will make her first three pointer on her first attempt. $\mathrm{P}(X = 2) = (0.6)(0.4) = 0.24$. If Maya does this every practice, then in about 24% of practices she will make her first three pointer on her second attempt. And so on.

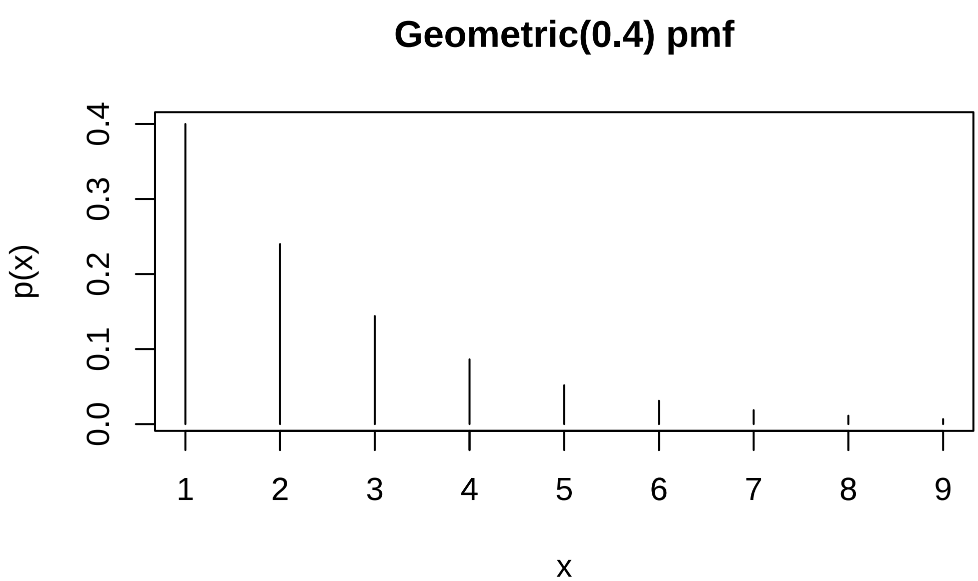
4. The key is to realize that Maya requires more than 5 attempts to obtain her first success if and only if the first 5 attempts are failures. Therefore,

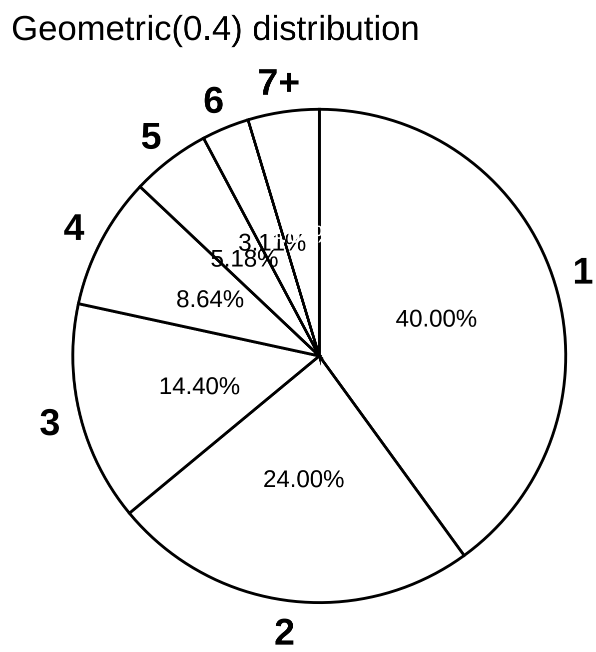
$$\mathrm{P}(X > 5) = (1 - 0.4)^5 = 0.078$$

5. Since X has a Geometric(0.4) distribution, $\mathrm{E}(X) = 1/0.4 = 2.5$. Over many practice sessions, she attempts on average 2.5 three pointers until she makes one per practice.

6. Since X has a Geometric(0.4) distribution, $\mathrm{Var}(X) = (1-0.4)/0.4^2 = 3.75$ so SD is 1.94. Over many practice sessions, the number of attempts until making one varies by about 1.94 attempts per practice on average.

x	p(x)
1	0.4000
2	0.2400
3	0.1440
4	0.0864
5	0.0518
6	0.0311
7	0.0187
8	0.0112
9	0.0067





Problem 2

Ron and Leslie agree to the following bet. They'll ask Professor Ross if he saw the Eras Tour live. If he did, Leslie will pay Ron \$200; if not, Ron will pay Leslie \$100. (Neither has any direct information about whether or not Prof Ross saw the Eras Tour live.)

1. Given this setup, which of the following is being judged as more likely: that Prof Ross saw the Eras Tour, or that he did not? Why?
2. Ron and Leslie agree that this is a fair bet, and neither would accept worse odds. What is their subjective probability that Professor Ross saw the Eras Tour?
3. Suppose they were to hypothetically repeat this bet many times, say 3000 times. Given the probability from the previous part, how many times would you expect Leslie to win? To lose? What would you expect Leslie's net dollar winnings to be? In what sense is this bet "fair"?
4. Let X be Leslie's net winnings. Identify the distribution of X. (Remember: Leslie's winnings are Ron's losses and vice versa.)
5. Compute and interpret $\mathrm{E}(X)$.

Solution

1. The larger potential payout corresponds to the *less* likely event. So they think Professor Ross is more likely to *not* have a TikTok account than to have one.
2. The payouts are in a 2 to 1 ratio, so the odds that Professor Ross has a TikTok account are 2 to 1 against. So Professor Ross is twice as likely to not have a TikTok account than to have one. This corresponds to a subjective probability[6] that Professor Ross has a TikTok account of 1/3 (and a probability that he does not have one of 2/3).
3. The probability that Leslie wins is 2/3, so you would expect her to win in 2000 of the 3000 repetitions. She wins \$100 each time she wins, so you would expect her to win a total of \$200,000 on games she wins. The probability that she loses is 1/3, so you would expect her to lose in 1000 of the 3000 repetions. She loses \$200 each time, so you would expect her to lose a total of \$200,000 on the games she loses. So you would expect Leslie's net winnings to be 0, and likewise for Ron.

Winner	Expected number of repetitions	Leslie's winnings per repetition (\$)	Leslie's expected total winnings (\$)
Leslie	2000	100	200,000
Ron	1000	-200	-200,000
Total	3000	NA	0

4. The bet is fair in the sense that neither party is expected to profit or lose in the long run.
5. X is 100 with probability 2/3 and -200 with probability 1/3, so $\mathrm{E}(X) = (100)(2/3) + (-200)(1/3) = 0$. Over many such bets, Leslie's net winnings per bet would be 0 on average.

Problem 3

Suppose that a total of 350 students at a college are taking a particular statistics course. The college offers five sections of the course, each taught by a different instructor. The class sizes are shown in the following table.

Section	A	B	C	D	E

[6] Technically, Ron and Leslie could still have different subjective probabilities. Leslie would not agree to worse odds, but she would accept better if Ron offered them. For example, given a potential loss of \$200, Leslie would also agree to a potential payout from Ron of \$125 rather than \$100. That is, Leslie would accept odds of 1.6 to 1 against ($200/125 = 1.6$), corresponding to a subjective probability of $1/(1 + 1.6) = 0.385$. So Leslie's subjective probability that Professor Ross has a TikTok account is *at least* 1/3. Similarly, Ron's subjective probability that Professor Ross has a TikTok account is *at most* 1/3.

Number of students	35	35	35	35	210

We are interested in: What is the average class size?

1. Suppose we randomly select one of the 5 *instructors*. Let X be the class size for the selected instructor. Specify the distribution of X. (A table is fine.)
2. Compute and interpret $\mathrm{E}(X)$.
3. Compute and interpret $\mathrm{P}(X = \mathrm{E}(X))$.
4. Suppose we randomly select one of the 350 *students*. Let Y be the class size for the selected student. Specify the distribution of Y. (A table is fine.)
5. Compute and interpret $\mathrm{E}(Y)$.
6. Compute and interpret $\mathrm{P}(Y = \mathrm{E}(Y))$.
7. Comment on how these two expected values compare, and explain why they differ as they do. Which average would you say is more relevant?

Solution

1. X takes the value 210 with probability 1/5 and 35 with probability 4/5.
2. $\mathrm{E}(X) = 210(1/5) + 35(4/5) = 70$ is the average number of students per class from the instructors' perspective. If we randomly select an instructor, record the number of students in the instructor's class, and repeat, the long run average number of students will be 70.
3. $\mathrm{P}(X = \mathrm{E}(X)) = \mathrm{P}(X = 70) = 0$. No instructor has a class whose size is equal to the average class size (from the instructor's perspective).
4. Y takes the value 210 with probability 210/350 and 35 with probability 140/350.
5. $\mathrm{E}(Y) = 210(210/350) + 35(140/350) = 140$ is the average number of students per class from the students' perspective. If we randomly select a student, record the number of students in the selected student's class, and repeat, the long run average number of students will be 140.
6. $\mathrm{P}(Y = \mathrm{E}(Y)) = \mathrm{P}(Y = 140) = 0$. No student is in a class whose size is equal to the average class size (from the students' perspective).
7. Colleges usually report average class size from the instructor/class perspective, which in this case would be 70 students. From the students' perspective, the average class size is not even close to 70! In fact, it's twice that size. Some students (140 of them, which is 40% of the total of 350 students) have the benefit of a small class size of 35. But most students (210 of them, which is 60% of the students) are stuck in a large class of 210 students. In other words, most students would be pretty seriously misled if they chose this college based on the advertised average class size of 35 students per class. From the students' perspective, it seems that 140 is the more relevant average to report. However, neither average really adequately represents what is happening here: there is wide variability in class sizes between large and small.

Problem 4

(This is an overly simplified example from actuarial science.) A life insurance company sells a term insurance policy to a 21 year old man (the "insured") that pays \$100,000 to a designated beneficiary if the insured dies within the next 5 years, and pays nothing if the insured does not die before age 26. The probability that a randomly chosen 21-year-old man will die each year can be found in mortality tables. The company collects a premium of \$250 at the beginning of each year of the 5 years of the term, as long as the insured is alive, as payment for the insurance. The amount Y (in dollars) that the company earns on this policy is \$250 per year, less the \$100,000 that it pays out in the event the insured dies. The company doesn't pay out anything if the insured does not die in the 5 year term.

Age at death	Earnings Y	Probability
21	−99,750	0.00183
22		0.00186
23		0.00189
24		0.00191
25		0.00193
26 or older		

1. Find the distribution of Y by completing the above table. For example, the value -99,750 provides an example of how Y is calculated when the insured dies in the first year of the policy.
2. Compute $\mathrm{E}(Y)$.
3. Write a sentence explaining in this context in what sense the number from the previous part is "expected".
4. Would you be willing to sell such an insurance policy to one of your 21 year old friends? Explain why this is good business for the insurance company, but not for you.
5. Compute $\mathrm{Var}(Y)$
6. Compute $\mathrm{SD}(Y)$.
7. Compute the probability that Y is more than 1 standard deviation above its mean.

Solution

Age at death	Earnings Y	Probability
21	99,750	0.00183
22	-99,500	0.00186
23	-99,250	0.00189
24	-99,000	0.00191
25	-98,750	0.00193

Age at death	Earnings Y	Probability
26 or older	1,250	0.99058

1. See above. The earnings are computed according to the contract; the insurance company only needs to pay out the 100000 if the insured dies before age 26. Since all the probabilities must add up to 1, the probability in the blank is $1 - (0.00183 + 0.00186 + 0.00189 + 0.00191 + 0.00193) = 0.99058$

2. $\mathrm{E}(Y) = (-99750)(0.00183) + \cdots + (-98750)(0.00193) + (1250)(0.99058) = 303.23$

3. If the insurance company sells a large number of these policies to 21 year-old men then the insurance company will expect to make, on average, a profit of \$303.36 per policy. For example, if they sell 1 million policies, their total profit would be expected to be close to \$303 million.

4. If you sell one policy, you are either going to make a profit of \$1250, or you will pay out about \$100,000. The chances are high you will make a profit of \$1250, but there is a risk that you will have to pay out around \$100,000. This risk is probably too great to take on considering the relatively small (in comparison) reward.

   On the other hand, the insurance company sells *many* policies. They operate in the long run. Over many policies they make a profit, on average, of about \$300 per policy. For example, for a million policies this equates to a profit of \$300 million. There would be some variability around this, but in the long run the average will win out. The company could be sure that they would only pay out the \$100,000 benefit for less than 1% of policyholders, and they will be earning \$1,250 on many other policies, more than enough to offset the relatively few \$100,000 payouts there are.

5. $\mathrm{E}(Y^2) = (-99750)^2(0.00183) + \cdots + (-98750)^2(0.00193) + (1250)^2(0.99058) = 94328849$. Then $\mathrm{Var}(Y) = \mathrm{E}(Y^2) - (\mathrm{E}(Y))^2 = 94328849 - (303)^2 = 94236827$. This is a giant number because it's measured in dollars^2.

6. $\mathrm{SD}(Y) = \sqrt{\mathrm{Var}(Y)} = 9707$ dollars.

7. 0. One SD above the mean here is $303 + 9707 = 10000$ and $\mathrm{P}(Y > 10000) = 0$ since it's not possible to have a value of this Y which is more than 1 SD above its mean.

Problem 5

In a sample of 50 Olympic women's gymnasts the heights have a symmetric bell-shaped distribution with mean 60 inches and standard deviation 3 inches. In a sample of 50 Olympic men's basketball players the heights have symmetric bell-shaped distribution with mean 76 inches and the standard deviation 5 inches. Suppose the two samples are combined to obtain a sample with 100 heights. Choose one of the options below to complete the following sentence;

explain your reasoning without doing any calculations (hint: drawing a picture might help; also, what is the mean of the combined sample?). The standard deviation of the combined sample is:

a. Less than 3 inches
b. Equal to 3 inches
c. Between 3 and 5 inches
d. Equal to 5 inches
e. Greater than 5 inches

Solution

Greater than 5 inches. As a very rough measure of variability, think about the overall range. The range of heights for both groups combined would be greater than the range of heights for either of the two groups separately, so you would expect more variability when the groups are combined. To see what happens with SD, it's easiest to draw a picture. Looking at the gymnasts alone, the SD of 3 is the average distance from the mean of 60 inches. Looking at the men's bball players alone, the SD of 5 is the average distance from the mean of 76. If the two samples are combined, the sample mean height would be 68 inches (right in the middle of 60 and 76 because the groups have the same size). The SD in the combined sample measures the average distance from the mean of 68. The gymnasts heights tend to be closer to their mean of 60 than the overall mean of 68; similarly, the men's bball player heights tend to be closer to their mean of 76 than the overall mean of 68. So the average distance from the mean will be greater for the combined sample than for either of the two separate samples.

Homework 4 Solutions

Problem 1

Xavier bets on red on each of 3 spins of a roulette wheel. After the third spin, Xavier leaves and Zander arrives and bets on black on the next two spins. Let X be the number of bets that Xavier wins and let Z be the number that Zander wins.

1. Identify the distribution of X and the distribution of Z.
2. Are X and Z independent? Explain.
3. Compute and interpret $\mathrm{P}(X = 2, Z = 1)$.
4. What does $X + Z$ represent in this context? Identify the distribution of $X + Z$ without doing any calculations. Be sure to specify the values of any parameters. Explain your reasoning.
5. Compute $\mathrm{P}(X + Z = 3)$ without summing many terms.
6. Suppose instead that Yolanda bets on black on all 5 spins of the wheel. Does $X + Y$ have a Binomial distribution? Answer yes or no and explain.
7. Suppose instead that Zander bet on the number 7 in his two bets. Does $X + Z$ have a Binomial distribution in this case? Answer yes or no and explain.

Solution

1. X has a Binomial(3, 18/38) distribution and Z has a Binomial(2, 18/38) distribution.
2. Yes, X only depends on the first three spins, and Z only depends on the last two, and the spins are (physically) independent, so X and Z are independent.
3. Since X and X are independent $\mathrm{P}(X = 2, Z = 1) = \mathrm{P}(X = 2)\mathrm{P}(Z = 1) = \binom{3}{2}(18/38)^2(20/38)^2\binom{2}{1}(18/38)^1(20/38)^1 = (0.354)(0.499) = 0.177$. Over many sets of 5 spins, 2 of the first 3 spins land on red and 1 of the last 2 spins land on black in about 18% of sets.
4. $X + Z$ is the total number of bets won by either of these players. The five spins satisfy the Binomial situation. Each trial results in either “win” or not. The probability of a win on any bet is 18/38; it doesn’t matter if they’re betting on black or red because each has the same probability of winning. The spins are independent. Between the two players there is a fixed number of 5 bets and we are counting the number of wins. So $X + Z$ has a Binomial(5, 18/38) distribution.

5. $\mathrm{P}(X+Z=3) = \binom{5}{3}(18/38)^3(20/38)^2 = 0.294$. Note that $X=2, Z=1$ is only one way $X+Z=3$.
6. $X+Y$ does not have a Binomial distribution, because the trials that determine X are not independent of those that determine Y. The random variables X and Y are not independent.
7. $X+Z$ would not have a Binomial distribution, because there is not the same probability of win (success) on all trials (18/38 for the first three but 1/38 for the last two).

Problem 2

Suppose the number of earthquakes per hour, for a certain range of magnitudes in a certain region, follows a Poisson distribution with parameter 0.7, independently from hour to hour.

1. Compute and interpret the probability that there is at least one earthquake of this size in the region in any given hour.
2. Compute and interpret the probability that there are exactly 3 earthquakes of this size in the region in any given hour.
3. Interpret the value 0.7 in context.
4. Construct a table, plot, and spinner corresponding to a Poisson(0.7) distribution.
5. Let X be the number of earthquakes in a 2 hour period. Identify by name the distribution of X. Be sure to specify the values of any relevant parameters.
6. Compute and interpret $\mathrm{P}(X=3)$.

Solution

Let X be the number of earthquakes in the region in a randomly selected hour. The pmf is

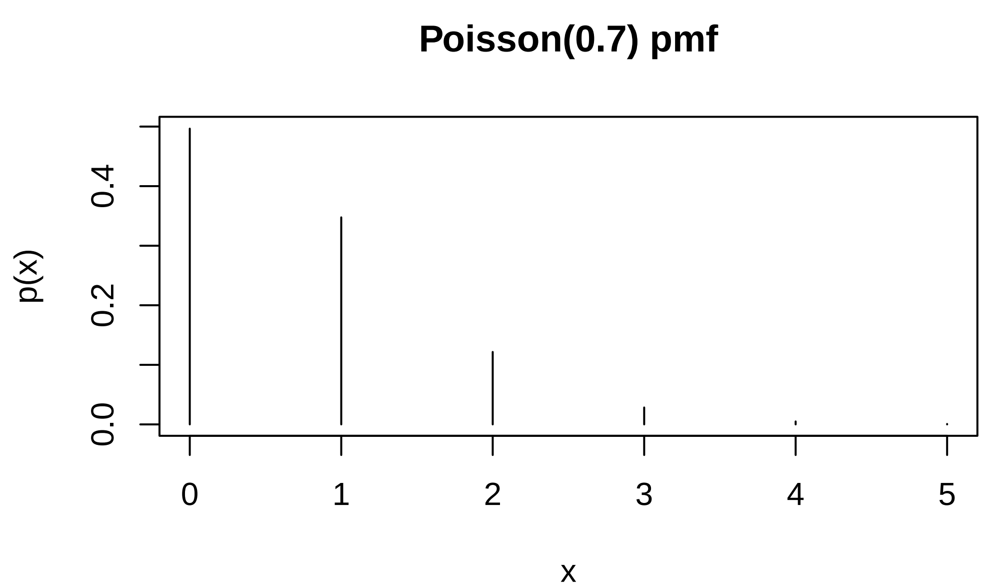
$$p_X(x) = e^{-0.7}\frac{0.7^x}{x!}, \quad x = 0, 1, 2, ...$$

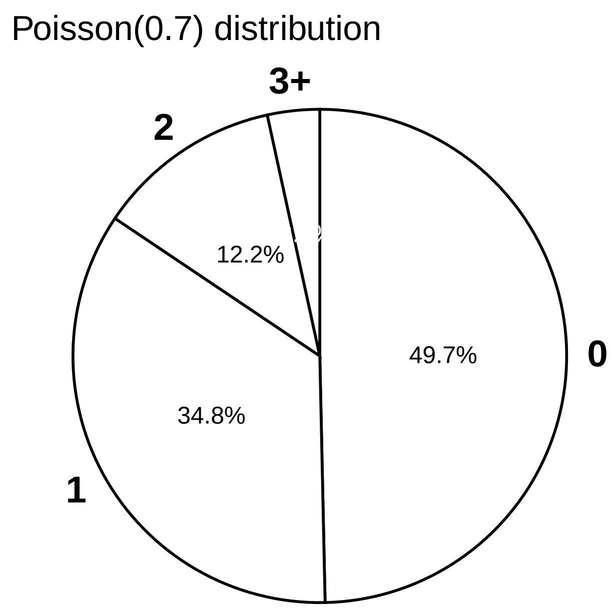
1. $\mathrm{P}(X \geq 1) = 1 - \mathrm{P}(X=0) = 1 - e^{-0.7}\frac{0.7^0}{0!} = 1 - e^{-0.7} = 1 - 0.497 = 0.503$. In about 50% of hour long intervals there is at least one quake.
2. $\mathrm{P}(X=3) = e^{-0.7}\frac{0.7^3}{3!} = 0.028$. In about 3% of hour long intervals there are exactly 3 quakes.
3. Over many hour-long time intervals, there are 0.7 earthquakes per hour on average.
4. See below.
5. Let X be the number of earthquakes in a 2 hour period then X has a Poisson(1.4) distribution. (This is similar to the car accident problem.)
6. $\mathrm{P}(X=3) = e^{-1.4}1.4^3/3! = 0.113$. Over many 2-hour periods, there are 3 earthquakes in about 11.3% of 2-hour periods.

```r
x <- 0:5

mu <- 0.7

px <- dpois(x, mu)

df <- data.frame(x, px)

df |>
  kable(col.names = c("x", "p(x)"), digits = 4)
```

x	p(x)
0	0.4966
1	0.3476
2	0.1217
3	0.0284
4	0.0050
5	0.0007

```r
plot(x, px, type = "h", xlab = "x", ylab = "p(x)", main = "Poisson(0.7) pmf")
```





Problem 3

Recall Example 2.6. Regina and Cady are meeting for lunch. Suppose they each arrive uniformly at random at a time between noon and 1:00, independently of each other. Record their arrival times as minutes after noon, so noon corresponds to 0 and 1:00 to 60. Recall the sample space from Example 2.2 and assume a uniform probability measure P.

Let R be Regina's arrival time and Y Cady's. Then $W = |R - Y|$ is the amount of time (minutes) the first person to arrive waits for the second person to arrive.

1. Compute and interpret $\mathrm{P}(W < 15)$. (Hint: we computed this already in Example 2.6.)
2. Compute and interpret $\mathrm{P}(W > 45)$. (Hint: use a method similar to the previous part.)
3. It can be shown that W has pdf

$$f_W(w) = (2/3600)(60 - w), \qquad 0 < w < 60.$$

   Sketch a plot of the pdf. Describe roughly what this means in context.

4. Use the pdf to compute $\mathrm{P}(W < 15)$. Set up an integral, but sketch a picture and use geometry to compute.
5. Use the pdf to compute $\mathrm{P}(W > 45)$. Set up an integral, but sketch a picture and use geometry to compute.

Solution

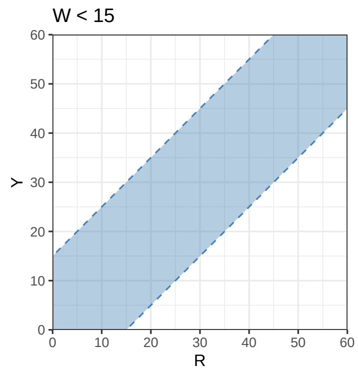


Figure 29.4

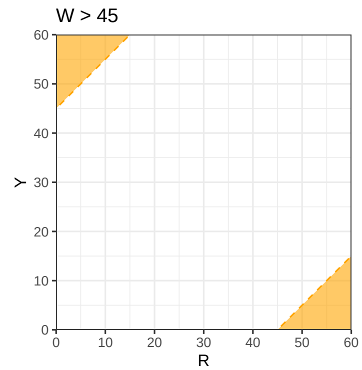


Figure 29.5

1. See blue shaded region above and use geometry to compute the area of the blue shaded region (it's easier to compute the area of the unshaded region first) divided by the total area of the square, which yields $\mathrm{P}(W < 15) = 0.4375$. If they meet like this over many days, on about 44% of days the first person will wait less than 15 minutes for the second person to arrive.
2. See orange shaded region above and use geometry to compute the area of the orange shaded region divided by the total area of the square, which yields $\mathrm{P}(W > 45) = 0.0625$. If they meet like this over many days, on about 6% of days the first person will wait more than 45 minutes for the second person to arrive.
3. See plot below. The density is highest at $w = 0$ and decreases linearly to 0 at $w = 60$. Waiting time is most likely to be close to 0 minutes, and likelihood of waiting close to w minutes decreases as w increases.
4. Integrate the pdf over $(0, 15)$; see the shaded blue area below. This gives an area of a trapezoid; the complementary unshaded area is a triangle with a base of (60 - 15), a height of $f(15) = (2/3600)(60 - 15)$, so the probability we want is $\mathrm{P}(W < 15) = 1 - (1/2)(60 - 15)(2/3600)(60 - 15) = 1 - (45/60)^2 = 0.4375$. As an integral

$$\mathrm{P}(W < 15) = \int_0^{15} (2/3600)(60 - w)dw = 0.4375$$

5. Integrate the pdf over $(45, 60)$; see the shaded orange area below. This gives an area of a triangle with a base of (60 - 45), a height of $f(45) = (2/3600)(60-45)$, so the probability we want is $\mathrm{P}(T < 45) = (1/2)(60 - 45)(2/3600)(60 - 45) = (15/60)^2 = 0.0625$. As an integral

$$\mathrm{P}(W < 45) = \int_{45}^{60} (2/3600)(60 - w)dw = 0.0625$$

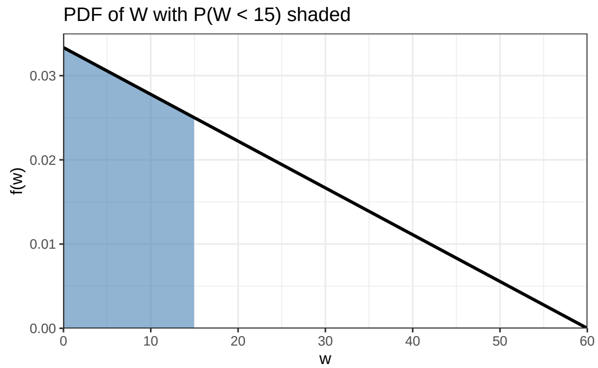


Figure 29.6

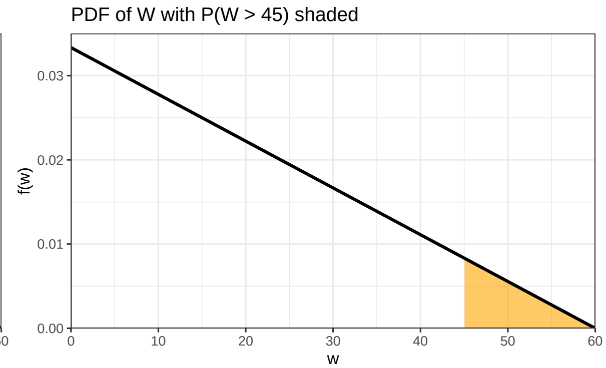


Figure 29.7

Problem 4

Continuing with the setup of Problem 3. Consider $T = \min(R, Y)$ is the time (minutes after noon) at which the first person arrives.

1. Is T the same random variable as W from Problem 3? Explain.
2. Compute and interpret $\mathrm{P}(T < 15)$. (Hint: we computed something similar in Example 2.6.)
3. Compute and interpret $\mathrm{P}(T > 45)$. (Hint: use a method similar to the previous part.)
4. It can be shown that T has pdf

$$f_T(t) = (2/3600)(60 - t), \qquad 0 < t < 60.$$

   Does T have the same distribution as W from Problem 3? Explain what this means.

Solution

1. No. T and W are different random variables. They are measuring different things. For example, if Regina arrives at time 10 and Cady arrives and time 42, then the first arrival time T is 10 and the waiting time W is 32.
2. See blue shaded region above and use geometry to compute the area of the blue shaded region (it's easier to compute the area of the unshaded region first) divided by the total area of the square, which yields $\mathrm{P}(T < 15) = 0.4375$. If they meet like this over many days, on about 44% of days the first person will arrive at most 15 minutes after noon.

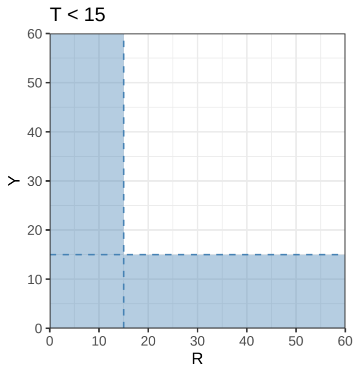


Figure 29.8

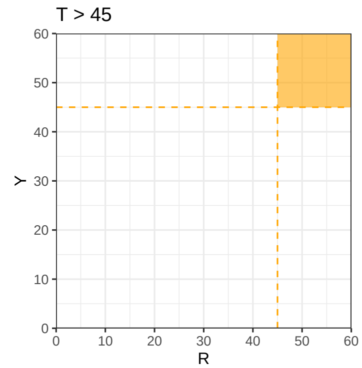


Figure 29.9

3. See orange shaded region above and use geometry to compute the area of the orange shaded region divided by the total area of the square, which yields $\mathrm{P}(T > 45) = 0.0625$. If they meet like this over many days, on about 6% of days the first person will arrive more than 45 minutes after noon.
4. Yes, T and W have the same distribution; they have the same pattern of variability over many days. Over many reps, 43.75% of reps have $W < 15$ and 43.75% of reps have $T < 15$ (though they're not the same reps). And so on. You could use the same spinner to simulates values of W as you could T.

Homework 5 solutions

Problem 1

In a certain region, times (minutes) between occurrences of earthquakes (of any magnitude) have a distribution with percentiles displayed in the table below.

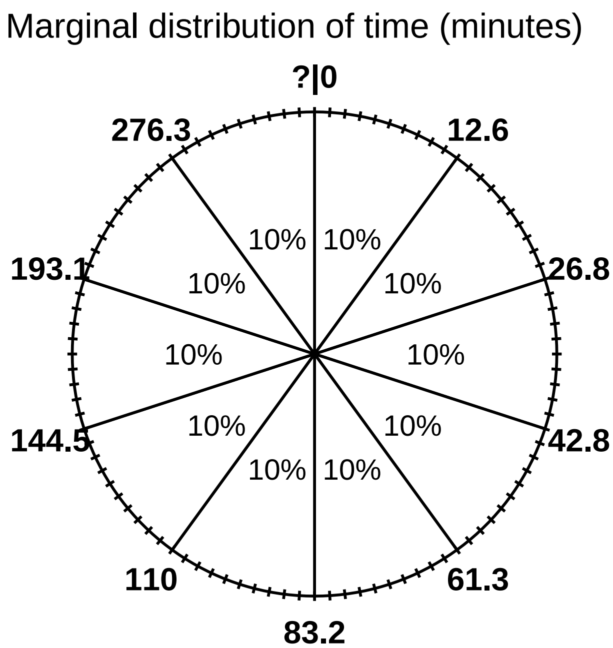
1. Construct a spinner corresponding to this distribution.
2. What percent of times are between 26.8 and 110.0 minutes?
3. Let F be the cdf. Evaluate and interpret $F(42.8)$.
4. Let Q be the quantile function. Evaluate and interpret $Q(0.9)$.
5. Sketch (by hand) a histogram of this distribution.

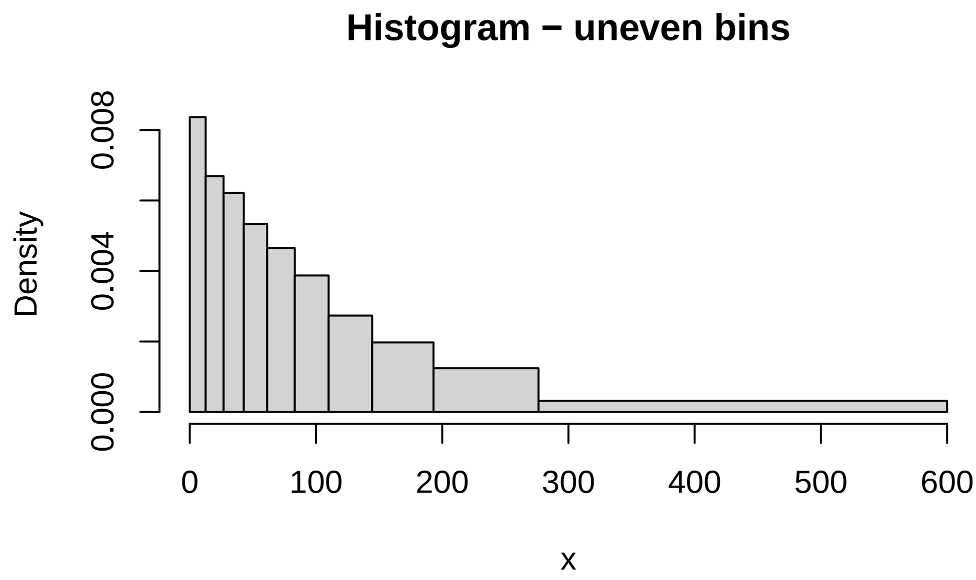
Solution

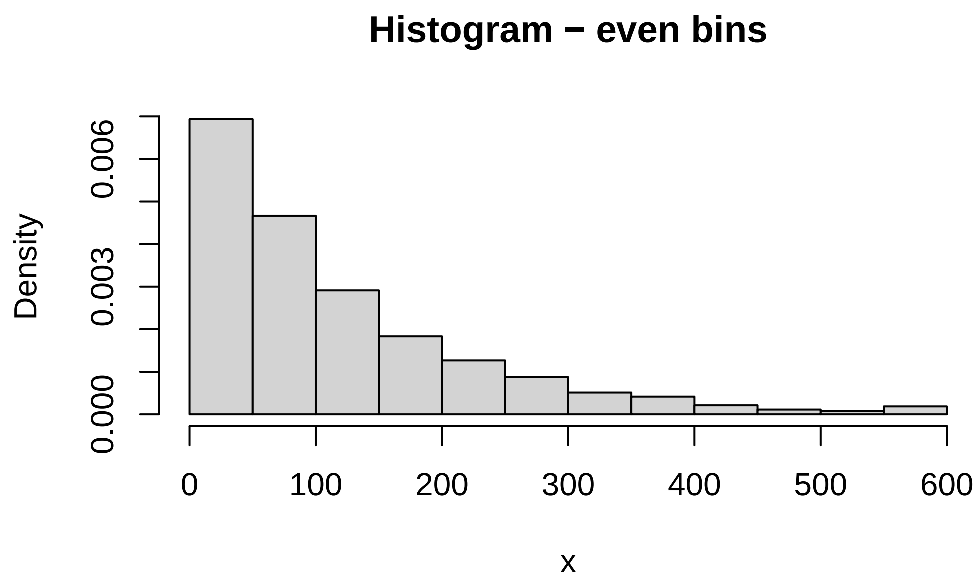
1. See below. The 50th percentile goes at "6 o'clock" (50% of the way around), etc.
2. 60% are below 110.0 and 20% are below 26.8, so 40% are between 26.8 and 110.0 minutes.
3. $F(42.8) = \mathrm{P}(X \leq 42.8) = 0.3$, since 42.8 minutes is the 30th percentile.
4. $Q(0.9) = 276.3$, since 276.3 minutes is the 90th percentile.
5. We always end with a histogram with even bins, but sometimes it helps to start sketching a histogram with unever bins. We know that 10% of the values are between 0 and 12.6, 10% are between 12.6 and 26.8, 10% are between 26.8 and 42.8, etc. So if we split the bins of the histogram based on these percentiles, each bin would correspond to a relative frequency of 10%. HOWEVER, AREA in a histogram represents relative frequency. So

Percentile	Value (minutes)
10th	12.6
20th	26.8
30th	42.8
40th	61.3
50th	83.2
60th	110.0
70th	144.5
80th	193.1
90th	276.3

each of the bins would have the same area, but they are unequally spaced, so they would not have the same height. For example, the first bin has area of 10% over a base of (0, 12.6), while the 80-90 percentile bin has area 10% spread over the interval (193.1, 276.3), so the bar for (193.1, 276.3) will have a smaller height than the bar for (0, 12.6). Sketching a histogram with these unevenly spaced bins first gives an idea of the shape; then you can sketch a histogram with a similar shape but equally spaced bins. We can see that density is highest near 0 and decreases as x increases; notice the stretching and shrinking on the spinner. (The highest percentile is the 90th, so we don't know what the upper bound is that almost all values fall below.)




Histogram − uneven bins
Density
0.000
0.004
0.008
0
100
200
300
400
500
600
x


Histogram − even bins
Density
0.000
0.003
0.006
0
100
200
300
400
500
600
x

Problem 2

Recall Example 2.6. Regina and Cady are meeting for lunch. Suppose they each arrive uniformly at random at a time between noon and 1:00, independently of each other. Record their arrival times as minutes after noon, so noon corresponds to 0 and 1:00 to 60. Recall the sample space from Example 2.2 and assume a uniform probability measure P.

Let R be Regina's arrival time and Y Cady's. Then $W = |R - Y|$ is the amount of time (minutes) the first person to arrive waits for the second person to arrive. It can be shown that W has pdf

$$f_W(w) = (2/3600)(60 - w), \qquad 0 < w < 60.$$

1. Let F be the cdf. Evaluate and interpret $F(15)$.
2. Find an expression for the cdf of W. Set up an integral, but sketch a picture and use geometry to compute.
3. Find and interpret the 25th percentile of W. (You can do the next part first if you want, but it might help to start with a specific number like in this part.)
4. Find the quantile function of W.
5. Sketch a spinner corresponding to the distribution of W. Label the 25th, 50th, and 75th percentiles.

Solution

1. $F(15) = \mathrm{P}(W \le 15)$, which we computed on a previous HW to be $1-(1-15/60)^2 = 0.4375$. If they meet like this over many days, on about 44% of days the first person needs to wait at most 15 minutes for the second person to arrive.
2. Similar to computing $F_W(15) = \mathrm{P}(W \le 15)$ from an earlier problem, but replace 15 with a generic w (and remember to change the dummy variable of integration). For $0 < w < 60$,

   $$F_W(w) = \mathrm{P}(W \le w) = \int_0^w (2/3600)(60 - u)du = 1 - (1 - w/60)^2$$

   Technically the cdf is defined for any value of w, so it's really

   $$F_W(w) = \begin{cases} 0, & w \le 0 \\ 1 - (1 - w/60)^2, & 0 < w < 60 \\ 1, & w \ge 60 \end{cases}$$

3. The 25th percentile satisfies

   $$0.25 = \mathrm{P}(W \le w) = F_W(w) = 1 - (1 - w/60)^2$$

   Solve to get $w = 60(1-\sqrt{1 - 0.25}) = 8.03$. The waiting time is at most (about) 8 minutes on 25% of days.

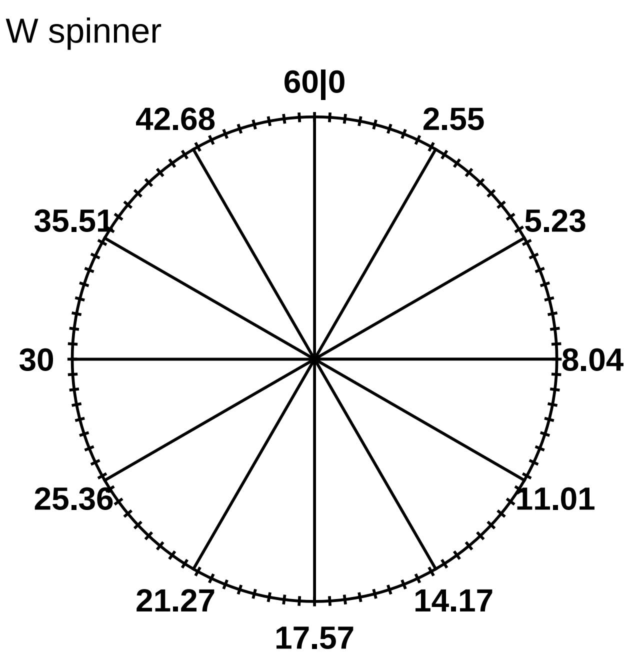
4. To find the p percentile, use the same process as in in the previous part.

$$p = \mathrm{P}(W \leq w) = F_W(w) = 1 - (1 - w/60)^2$$

Solve to get $w = 60(1 - \sqrt{1-p})$. So the quantile function is $Q_W(p) = 60(1 - \sqrt{1-p}), 0 \leq p \leq 1$.

5. See spinner below. We can use the quantile function to label the spinner.

   - The 25th percentile of $Q_W(0.25) = 60(1 - \sqrt{1 - 0.25}) = 8.02$ minutes would go at the "3 o'clock" point.
   - The 50th percentile of $Q_W(0.5) = 60(1 - \sqrt{1 - 0.5}) = 17.57$ minutes would go at the "6 o'clock" point.
   - The 75th percentile of $Q_W(0.75) = 60(1 - \sqrt{1 - 0.75}) = 30$ minutes would go at the "9 o'clock" point.
   - Notice that the values around the spinner are unequally spaced. The density of W is higher near 0 so intervals near 0 are stretched out; the density of W is lower near 60 minutes so intervals near 60 are shrunk.



Problem 3

Continuing Problem 2. For this problem you should set up integrals when appropriate but you can use software to compute them.

1. Explain in full detail how you could use simulation to approximate
    a. $\mathrm{E}(W)$
    b. $\mathrm{E}(W^2)$
    c. $\mathrm{Var}(W)$ (explain how you can do it directly without using the previous part.)
2. Compute and interpret $\mathrm{E}(W)$.
3. Compute $\mathrm{E}(W^2)$.
4. Compute $\mathrm{Var}(W)$
5. Compute and interpret $\mathrm{SD}(W)$.
6. Compute the probability that W is more than 1 SD away from its mean.

Solution

1. Simulate many values of W (e.g., using the spinner from the previous part).
    a. Average the values of W to approximate $E(W)$.
    b. Square each simulated value of W and average the W^2 values to approximate $E(W^2)$.
    c. For each simulated value of W, subtract the average of W and square to get the squared deviation and average the squared deviations to approximate $\mathrm{Var}(W)$.
2. Use the definition of EV for continuous RV
$$\mathrm{E}(W) = \int_0^{60} w f_W(w) dw = \int_0^1 w\left(2/3600(60-w)\right) dw = 20$$
If they were to meet like this over many days, the long run average time the first person has to wait for the second person to arrive is 20 minutes.
3. Use LOTUS
$$\mathrm{E}(W^2) = \int_0^{60} w^2 f_W(w) dw = \int_0^{60} w^2\left(2/3600(60-w)\right) dw = 600$$
4. We did all the work in the previous parts. $\mathrm{Var}(W) = \mathrm{E}(W^2) - \mathrm{E}(W)^2 = 600 - (20)^2 = 200$ square-minutes.
5. $\mathrm{SD}(W) = 14.1$ minutes
6. $\mathrm{P}(|W - \mathrm{E}(W)| > \mathrm{SD}(W)) = \mathrm{P}(W > 20 + 14.1) + \mathrm{P}(W < 20 - 14.1)$. In this case you can find the corresponding probability/area in this case with geometry. Also, integrating gives
$$\int_0^{20-14.1} 2/3600(60-w)dw + \int_{20+14.1}^1 2/3600(60-w)dw = 0.37$$

Problem 4

Lifetimes of a certain type of component are Exponentially distributed with mean 20 thousand hours.

1. Approximate the probability that the lifetime, rounded to the nearest thousand hours, of a randomly selected component is 30 thousand hours.
2. Compute the probability that a randomly selected component has a lifetime greater than 30 thousand hours.
3. Compute the standard deviation of component lifetimes.
4. Suppose a device contains 2 such components connected in series, so that the system functions only if both of the components are functioning. Suppose the component lifetimes are independent of each other. Compute the probability that the *device* has a lifetime greater than 30 thousand hours.
5. Let T be the lifetime of the *device*. Compute $\mathrm{P}(T > t)$.
6. Identify the distribution of T by name, including the values of any relevant parameters.

Solution

For a single component the pdf of lifetime X (thousand hours) is $f(x) =(1/20)e^{-x/20}, x>0 $

1. The probability that the lifetime, rounded to the nearest thousand hours, of a randomly selected component is 30 thousand hours is approximately $f(30)(1)$; the (1) is because we're rounding to the nearest 1 thousand hours. $f(30)(1) = (1/20)e^{-30/20}(1) = 0.011$.
2. For Exponential $P(X > 30) = e^{-30/20} = 0.223$.
3. For Exponential the SD and the EV are the same, 20 thousand hours.
4. Let X_1, X_2 be the lifetime of the two components. The device functions only in both components function

$$
\begin{aligned}
P(T > 30) &= P(X_1 > 30, X_2 > 30) \\
&= P(X_1 > 30)P(X_2 > 30) && \text{independence} \\
&= e^{-30/20}e^{-30/20} && \text{Exponential} \\
&= e^{-(2/20)30} = 0.05
\end{aligned}
$$

5. Same as the previous calculation with 30 replaced by $t > 0$.

$$
\begin{aligned}
P(T > t) &= P(X_1 > t, X_2 > t) \\
&= P(X_1 > t)P(X_2 > t) && \text{independence} \\
&= e^{-t/20}e^{-t/20} && \text{Exponential} \\
&= e^{-(2/20)t}
\end{aligned}
$$

6. The previous calculation shows that the cdf of T follows an Exponential form, so T has an Exponential distribution with rate parameter 2/20=0.1 and mean 10 thousand hours.

Summary of Some Common Distributions

Name and parameters	Possible values	pmf/pdf	cdf	Mean	SD
Discrete					
Poisson(μ)	$x = 0, 1, 2, \ldots$	$e^{-\mu} \frac{\mu^x}{x!}$	No closed form	μ	$\sqrt{\mu}$
Binomial(n, p)	$x = 0, 1, 2, \ldots, n$	$\binom{n}{x} p^x (1-p)^{n-x}$	No closed form	np	$\sqrt{np(1-p)}$
Geometric(p)	$x = 1, 2, 3, \ldots$	$p(1-p)^{x-1}$	$1 - (1-p)^x$	$\frac{1}{p}$	$\sqrt{\frac{1-p}{p^2}}$
Continuous					
Uniform(a, b)	$a < x < b$	$\frac{1}{b-a}$	$\frac{x-a}{b-a}$	$\frac{a+b}{2}$	$\frac{b-a}{\sqrt{12}}$
Exponential(λ)	$x > 0$	$\lambda e^{-\lambda x}$	$1 - e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda}$
Normal(μ, σ)	$-\infty < x < \infty$	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$	No closed form (use tables)	μ	σ

References